

# CHEM 356: Quantum Chemistry

## Student Course Notes

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# Contents

|                                                                        |           |
|------------------------------------------------------------------------|-----------|
| <b>Document Key</b>                                                    | <b>13</b> |
| <b>Changelog</b>                                                       | <b>13</b> |
| <b>1 What is Quantum Mechanics?</b>                                    | <b>14</b> |
| 1.1 Introduction . . . . .                                             | 14        |
| 1.2 Schrödinger Equation . . . . .                                     | 14        |
| 1.2.1 Electron Models . . . . .                                        | 14        |
| 1.3 QM Software . . . . .                                              | 15        |
| <b>2 Waves and Distributions</b>                                       | <b>16</b> |
| 2.1 Physical Matter . . . . .                                          | 16        |
| 2.2 Light . . . . .                                                    | 17        |
| 2.3 Electrons . . . . .                                                | 18        |
| 2.4 SQUID . . . . .                                                    | 19        |
| <b>3 A Timeline of Physics</b>                                         | <b>21</b> |
| 3.1 Puzzlers . . . . .                                                 | 21        |
| 3.2 A Very Brief Note on Relativity . . . . .                          | 23        |
| 3.2.1 Ball . . . . .                                                   | 23        |
| 3.2.2 Light . . . . .                                                  | 23        |
| 3.2.3 Special Relativity . . . . .                                     | 24        |
| <b>4 Blackbody Radiation and First Evidence of Energy Quantization</b> | <b>25</b> |
| 4.1 Blackbody Radiation . . . . .                                      | 25        |
| 4.2 Planck . . . . .                                                   | 26        |
| 4.3 Einstein . . . . .                                                 | 27        |
| <b>5 Emission Spectra of Atoms</b>                                     | <b>30</b> |
| 5.1 Spectral Series of Hydrogen . . . . .                              | 30        |
| 5.2 Bohr . . . . .                                                     | 31        |
| 5.2.1 How Bohr Solved the Problem . . . . .                            | 31        |
| <b>6 Light and Electron Waves and Differential Equations</b>           | <b>32</b> |

|           |                                                                         |           |
|-----------|-------------------------------------------------------------------------|-----------|
| 6.1       | De Broglie . . . . .                                                    | 32        |
| 6.2       | Thermal Electrons and Electron Microscopy . . . . .                     | 33        |
| 6.3       | Plane Wave . . . . .                                                    | 33        |
| 6.4       | Introduction to Differential Equations . . . . .                        | 34        |
| <b>7</b>  | <b>Analyzing a General Time-Dependent Wave Equation</b>                 | <b>38</b> |
| 7.1       | Setup . . . . .                                                         | 38        |
| 7.2       | Boundary Conditions . . . . .                                           | 39        |
| 7.3       | Rewriting . . . . .                                                     | 39        |
| 7.3.1     | Consequences . . . . .                                                  | 40        |
| 7.4       | Integrating $X(x)$ . . . . .                                            | 40        |
| 7.4.1     | Initial Values for $X(x)$ . . . . .                                     | 41        |
| 7.4.1.1   | $X(0) = 0$ . . . . .                                                    | 41        |
| 7.4.1.2   | $X(a) = 0$ . . . . .                                                    | 41        |
| 7.5       | Cases of $n$ . . . . .                                                  | 42        |
| 7.6       | Integrating $T(t)$ . . . . .                                            | 44        |
| 7.7       | Combining Solutions . . . . .                                           | 44        |
| <b>8</b>  | <b>Introduction to Statistics</b>                                       | <b>46</b> |
| 8.1       | Discrete Example: Die Throw . . . . .                                   | 46        |
| 8.1.1     | Fraction . . . . .                                                      | 47        |
| 8.1.2     | Probability . . . . .                                                   | 47        |
| 8.1.3     | Average . . . . .                                                       | 47        |
| 8.1.4     | Variance/Standard Deviation . . . . .                                   | 48        |
| 8.1.4.1   | $\sigma^2 = 0$ . . . . .                                                | 48        |
| 8.1.4.2   | Rewriting Variance . . . . .                                            | 49        |
| 8.1.4.2.1 | Die Example . . . . .                                                   | 50        |
| 8.2       | Continuous Example: Mass Density . . . . .                              | 50        |
| 8.2.1     | Moments of the Distribution . . . . .                                   | 50        |
| 8.2.2     | Example . . . . .                                                       | 51        |
| 8.3       | Discrete vs. Continuous . . . . .                                       | 51        |
| <b>9</b>  | <b>A Plausible Argument to Derive the Schrödinger Equation</b>          | <b>52</b> |
| 9.1       | Use of de Broglie and Classical Wave Equation . . . . .                 | 52        |
| 9.2       | Derivation of Expression for Momentum From Wave Equation and de Broglie | 53        |
| 9.3       | From Classical Mechanics to the Schrödinger Equation . . . . .          | 54        |
| 9.4       | Operators . . . . .                                                     | 55        |

|           |                                                                        |           |
|-----------|------------------------------------------------------------------------|-----------|
| 9.4.1     | Previous Operators . . . . .                                           | 55        |
| 9.4.2     | Operator Examples . . . . .                                            | 55        |
| 9.4.3     | Linear Operators . . . . .                                             | 57        |
| 9.4.4     | Commutator . . . . .                                                   | 58        |
| 9.4.4.1   | Commutator Example . . . . .                                           | 59        |
| 9.5       | Eigenvalue Equations . . . . .                                         | 59        |
| 9.6       | Example . . . . .                                                      | 60        |
| 9.6.1     | Counterexample . . . . .                                               | 62        |
| 9.6.2     | Kinetic Energy Operator . . . . .                                      | 62        |
| <b>10</b> | <b>Particle in a Box</b>                                               | <b>64</b> |
| 10.1      | Operators . . . . .                                                    | 67        |
| 10.2      | Boundary Conditions . . . . .                                          | 67        |
| 10.3      | Eigenfunction or Not? . . . . .                                        | 68        |
| 10.4      | Further Discussion: Discrete Energy Levels . . . . .                   | 68        |
| <b>11</b> | <b>Wavefunctions as Probability Distributions</b>                      | <b>70</b> |
| 11.1      | Probability Distributions . . . . .                                    | 70        |
| 11.1.1    | Wave Functions as Probability Distributions . . . . .                  | 70        |
| 11.1.2    | Constants of Normalization . . . . .                                   | 71        |
| 11.1.3    | Example: Normalizing the Solutions for a Particle in the Box . . . . . | 73        |
| 11.2      | Definition of Mean Values or <i>Expectation Values</i> . . . . .       | 75        |
| 11.2.1    | Special Example of an Expectation Value of an Eigenfunction . . . . .  | 76        |
| <b>12</b> | <b>2D Particle in a Box and Degeneracies</b>                           | <b>78</b> |
| 12.1      | 2D Particle in a Box . . . . .                                         | 78        |
| 12.1.1    | Setup . . . . .                                                        | 78        |
| 12.1.2    | Separation of Variables . . . . .                                      | 80        |
| 12.1.3    | Boundary Conditions . . . . .                                          | 81        |
| 12.1.4    | Solving . . . . .                                                      | 81        |
| 12.1.5    | Connecting Together . . . . .                                          | 81        |
| 12.2      | Degeneracies . . . . .                                                 | 82        |
| 12.2.1    | Particle in a 3D Box . . . . .                                         | 83        |
| <b>13</b> | <b>Particle on a Ring</b>                                              | <b>85</b> |
| 13.1      | Setup . . . . .                                                        | 85        |
| 13.2      | Kinetic Energy (with respect to) Angular Momentum . . . . .            | 86        |



|            |                                                                                                |            |
|------------|------------------------------------------------------------------------------------------------|------------|
| 13.3       | Periodicity . . . . .                                                                          | 87         |
| 13.4       | Solving . . . . .                                                                              | 87         |
| 13.4.1     | Generalization: Degenerate Eigenstates of the Schrödinger Equation<br>are Not Unique . . . . . | 90         |
| 13.5       | Normalization . . . . .                                                                        | 91         |
| 13.6       | Summary of Particle on a Ring . . . . .                                                        | 91         |
| <b>14</b>  | <b>Postulates of Quantum Mechanics and Hermitian Operators</b>                                 | <b>92</b>  |
| 14.1       | Postulates of Quantum Mechanics . . . . .                                                      | 92         |
| 14.2       | Measurement . . . . .                                                                          | 93         |
| 14.3       | Hermiticity . . . . .                                                                          | 94         |
| 14.3.1     | Useful properties of Hermitian Operators . . . . .                                             | 94         |
| 14.3.1.1   | 1 <sup>st</sup> Condition . . . . .                                                            | 94         |
| 14.3.1.2   | 2 <sup>nd</sup> Condition . . . . .                                                            | 95         |
| 14.3.1.2.1 | Orthogonalization . . . . .                                                                    | 95         |
| 14.3.1.2.2 | Gram-Schmidt . . . . .                                                                         | 96         |
| 14.3.1.3   | 3 <sup>rd</sup> Condition . . . . .                                                            | 97         |
| 14.3.2     | Formal Definition of Hermiticity . . . . .                                                     | 98         |
| 14.3.3     | Examples of Hermitian Operators . . . . .                                                      | 98         |
| 14.3.3.1   | Proofs of Hermiticity . . . . .                                                                | 99         |
| 14.3.3.1.1 | Potential Energy . . . . .                                                                     | 99         |
| 14.3.3.1.2 | Momentum . . . . .                                                                             | 99         |
| <b>15</b>  | <b>Proving Hermitian Properties</b>                                                            | <b>101</b> |
| 15.1       | Summary of Hermitian Properties . . . . .                                                      | 101        |
| 15.2       | Proof and Discussion of Hermitian Properties . . . . .                                         | 102        |
| 15.2.1     | Real Eigenvalues . . . . .                                                                     | 102        |
| 15.2.2     | Orthonormal Eigenfunctions . . . . .                                                           | 103        |
| 15.2.2.1   | Normality . . . . .                                                                            | 103        |
| 15.2.2.2   | Orthogonality . . . . .                                                                        | 103        |
| 15.2.3     | Degeneracy . . . . .                                                                           | 103        |
| 15.2.3.1   | Orthonormality . . . . .                                                                       | 104        |
| 15.2.3.1.1 | Normality . . . . .                                                                            | 104        |
| 15.2.3.1.2 | Orthogonality . . . . .                                                                        | 104        |
| 15.2.4     | Orthonormal Set of Eigenfunctions . . . . .                                                    | 106        |
| 15.2.5     | Completeness . . . . .                                                                         | 106        |

|            |                                                                           |            |
|------------|---------------------------------------------------------------------------|------------|
| 15.2.5.1   | Normalization . . . . .                                                   | 107        |
| 15.3       | Probability . . . . .                                                     | 108        |
| 15.3.1     | Calculation of Moments . . . . .                                          | 108        |
| 15.3.2     | Interpretation . . . . .                                                  | 108        |
| <b>16</b>  | <b>Probability and Measurement</b>                                        | <b>109</b> |
| 16.1       | Proving Moments . . . . .                                                 | 109        |
| 16.2       | Measurement . . . . .                                                     | 110        |
| 16.2.1     | Controlling $\psi$ . . . . .                                              | 110        |
| 16.2.2     | Measuring Consecutively . . . . .                                         | 112        |
| <b>17</b>  | <b>Simultaneous Measurements, Heisenberg Uncertainty, and Dirac Delta</b> | <b>113</b> |
| 17.1       | Simultaneous . . . . .                                                    | 113        |
| 17.2       | Example: Particle on a Ring . . . . .                                     | 114        |
| 17.3       | General Proof . . . . .                                                   | 114        |
| 17.4       | Sharp Values for $\hat{A}$ and $\hat{B}$ . . . . .                        | 115        |
| 17.5       | What if they don't commute? . . . . .                                     | 116        |
| 17.6       | Heisenberg Uncertainty . . . . .                                          | 117        |
| 17.7       | Continuous Spectra . . . . .                                              | 117        |
| 17.7.1     | Another Continuous Spectrum: Operator $\hat{x}$ . . . . .                 | 118        |
| 17.7.1.1   | Dirac Delta function . . . . .                                            | 119        |
| 17.7.1.1.1 | Probabilities from Dirac Delta . . . . .                                  | 120        |
| <b>18</b>  | <b>Time Dependent Schrödinger Equation</b>                                | <b>121</b> |
| 18.1       | Time Dependent Schrödinger Equation . . . . .                             | 121        |
| 18.2       | Special Case: $\hat{H}$ is Time Independent . . . . .                     | 121        |
| 18.2.1     | Special Solutions . . . . .                                               | 122        |
| 18.2.1.1   | Discussion of Special Solutions . . . . .                                 | 123        |
| 18.2.1.1.1 | Time Independent Operators . . . . .                                      | 125        |
| 18.3       | General Solution . . . . .                                                | 125        |
| <b>19</b>  | <b>Spin and Quantum Entanglement</b>                                      | <b>129</b> |
| 19.1       | Measuring Spin . . . . .                                                  | 129        |
| 19.2       | Lunch Box Analogy . . . . .                                               | 130        |
| 19.3       | Spin . . . . .                                                            | 131        |
| 19.4       | Quantum Entanglement . . . . .                                            | 133        |
| 19.5       | Schrödinger's Cat . . . . .                                               | 134        |

|                                                                                                    |            |
|----------------------------------------------------------------------------------------------------|------------|
| <b>20 Hamiltonian of a Molecule, Born-Oppenheimer Approximation, and Potential Energy Surfaces</b> | <b>136</b> |
| 20.1 Kinetic Energy . . . . .                                                                      | 136        |
| 20.1.1 Multiple Atoms/Electrons . . . . .                                                          | 137        |
| 20.2 Potential . . . . .                                                                           | 137        |
| 20.2.1 Nuclear–Nuclear . . . . .                                                                   | 137        |
| 20.3 Combining Terms . . . . .                                                                     | 138        |
| 20.4 Born-Oppenheimer Approximation . . . . .                                                      | 139        |
| 20.5 Potential Energy Surfaces . . . . .                                                           | 140        |
| 20.5.1 More Approximations . . . . .                                                               | 141        |
| <b>21 Discussion of Vibrational Motion Using the Harmonic Oscillator Model</b>                     | <b>143</b> |
| 21.1 Summary . . . . .                                                                             | 143        |
| 21.2 Harmonic Oscillator Model for Diatomic molecules . . . . .                                    | 143        |
| 21.2.1 Nondimensionalizing and Standard Form of the Harmonic Oscillator . . . . .                  | 147        |
| 21.3 Ladder Operators . . . . .                                                                    | 150        |
| 21.3.1 Some Identities . . . . .                                                                   | 152        |
| 21.3.2 Use of the Ladder Operators to Construct New Eigenfunctions . . . . .                       | 152        |
| 21.3.2.1 Proving Ladder Operators Relations . . . . .                                              | 154        |
| <b>22 2-Body Problem and Spherical Coordinates</b>                                                 | <b>157</b> |
| 22.1 2-Body Problem . . . . .                                                                      | 157        |
| 22.2 Classical Energy . . . . .                                                                    | 158        |
| 22.2.1 Energy and Momentum . . . . .                                                               | 158        |
| 22.2.2 Defining Relative/Simpler Variables . . . . .                                               | 158        |
| 22.2.3 Rewriting . . . . .                                                                         | 159        |
| 22.3 Hamiltonian and Schrödinger Equation . . . . .                                                | 160        |
| 22.3.1 Translational Energy . . . . .                                                              | 160        |
| 22.3.2 Relative Motion . . . . .                                                                   | 161        |
| 22.4 Spherical Coordinates . . . . .                                                               | 161        |
| <b>23 Angular Hamiltonian and Spherical Harmonics</b>                                              | <b>166</b> |
| 23.1 Converting Hamiltonian . . . . .                                                              | 166        |
| 23.1.1 Chain Rule Sketch . . . . .                                                                 | 166        |
| 23.1.2 Angular Momentum Operator . . . . .                                                         | 167        |
| 23.2 Angular Solutions . . . . .                                                                   | 167        |
| 23.3 Exact Solution . . . . .                                                                      | 168        |

|           |                                                                                           |            |
|-----------|-------------------------------------------------------------------------------------------|------------|
| 23.4      | Harmonic Oscillator and Rigid Rotor . . . . .                                             | 169        |
| 23.4.1    | $r$ to $x$ . . . . .                                                                      | 169        |
| 23.4.2    | $l$ to $J$ . . . . .                                                                      | 170        |
| 23.4.3    | Final Rovibrational Energies . . . . .                                                    | 171        |
| <b>24</b> | <b>Rovibrational Energy</b>                                                               | <b>172</b> |
| 24.1      | Overview . . . . .                                                                        | 172        |
| 24.2      | Solving for Energy . . . . .                                                              | 173        |
| 24.3      | Possible Transitions . . . . .                                                            | 173        |
| 24.3.1    | Pure Rotational . . . . .                                                                 | 174        |
| 24.3.2    | Rovibrational . . . . .                                                                   | 175        |
| 24.3.2.1  | Not Assuming Constants are the Same . . . . .                                             | 176        |
| 24.3.3    | Low Temperatures . . . . .                                                                | 176        |
| <b>25</b> | <b>Rotational, Vibrational, Translational Degrees of Freedom for Polyatomic Molecules</b> | <b>178</b> |
| 25.1      | Overview . . . . .                                                                        | 178        |
| 25.2      | Energy Types and Degrees of Freedom . . . . .                                             | 179        |
| 25.2.1    | Discussion on IR/Raman Spectroscopy . . . . .                                             | 181        |
| 25.3      | Quantum Chemistry Calculations and Nuclear Motion . . . . .                               | 181        |
| 25.3.1    | Vibrations . . . . .                                                                      | 183        |
| 25.3.2    | Vibrational Hamiltonian . . . . .                                                         | 184        |
| <b>26</b> | <b>Angular Momentum Theory</b>                                                            | <b>186</b> |
| 26.1      | The Purpose of This Theory . . . . .                                                      | 186        |
| 26.2      | Useful Preliminary Definitions . . . . .                                                  | 186        |
| 26.2.1    | Useful Commutation Relations Between Position and Momentum Operators in 3D . . . . .      | 187        |
| 26.2.2    | Results That Can Be Derived From This . . . . .                                           | 188        |
| 26.3      | Interlude: Useful Identities with Commutators) . . . . .                                  | 189        |
| 26.3.1    | Swapped Order . . . . .                                                                   | 189        |
| 26.3.1.1  | Proof . . . . .                                                                           | 189        |
| 26.3.2    | Constant: Scalar That Can Be Pulled Out . . . . .                                         | 189        |
| 26.3.2.1  | Proof . . . . .                                                                           | 190        |
| 26.3.3    | Operator Commutation . . . . .                                                            | 190        |
| 26.3.4    | Simplification of Commutators: Products of Operators . . . . .                            | 190        |
| 26.3.4.1  | Proof . . . . .                                                                           | 191        |

|           |                                                                            |            |
|-----------|----------------------------------------------------------------------------|------------|
| 26.3.5    | Simplification of Commutators: Sums of Operators . . . . .                 | 191        |
| 26.3.5.1  | Proof . . . . .                                                            | 191        |
| 26.4      | Derivations of Angular Momentum Relations . . . . .                        | 192        |
| 26.4.1    | Directional Angular Momentum . . . . .                                     | 192        |
| 26.4.2    | Commutation of $\hat{L}^2$ with Directional Angular Momentum . . . . .     | 192        |
| 26.4.3    | $\hat{L}^2$ in Terms of Ladder Operators . . . . .                         | 193        |
| 26.4.4    | Commutation of $\hat{L}_z$ with Ladder Operators . . . . .                 | 194        |
| 26.4.5    | Commutation of $\hat{L}^2$ with Ladder Operators . . . . .                 | 194        |
| 26.5      | Crux . . . . .                                                             | 194        |
| 26.5.1    | Key Result $a \geq b^2$ . . . . .                                          | 195        |
| 26.5.2    | Crux: Using Ladder Operators to Create Other Eigenfunctions . . . . .      | 196        |
| <b>27</b> | <b>Understanding Angular Momentum and Orbitals</b>                         | <b>200</b> |
| 27.1      | Quick Review . . . . .                                                     | 200        |
| 27.1.1    | Useful Forms . . . . .                                                     | 200        |
| 27.1.2    | Laddering . . . . .                                                        | 200        |
| 27.1.3    | Key Result . . . . .                                                       | 201        |
| 27.2      | Bounds . . . . .                                                           | 201        |
| 27.2.1    | Upper Bound . . . . .                                                      | 201        |
| 27.2.2    | Lower Bound . . . . .                                                      | 202        |
| 27.2.3    | Connecting the bounds . . . . .                                            | 203        |
| 27.3      | Summary of angular momentum analysis . . . . .                             | 203        |
| 27.3.1    | Meaning of it All . . . . .                                                | 203        |
| 27.4      | Final Remarks on Formal Angular Momentum Theory . . . . .                  | 204        |
| 27.5      | Angular Momentum in Spherical Form . . . . .                               | 205        |
| 27.5.1    | Eigenfunctions of $\hat{L}_z$ , with respect to $\varphi$ . . . . .        | 206        |
| 27.5.2    | Difficulty to Represent Half-Integer Spin in Angular Coordinates . . . . . | 207        |
| 27.5.3    | Discussion of $\theta$ Form of the Spherical Harmonics . . . . .           | 208        |
| 27.5.3.1  | Ladder Operator Construction of Spherical Harmonics . . . . .              | 209        |
| 27.5.3.2  | Real Cartesian Orbitals . . . . .                                          | 210        |
| <b>28</b> | <b>Radial Equation</b>                                                     | <b>212</b> |
| 28.1      | Solving Radial Equations . . . . .                                         | 212        |
| 28.2      | Introduction . . . . .                                                     | 212        |
| 28.2.1    | Cleaning Up . . . . .                                                      | 213        |
| 28.2.2    | Trial Solution . . . . .                                                   | 213        |

|            |                                                            |            |
|------------|------------------------------------------------------------|------------|
| 28.2.3     | Substitute Trial solution into radial equation . . . . .   | 215        |
| 28.2.4     | Solutions . . . . .                                        | 215        |
| 28.2.4.1   | $s$ -Orbitals, Simplest Example . . . . .                  | 215        |
| 28.2.4.2   | $s$ -Orbitals, Another Example . . . . .                   | 216        |
| 28.2.4.3   | General Solution . . . . .                                 | 219        |
| 28.2.4.3.1 | Highest Power . . . . .                                    | 219        |
| 28.2.4.3.2 | $n$ , Principal Quantum Number . . . . .                   | 220        |
| 28.2.4.3.3 | Lowest Power . . . . .                                     | 220        |
| 28.2.4.4   | Interpretation . . . . .                                   | 220        |
| 28.2.5     | Table of Solutions . . . . .                               | 221        |
| 28.3       | Summary of Angular and Radial Solutions . . . . .          | 222        |
| 28.3.1     | Energy Degeneracy . . . . .                                | 222        |
| <b>29</b>  | <b>Variational Principle</b>                               | <b>224</b> |
| 29.1       | Exact Solutions . . . . .                                  | 224        |
| 29.2       | Variational Principle . . . . .                            | 224        |
| 29.2.1     | Unnormalized Function . . . . .                            | 228        |
| 29.2.2     | Examples . . . . .                                         | 229        |
| 29.2.2.1   | Unworked: Harmonic Oscillator . . . . .                    | 229        |
| 29.2.2.2   | Partially Worked: $1s$ Orbital . . . . .                   | 230        |
| 29.2.2.3   | Unworked: $1s$ Orbital (Non-Trivial) . . . . .             | 232        |
| <b>30</b>  | <b>Variational Principle: More Concrete Examples</b>       | <b>234</b> |
| 30.1       | Variational Principle . . . . .                            | 234        |
| 30.1.1     | Parameter . . . . .                                        | 234        |
| <b>31</b>  | <b>Linear Variational Principle</b>                        | <b>238</b> |
| 31.1       | The Essence . . . . .                                      | 238        |
| 31.2       | Deriving the Linear Form . . . . .                         | 238        |
| 31.2.1     | Derivative . . . . .                                       | 239        |
| 31.2.1.1   | Numerator . . . . .                                        | 240        |
| 31.2.1.2   | Denominator . . . . .                                      | 241        |
| 31.2.1.3   | Combination . . . . .                                      | 241        |
| 31.2.2     | Matrix Form . . . . .                                      | 241        |
| 31.2.2.1   | Deep Connection to Early Quantum Mechanics . . . . .       | 242        |
| 31.2.2.2   | $2 \times 2$ Problem: Discussion of General Case . . . . . | 243        |

|            |                                                                                                     |            |
|------------|-----------------------------------------------------------------------------------------------------|------------|
| 31.2.3     | Complicated Specific Example: Particle on a Ring with a Complex Perturbation . . . . .              | 246        |
| 31.2.3.1   | Calculation of Matrix Elements . . . . .                                                            | 247        |
| 31.2.3.1.1 | Hamiltonian . . . . .                                                                               | 247        |
| 31.2.3.1.2 | Worked Out: Hamiltonian Math . . . . .                                                              | 248        |
| 31.2.3.1.3 | Potential Energy . . . . .                                                                          | 249        |
| 31.2.3.1.4 | Eigenvalues . . . . .                                                                               | 251        |
| 31.2.3.2   | Eigenfunctions . . . . .                                                                            | 253        |
| 31.3       | Further Discussion . . . . .                                                                        | 254        |
| <b>32</b>  | <b>Atomic Term Symbols</b>                                                                          | <b>255</b> |
| 32.1       | Introduction . . . . .                                                                              | 255        |
| 32.2       | Configuration Interaction . . . . .                                                                 | 256        |
| 32.2.1     | Predicting Eigenvalue Patterns From Considerations of Angular Momentum and Spin operators . . . . . | 256        |
| 32.2.2     | Eigenvalues of $\hat{L}_z$ , $\hat{S}_z$ . . . . .                                                  | 257        |
| 32.2.3     | Commutation relations with Molecular Hamiltonian . . . . .                                          | 257        |
| 32.2.3.1   | Inclusion of Spin-Orbit Coupling . . . . .                                                          | 258        |
| 32.3       | Recipe . . . . .                                                                                    | 260        |
| 32.4       | Fully Worked Example: Carbon . . . . .                                                              | 260        |
| 32.4.1     | Orbital Filling/Open Shell . . . . .                                                                | 260        |
| 32.4.2     | Number of Microstates . . . . .                                                                     | 261        |
| 32.4.3     | Maximizing $M_S$ . . . . .                                                                          | 261        |
| 32.4.4     | Maximizing $M_L$ . . . . .                                                                          | 263        |
| 32.4.5     | Accounting Energy Levels . . . . .                                                                  | 264        |
| 32.4.6     | Listing first two term symbols . . . . .                                                            | 264        |
| 32.4.7     | Sketching . . . . .                                                                                 | 264        |
| <b>33</b>  | <b>Additional Atomic Term Examples</b>                                                              | <b>265</b> |
| 33.1       | Nitrogen . . . . .                                                                                  | 265        |
| 33.1.1     | Filling Orbitals/Open Shell . . . . .                                                               | 265        |
| 33.1.2     | Number of Microstates . . . . .                                                                     | 265        |
| 33.1.3     | Maximizing $M_S$ . . . . .                                                                          | 265        |
| 33.1.4     | Maximizing $M_L$ . . . . .                                                                          | 267        |
| 33.1.5     | Accounting . . . . .                                                                                | 268        |
| 33.1.6     | Finding Missing State: Thought Exercise . . . . .                                                   | 268        |

|                                              |     |
|----------------------------------------------|-----|
| 33.1.7 Listing Out . . . . .                 | 269 |
| 33.2 Iron . . . . .                          | 269 |
| 33.2.1 Filling Orbitals/Open Shell . . . . . | 269 |
| 33.2.2 Number of Microstates . . . . .       | 270 |
| 33.2.3 Maximizing $M_S$ . . . . .            | 270 |
| 33.2.4 Maximizing $M_L$ . . . . .            | 271 |
| 33.2.5 Accounting . . . . .                  | 272 |
| 33.2.6 Listing Out . . . . .                 | 273 |
| 33.2.7 Sketching . . . . .                   | 274 |



# Document Key

There are specific callouts throughout the document:

-  refers to a math derivation.
-  refers to a worked in-class example.
-  refers to an aside. Something in-depth to help understanding but not required reading.
-  refers to a tip.
-  refers to a summary.
-  refers to a warning.

## Changelog

**1.0** Kathryn Froese (2024-09 – 2024-12) — Initial digital notes created in Obsidian from fountain pen and paper lecture notes. Graphics created in Excalidraw.

**2.0** Kathryn Froese (2025-02 – 2025-08) — Converted to  $\text{\LaTeX}$ !

# 1 What is Quantum Mechanics?

## 1.1 Introduction

Quantum mechanics is the fundamental theory of matter that describes the behaviour of both nuclei and electrons. It is an abstract mathematical framework, based on a few basic principles. It is hard to interpret physically and this aspect is still a topic of research and discussion.

In chemistry, quantum mechanics helps understand e.g.:

- Molecular structure
- Spectroscopic properties
- Thermochemistry
- Rates of reaction

## 1.2 Schrödinger Equation

The famous equation is

$$\hat{H}\psi_\lambda = E_\lambda\psi_\lambda$$

with  $\hat{H}$  referring to the Hamiltonian and  $\lambda$  refers to the index of a state and energy level.

This equation describes the energy levels of any system. It uses multiple inputs to achieve an answer.

$$\text{Input: } \begin{cases} \text{Which nuclei?} \\ \text{How many electrons?} \end{cases}$$

Solving the equation is hard and for multi-electron systems is impossible to do by hand. Computing an approximate but precise solution is easier using sophisticated software that has been developed over the years (e.g. Gaussian).

### 1.2.1 Electron Models

Different models help understand the behaviour of nuclear motions. These models are relatively simple:

- Translational
- Rotational
- Vibrational

but *electronic structure* is more difficult and cannot be modeled easily.

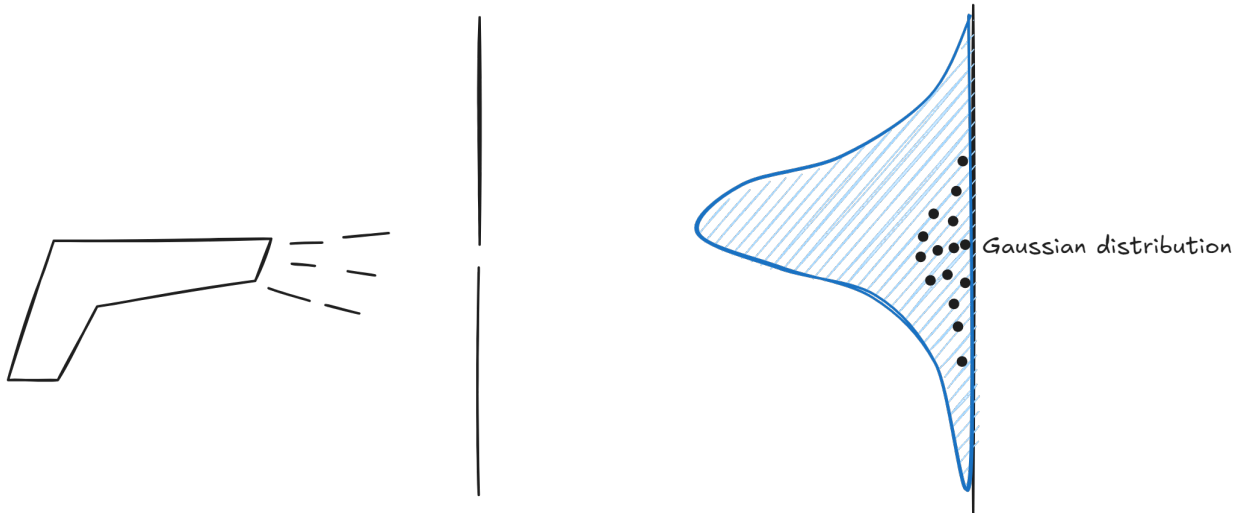
## 1.3 QM Software

There are many standard ways of calculating chemical systems using software:

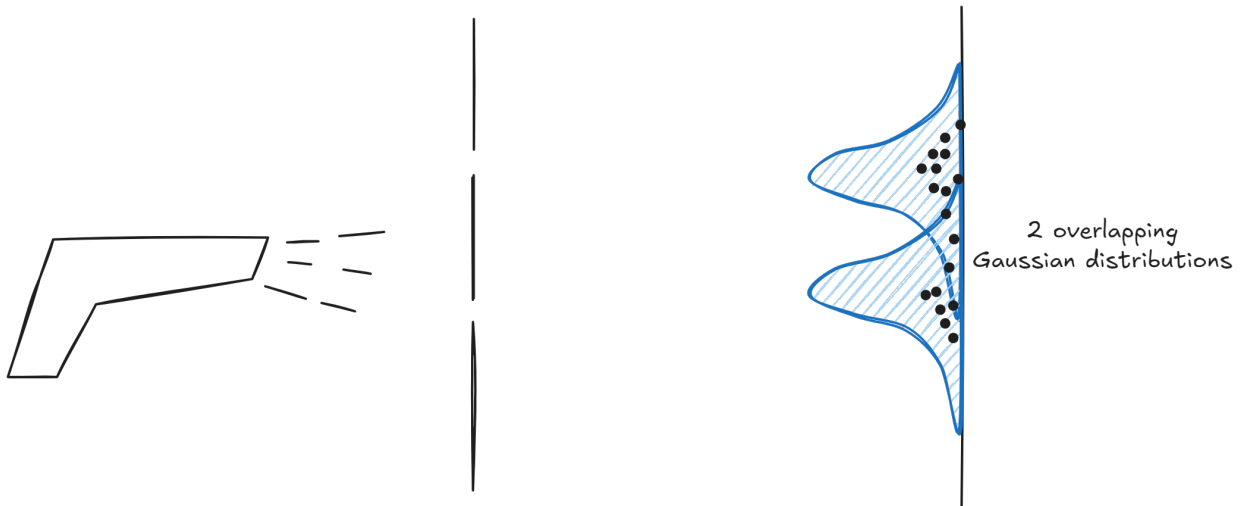
- Gas phase molecules; routine
- Crystals; possible, by making use of periodic properties
- Solution chemistry; more difficult, by using solvation models to mimic solvents
- Liquids; difficult, by combining quantum and classical Newtonian models

## 2 Waves and Distributions

### 2.1 Physical Matter



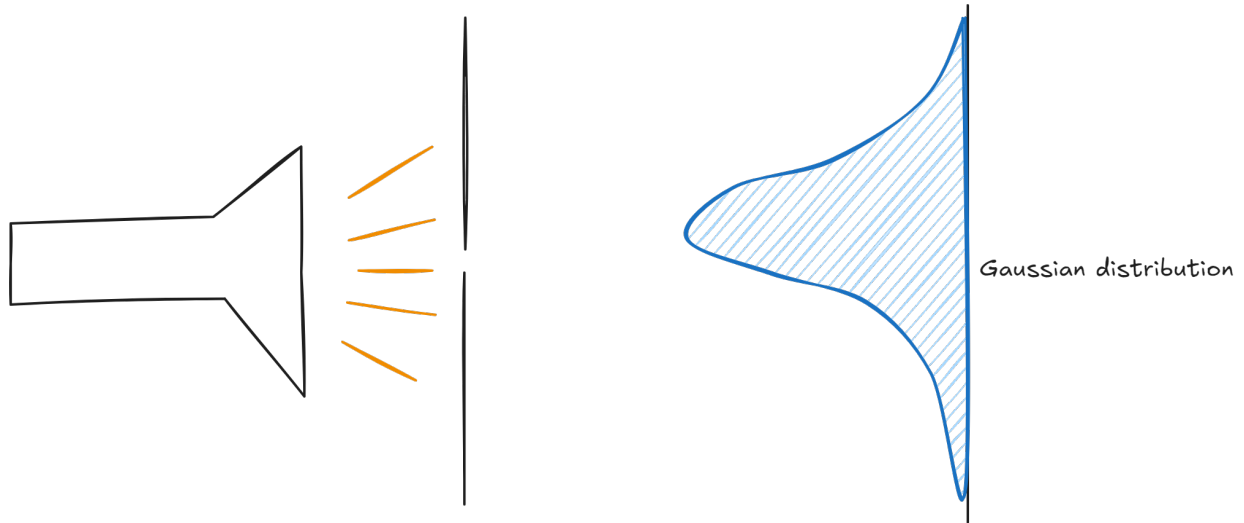
**Figure 2.1:** Physical matter (bullets) passing through a single slit, producing a Gaussian distribution. Each bullet arrives at a specific spot on the screen, and the collection makes a distribution.



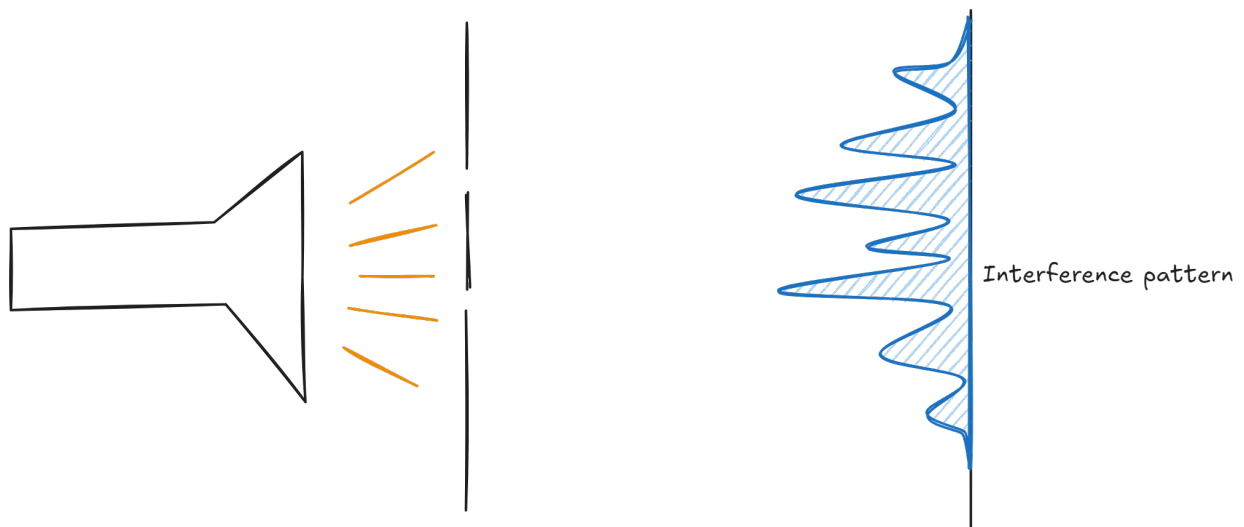
**Figure 2.2:** Physical matter (bullets) passing through double slits, producing two overlapping Gaussian distributions, which are just the combined solution for individual slits.

## 2.2 Light

Unlike bulk physical matter, light passing through double slits produces a curious result.

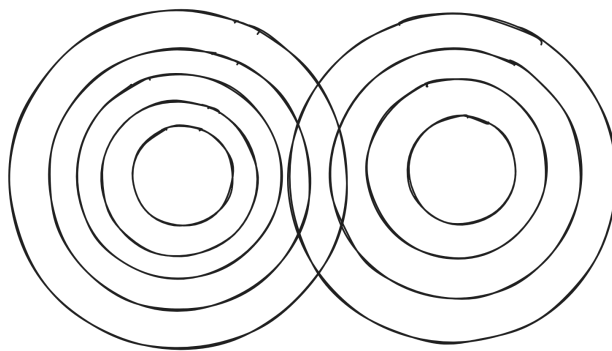


**Figure 2.3:** Light passing through a single slit, producing a Gaussian distribution



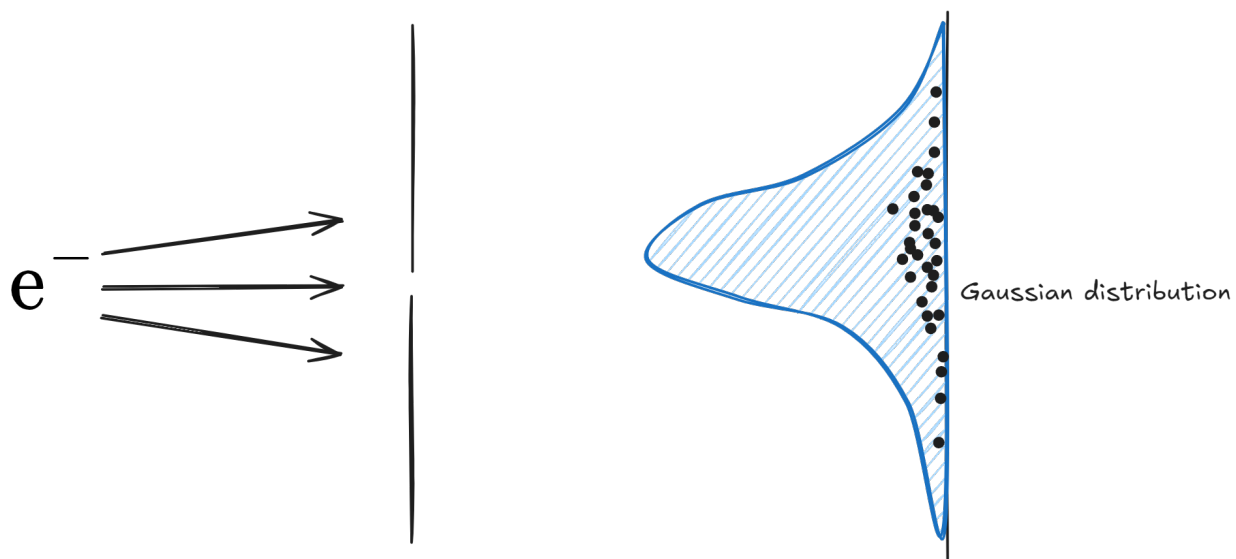
**Figure 2.4:** Light passing through double slits, producing an **interference pattern**

These wave interference patterns occur from constructive/destructive interference, just like waves of water with their peaks and troughs.

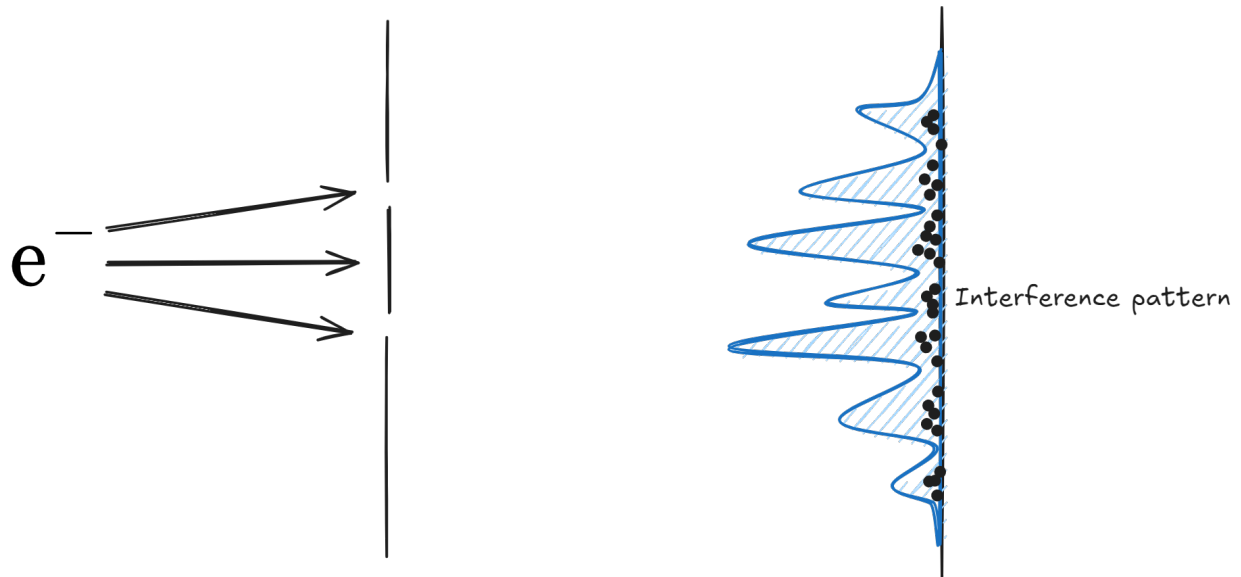


**Figure 2.5**

## 2.3 Electrons



**Figure 2.6:** Electrons passing through a single slit, producing a Gaussian distribution. This looks similar to the shotgun picture from before.

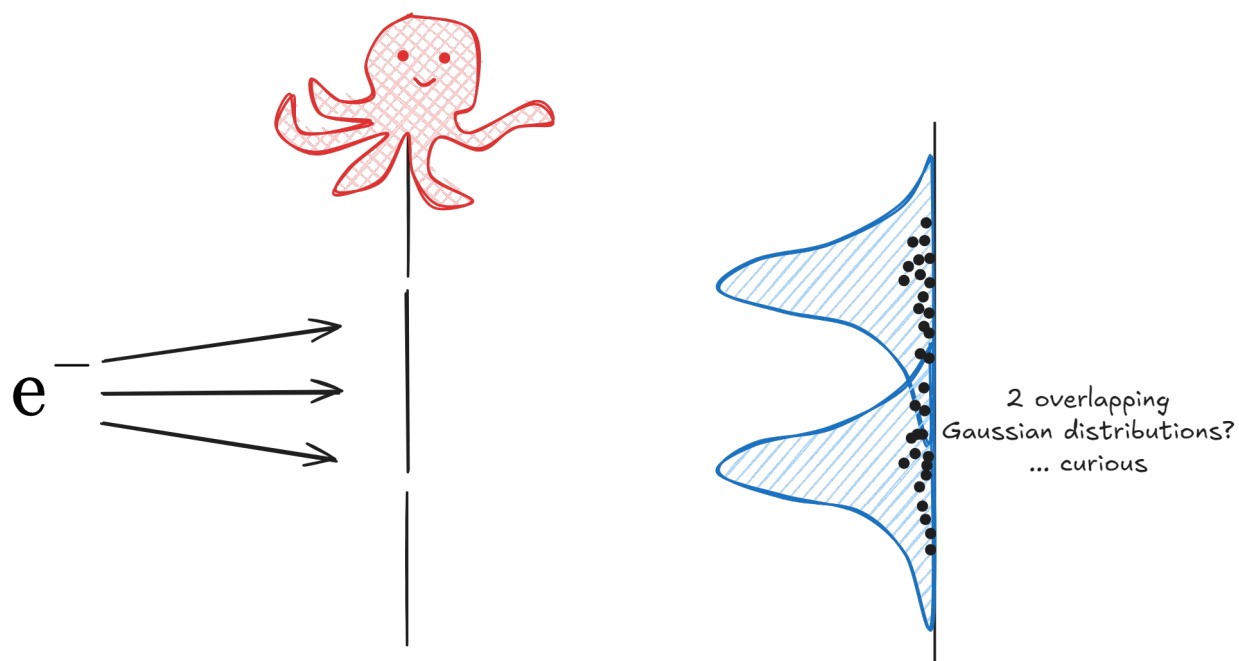


•

**Figure 2.7:** Electrons passing through double slits, producing the same interference pattern as light. However, each electron is recorded as a single spot on the screen. The collection yields the distribution.

## 2.4 SQUID

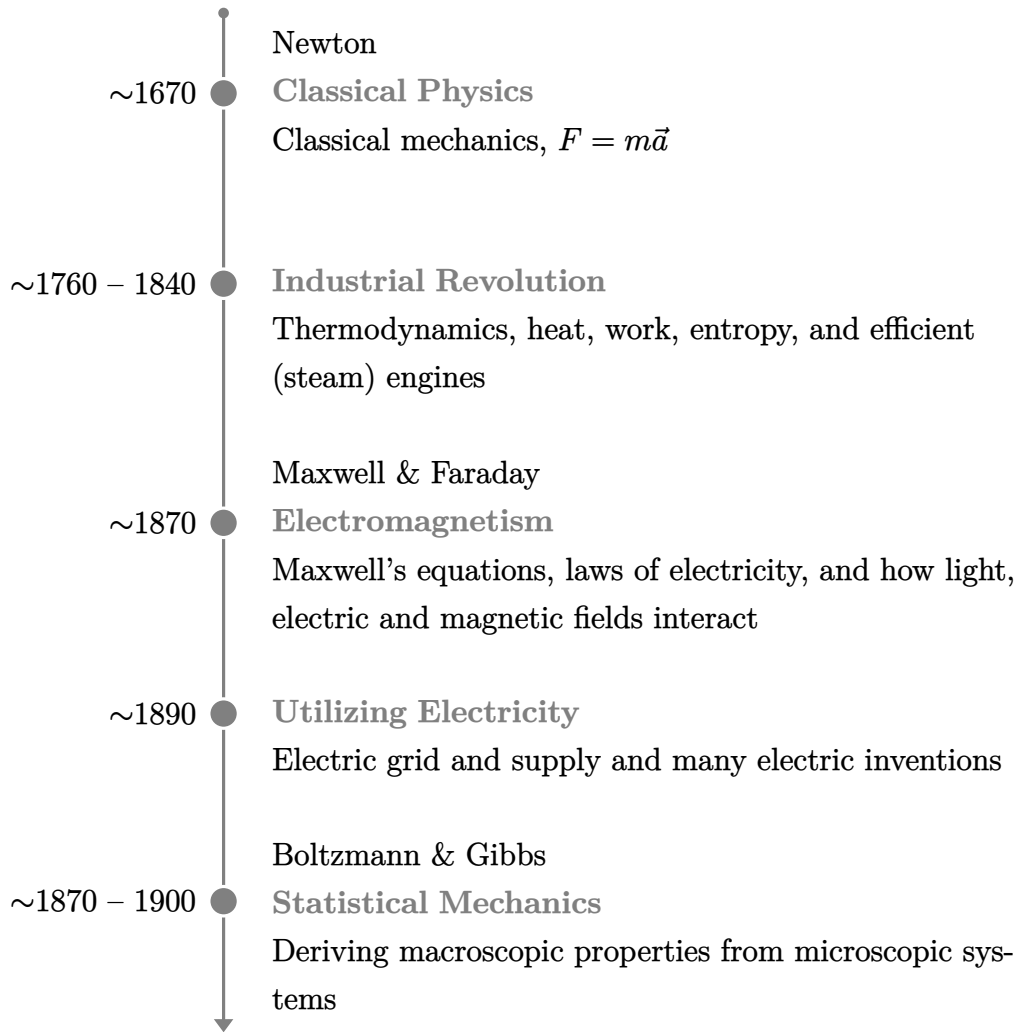
A superconducting quantum interference device (SQUID) can be used to track which slit electrons go through... what happens? The SQUID is acting as an **observer** of the electrons.



**Figure 2.8:** Observing through which slit the electron goes destroys the interference pattern! We see the same picture as a shotgun with double slit.



## 3 A Timeline of Physics



**Figure 3.1:** An approximate timeline of major innovations in physics. All dates are approximate (and could be somewhat wrong/contentious, not a history class).

### 3.1 Puzzlers

Certain problems defied physical explanation... until they were solved by radical new theories

- Velocity of light is constant, independent of the source and inertial mass is equal to

gravitational mass.

- This was derived from Maxwell's equations.
- This was solved by Einstein's theory of relativity (special and general).
- The General Theory of Relativity was developed from 1907–1915 and was very different from Newtonian gravity (see below).

$$v_g = g \frac{m_1 m_2}{|\vec{r}_1 - \vec{r}_2|}$$

- Blackbody radiation. In the classical theory, light would emit infinitely into the ultra-violet range.
  - This was solved by Planck in 1900.
- Photoelectric effect. In the classical theory, light would continually transfer energy to the metal and continuous emission of electrons would occur.
  - This was solved by Einstein.
- Emission spectra of atoms (and stars).
  - This was introduced by Bohr and was the first quantum theory, that being the idea of discrete energy levels.
- All of chemistry...!
  - Solved (at least theoretically) 1925 onwards.
- X-rays.
  - Discovered by Röntgen.
- Radioactivity.
  - Explored by the Curies and Becquerel.
- $\gamma$ -Rays.
- Superconductivity and superfluidity (Kamerlingh Onnes).
- The electron and nucleus of an atom.
  - Discovered by Compton in 1899 and Rutherford in 1911 respectively.
- Quantum mechanics as we know it by Schrödinger, Heisenberg and Dirac! (The latter Marcel Nooijen's personal favourite)
  - Paul Dirac prepared his seminal mathematical quantum theory in 1927.

## 3.2 A Very Brief Note on Relativity

Einstein derived some very interesting consequences from the fact that the speed of light is independent of the velocity of the source. This was verified by the famous Michelson-Morley experiment. Let us try to understand what this *constancy of the speed of light* means. To this end we first look at a simple “real life” experiment. Say you ride a skateboard and throw a ball in the air to catch it again. How do you throw (assuming no wind)?

Next you are sitting in a fast moving train. You throw the ball straight up and catch it again. An observer on the ground watches this. How would you compare notes?

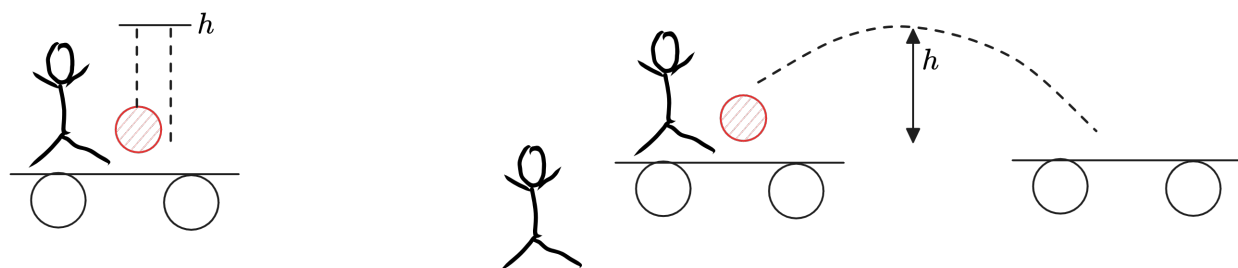


Figure 3.2

### 3.2.1 Ball

For the observer on the train the ball travels  $2h$  in some time  $\Delta t$ , with an average velocity  $v_{\text{train}} = 2h/\Delta t$ . The observer on the ground has a different perspective. For them the ball travels further (approximately  $2\sqrt{h^2 + (V\Delta t)^2}$ ). The time of travel is the same. Therefore, the average velocity is higher for the person on the ground. Of note, we must include the velocity of the moving train when making calculations. Hence, velocity depends on the observer (when throwing balls).

If the bottle, thrown out of a fast moving train, would hit you, an innocent bystander next to the tracks, you quickly realize the reality of this “difference in perspective”.

### 3.2.2 Light

Now do a similar experiment with light. The person on the train shines a flashlight on to a (very high) ceiling that has a mirror. They can measure the time for the light to return and the average velocity. Now for the observer on the ground the light travels further (again, motion of the train). However, the velocity of light is the *same*, independent of the velocity of the source. Very different from throwing a ball. If the velocity is the same, but the distance

traveled is longer, it follows that the *time* it takes for the person on the ground takes longer! They could shine a similar flashlight on the ground, with the same high ceiling, and see the light on the ground return faster than the light on the train.

Conversely, the person on the train would say the ground light travels further (speeding away from the train), and hence the (light) clock on the ground runs slow. This very surprising conclusion (time depends on perspective, or the motion of an observer) is a simple consequence of the constant velocity of light (independent of the source, or, for all observers). This is the essence.

The detailed math is not so hard, but it is hard to have an intuition for relativity as it is so different from everyday experience. The effects are hard to observe for us, as the speed of light is so high.

Let us express the math, if we denote by  $L$  the scalar distance the light travels on its return trip on the train we get

$$L^{\text{ground}} > L^{\text{train}}$$

$$c = \begin{cases} v_{\text{avg}}^{\text{ground}} = \frac{L^{\text{ground}}}{\Delta t^{\text{ground}}} \\ v_{\text{avg}}^{\text{train}} = \frac{L^{\text{train}}}{\Delta t^{\text{train}}} \end{cases}$$

$L^{\text{train}}$  is smaller than  $L^{\text{ground}}$ , so in order for their velocity to be equal the  $\Delta t$  must be different. Therefore

$$\Delta t^{\text{train}} < \Delta t^{\text{ground}}$$

The time is different, why? Special relativity, and how it arises out of the constancy of the speed of light.

### 3.2.3 Special Relativity

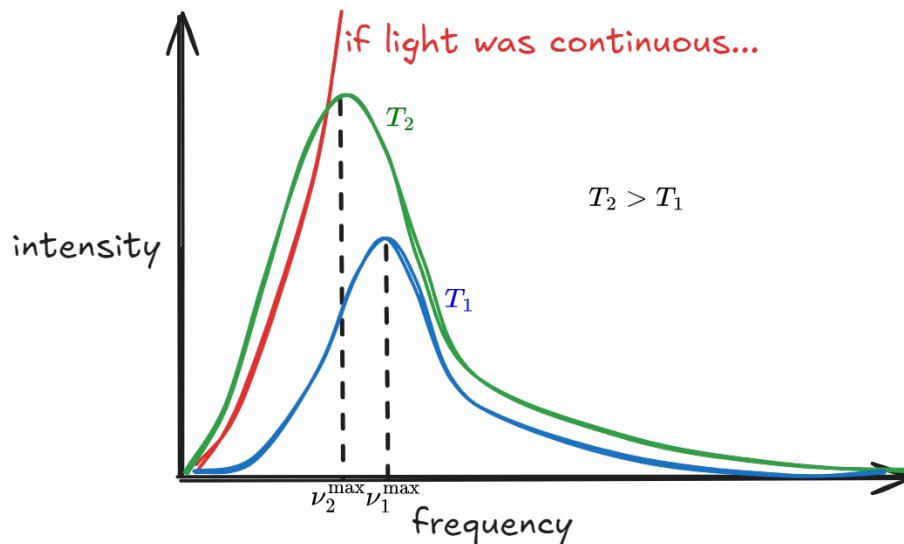
Later, special relativity arrived at the famous  $E = mc^2$ , as another consequence of the constancy of the speed of light.

## 4 Blackbody Radiation and First Evidence of Energy Quantization

### 4.1 Blackbody Radiation

**Blackbody radiation** is the first evidence of **quantized** energy, that is light being split into small, discrete units of energy in contrast to the assumption that light is continuous.

The observed phenomenon is that a blackbody (an object made out of a nonreflective, fully absorbent, idealized material) when heated, releases energy in the form of light. The light emitted, depending on the temperature, forms a distribution of frequencies.

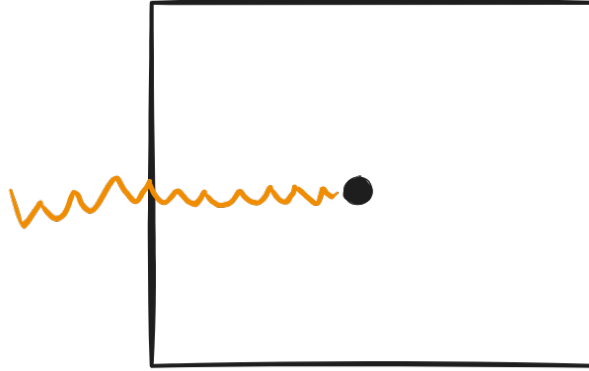


**Figure 4.1:** Comparison of blackbody radiation at different temperatures

The light released is dependent on the distribution function  $\rho_\nu(T) d\nu$ , with (SI) units of  $\text{J m}^{-3}$ . It might be more intuitive to write  $\rho_T(\nu)$  (a function of frequency). Black body radiation can be measured carefully in the laboratory but it can also be seen in light emitted by stars, which are approximate black bodies.

The experimental setup uses a tiny hole that emits radiation, which could be measured precisely, with the object being heated at some temperature  $T$ , at equilibrium with radiation inside. Real (and imperfect) examples of blackbodies include metal being heated, furnaces,

etc. The curves do not depend on material.



**Figure 4.2**

The blackbody radiation can be measured for stars as well, which aided in measuring very high temperatures. In 1870 an experiment discovered the Wien displacement where  $\lambda_{\max}T = 2.9 \times 10^{-3} \text{ m K}$ , which is the difference in maximum wavelength based on the temperature of the blackbody.

## 4.2 Planck

Planck used experimental data to derive a general formula

$$\rho_{\nu}(T) = \frac{8\pi h}{c^3} \frac{\nu^3}{e^{h\nu/k_B T} - 1}$$

He used an assumption that defied physics at the time, with light having to come in discrete units of  $n$  where  $n = 0, 1, 2, \dots$  of “quantized” energy. Light is an oscillator with frequency  $\nu$  and discrete energies  $n h \nu$ .

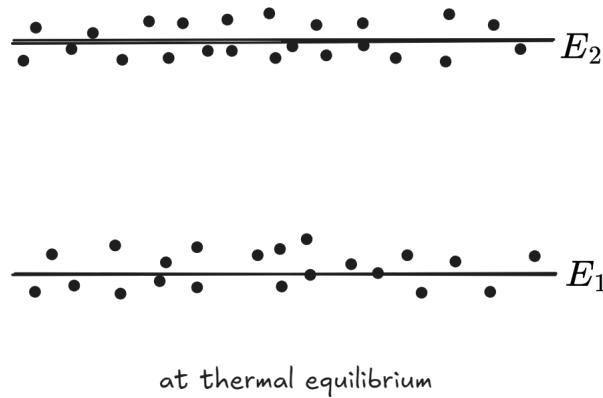
Taking this formula and either having  $\nu \rightarrow 0$  or  $h \rightarrow 0$  we arrive at the classical physics formula (hard to derive!)

$$\rho_{\nu}(T) = \frac{8\pi k_B T}{c^3} \nu^2$$

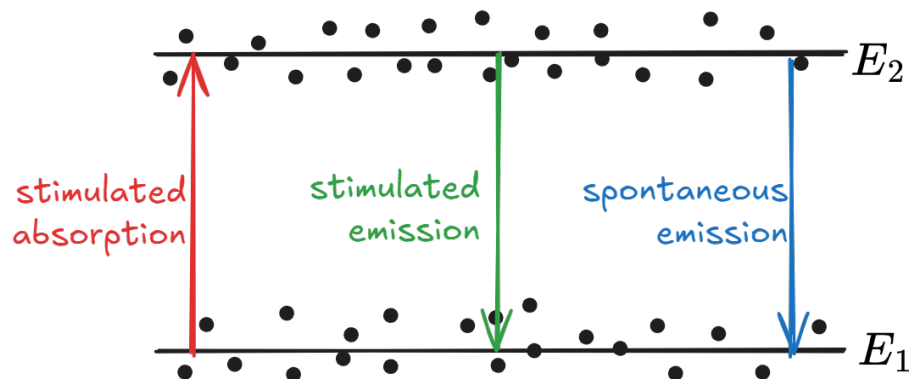
While  $h$  is now considered a constant, it is the value that adds “discreteness”. By having it go to 0, mathematically that means that light is continuous. It must be very small, close to 0, otherwise the discreteness of energy would have been noticed much earlier and easier.

## 4.3 Einstein

Einstein used this fact of discrete energy levels to derive differential equations to model spontaneous emission, stimulated emission, and stimulated absorption.



**Figure 4.3:** The energy difference between these two energy levels is  $\Delta E = E_2 - E_1 = h\nu$ .



**Figure 4.4:** Spontaneous emission is when an electron moves from a higher to a lower energy level, releasing light energy. It does this independent of a light source.

Stimulated absorption is when an electron is excited by an outside light source, moving from a lower to a higher energy level.

Stimulated emission occurs when an electron releases light as it falls back from the higher to the lower energy level, but in the presence of radiation of the same frequency. The stimulated emission is much stronger than the spontaneous emission.

At thermal equilibrium the population of electrons in the first and second states are

$N_1 \propto e^{-E_1/k_B T}$  and  $N_2 \propto e^{-E_2/k_B T}$  respectively. The ratio between them is

$$\begin{aligned}\frac{N_1}{N_2} &= e^{-(E_1-E_2)/k_B T} \\ &= e^{(E_2-E_1)/k_B T} \\ &= e^{\Delta E/k_B T} \quad (\text{particles or molecules}) \\ &= e^{\Delta E/RT} \quad (\text{moles})\end{aligned}$$

Note that  $R = k_B N_A$ .

## Einstein's Derivations

— We have a set of three different differential equations, each representing a different type of energy transition

$$\begin{cases} \frac{dN_2}{dt} = BN_1\rho_\nu(T) & (\text{stimulated absorption}) \\ \frac{dN_2}{dt} = -BN_2\rho_\nu(T) & (\text{stimulated emission}) \\ \frac{dN_2}{dt} = -AN_2 & (\text{spontaneous emission}) \end{cases}$$

where  $B$  and  $A$  are constants with no  $t$  dependence.

From here we combine into one equation

$$\begin{aligned}\frac{dN_2}{dt} &= BN_1\rho_\nu(T) - BN_2\rho_\nu(T) - AN_2 = 0 \\ &= B\rho_\nu(T)(N_1 - N_2) = AN_2 \\ \Rightarrow A &= B\rho_\nu(T) \left( \frac{N_1}{N_2} - 1 \right) \\ \frac{A}{B} &= \rho_\nu(T) \left( \frac{N_1}{N_2} - 1 \right) \quad \text{Subbing in the population formula} \\ \Rightarrow \frac{A}{B} &= \rho_\nu(T)(e^{\Delta E/k_B T} - 1) \\ \rho_\nu(T) &= \frac{A}{B} \frac{1}{e^{\Delta E/k_B T} - 1} \quad \text{Subbing in } \Delta E = h\nu \\ \rho_\nu(T) &= \frac{A}{B} \frac{1}{e^{h\nu/k_B T} - 1}\end{aligned}$$

With  $\frac{A}{B}$  being the ratio of spontaneous emission ( $A$ ) to stimulated emission ( $B$ ), it also a *property of radiation, independent of material*. This was derived by



Dirac in 1928.

$$\frac{A}{B} = \frac{8\pi h}{c^3} \nu^3$$
$$\implies \rho_\nu(T) = \frac{8\pi h}{c^3} \frac{\nu^3}{e^{h\nu/k_B T} - 1}$$

Einstein postulated this ratio of his  $A$  and  $B$  coefficients. He essentially reverse-engineered this so that he could derive the proper Planck formula for blackbody radiation. The important point is that this derivation sheds light on what is really involved in blackbody radiation.

Light is in thermal equilibrium with matter described by various quantum energy levels. Every pair of levels can be in equilibrium with the corresponding frequency  $\nu = \frac{E_2 - E_1}{h}$ . In a blackbody very many frequencies are present (energy levels). This property only depends on the frequency of light and the presence of the corresponding (or nearby) energy difference in a material.

# 5 Emission Spectra of Atoms

## 5.1 Spectral Series of Hydrogen

The first atom whose emission spectrum was investigated was hydrogen. These appear in the emission spectra of stars frequently, as hydrogen is a very common element throughout the universe in different solar bodies.

At different and specific wavelengths, emission lines were noticed. In order, the following series were discovered:

- Balmer,  $n = 2$ , in 1885
- Lyman,  $n = 1$ , in 1904–1914
- Paschen,  $n = 3$ , in 1908
- Brackett,  $n = 4$ , in 1922
- Pfund,  $n = 5$ , in 1924
- Humphreys,  $n = 6$ , in 1953

All can be expressed in terms of the Rydberg formula

$$\frac{1}{\lambda} = R_{\infty} \left| \frac{1}{n_1^2} - \frac{1}{n_2^2} \right| Z^2$$

Note that  $R_{\infty}$  refers to the Rydberg constant for an atom assumed to be infinitely heavy (this only works because the mass of an electron is much, much lighter than the mass of the nucleus).

While each of these were often experimentally observed in terms of wavelength  $\lambda$ , they are all just expressions of energy and more useful theoretically in that form, and can be converted using Planck's constant  $\Delta E = h\nu = h\frac{c}{\lambda}$ . Note here that Planck's constant has units of J s.

In spectroscopy the preferred unit is wavenumber, with units of  $\text{cm}^{-1}$ . Conceptually, wavenumber can be thought of as number of crests per centimeter. By convention,  $\tilde{\nu}$  refers to wavenumber and  $\tilde{R}_{\text{H}}$  referring to the wavenumber form of the Rydberg constant, calculated by  $\tilde{R}_{\text{H}} = \frac{R_{\text{H}}}{hc} [\text{cm}^{-1}]$ .

## 5.2 Bohr

Rutherford discovered the positive nucleus in 1911. Bohr proposed that atoms have discrete energy levels, which can be calculated using Rydberg's formula.

$$h\nu = |\Delta E| = |E_f - E_i|$$

All of the previous series observed empirically could now be explained and calculated theoretically. Bohr's model of the electron had electrons orbit circularly about the nucleus, at particular radii with stationary orbits.

There were some problems with this at first. Thinking of electrons as accelerating charges, they could be explained by the Coulomb force  $F = \frac{q_1 q_2}{h\pi\epsilon_0 |R|^2}$ . An accelerating charge had angular momentum, with it losing energy as radiation as it spirals into the center... therefore the electron should spiral toward the nucleus. This is empirically false.

### 5.2.1 How Bohr Solved the Problem

Bohr assumed that angular momentum,  $L$  was quantized with

$$L = mvR = n\hbar = n\frac{h}{2\pi}$$

He then combined the Coulomb force equation with angular momentum to arrive at

$$E_k - E_n = h\nu = hc\tilde{\nu} = -\frac{1}{8}\frac{me^4}{\epsilon_0 ch^3} \left( \frac{1}{k^2} - \frac{1}{n^2} \right)$$

with  $R_H = \frac{1}{8}\frac{me^4}{\epsilon_0^2 ch^3}$

Later he used the reduced mass of the electron  $\mu = m_e \frac{m_p}{m_e + m_p}$ .

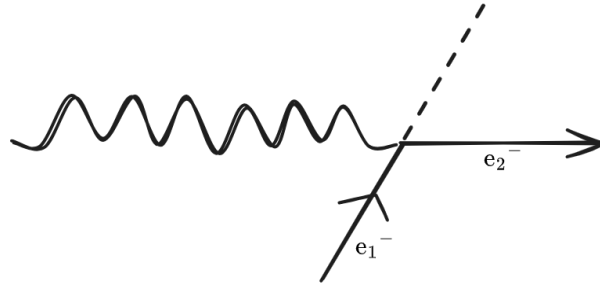
Later analysis was repeated but using a changed Coulomb term to account for the nuclear charge  $Z$ . This could then be used for later elements by accounting for the mass of the nucleus, to predict  $\text{He}^+$ ,  $\text{Li}^{2+}$ ,  $\text{Be}^{3+}$ , and  $\text{Ne}^{9+}$ . Additionally, this allowed them to distinguish between the spectra of deuterium ( $^2\text{H}$ ) and tritium ( $^3\text{H}$ ), as their spectra was shifted relative to hydrogen due to the increased nuclear mass.

# 6 Light and Electron Waves and Differential Equations

## 6.1 De Broglie

The physicist Louis de Broglie was supervised by Langevin and examined by Einstein.

He developed the idea that electrons have wave character and a new idea of thinking about momentum.



**Figure 6.1:** Compton discovered that light has momentum, by using it to deflect electrons. How can light have momentum if it has no mass?

De Broglie set out to solve this. His starting point was *relativistic kinetic energy*.

$$E_{\text{kin}} = \sqrt{m_0^2 c^4 + p^2 c^2} = m_0 c^2 \sqrt{1 + \frac{p^2 c^2}{m_0^2 c^4}}$$

$$\frac{1}{2} m_0 v^2 = \frac{1}{2} \frac{(m_0 v)^2}{m_0} \quad \text{note that } p = m_0 v$$

$$\text{if } p = 0 \text{ then } E_{\text{kin}} \approx m_0 c^2 \quad (\text{Einstein!})$$

$$\text{Newton : } E_{\text{kin}} = \frac{1}{2} \frac{(mv)^2}{m} = \frac{p^2}{2m_0}$$

$$\text{if } p \ll m_0 c^2 \text{ then } E_{\text{kin}} \approx \underbrace{m_0 c^2}_{\text{constant, new}} + \underbrace{\frac{p^2}{2m_0}}_{\text{Newtonian, old}}$$

Light has no mass,  $m_0 = 0$ . Kinetic energy of light can be described by  $E_{\text{kin}} = h\nu$ , therefore

substituting that into  $E_{\text{kin}} = \sqrt{m_0^2 c^4 + p^2 c^2}$  we get

$$\begin{aligned} E_{\text{kin}} &= \sqrt{p^2 c^2} \\ &= pc = h\nu \\ \implies p &= \frac{h\nu}{c} = \frac{h}{\lambda} \end{aligned}$$

This formula shows that light with zero mass, still has momentum. The momentum of any wave can be described by its frequency or wavelength. The bold new idea of de Broglie was that he would associate a wave with wavelength  $\lambda = \frac{h}{p}$  with particles that were moving with momentum ( $p$ ), particularly electrons.

## 6.2 Thermal Electrons and Electron Microscopy

We can use the translational thermal energy of an electron to calculate the smallest wavelength it can resolve by equating kinetic and thermal energy,

$$\begin{aligned} \frac{1}{2} m_e v^2 &= \frac{3}{2} k_B T \quad T = 300 \text{ K} \\ \implies v &= 10^4 \text{ m s}^{-1} \\ p = mv, \lambda &= \frac{h}{p} \\ \implies \lambda &= 60 \text{ nm} \end{aligned}$$

We see that electrons can resolve much smaller distances than light can, which is the basis of electron microscopy.

## 6.3 Plane Wave

In 1-dimension, a plane wave can be described by

$$e^{ikx} = e^{i\frac{2\pi}{\lambda}x} \quad k = \frac{2\pi}{\lambda} \text{ (wavenumber)}$$

From here, we can arrive at a relation between  $k$  and  $p$ . This also leads to the definition of  $\hbar$ , which we will use often in quantum mechanics.

$$\begin{aligned}\frac{1}{\lambda} &= \frac{k}{2\pi} \\ p &= \frac{h}{\lambda} = \frac{h}{2\pi} k = \hbar k \\ \implies \hbar &= \frac{h}{2\pi} \quad (\text{will see often!})\end{aligned}$$

For a plane wave in 3-dimensions, it can be described by  $e^{i(\vec{k}\cdot\vec{x}-\omega t)}$  with  $\omega = 2\pi\nu$ , which leads into the similar vector equation  $\vec{p} = \hbar\vec{k}$ . Note that  $\vec{k} = (k_x, k_y, k_z)$  and therefore its magnitude can be described by  $\sqrt{k_x^2 + k_y^2 + k_z^2}$ . Similarly,  $\vec{x} = (x, y, z)$  and its magnitude  $\sqrt{x^2 + y^2 + z^2}$ .

This means that 1-dimensional waves can be generalized to 3-dimensions.

Additionally, these waves have energy as well, which can be described using similar variables, like

$$E = h\nu = \hbar\omega = \left(\frac{h}{2\pi}\right)(2\pi\nu)$$

## 6.4 Introduction to Differential Equations

The mathematical topic of differential equations is very large. Quantum mechanics is often described in terms of differential equations (for example the Schrödinger equation itself). We will introduce the topic using simple examples. The idea is that one can provide an equation for a function, e.g.  $y(x)$  in terms of  $y(x)$  its derivatives such as  $y'(x), y''(x), \dots$ . The solution to the equation is the actual function  $y(x)$  that satisfies the starting differential equation. Let's start with an example.

### Solving with all real roots



Take the equation  $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = 0$ . We assume the solution  $e^{\lambda x}$  works with  $\lambda$  being constant.

$$\begin{cases} y = e^{\lambda x} \\ \frac{dy}{dx} = \lambda e^{\lambda x} \\ \frac{d^2y}{dx^2} = \lambda^2 e^{\lambda x} \end{cases}$$

We now sub these into the equation

$$\begin{aligned}\lambda^2 e^{\lambda x} - 5\lambda e^{\lambda x} + 6e^{\lambda x} &= 0 \\ \frac{\lambda^2 e^{\lambda x} - 5\lambda e^{\lambda x} + 6e^{\lambda x}}{e^{\lambda x}} &= \frac{0}{e^{\lambda x}} \\ \lambda^2 - 5\lambda + 6 &= 0 \quad (\text{characteristic polynomial}) \\ (\lambda - 3)(\lambda - 2) &= 0 \\ \implies \lambda &= 2, 3\end{aligned}$$

We now add these constants back in as  $y(x) = c_1 e^{3x} + c_2 e^{2x}$  with  $c_1, c_2 \in \mathbb{R}$ .

These can also be tested, with

$$\begin{cases} \frac{d^2 y}{dx^2} = 9e^{3x} \\ \frac{dy}{dx} = 3e^{3x} \\ y = e^{3x} \end{cases}$$

Now subbing back in as  $c_1(9e^{3x} - 5(3e^{3x}) + 6e^{3x}) = c_1(9e^{3x} - 15e^{3x} + 6e^{3x}) = 0$ .  
Similarly for  $c_2$ .

## Solving with imaginary roots



The differential equation is  $\frac{d^2 y}{dx^2} + y = 0$ . Similar steps as before, we arrive at

$$\begin{aligned}\lambda^2 + 1 &= 0 \\ \implies \lambda &= \pm i\end{aligned}$$

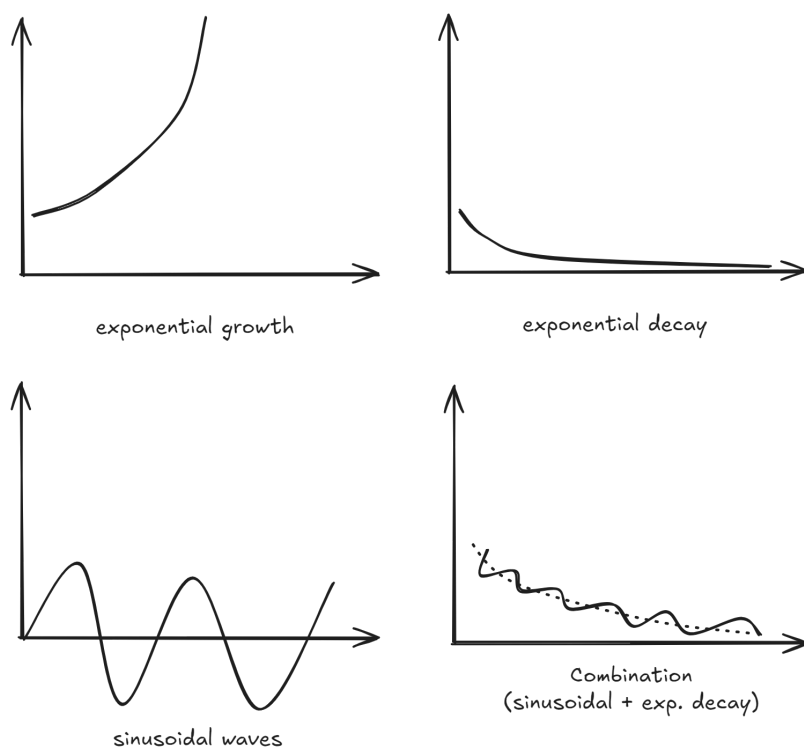
Substituting these in, we get

$$\begin{aligned}y(x) &= c_1 e^{ix} + c_2 e^{-ix} \quad \text{Using Euler's formula} \\ &= c_1 (\cos x + i \sin x) + c_2 (\cos x - i \sin x) \\ &= \underbrace{(c_1 + c_2)}_{=d_1} \cos x + \underbrace{(ic_1 - ic_2)}_{=d_2} \sin x \\ &= d_1 \cos x + d_2 \sin x\end{aligned}$$

In the exercises you will find an example where  $\lambda = a + ib$  is complex. Those solutions

can be meaningful in the right context.

This method of solving differential equations with a trial solution (where we chose  $y(x) = e^{\lambda x}$ ) only works when there are constant coefficients, the equation is linear in  $y$  (see below aside), and the equation is homogeneous ( $= 0$ ). For further discussion of solving differential equations such as justification/derivation or how we might solve different types, try a textbook dedicated to them, but in quantum mechanics we generally stick to specific types that are solvable with this easy method.



**Figure 6.2:** These represent the equations  $e^{\alpha x}$ ,  $e^{-\alpha x}$ ,  $\sin x$ , and  $e^{\alpha x + i\beta x}$ , where  $\alpha > 0$ .

### Linearity in $y$



The requirement that “the equation is linear  $y$ ” is key. Thankfully it is easy to tell by visual inspection. The best way to understand linearity in  $y$  is by constructing some examples. To be linear, the differential equation must take the form

$$A_1(x)y^{(n)} + \dots + A_n(x)y' + A_{n+1}(x)y = b(x)$$



So a second order differential equation must look like

$$A_1(x)y'' + A_2(x)y' + A_3(x)y = b(x)$$

where  $b(x)$  and each  $A_n(x)$  can be anything. Functions, constant coefficients, zero.

But we cannot have something like

$$A_1(x)\frac{1}{y''} + A_2(x)\ln(y)y' + A_3(x)y = b(x)$$

because neither the first or second derivative in the equation are linear.

Why is it that a crazy differential equation like

$$(\sin(x^2) + e^x)y'' + \ln(x + \operatorname{arcsec}(x))y' + y = x^x$$

is considered linear? We wouldn't wish to solve it, but the reason it is linear is that we are always talking about our linear equation with respect to our dependent variable  $y$  and not our independent variable  $x$ . We can input any possible value of  $x$  into that equation and produce something with constant coefficients. Each term of the differential equation, those with  $y(x)$ ,  $y'(x)$ ,  $y''(x)$ , are transformed linearly for all possible values of  $x$  and thus the equation is considered linear.

We like linear differential equations. They are solvable and easy to do on a computer. All linear, one-variable (ordinary) different equations are solvable on at least part of the interval where they are defined. That is not true for many differential equations.

## 7 Analyzing a General Time-Dependent Wave Equation

### 7.1 Setup

Consider a wave equation, resulting from the vibration of something such as a string (or electron).

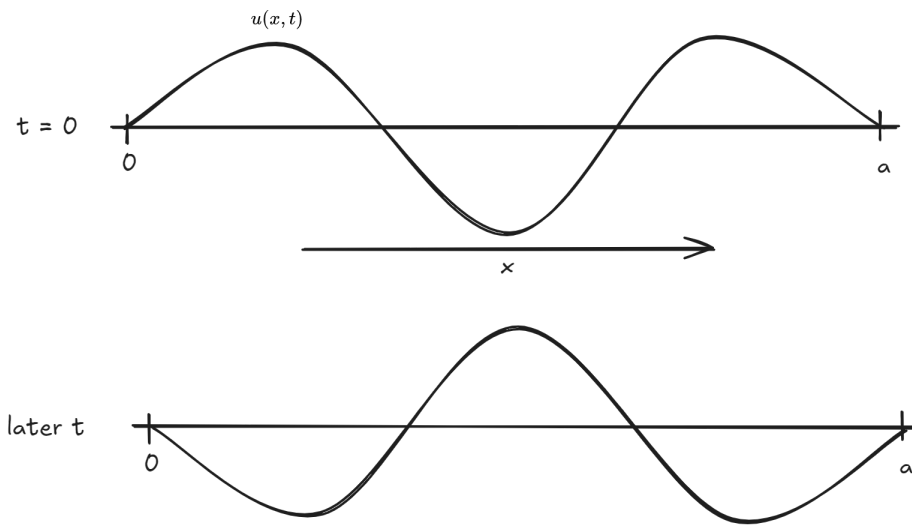


Figure 7.1

We have the differential equation

$$\frac{\partial^2 u(x, t)}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 u(x, t)}{\partial t^2}$$

where  $v$  is the phase velocity or velocity of wave propagation. It is a function of two variables,  $x$ , position, and  $t$ , time. Because there is more than one variable we use partial derivatives, expressed by the  $\partial$  symbol.

## 7.2 Boundary Conditions

We know the function at two values,  $a$ , which is arbitrary, and 0.

$$u(0, t) = u(a, t) = 0$$

## 7.3 Rewriting

We rewrite the equation, splitting it into two parts:

- $X(x)$ , the positional part
- $T(t)$ , the time part

We write the solution in the form  $u(x, t) = X(x)T(t)$ . This *only works when the functions and variables are independent of each other*, we are imposing this *assumption*.

Despite the two functions being independent, we cannot just solve for one and ignore the other. Instead, we substitute  $u(x, t) = X(x)T(t)$  into our differential equation, and set things up on the  $x$  side such that we have our derivative with respect to  $x$  on the left, with some coefficient function based on  $t$  that is constant with respect to  $x$ . We then do the opposite.

$$\frac{\partial^2 X}{\partial x^2} T(t) = \frac{1}{v^2} X(x) \frac{\partial^2 T}{\partial t^2}$$

We then separate our  $x$  and  $t$  based functions on each side by dividing by  $X(x)T(t)$

$$\frac{\partial^2 X}{\partial x^2} \frac{1}{X(x)} = \frac{1}{v^2} \frac{\partial^2}{\partial t^2} \frac{1}{T(t)}$$

We now have two functions

$$\begin{cases} f(x) = \frac{\partial^2 X}{\partial x^2} \frac{1}{X(x)} \\ g(t) = \frac{\partial^2 T}{\partial t^2} \frac{1}{T(t)} \\ f(x) = g(t) \end{cases}$$

### 7.3.1 Consequences

$$\begin{aligned} f(x) = \begin{cases} g(t_0) \\ g(t_1) \end{cases} &\implies f(x) \text{ is constant} \\ g(t) = \begin{cases} f(x_0) \\ f(x_1) \end{cases} &\implies g(x) \text{ is constant} \end{aligned}$$

This is only true if *both functions are constant*, and importantly, *equal to the same constant*.

We call this constant  $-k^2$  (with hindsight), known as the **constant of separation**. We have some constant  $C$ , but set it equal to  $-k^2$  as this produces a nicer form<sup>1</sup>.

$$\begin{aligned} \frac{d^2 X}{dx^2} &= -k^2 X(x) & X(0) = X(a) &= 0 \\ \frac{1}{v^2} \frac{dT^2}{dt^2} &= -k^2 T(t) & \text{no boundary conditions} \end{aligned}$$

## 7.4 Integrating $X(x)$

We now assume the solution takes the form of  $X(x) = e^{\lambda x}$ . This is a reasonable guess, given we have a wave equation that can likely be expressed as something similar to  $e^{ix} = \cos x + i \sin x$ .

$$\begin{cases} X(x) = e^{\lambda x} \\ \frac{dX}{dx} = \lambda e^{\lambda x} \\ \frac{d^2 X}{dx^2} = \lambda^2 e^{\lambda x} \end{cases}$$

$$\begin{aligned} \lambda^2 e^{\lambda x} &= -k^2 e^{\lambda x} \\ \lambda^2 &= -k^2 \\ \implies \lambda &= \sqrt{-k^2} \\ &= \sqrt{-1} \sqrt{k^2} \\ &= \pm ik \quad (\text{not just taking the positive root of } k) \end{aligned}$$

---

<sup>1</sup>This allows us to produce the **characteristic solution** as  $x^2 - C = x^2 + k^2 = (x - ik)(x + ik) = 0$ .

### 7.4.1 Initial Values for $X(x)$

We now have the formula  $X(x) = c_1 e^{ikx} + c_2 e^{-ikx}$  which can be expanded using Euler's formula to

$$\begin{aligned} c_1 e^{ikx} + c_2 e^{-ikx} &= c_1 (\cos kx + i \sin kx) + c_2 (\cos kx - i \sin kx) \\ &= \underbrace{(c_1 + c_2)}_{d_1} \cos kx + \underbrace{i(c_1 - c_2)}_{d_2} \sin kx \\ &= d_1 \cos kx + d_2 \sin kx \end{aligned}$$

#### 7.4.1.1 $X(0) = 0$

We now substitute in  $X(0) = 0$

$$\begin{aligned} X(0) &= d_1 \underbrace{\cos(k(0))}_1 + d_2 \underbrace{\sin(k(0))}_0 = 0 \\ d_1 &= 0 \end{aligned}$$

Our equation is now

$$X(x) = d_2 \sin kx$$

#### 7.4.1.2 $X(a) = 0$

We now use our second initial value  $X(a) = 0$ .

$$X(a) = d_2 \sin ka = 0$$

When is  $\sin x = 0$ ? The trivial solution is when  $d_2 = 0$ , but we ignore it because it is not useful. Consider the unit circle, where  $\pi n$  ( $n = 0, \pm 1, \pm 2, \dots$ ) results in  $\sin x = 0$ , and these are all of the zeros of  $\sin x$ . Now considering the  $ka$  term we have

$$\begin{aligned} \sin ka &= 0 \\ \implies ka &= \pi n \\ k &= \frac{\pi n}{a} \\ \implies -k^2 &= -\frac{\pi^2 n^2}{a^2} \end{aligned}$$

## 7.5 Cases of $n$

Substituting in the expanded  $k$  to  $d_2 \sin kx$  we have

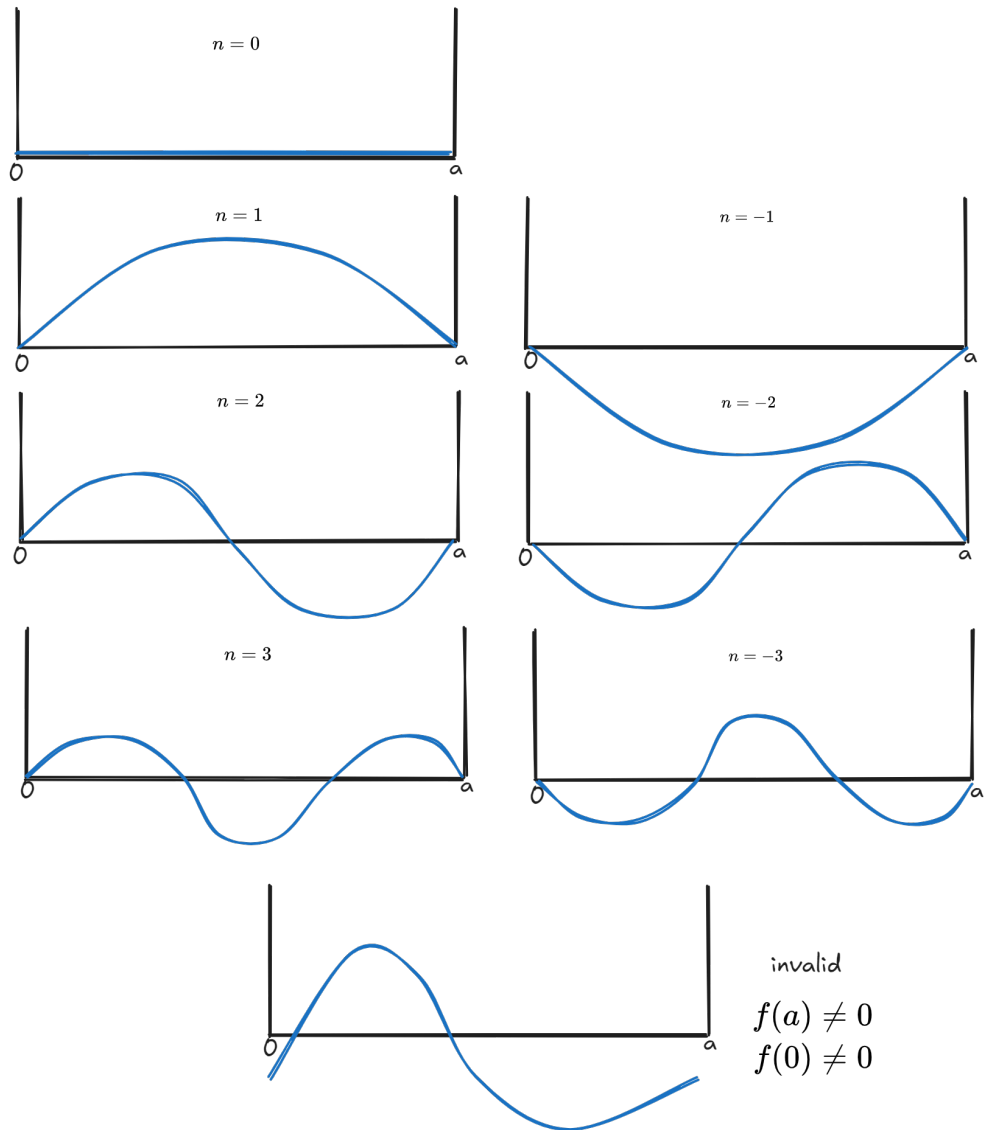
$$X(x) = d_2 \sin \left( \frac{\pi n}{a} x \right)$$

Considering the cases of  $n$  when  $\sin \left( \frac{\pi n}{a} x \right) = 0$ ,

$$\begin{cases} n = 0; \text{ not of interest, trivial solution} \\ n = 1, 2, 3, \dots; \text{ of interest} \\ n = -1, -2, -3 \dots; \text{ not of interest, see below} \end{cases}$$

The third case is neglected as  $\sin(-x) = -\sin x$ , just the negative of second case, whose negative is absorbed into the  $d_2$  constant.

The value of  $n$  can be physically represented.



**Figure 7.2**

## 7.6 Integrating $T(t)$

Now that we know the value of  $-k^2$  we can solve the other side

$$\begin{aligned}\frac{1}{v^2} \frac{d^2 T}{dt^2} &= -\frac{\pi^2 n^2}{a^2} T(t) \\ \frac{d^2 T}{dt^2} &= -\frac{\pi^2 n^2 v^2}{a^2} T(t) \\ \frac{d^2 T}{dt^2} &= -\underbrace{\left(\frac{\pi n v}{a}\right)^2}_{\omega_n^2} T(t) \\ \frac{d^2 T}{dt^2} &= -\omega_n^2 T(t)\end{aligned}$$

Note here that  $\omega_n = \omega_0 n$  and  $\omega_0 = \frac{\pi v}{a}$ .

This equation is solved similarly as before, but with  $-\omega^2$  being the constant of separation.

$$T(t) = e_1 \cos \omega_n t + e_2 \sin \omega_n t$$

Note that  $i$  is absorbed into the constant.

## 7.7 Combining Solutions

$$\begin{cases} X(t) = d_2 \sin kx \\ T(t) = e_1 \cos \omega_n t + e_2 \sin \omega_n t \\ k = \frac{\pi n}{a} \\ \omega_n = \frac{\pi n v}{a} \end{cases}$$

The full equation has the form

$$\begin{aligned}u(x, t) &= X(x)T(t) \\ &= (d_2 \sin kx)(e_1 \cos \omega_n t + e_2 \sin \omega_n t) \\ &= \underbrace{d_2 e_1}_{A_n} \sin kx \cos \omega_n t + \underbrace{d_2 e_2}_{B_n} \sin kx \sin \omega_n t \\ &= A_n \sin \left(\frac{\pi n}{a} x\right) \cos \left(\frac{\pi n v}{a} t\right) + B_n \sin \left(\frac{\pi n}{a} x\right) \sin \left(\frac{\pi n v}{a} t\right)\end{aligned}$$



This is a solution for arbitrary  $n$ , and arbitrary coefficients  $A_n, B_n$ . The full wave equation is defined by

$$u(x, t) = \sum_n A_n \sin\left(\frac{\pi n}{a}x\right) \cos\left(\frac{\pi n v}{a}t\right) + B_n \sin\left(\frac{\pi n}{a}x\right) \sin\left(\frac{\pi n v}{a}t\right)$$

We can discuss the meaning of the terms in this equation, for a vibrating string like on a violin. The lowest frequency  $\omega_1 = \frac{\pi v}{a}$  determines the pitch of the note. It depends on the length of the string (the position of a finger on the snare). It also depends on properties of the snare (thickness, which the pitch and  $v$ ). The other terms oscillating with frequency  $\omega_n = n\omega_1$  are called the overtones. Their mixing determines the sound we hear, distinguishing a guitar, cello, or trumpet from a violin. For a voice it determines the vowel sound (a, e, i, u, o, eu, ui, euw and so on). The initial shape of the wave (the coefficients  $A_n$ , or speed,  $B_n$ ) determine the attack of a note (the consonants of the voice). In this case, our wave equation is simple and undamped, which means the waves will oscillate forever and never dissipate.

The math we did here is relevant for musical strings, but also for quantum mechanics!

## 8 Introduction to Statistics

QM is a statistical theory, with the results only being understood with repeated experiments. We can't predict the position of an individual electron, or the outcome of an individual experiment, even in principle.

The outcome can be predicted by the product of possible values and the probability of that outcome.

### 8.1 Discrete Example: Die Throw

Consider a die throw, with 6 possible outcomes ( $a_i$ ): 1, 2, 3, 4, 5, 6. We track the results over 6000 separate dice throws.

**Table 8.1**

| Outcome ( $a_i$ ) | # Throws ( $n_i$ ) |
|-------------------|--------------------|
| 1                 | 1010               |
| 2                 | 980                |
| 3                 | 995                |
| 4                 | 2025               |
| 5                 | 1030               |
| 6                 | 960                |

The total is calculated by

$$n_{\text{tot}} = \sum_i n_i = 6000$$

### 8.1.1 Fraction

We also have a fraction  $f_i$  for each outcome, which can be calculated. Additionally the fraction of all possibilities should equal 1, which is fairly intuitive.

$$\begin{aligned}f_i &= \frac{n_i}{n_{\text{tot}}} \\ \sum_i f_i &= 1 \\ \sum_i \frac{n_i}{n_{\text{tot}}} &= \frac{1}{n_{\text{tot}}} \sum_i n_i = \frac{n_{\text{tot}}}{n_{\text{tot}}} = 1\end{aligned}$$

### 8.1.2 Probability

We can calculate the individual probability,  $P_i$ , based on the number of each outcome. This result is only meaningful over many *repeated experiments*.

$$P_i = \frac{n_i}{n_{\text{tot}}}$$

For each outcome, we have a *theoretical outcome* as well. For a die throw, for each result the probability should be  $\frac{1}{6}$ . With repeated experiments, the probability should converge to this number.

$$P_{i,\text{theo}} = P_i = \frac{1}{6}$$

The probability over all outcomes should be equal to 1, as in

$$\sum_{\text{all}} P_i = 1$$

### 8.1.3 Average

From this probability information we can calculate the average outcome, which *doesn't need to be a possible outcome* (i.e. for the dice, the average outcome doesn't need to be a whole number)

$$\begin{aligned}\langle A \rangle &= \bar{A} = \sum_i f_i a_i \\ &= \frac{1}{n_{\text{tot}}} \sum_i n_i a_i \\ &\approx \sum_i P_i a_i\end{aligned}$$

For the dice, the theoretical average is

$$\begin{aligned}
 \bar{A} &= \sum_i \frac{1}{6} a_i \\
 &= \frac{1}{6} \sum_i a_i \\
 &= \frac{1}{6} (1 + 2 + 3 + 4 + 5 + 6) \\
 &= \frac{21}{6} \\
 &= 3\frac{1}{2}
 \end{aligned}$$

### 8.1.4 Variance/Standard Deviation

The variance is denoted as  $\sigma^2$ . We take the positive root and to convert it to a more useful form, with  $\sigma$  standard deviation.

$$\sigma^2 = \sum_i P_i (a_i - \bar{A})^2 \geq 0$$

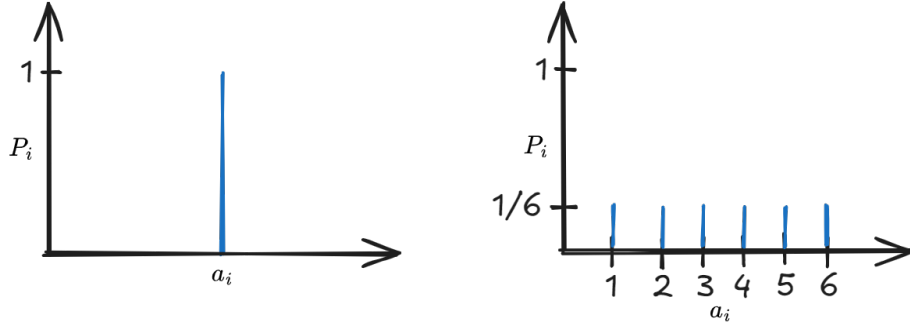
The standard deviation can be calculated manually (consider a 5-sided die for simplicity)

$$\begin{aligned}
 \bar{A} &= 3 \\
 \sigma^2 &= \frac{1}{5} \left( (1-3)^2 + (2-3)^2 + (3-3)^2 + (4-3)^2 + (5-3)^2 \right) \\
 &= \frac{1}{5} (4 + 1 + 0 + 1 + 4) \\
 &= 2 \\
 \implies \sigma &= \sqrt{2}
 \end{aligned}$$

#### 8.1.4.1 $\sigma^2 = 0$

If the variance was equal to 0, what would that mean? It implies that

$$\begin{aligned}
 a_i &= a_1 = a_2 = a_3 = \dots \\
 \implies \bar{A} &= a_i \\
 \implies P_i &= 1
 \end{aligned}$$



**Figure 8.1**

If real data would have a variance equal to 0, as it would imply outcome is always the same. In quantum mechanics this can happen.

#### 8.1.4.2 Rewriting Variance

$$\begin{aligned}
 \sigma^2 &= \sum_i P_i (a_i - \bar{A})^2 \\
 &= \sum_i P_i (a_i^2 - 2a_i \bar{A} + \bar{A}^2) \\
 &= \sum_i P_i a_i^2 - 2 \sum_i P_i a_i \bar{A} + \sum_i P_i \bar{A}^2 \\
 &= \underbrace{\sum_i P_i a_i^2}_{\bar{A}^2} - 2 \underbrace{\bar{A} \sum_i P_i a_i}_{\bar{A}} + \underbrace{\bar{A}^2 \sum_i P_i}_1 \\
 &= \bar{A}^2 - 2\bar{A}^2 + \bar{A}^2 \\
 &= \langle A^2 \rangle - \langle A \rangle^2
 \end{aligned}$$

#### 8.1.4.2.1 Die Example

This is a more convenient way to calculate variance/standard deviation. For example (with the die from earlier)

$$\begin{aligned}\sigma^2 &= \langle A^2 \rangle - \langle A \rangle^2 \\ &= \frac{1}{5}(1^2 + 2^2 + 3^2 + 4^2 + 5^2) - \left( \frac{1}{5}(1 + 2 + 3 + 4 + 5) \right)^2 \\ &= \frac{1}{5}(1 + 4 + 9 + 16 + 25) - 3^2 \\ &= 11 - 9 \\ &= 2\end{aligned}$$

## 8.2 Continuous Example: Mass Density

Consider a continuous distribution, for example of density of mass between two points. Let  $\rho$  be the density,  $x$  the position, and  $m$  the mass.

$$\rho(x) dx = dm$$

(the mass between  $x$  and  $x + dx$ , in 1D)

$$\begin{aligned}\int_{-\infty}^{\infty} \rho(x) dx &= M \quad (\text{total mass}) \\ \int_a^b \rho(x) dx &= m \quad (\text{mass between points } a \text{ and } b)\end{aligned}$$

This can now be normalized, to fit a probability distribution

$$P(x) dx = \frac{1}{M} \rho(x) dx$$

(probability to find a fraction of mass between  $x$  and  $x + dx$ )

### 8.2.1 Moments of the Distribution

The average position, also known as the **first moment** in this distribution is defined as

$$\langle x \rangle = \int_{-\infty}^{\infty} x P(x) dx$$

Similarly, the second moment is

$$\langle x^2 \rangle = \int_{-\infty}^{\infty} x^2 P(x) dx$$

One could also calculate the variance and mean of some function  $f(x)$ , rather than  $x$

### 8.2.2 Example

Consider a box, with positions between 0 and  $a$ . Its probability distribution is

$$P(x) = \begin{cases} \frac{1}{a} & 0 \leq x \leq a \\ 0 & \text{elsewhere} \end{cases}$$

$$\begin{aligned} \int_{-\infty}^{\infty} P(x) dx &= \int_0^a \frac{1}{a} dx = \frac{x}{a} \Big|_0^a = 1 \quad (\text{normalized}) \\ \int_{-\infty}^{\infty} x P(x) dx &= \int_0^a x \frac{1}{a} dx = \frac{x^2}{2a} \Big|_0^a = \frac{a}{2} \\ \int_{-\infty}^{\infty} x^2 P(x) dx &= \int_0^a x^2 \frac{1}{a} dx = \frac{x^3}{3a} \Big|_0^a = \frac{a^2}{3} \\ \sigma^2 &= \langle x^2 \rangle - \langle x \rangle^2 = \frac{a^2}{3} - \left(\frac{a}{2}\right)^2 \\ &= \frac{a^2}{12} \geq 0 \end{aligned}$$

## 8.3 Discrete vs. Continuous

Table 8.2

| Discrete                                               | Continuous                                             |
|--------------------------------------------------------|--------------------------------------------------------|
| $i$                                                    | $x$                                                    |
| $a_i$                                                  | $f(x)$                                                 |
| $P_i$                                                  | $P(x) dx$                                              |
| $\langle A \rangle = \sum_i P_i a_i$                   | $\langle f \rangle = \int f(x) P(x) dx$                |
| $\langle A^2 \rangle = \sum_i P_i a_i^2$               | $\langle f^2 \rangle = \int f(x)^2 P(x) dx$            |
| $\sigma^2 = \langle A^2 \rangle - \langle A \rangle^2$ | $\sigma^2 = \langle f^2 \rangle - \langle f \rangle^2$ |

## 9 A Plausible Argument to Derive the Schrödinger Equation

### 9.1 Use of de Broglie and Classical Wave Equation

In this section we will use a somewhat plausible argument to derive the Schrödinger equation from a consideration of the wave equation, and the de Broglie relation. This is not a rigorous derivation, but it gives some sense of where it comes from.

De Broglie's work lead to the equation  $\lambda = \frac{h}{p} \implies p = \frac{h}{\lambda}$ . His work lead to the understanding that

- particles, such as electrons, have wavelength and consequently wavelike behaviour
- waves (such as light) have momentum, even without having mass, generalizing the earlier conception of momentum  $p = mv$  to waves.

We used the wave equation earlier (known from classical wave theory), which started from the differential equation

$$\frac{\partial^2 u}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 u}{\partial t^2}$$

It had the general solution

$$u(x, t) = \sum_n A_n \sin(k_n x) \cos(\omega_n t) + B_n \sin(k_n x) \sin(\omega_n t)$$

To simplify the analysis we will consider one particular term of the general solution, which gives a particular solution  $u(x, t)$  we will work with, calling the spatial term  $\psi(x)$ . We use that the wave is periodic in  $t$ ,

$$u(x, t) = \psi(x) \cos(\omega t)$$

Again, this is not a mathematical proof, so choosing this solution is somewhat arbitrary, but it will demonstrate the idea and plausibility well.



## 9.2 Derivation of Expression for Momentum From Wave Equation and de Broglie

We start by taking the 2<sup>nd</sup> derivative of our specific solution, both with respect to position  $x$  and time  $t$ .

$$\begin{aligned}\frac{\partial^2}{\partial x^2} \left( \psi(x) \cos(\omega t) \right) &= \frac{\partial}{\partial x} \left( \frac{\partial \psi}{\partial x} \cos(\omega t) \right) \\ &= \frac{\partial^2 \psi}{\partial x^2} \cos(\omega t) \\ \frac{\partial^2}{\partial t^2} \left( \psi(x) \cos(\omega t) \right) &= \frac{\partial}{\partial t} \left( -\psi(x) \omega \sin(\omega t) \right) \\ &= -\psi(x) \omega^2 \cos(\omega t)\end{aligned}$$

We now substitute these back into  $\frac{\partial^2 u}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 u}{\partial t^2}$

$$\frac{\partial^2 \psi}{\partial x^2} \cos(\omega t) = -\frac{1}{v^2} \psi(x) \omega^2 \cos(\omega t)$$

Now dividing both sides by  $\cos(\omega t)$ , we get

$$\begin{aligned}\frac{\partial^2 \psi}{\partial x^2} &= -\frac{1}{v^2} \psi(x) \omega^2 \\ &= -\left(\frac{\omega}{v}\right)^2 \psi(x) & (\omega = 2\pi\nu) \\ &= -\left(\frac{2\pi\nu}{v}\right)^2 \psi(x) & (v = \nu\lambda) \\ &= -\left(\frac{2\pi}{\lambda}\right)^2 \psi(x) & \left(\lambda = \frac{h}{p}, \text{ de Broglie}\right) \\ &= -\left(\frac{2\pi p}{h}\right)^2 \psi(x) & \left(\hbar = \frac{h}{2\pi}\right) \\ \frac{\partial^2 \psi}{\partial x^2} &= -\hbar^{-2} p^2 \psi(x) \\ -\hbar^2 \frac{\partial^2 \psi}{\partial x^2} &= p^2 \psi(x)\end{aligned}$$

This is our first partial result:

$$p^2 \psi(x) = -\hbar^2 \frac{\partial^2 \psi}{\partial x^2}$$

## 9.3 From Classical Mechanics to the Schrödinger Equation

In classical mechanics, we have the total energy formula

$$E_{\text{tot}} = E_{\text{kin}} + E_{\text{pot}} \iff E = T + V$$

Kinetic energy ( $T = \frac{1}{2}mv^2$ ) can be expressed in terms of momentum ( $p = mv \implies p^2 = m^2v^2$ ), or

$$T = \frac{p^2}{2m}$$

Now our kinetic energy can be subbed in, and with the potential energy denoted  $V(x)$  (with respect to the position  $x$ ), we get

$$\begin{aligned} E &= \frac{p^2}{2m} + V(x) \\ \frac{p^2}{2m} &= E - V(x) \\ p^2 &= 2m(E - V(x)) \end{aligned}$$

Now we make a big jump, and say this equation is valid if we act on a wavefunction

$$p^2\psi(x) = 2m(E - V(x))\psi(x)$$

We can now substitute the momentum term derived before into  $-\hbar^2 \frac{\partial^2 \psi}{\partial x^2} = p^2\psi(x)$ , to get

$$\begin{aligned} -\hbar^2 \frac{\partial^2 \psi}{\partial x^2} &= -2m(E - V(x))\psi(x) \\ -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} &= (E - V(x))\psi(x) \\ -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} &= E\psi(x) - V(x)\psi(x) \\ -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V(x)\psi(x) &= E\psi(x) \end{aligned}$$

Now we interpret the left hand side as an operator acting on  $\psi(x)$ . It is called the Hamiltonian operator.

$$\hat{H} = -\frac{\hbar}{2m} \frac{d^2}{dx^2} + V(x)$$

We can then simplify the above equation to get the classical Schrödinger equation

$$\begin{aligned} -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V(x)\psi(x) &= E\psi(x) \\ \implies \hat{H}\psi(x) &= E\psi(x) \end{aligned}$$

Let us emphasize again that this is a hand-waving argument. In fact, the Schrödinger equation cannot be rigorously derived. We have to assume its validity (a postulate). It does lead to an important mathematical concept, the notion of an operator. This we will discuss next.

## 9.4 Operators

An operator is something that takes a function as input and outputs another function, similar in concept<sup>1</sup> to a function taking a value as input and outputting another value. We denote operators with hats, as in

$$\hat{A}f(x) = g(x)$$

### 9.4.1 Previous Operators

We have seen operators before, in the form of a matrix, such as

$$\begin{bmatrix} 3 & 2 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3x_1 + 2x_2 \\ 4x_1 + x_2 \end{bmatrix}$$

where  $(x_1, x_2) \mapsto (3x_1 + 2x_2, 4x_1 + x_2)$ . This represents a transformation  $\mathbb{R}^2 \rightarrow \mathbb{R}^2$ . In linear form we have the formula  $A\vec{x} = \vec{b}$ . In this example, we can see that  $A$  is our operator,  $\vec{x}$  is our function (/vector) and  $\vec{b}$  is our output function (again, /vector).

### 9.4.2 Operator Examples

Using the format  $\hat{A}f(x) = g(x)$ , below is a list of example operators and their action on a function.

---

<sup>1</sup>And in fact, to a mathematician, operators are just a special type of function whose inputs are other functions.

**Table 9.1**

| $\hat{A}$                              | $f(x)$    | $g(x)$                                   |
|----------------------------------------|-----------|------------------------------------------|
| $\frac{d^2}{dx^2} + 2\frac{d}{dx} + 3$ | $x^3$     | $6x + 6x^2 + 3x^3$                       |
| $x\frac{d}{dx}$                        | $x^2$     | $2x^3$                                   |
| $\frac{d}{dx}x$                        | $x^2$     | $3x^2$                                   |
| $-i\hbar\frac{d}{dx}$                  | $e^{ikx}$ | $(-i\hbar)(ik)e^{ikx} = \hbar k e^{ikx}$ |

Working explicitly through the first example we have

$$\begin{aligned}
 \hat{A} &= \frac{d^2}{dx^2} + 2\frac{d}{dx} + 3 \\
 f(x) &= x^3 \\
 \hat{A}f(x) &= \left( \frac{d^2}{dx^2} + 2\frac{d}{dx} + 3 \right) x^3 \\
 &= \frac{d^2}{dx^2}(x^3) + 2\left( \frac{d}{dx}(x^3) \right) + 3x^3 \\
 &= \frac{d}{dx}(3x^2) + 2(3x^2) + 3x^3 \\
 &= 6x + 6x^2 + 3x^3
 \end{aligned}$$

We can also make compound operators, or products of operators, e.g.

$$\begin{aligned}
 \hat{A} &= x, \hat{B} = \frac{d}{dx} \\
 \hat{A}\hat{B} &= x\frac{d}{dx}
 \end{aligned}$$

and

$$\begin{aligned}
 \hat{A}\hat{B}f(x) &= \hat{A}(\hat{B}f(x)) \rightarrow \hat{A}g(x) \\
 x\frac{d}{dx}f(x) &= x\left( \frac{d}{dx}f(x) \right) = x\frac{df}{dx}
 \end{aligned}$$

In general the order of operators matters and  $\hat{A}\hat{B} \neq \hat{B}\hat{A}$ . In the above example,

$$\hat{B}\hat{A}f(x) = \frac{d}{dx}xf(x) = \frac{d}{dx}(xf(x)) = 1 \cdot f(x) + x\frac{df}{dx} = f(x) + x\frac{df}{dx}$$

It is said that operators are (generally) not commutative.<sup>2</sup> Our example shows  $x \frac{d}{dx} \neq \frac{d}{dx} x$ , by acting on a general function  $f(x)$ .

In quantum mechanics we always need to pay careful attention to the order of operators. When writing operators in a given order, they act from the inside to the outside, so the rightmost operator is the one acting on the input function first. For example, in the case of the  $x \frac{d}{dx}$  operator, we take the derivative first. You can always remember this by bracketing carefully:

$$x \frac{d}{dx}(f(x)) = x \left( \frac{d}{dx}(f(x)) \right)$$

### 9.4.3 Linear Operators

An operator is linear if

$$\hat{A}(c_1 f(x) + c_2 g(x)) = c_1 \hat{A}f(x) + c_2 \hat{A}g(x)$$

and

$$\hat{A}(c_1 f(x)) = c_1 \hat{A}f(x)$$

for all constants  $c_1, c_2$ , and functions  $f(x), g(x)$ .

The operators shown above were all linear (including  $x \frac{d}{dx}$ ). As an example of a non-linear operator consider  $\hat{A}(f(x)) = f(x) + 3$ , this is not linear. We work through this example

$$\hat{A}(c_1 h(x) + c_2 g(x)) = c_1 h(x) + c_2 g(x) + 3$$

$$c_1 \hat{A}h(x) + c_2 \hat{A}g(x) = c_1 f(x) + 3 + c_2 g(x) + 3 = c_1 h(x) + c_2 g(x) + 6$$

$$c_1 h(x) + c_2 g(x) + 3 \neq c_1 h(x) + c_2 g(x) + 6$$

Therefore, we see  $\hat{A}$  is not linear because  $\hat{A}(c_1 f(x) + c_2 g(x)) \neq c_1 \hat{A}f(x) + c_2 \hat{A}g(x)$ .

Note here that when saying  $\hat{A}(f(x)) = f(x) + 3$ , that  $f(x)$  is our entire input. It is a general definition for any function  $f(x)$ . Don't be confused and think that refers to a specific function. Therefore, with an input of  $c_1 h(x) + c_2 g(x)$ , the result is just  $(c_1 h(x) + c_2 h(x)) + 3$ .

---

<sup>2</sup>Commutative means that order *doesn't* matter. For example, addition and multiplication are commutative, as in  $2 + 3 = 3 + 2$  or  $2 \times 3 = 3 \times 2$ . Subtraction and division aren't, as in  $2 - 3 \neq 3 - 2$  or  $2/3 \neq 3/2$ . We say something isn't commutative if it's not *generally* so, obviously there are special subtraction/division operations we could devise where the order doesn't matter.

### 9.4.4 Commutator

Before we saw that the order of operators matters in general. However in quantum mechanics there are important pairs of operators in which the order does not matter.

Say we have two operators  $\hat{A}$  and  $\hat{B}$ , they can be used in sequence

$$\hat{A}\hat{B}f(x) = \hat{A}(\hat{B}f(x))$$

$$\hat{B}\hat{A}f(x) = \hat{B}(\hat{A}f(x))$$

$$\text{but generally } \hat{A}\hat{B}f(x) \neq \hat{B}\hat{A}f(x)$$

We can define the commutator of two operators as

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$$

Two operators commute if and only if the commutator is equal to 0, as

$$[\hat{A}, \hat{B}] = 0$$

To evaluate the commutator one should always act on a generic function  $f(x)$ . Hence if operators commute, for any function  $f(x)$  it should be true that

$$[\hat{A}, \hat{B}]f(x) = \hat{A}(\hat{B}f(x)) - \hat{B}(\hat{A}f(x)) = 0$$

The commutator expression can be converted into one operator, as in

$$[\hat{A}, \hat{B}]f(x) = (\hat{A}\hat{B} - \hat{B}\hat{A})f(x)$$

$$= \hat{C}f(x)$$

$$[\hat{A}, \hat{B}] = \hat{C}$$

$$\hat{C} \begin{cases} = 0 & \hat{A}, \hat{B} \text{ are commutative} \\ \neq 0 & \hat{A}, \hat{B} \text{ are not commutative} \end{cases}$$

#### 9.4.4.1 Commutator Example

For example, consider the operators  $[x, \frac{d}{dx}]$  and the resulting commutator

$$\begin{aligned}\left(x \frac{d}{dx} - \frac{d}{dx} x\right) f(x) &= x \frac{d}{dx} (f(x)) - \frac{d}{dx} x (f(x)) \\ &= x \frac{df}{dx} - \left(f(x) + x \frac{df}{dx}\right) \\ &= -f(x) \\ \Rightarrow \left[x, \frac{d}{dx}\right] &= -1\end{aligned}$$

## 9.5 Eigenvalue Equations

If we have some function  $f(x)$  and some operator  $\hat{A}$ , the function is an **eigenfunction** if the result  $\hat{A}f(x)$  is a scalar multiple of the original function.

$$\begin{aligned}\hat{A}f(x) &= af(x) \\ (a \text{ must not depend on } x)\end{aligned}$$

**Eigen** is a German word that means “belongs to”. In this case we have an eigenfunction, where  $a$  is the **eigenvalue** which belongs to that specific function to form an **eigenpair**.

### Understanding operators, from a linear algebra perspective



We can use a linear algebra understanding to understand eigenfunctions. If you are not well acquainted in linear algebra, treat this as a metaphor. If you are familiar with linear algebra, this metaphor is in fact not a metaphor, and just a direct application of one of the best fields of math to quantum mechanics.

An eigenfunction is just an eigenvector that also happens to be a function, but we are now working in a domain of functions instead of traditional vectors.

An eigenvector is any vector, that when transformed by a linear transformation, retains its original direction but is now scaled by a constant scalar coefficient, with this coefficient being the paired eigenvalue.

For example, the linear transformation (represented by the matrix) of a 2D vector (with components  $x_1, x_2$ )

$$\begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

has eigenvectors  $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$  with eigenvalue 3 and  $\begin{bmatrix} -1 \\ 0 \end{bmatrix}$  with eigenvalue 2.

A note here, the vector space this transformation is operating in has limited dimensions (2), and therefore there are limited eigenvalues. Not so for the vector space we choose to work with in quantum mechanics. The number of different eigenfunctions and corresponding eigenvalues is often infinite. For simple examples we can still characterize them compactly.

We could also represent this transformation using operator notation, with the same linear transformation being represented by linear operator  $\hat{K}$  and the same vector represented by function  $h(x)$  to write

$$\hat{K}h(x)$$

to display the *same* mathematical operation.

With this, we see that a linear operator is akin to a linear transformation<sup>a</sup> and that two different notations can represent the same math.

A note here, the vector space this transformation is operating in has limited dimensions (2), and therefore there are limited eigenvalues. The vector spaces we work with in quantum mechanics are all infinite dimensional, and so a linear operator may have infinitely many eigenvalues.

---

<sup>a</sup>In fact, linear operators are a type of linear transformation!

## 9.6 Example

We will consider an example which has great relevance to quantum mechanics. We have the operator

$$\hat{A} = -i\hbar \frac{d}{dx}$$



If we apply this to the specific function

$$f(x) = e^{ikx}$$

We get

$$\begin{aligned}\hat{A}f(x) &= -i\hbar \frac{d}{dx}(e^{ikx}) \\ &= (-i\hbar)(ik e^{ikx}) \\ &= \hbar k e^{ikx}\end{aligned}$$

The original function was an eigenfunction, because the result of operation was just a scalar multiple of the original function where

$$\hat{A}f(x) = \underbrace{(\hbar k)}_{\text{eigenvalue}} f(x)$$

This operator is called the momentum operator. Let's see why by taking the eigenvalue

$$\begin{aligned}\hbar k &= \frac{h}{2\pi} \frac{2\pi}{\lambda} \\ &= \frac{h}{\lambda} \\ &= p_x \quad (1D \text{ wave}) \\ \implies \hat{A}f(x) &= p_x f(x)\end{aligned}$$

Therefore, the function  $e^{ikx}$  is an eigenfunction of the operator, designated  $\hat{p}_x$  for clarity.

$$\begin{aligned}\hat{p}_x f(x) &= p_x f(x) \\ \hat{p}_x e^{ikx} &= p_x e^{ikx}, p_x = \hbar k\end{aligned}$$

### 9.6.1 Counterexample

What if we were to take some other function? What effect would the operator have on it? Take the function  $g(x) = e^{-\lambda x^2}$  where  $\lambda$  is some constant.

$$\begin{aligned}\hat{A}g(x) &= i\hbar \frac{d}{dx} (e^{-\lambda x^2}) \\ &= i\hbar (-2\lambda x e^{-\lambda x^2}) \\ &= -2\lambda i\hbar x e^{-\lambda x^2}\end{aligned}$$

We see that the resulting function is not a scalar multiple of the original, as the coefficient depends on  $x$ . Therefore, this function is not an eigenfunction of the operator and there is likewise no associated eigenvalue. Eigenfunctions are very special functions and they depend on the operator. In general there can be many different eigenfunctions and eigenvalues. In this case  $e^{ikx}$  is an eigenfunction for every value of  $k$ , even complex  $k$ .

### 9.6.2 Kinetic Energy Operator

If we use the relationship between momentum and kinetic energy, we get

$$\begin{aligned}T &= \frac{p_x^2}{2m} \\ \hat{p}_x e^{ikx} &= p_x e^{ikx} \\ \implies p_x \hat{p}_x &= i\hbar \frac{d}{dx} \\ \text{Let } T(\hat{p}_x) &= \hat{T}_x \\ &= \frac{\hat{p}_x^2}{2m} \\ &= \frac{\left(-i\hbar \frac{d}{dx}\right)^2}{2m} \\ &= -\frac{\hbar^2}{2m} \frac{d^2}{dx^2}\end{aligned}$$

We have argued that the kinetic energy operator  $\hat{T}_x$  is represented by

$$\hat{T}_x = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2}$$

We can show that  $e^{ikx}$  is also an eigenfunction of  $\hat{T}$

$$\hat{T}e^{ikx} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2}(e^{ikx}) = -\frac{\hbar^2}{2m}(ik)(ik)e^{ikx} = +\frac{\hbar^2 k^2}{2m}e^{ikx}$$

This eigenvalue is the same as  $\frac{p_x^2}{2m}$ . This shows the connection between the quantum mechanical kinetic energy and the classical expression. In quantum mechanics we act on an eigenfunction to get the classical counterpart.

We can now also consider the operator corresponding to the total classical energy and compare it to the corresponding operator.

$$E = T + V(x)$$

$$\hat{H} = \hat{T} + \hat{V} \hat{H}\psi(x) = \left[ -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x) \right] \psi(x)$$

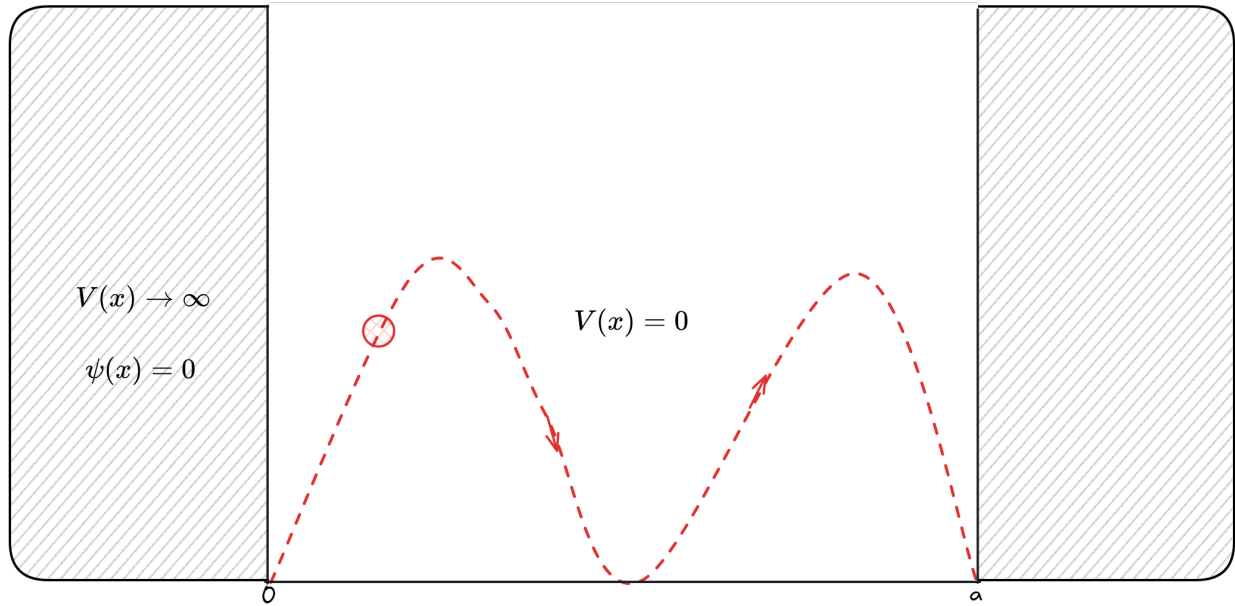
The operator  $\hat{V}$  acting on  $\psi(x)$  just multiplies the function by  $V(x)$ .

If we solve the eigenvalue equation  $\hat{H}\psi(x) = E\psi(x)$ , we get the eigenvalue which we can associate with the energy of the system. The eigenfunctions are in general hard to find and depend on the form of the potential. The procedure is analogous however, to what we did for  $\hat{p}_x = i\hbar \frac{d}{dx}$ , with eigenvalue  $p_x = \hbar k$ , or the kinetic energy with eigenvalue  $\frac{\hbar^2 k^2}{2m} = \frac{p_x^2}{2m}$ .

$\hat{H} = \hat{T} + \hat{V}$ . For the particle in the box (and some other models), we assume  $\hat{V} = 0$  therefore  $\hat{H} = \hat{T}$ . We will discuss this problem next.

## 10 Particle in a Box

One of the “simplest”<sup>1</sup> models in quantum mechanics is the particle in a box in one dimension.



**Figure 10.1**

The position of the particle is described by  $x$ , with the box having bounds  $0 \leq x \leq a$ . The potential energy is described by

$$V(x) = \begin{cases} 0, & 0 \leq x \leq a & \text{(inside box)} \\ \infty, & \text{elsewhere} & \text{(outside box)} \end{cases}$$

$$\psi(x) = \begin{cases} 0, & \text{(outside box)} \\ \text{continuous, everywhere} \end{cases}$$

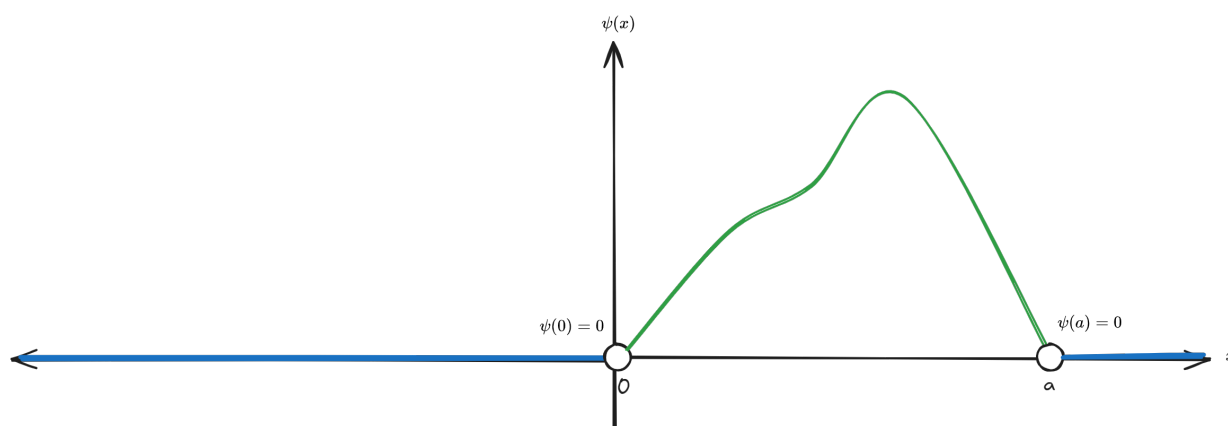
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<sup>1</sup>No model in quantum mechanics is simple though ☺

In order for the function to remain continuous, we must set  $\psi(0) = 0$  and  $\psi(a) = 0$ .

$$\begin{aligned}\psi(0) &= \underbrace{\lim_{x \rightarrow 0} \psi(x)}_{:= \text{continuity}} = \underbrace{\lim_{x \rightarrow 0^-} \psi(x)}_{\text{limit from 1 side}} = 0 \\ \psi(a) &= \lim_{x \rightarrow a} \psi(x) = \lim_{x \rightarrow a^+} \psi(x) = 0\end{aligned}$$

The requirement for  $\psi(x) = 0$  outside the box comes from the fact that we want finite descriptions of energy. So when  $V(x) \rightarrow \infty$  outside the box we need to cancel out that term in the equation  $T\psi(x) + V(x)\psi(x)$ , so  $\psi(x)$  must be 0.



**Figure 10.2:** Visually it is easy to see that  $\psi(0 \text{ or } a) = 0$ , in order for the function to remain continuous.

### Boundary conditions on $\psi(x)$



We have boundary conditions such that  $\psi(0 \text{ or } a) = 0$ . Why? It has to do with the continuity and structure of the function.

The  $V(x)$  term drastically changes, we have it equal to 0 inside the box and suddenly equal to  $\infty$  outside the box. This is not meaningful physics, it is a mathematical model. The wavefunction  $\psi(x)$  has to be 0 when  $V(x)$  is infinite, but we still want the wavefunction to be continuous at the boundary of the box (because continuous functions are useful). It is not differentiable there, because it forms a cusp at the boundary points between being constantly zero outside and some interesting function inside (see Figure 10.2). Inside the box, the wavefunction  $\psi(x)$  must at least be twice-differentiable, as we need to take two derivatives for it to obey the Schrödinger equation! This drastic setup impacts the function greatly, as  $V(x)$  must explode rapidly to  $\infty$  and decay

rapidly to 0. The result is that the wavefunction  $\psi(x)$  is continuous overall, and is twice differentiable on the interior of the box but not at its boundary points.

This discussion is an example of the mathematical phenomenon of **regularity**, which is a measure of how nice/smooth a given function is. Our drastic boundary conditions force our wavefunction's regularity to be as low as possible while still being high enough to take second derivatives on the interior of the box.

One can write down a (partial) list of possible regularities, starting from low regularity and going towards higher regularity:

- *continuous*
  - (our function if we must include the oddity of our boundary points)
- differentiable
- differentiable with continuous derivative
- twice differentiable
  - (our function on the interior of the box)
- twice differentiable with continuous second derivative
- ...
- smooth (infinitely differentiable)
- analytic (convergent Taylor series)

## 10.1 Operators

We focus our work on the inside of the box, which is described by the kinetic energy operator and the potential energy, as in

$$\begin{aligned}
 \left( -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + \underbrace{V(x)}_{\text{inside box}} \right) \psi(x) &= E\psi(x) \\
 -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} (\psi(x)) &= E\psi(x) \\
 -\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} &= \underbrace{E}_{\text{constant}} \psi(x) \\
 \frac{d^2\psi}{dx^2} &= -\frac{2Em}{\hbar^2} \psi(x)
 \end{aligned}$$

This result is just a variation of the vibrating string wave equation ( $\frac{d^2f}{dx^2} = cf(x)$ ). The general solution (for the spatial portion) was

$$\psi(x) = c_1 \sin(kx) + c_2 \cos(kx)$$

## 10.2 Boundary Conditions

We have the boundary conditions  $\psi(0) = \psi(a) = 0$ , with a similar process previously seen when deriving a general wave equation.

$$\begin{aligned}
 x = 0 &\implies \psi(0) = 0 \implies c_2 = 0 \implies \psi(x) = c_1 \sin(kx) \\
 x = a &\implies \psi(a) = c_1 \sin(ka) = 0 \implies k = \frac{\pi n}{a}, n = 1, 2, 3, \dots, c_1 \neq 0
 \end{aligned}$$

Let us note that also functions  $\psi(x) = d_1 e^{\lambda x} + d_2 e^{-\lambda x}$ , with  $\lambda$  real would satisfy the Schrödinger equation. However the eigenvalue would be negative, which would be funny for kinetic energy (even in quantum mechanics). The better mathematical reason is that such functions can never satisfy both boundary conditions, and therefore they are not valid solutions. The boundary conditions lead to the sine wave solutions, and they lead to the quantization of energy,  $n = 1, 2, 3, 4, \dots$

## 10.3 Eigenfunction or Not?

From here we have the function  $c_1 \sin\left(\frac{\pi nx}{a}\right)$ , but is it an eigenfunction of operator  $\hat{T}_x$ ? Let's see

$$\begin{aligned} -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \left( \sin\left(\frac{\pi nx}{a}\right) \right) &= -\frac{\hbar^2}{2m} \frac{d}{dx} \left( \left(\frac{\pi n}{a}\right) \cos\left(\frac{\pi nx}{a}\right) \right) \\ &= -\frac{\hbar^2}{2m} \left( -\left(\frac{\pi n}{a}\right)^2 \sin\left(\frac{\pi nx}{a}\right) \right) \\ &= \frac{\hbar^2}{2m} \left(\frac{\pi n}{a}\right)^2 \sin\left(\frac{\pi nx}{a}\right) \\ &= \underbrace{\frac{\hbar^2 \pi^2 n^2}{2ma^2}}_{\text{eigenvalue}} \sin\left(\frac{\pi nx}{a}\right) \end{aligned}$$

## 10.4 Further Discussion: Discrete Energy Levels

The Schrödinger equation is described by

$$\hat{H}\psi_n(x) = E_n\psi(x)$$

where the wavefunction,  $\psi_n(x)$ , is eigenfunction of the Hamiltonian,  $\hat{H}$ . Importantly it is the *boundary conditions that allow for discrete solutions*.

### How infinity can be discrete



For a linear transformation, for example  $\mathbb{R}^3 \rightarrow \mathbb{R}^3$ , we have a limited number of eigenvalues, 3 in this example. This is based on the vector space we are working in. For the Hamiltonian, the vector space we are working in is infinite-dimensional, and therefore we have infinitely many eigenvalues. Importantly though it is not “continuously infinite” and we have discrete eigenvalues.

Compare the infinity of counting  $1, 2, 3, \dots$  and counting all the values between 1 and 2. For the second example we can count  $1.1, 1.2, 1.3, \dots$  but we can always go finer with  $1.01, 1.02, 1.03, \dots$  and  $1.001, 1.002, 1.003, \dots$  and further on. Even though both are infinite, the sizes of infinity are different: one is discrete, and the other is continuous.



The values of these discrete eigenvalues correspond to and mathematically represent our discrete energy levels. We have infinite energy levels,  $n = 1, 2, 3, \dots$  but those levels are not a continuum, energy is quantized, so mathematically it must be quantized as well. The inclusion of  $n$  or  $k$  in our solution allows for us to make sure our solution is discrete.

For the value of our discrete solutions we have

$$\begin{aligned}\psi_n(x) &= \sin\left(\frac{\pi n x}{a}\right) \\ E_n &= \frac{\hbar^2 \pi^2}{2ma^2} n^2 = \frac{\hbar^2}{8ma^2} n^2 \\ n &= 0, 1, 2, 3, \dots\end{aligned}$$

For calculating energy levels, the expression  $\frac{\hbar^2 \pi^2}{2ma^2}$  (or its alternate form with  $h$ ) is just some constant, so for simplicity we often calculate energy levels based on the unscaled  $n^2$ .

In the assignments and tutorial we will consider another problem, the particle on the ring. The wavefunction is now a function of the angle  $\varphi$ , and the Hamiltonian is given by

$$\hat{H} = -\frac{\hbar^2}{2m} \frac{d^2}{d\varphi^2}$$

and we will have periodic boundary conditions  $\psi(\varphi + 2\pi) = \psi(\varphi)$ . The solutions are

$$\begin{aligned}\psi_n(\varphi) &= e^{ik\varphi} \\ E_k &= \frac{\hbar^2 k^2}{2mR^2} \\ e^{ik\varphi} &= \cos(k\varphi) + i \sin(k\varphi) \\ k &= 0, \pm 1, \pm 2, \pm 3, \dots\end{aligned}$$

The solution for  $+k$  and  $-k$  have the same energies, and the general solution with energy  $E_k = \frac{\hbar^2 k^2}{2mR^2}$  can be expressed as

$$\begin{aligned}\psi(\varphi) &= c_1 e^{ik\varphi} + c_2 e^{-ik\varphi} \\ &= d_1 \cos(k\varphi) + d_2 \sin(k\varphi)\end{aligned}$$

We will be considering more examples.

# 11 Wavefunctions as Probability Distributions

## 11.1 Probability Distributions

### 11.1.1 Wave Functions as Probability Distributions

Wavefunctions are associated with a probability distribution. Let's call it  $P(x)$ . As a probability distribution there are some requirements

$$\int P(x) dx = 1$$
$$P(x) \geq 0$$

Wavefunctions are some complex function

$$\psi(x) = f(x) + ig(x)$$

so we have a conjugate  $\psi^*(x)$  as well

$$\psi^*(x) = f(x) - ig(x)$$

As a complex number the wavefunction cannot be interpreted physically. Fortunately, by taking the **absolute square** it allows us to make our complex function into something fully real. To calculate the absolute square, denoted  $|\psi(x)|^2$ , simply take the complex number or function and multiply it with its conjugate, as below

$$|\psi(x)|^2 = \psi^*(x)\psi(x)$$

This new squared function can be *interpreted like a probability distribution*. Therefore

$$|\psi(x)|^2 = P(x)$$
$$= \psi^*(x)\psi(x)$$

But why does taking the absolute square make our complex function real? Let's work it out

$$\begin{aligned}
 P(x) &= (f(x) + ig(x))(f(x) - ig(x)) \\
 &= (f(x))^2 + (g(x))^2 + \cancel{if(x)g(x)} - \cancel{ig(x)f(x)} \\
 &= (f(x))^2 + (g(x))^2 \\
 &\geq 0
 \end{aligned}$$

No more  $i$ , so everything is real!

Theoretically a wavefunction could come from anywhere, but normally it results from an **eigenvalue equation** (for example the Schrödinger equation with its Hamiltonian operator and energy eigenvalues) and that's what's most useful to us, anyway.

### 11.1.2 Constants of Normalization

A probability distribution is only useful to us when it's normalized (equal to 1). The absolute square of the wavefunction doesn't come normalized, in general

$$\int_{-\infty}^{\infty} \psi^*(x)\psi(x)dx \neq 1$$

This can be fixed, by taking a constant  $C$  and using it as a scalar coefficient. But, does the constant interfere with our eigenvalue equation? No. Let's work it out. Suppose that  $a$  is some eigenvalue of the eigenfunction  $\psi(x)$ .

$$\begin{aligned}
 \hat{A}(C\psi(x)) &= C(\hat{A}\psi(x)) \\
 &= C(a\psi(x)) = a(C\psi(x))
 \end{aligned}$$

Here we use that  $\hat{A}$  in quantum mechanics is a linear operator.

Note that the conjugate of any real number is just that number

$$\underbrace{(a + i0)^*}_{\mathbb{R}} = (a - i0) = a$$

Something to note is when normalizing a function, the *bounds of the integral can depend on the problem*. For example a wavefunction may depend on an angular coordinate  $\varphi$ , and then we would integrate over one full period, such as between 0 and  $2\pi$ . Examples will follow.

So we know our probability distribution integral should equal 1. We know having a

scalar coefficient on our wavefunction doesn't interfere with it being an eigenfunction of some operator. Let's start with inserting  $C$  into our absolute square integral.

$$\begin{aligned}
 \int_{-\infty}^{\infty} P(x) dx &= \int (C\psi(x))(C\psi(x))^* dx \\
 &= \int (C\psi(x))(C^*\psi^*(x)) dx \quad \text{taking out constants} \\
 &= CC^* \int \psi(x)\psi^*(x) dx \\
 C \text{ can be chosen to be real, so } C^* &= C, CC^* = CC = C^2 \\
 &= C^2 \underbrace{\int \psi(x)\psi^*(x) dx}_I
 \end{aligned}$$

In order for this to work, we have a list of requirements for  $I$

$$I \begin{cases} \neq 0 \\ \in \mathbb{R} \\ > 0 \\ \nrightarrow \infty \end{cases}$$

If the integral satisfies these conditions (not 0, not  $\infty$ ), we can define a new normalized wavefunction  $\tilde{\psi}(x) = \frac{1}{\sqrt{I}}\psi(x)$ . With this new normalized wavefunction we can define a probability distribution

$$P(x) = \tilde{\psi}^*(x)\tilde{\psi}(x)$$

that satisfies the correct requirements:

$$\int P(x) dx = 1$$

$P(x)$  is real everywhere and  $P(x) \geq 0$  for all  $x$

### Conditions on $I$



The condition that the wavefunction can be normalized is a strong condition. It is said that wavefunctions have to be “square integrable”. This means that wavefunctions have to go to zero sufficiently fast as  $x$  becomes large. For example, the absolute squared of  $\Psi(x) = \frac{1}{\sqrt{|x|}}$  would integrate into  $\int \frac{1}{|x|} dx \rightarrow \ln(x)$ , and the limit goes to  $\infty$ . Therefore, this is not a proper wavefunction. The fact that we require wavefunctions to be “square integrable” is a strong additional

condition. It often leads to only discrete solutions and quantization of energy, when solving the Schrödinger equation. We will discuss more in chapter 4.

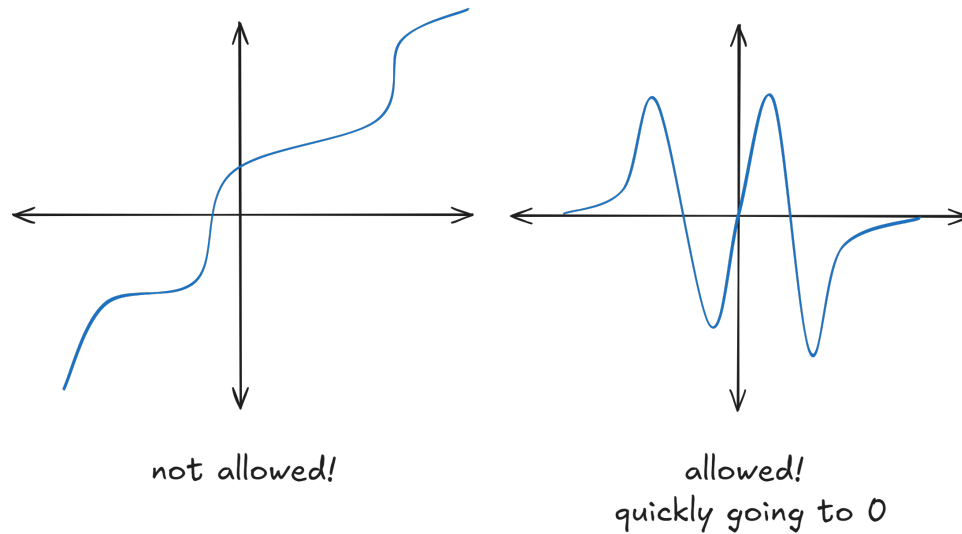


Figure 11.1

### 11.1.3 Example: Normalizing the Solutions for a Particle in the Box

We already know the solution for  $\psi_n(x) = \sin \frac{\pi n x}{a}$ . Notice that they are all real, but the conjugate of a real number is just that same real number, so it's valid to substitute in for

$\psi_n^*(x)$  as well.

$$\begin{aligned}
C^2 \int_0^a \psi_n^*(x) \psi_n(x) &\equiv 1 \\
C^2 \int_a^b \sin\left(\frac{\pi nx}{a}\right) \sin\left(\frac{\pi nx}{a}\right) dx \\
\sin^2 x &= \frac{1 - \cos 2x}{2} \\
&= C^2 \int_0^a \frac{1}{2} - \frac{1}{2} \cos\left(\frac{2\pi nx}{a}\right) dx \\
&= C^2 \left[ \frac{1}{2}x - \frac{a}{2\pi n} \sin\left(\frac{2\pi nx}{a}\right) \right]_0^a \\
\sin(2\pi n), n = 1, 2, 3, \dots : \sin(2\pi n) &= 0 \\
&= C^2 \left[ \frac{1}{2}x \right]_0^a \\
\Rightarrow 1 &= C^2 \frac{1}{2}a \\
\Rightarrow \frac{2}{a} &= C^2 \\
C &= \sqrt{\frac{2}{a}}
\end{aligned}$$

Our normalized solution is thus

$$\sqrt{\frac{2}{a}} \sin\left(\frac{\pi nx}{a}\right)$$

Now with our distribution normalized we have a nice way of thinking about it. Between any bounds  $x = b, c$  we have the result being the probability to find that electron (with a full period the probability to find it is 1) or the fraction of particles between those two bounds (with a full period having the full fraction of particles).

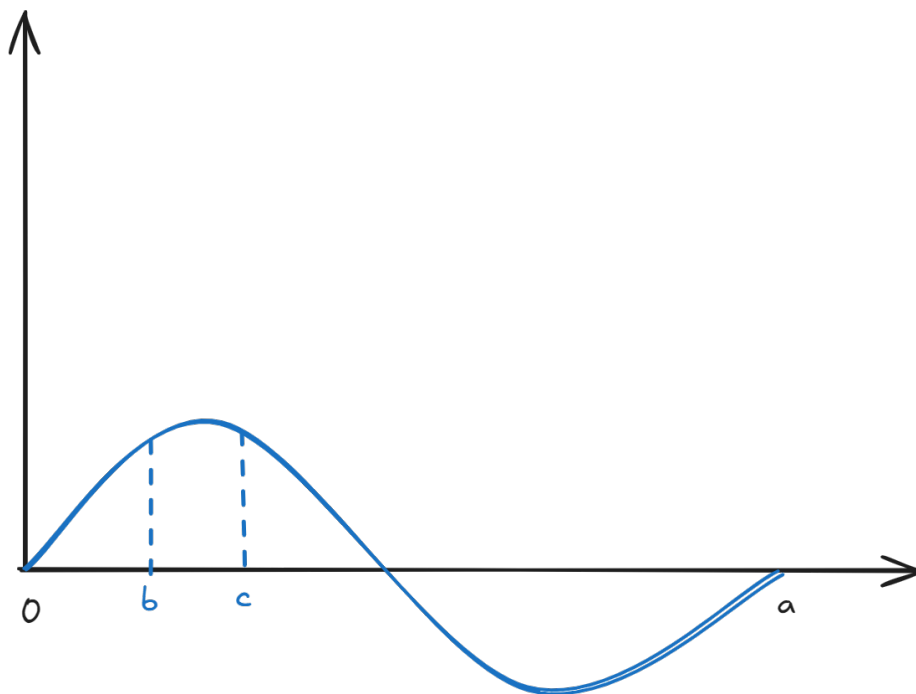


Figure 11.2

## 11.2 Definition of Mean Values or *Expectation Values*

The normalized wavefunction defines a probability distribution  $P(x) = \psi^*(x)\psi(x)$ . This means we can immediately calculate the moments of the wavefunction:

$$\begin{aligned}\langle x \rangle &= \int_{-\infty}^{\infty} xP(x) \, dx = \int_{-\infty}^{\infty} x\psi^*(x)\psi(x) \, dx = \int_{-\infty}^{\infty} \psi^*(x)x\psi(x) \, dx \\ \langle x^2 \rangle &= \int_{-\infty}^{\infty} x^2P(x) \, dx = \int_{-\infty}^{\infty} x^2\psi^*(x)\psi(x) \, dx = \int_{-\infty}^{\infty} \psi^*(x)x^2\psi(x) \, dx\end{aligned}$$

In the above we can reorder the terms in the integrand as it is just a product of numbers, which commute. The last form is the preferred way of writing it, as this generalizes. In particular, we would likewise like to define the averages of other operators, like momentum and energy, or some general operator  $\hat{A}$ .

$$\begin{aligned} \langle \hat{p}_x \rangle, \langle \hat{p}_x^2 \rangle \\ \langle \hat{H} \rangle, \langle \hat{H}^2 \rangle \\ \langle \hat{A} \rangle, \langle \hat{A}^2 \rangle \end{aligned}$$

In quantum mechanics these means are referred to as expectation values. The general definition of the expectation value of an operator  $\hat{A}$  for a normalized wavefunction is

$$\langle \hat{A} \rangle = \int_{-\infty}^{\infty} \psi^*(x) \hat{A} \psi(x) dx = \int_{-\infty}^{\infty} \psi^*(x) (\hat{A} \psi(x)) dx$$

Here we act with  $\hat{A}$  on the function  $\psi(x)$ , this gives a new function  $\hat{A}\psi(x) = \phi(x)$ , and the integrand becomes  $\psi^*(x)\phi(x)$ . This is how one would calculate it. We typically write it as the first definition, with the operator  $\hat{A}$  in the middle, acting on the function to its right. If a wavefunction is not normalized we can write a more general expression

$$\langle \hat{A} \rangle = \frac{\int_{-\infty}^{\infty} \psi^*(x) \hat{A} \psi(x) dx}{\int_{-\infty}^{\infty} \psi^*(x) \psi(x) dx}$$

### 11.2.1 Special Example of an Expectation Value of an Eigenfunction

Consider the special case that  $\psi(x)$  is an eigenfunction of  $\hat{A}$ ,  $\hat{A}\psi_a(x) = a\psi_a(x)$ . Here  $a$  is the eigenvalue (a real number). We also use this as a subscript of the wavefunction to indicate it is an eigenstate (another name for eigenfunction in this context) with eigenvalue  $a$ . We obtain an interesting result if we calculate both  $\langle \hat{A} \rangle$  and  $\langle \hat{A}^2 \rangle$ .

$$\begin{aligned} \langle \hat{A} \rangle &= \int_{-\infty}^{\infty} \psi_a^*(x) (\hat{A} \psi_a(x)) dx \\ &= \int_{-\infty}^{\infty} \psi_a^*(x) a \psi_a(x) dx \\ &= a \underbrace{\int_{-\infty}^{\infty} \psi_a^*(x) \psi_a(x) dx}_{=1} \\ &= a \end{aligned}$$



Similarly,

$$\begin{aligned}
\langle \hat{A}^2 \rangle &= \int_{-\infty}^{\infty} \psi_a^*(x) (\hat{A}^2 \psi(x)) \, dx \\
&= \int_{-\infty}^{\infty} \psi_a^*(x) (\hat{A}(\hat{A} \psi(x))) \, dx \\
&= \int_{-\infty}^{\infty} \psi_a^*(x) (a(a\psi(x))) \, dx \\
&= \int_{-\infty}^{\infty} \psi_a^*(x) (a^2 \psi(x)) \, dx \\
&= a^2 \underbrace{\int_{-\infty}^{\infty} \psi_a^*(x) (\psi(x)) \, dx}_{=1} \\
&= a^2
\end{aligned}$$

Therefore, using the definitions from the statistics section, the variance is

$$\begin{aligned}
\sigma_{\hat{A}}^2 &= \langle \hat{A}^2 \rangle - \langle \hat{A} \rangle^2 \\
&= a^2 - (a)^2 \\
&= 0
\end{aligned}$$

Let us try to understand what this means. If we would think of  $\hat{A}$  as position, the particle can be found only at the position  $a$ , as the mean position is  $a$ , with zero spread because the variance is zero. The same is true for any case of the operator  $\hat{A}$ , with position replaced by the appropriate quantity the operator  $\hat{A}$  measures. That is, if the wavefunction is described as an eigenfunction of  $\hat{A}$ , the mean value is  $a$ , and the spread around this mean is zero,  $\sigma_{\hat{A}}^2 = 0$ . This means the particle has the value  $a$  with 100% certainty. In general, for a particle described by a function that is not an eigenstate, there would be a spread of eigenvalues, each with their respective probabilities. More discussion in Chapter 4!

## 12 2D Particle in a Box and Degeneracies

### 12.1 2D Particle in a Box

#### 12.1.1 Setup

Say we have a box that is 2D, with dimensions  $x, y$ . The set up is similar to the 1-dimensional problem, but instead the box has two parameters  $a$  (in the  $x$  direction) and  $b$  (in the  $y$  direction).

$$V(x, y) = \begin{cases} 0 & 0 \leq x \leq a \\ 0 & 0 \leq y \leq b \\ \infty & \text{elsewhere (outside box)} \end{cases}$$

$$\psi(x, y) = \begin{cases} 0 & \text{(outside box)} \\ \text{continuous} & \text{everywhere, and hence zero at the edge of the box} \end{cases}$$

Our Hamiltonian is similar to before, aside from an extra dimension

$$\hat{H} = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} - \frac{\hbar^2}{2m} \frac{\partial^2}{\partial y^2} + \cancel{V(x, y)} \rightarrow 0$$

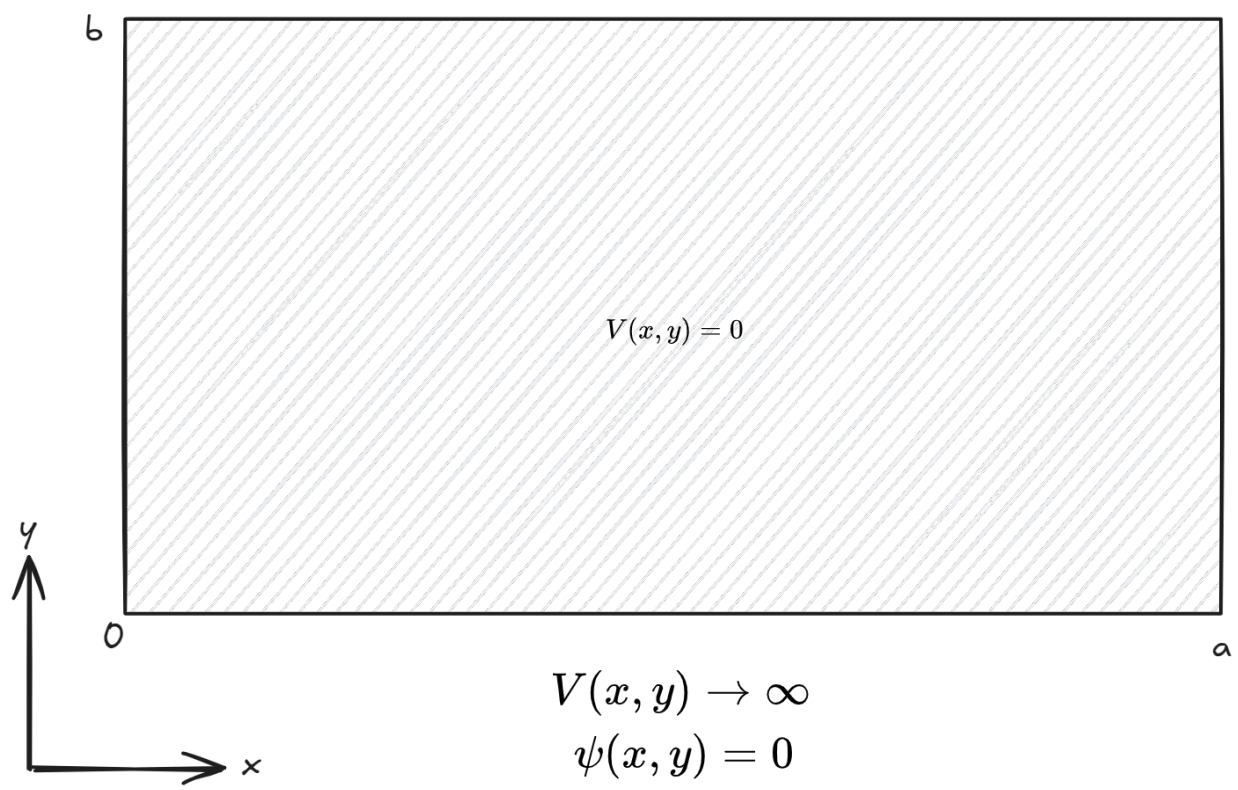
Therefore the resulting 2D Schrödinger equation is

$$\begin{aligned} \hat{H}\psi(x, y) &= E\psi(x, y) \\ \left( -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} - \frac{\hbar^2}{2m} \frac{\partial^2}{\partial y^2} \right) \psi(x, y) &= E\psi(x, y) \end{aligned}$$

Our equation is composed of 2 kinetic energy operators. These come from the momentum operators for  $x$  and  $y$  where

$$\begin{aligned} \hat{p}_x &= -i\hbar \frac{\partial}{\partial x} \\ \hat{p}_y &= -i\hbar \frac{\partial}{\partial y} \end{aligned}$$

If we were to square these momentum operators, to get the kinetic energy operators for our Hamiltonian ( $\hat{H} = (\hat{T}_x + \hat{T}_y) + (\hat{V}(x, y))$ ), they would both become real



**Figure 12.1**

$$i^2 = -1.$$

### 12.1.2 Separation of Variables

For this problem the separation of variables technique will work. The basic step consists of assuming that the solution is a product function  $\psi(x, y) = X(x)Y(y)$ . Then we can manipulate further, by substituting in S.E.

$$\begin{aligned}\hat{H}X(x)Y(y) &= EX(x)Y(y) \\ -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} X(x)Y(y) - \frac{\hbar^2}{2m} \frac{d^2}{dy^2} Y(y)X(x) &= EX(x)Y(y) \\ -\frac{\hbar^2}{2m} \frac{d^2 X}{dx^2} Y(y) - \frac{\hbar^2}{2m} \frac{d^2 Y}{dy^2} X(x) &= EX(x)Y(y) \quad \text{Divide both sides by } X(x)Y(y) \\ -\frac{\hbar^2}{2m} \frac{d^2 X}{dx^2} \frac{1}{X(x)} - \frac{\hbar^2}{2m} \frac{d^2 Y}{dy^2} \frac{1}{Y(y)} &= E\end{aligned}$$

We now have two independent functions that add up to the same constant for all values of  $x, y$

$$\begin{aligned}f(x) &= -\frac{\hbar^2}{2m} \frac{d^2 X}{dx^2} \frac{1}{X(x)} \\ g(y) &= -\frac{\hbar^2}{2m} \frac{d^2 Y}{dy^2} \frac{1}{Y(y)} \\ f(x) + g(y) &= E\end{aligned}$$

This can only be true if each function itself is constant. We can demonstrate this by picking arbitrary values  $y_0$  and  $x_0$  to input in each. Therefore

$$\begin{aligned}f(x) &= \underbrace{E - g(y_0)}_{\text{constant}} \equiv E_x \\ g(y) &= \underbrace{E - f(x_0)}_{\text{constant}} \equiv E_y\end{aligned}$$

and we obtain

We can split energy,  $E$  into two components  $E_x$  and  $E_y$  and write the equation compo-

nentwise

$$\begin{aligned}-\frac{\hbar^2}{2m} \frac{d^2 X}{dx^2} &= E_x X(x) \\ -\frac{\hbar^2}{2m} \frac{d^2 Y}{dy^2} &= E_y Y(y)\end{aligned}$$

So the 2-dimensional Schrödinger equation. separates into two 1-dimensional equations. This is a result of the technique of separation of variables (assuming the product form of the wavefunction). This does not always work, but here it does, and is very powerful.

### 12.1.3 Boundary Conditions

We have the boundary conditions, that the wavefunction  $\psi(x, y)$  is zero on the edges of the box:  $x = 0, x = a, y = 0, y = b$ . This translates to

$$\begin{aligned}Y(y) &= \begin{cases} 0 & \text{if } y = 0 \\ 0 & \text{if } y = b \end{cases} \\ X(x) &= \begin{cases} 0 & \text{if } x = 0 \\ 0 & \text{if } x = a \end{cases}\end{aligned}$$

### 12.1.4 Solving

Solving is just like doing 2 1-dimensional equations. Remember the solution for a 1-dimensional box was  $\sqrt{\frac{2}{a}} \sin\left(\frac{\pi n x}{a}\right)$  (normalized), so we just repeat that.

$$\begin{aligned}X_{n_x}(x) &= \sqrt{\frac{2}{a}} \sin\left(\frac{\pi n_x x}{a}\right) \\ E_{n_x} &= \frac{\hbar^2 \pi^2}{2ma^2} n_x^2 = \frac{h^2}{8ma^2} n_x^2\end{aligned}$$

$$\begin{aligned}Y_{n_y}(y) &= \sqrt{\frac{2}{b}} \sin\left(\frac{\pi n_y y}{b}\right) \\ E_{n_y} &= \frac{\hbar^2 \pi^2}{2mb^2} n_y^2 = \frac{h^2}{8mb^2} n_y^2\end{aligned}$$

### 12.1.5 Connecting Together

We now merge together our two independent solutions.

The total energy is just all the individual energy components summed together.

$$\begin{aligned}
 E_{\text{tot}} &= \sum E_{n_i} = E_x + E_y \\
 &= \frac{\hbar^2 \pi^2}{2ma^2} n_x^2 + \frac{\hbar^2 \pi^2}{2mb^2} n_y^2 \\
 E_{n_x, n_y} &= \frac{\hbar^2 \pi^2}{2m} \left( \frac{n_x^2}{a^2} + \frac{n_y^2}{b^2} \right)
 \end{aligned}$$

The total wavefunction is from multiplying all the wave components together.

$$\begin{aligned}
 \psi_{\text{tot}}(x, y) &= \prod \psi_{n_i} = \psi_{n_x} \psi_{n_y} \\
 \psi_{n_x, n_y} &= \sqrt{\frac{2}{a}} \sqrt{\frac{2}{b}} \sin\left(\frac{\pi n_x x}{a}\right) \sin\left(\frac{\pi n_y y}{b}\right)
 \end{aligned}$$

## 12.2 Degeneracies

Suppose we have a square box, where  $a = b$ . Our solution becomes

$$\begin{aligned}
 E_{n_x, n_y} &= \frac{\hbar^2 \pi^2}{2m} \left( \frac{n_x^2}{a^2} + \frac{n_y^2}{a^2} \right) \\
 &= \frac{\hbar^2 \pi^2}{2ma^2} (n_x^2 + n_y^2) \\
 \psi_{n_x, n_y} &= \sqrt{\frac{2}{a}} \sqrt{\frac{2}{a}} \sin\left(\frac{\pi n_x x}{a}\right) \sin\left(\frac{\pi n_y y}{a}\right) \\
 &= \frac{2}{a} \sin\left(\frac{\pi n_x x}{a}\right) \sin\left(\frac{\pi n_y y}{a}\right)
 \end{aligned}$$

For each value of  $n_x, n_y$  ( $1, 2, 3, \dots$ ) we have a different wavefunction. Importantly, *different wavefunctions can have the same energy*. The number of different wavefunctions (linearly independent, see later) with the same energy is called the **degeneracy** of the energy level. We can construct the following table, where the energy is given in multiples of  $\frac{\hbar^2 \pi^2}{2ma^2}$

**Table 12.1**

| $n_x$ | $n_y$ | $E_{n_x, n_y}$ | Degeneracy |
|-------|-------|----------------|------------|
| 1     | 1     | 2              | 1          |
| 1     | 2     | 5              | 2          |
| 2     | 1     | 5              |            |
| 2     | 2     | 8              | 1          |
| 1     | 3     | 10             | 2          |
| 3     | 1     | 10             |            |

### 12.2.1 Particle in a 3D Box

We can do a exactly the same analysis for a box in 3 dimensions,  $x, y, z$ , with box dimensions  $a, b, c$ . The energy now consists of three terms, and in the box where  $a = b = c$  this simplifies

$$\begin{aligned}
 E_{n_x, n_y, n_z} &= \frac{\hbar^2 \pi^2}{2ma^2} n_x^2 + \frac{\hbar^2 \pi^2}{2mb^2} n_y^2 + \frac{\hbar^2 \pi^2}{2mc^2} n_z^2 \\
 &= \frac{\hbar^2 \pi^2}{2ma^2} (n_x^2 + n_y^2 + n_z^2)
 \end{aligned}$$

We can form another table of degeneracies for the 3D case.

**Table 12.2**

| $n_x$ | $n_y$ | $n_z$ | $E_{n_x, n_y, n_z}$ | Degeneracy |
|-------|-------|-------|---------------------|------------|
| 1     | 1     | 1     | 3                   | 1          |
| 1     | 1     | 2     | 6                   | 3          |
| 1     | 2     | 1     | 6                   |            |
| 2     | 1     | 1     | 6                   |            |
| 2     | 2     | 1     | 9                   | 3          |
| 2     | 1     | 2     | 9                   |            |
| 1     | 2     | 2     | 9                   |            |
| 1     | 1     | 3     | 11                  | 3          |
| 1     | 3     | 1     | 11                  |            |
| 3     | 1     | 1     | 11                  |            |
| 2     | 2     | 2     | 12                  | 1          |
| 1     | 2     | 3     | 14                  | 6          |
| 1     | 3     | 2     | 14                  |            |
| 2     | 1     | 3     | 14                  |            |
| 2     | 3     | 1     | 14                  |            |
| 3     | 1     | 2     | 14                  |            |
| 3     | 2     | 1     | 14                  |            |

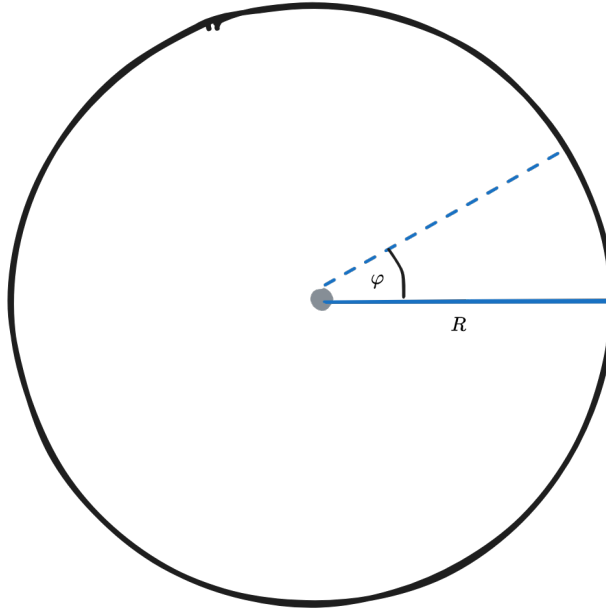
We see that the maximum degeneracy is 6, when all of the  $n_x, n_y, n_z$  are different, leading to 6 permutations. This can be compared to the 2D's box maximum of 2.



## 13 Particle on a Ring

### 13.1 Setup

Suppose we have a particle on a ring. Such a system could be realized by a metal ring, and the electron moves freely ( $V = 0$ ) along the ring. Leaving the ring would require a lot of energy and we can say the potential out of the ring is  $\infty$ . We start by defining parameters of the ring



**Figure 13.1:** We have an independent parameter  $\varphi$  and constant parameter  $R$  that are key to defining the ring.

We consider the wavefunction to be a function of the angle  $\psi(\varphi)$ . In this case, because we have a particle moving along a circular path, it has angular momentum, denoted as  $L$ . Importantly, *angular momentum is conserved*. An object moving in a circular path has angular momentum  $L = mvR$ .

A real world example of angular momentum is a figure skater. When they are spinning with their arms out they spin slower, but when they move their hands in, they start spinning faster (by moving their arms out,  $R$  becomes larger, so  $v$  must decrease since  $L$  is constant, and vice-versa).

## 13.2 Kinetic Energy (with respect to) Angular Momentum

We can derive the formula for kinetic energy in terms of angular momentum, where

$$\begin{aligned} T &= \frac{1}{2}mv^2 \\ &= \frac{1}{2} \frac{(mv)^2}{m} = \frac{L^2}{2mR^2} \end{aligned}$$

Taking the Hamiltonian  $\hat{H} = \hat{T} + \hat{V}$ , we can use the derived form of kinetic energy for angular momentum for the operator. Similar to the box, the potential energy inside of the ring can be ignored and  $\hat{H} = \hat{T}_\varphi$

$$\begin{aligned} \hat{L}_\varphi &= -i\hbar \frac{\partial}{\partial \varphi} \\ \hat{L}_\varphi^2 &= \left( -i\hbar \frac{\partial}{\partial \varphi} \right)^2 = -\hbar^2 \frac{\partial^2}{\partial \varphi^2} \end{aligned}$$

Obtaining the kinetic energy operator with respect to angular momentum, we get

$$\begin{aligned} \hat{T}_\varphi &= T(\hat{L}) \\ &= \frac{\hat{L}_\varphi^2}{2mR^2} \\ &= -\frac{\hbar^2}{2mR^2} \frac{\partial^2}{\partial \varphi^2} \end{aligned}$$

Therefore the Schrödinger equation takes the form

$$\begin{aligned} \hat{H}\psi(\varphi) &= E\psi(\varphi) \\ -\frac{\hbar^2}{2mR^2} \frac{d^2\psi}{d\varphi^2} &= E\psi(\varphi) \end{aligned}$$

This derivation has a handwavy character. It can be made rigorous by using that classically  $L = L_z = xp_y - yp_x$  in Cartesian coordinates. The quantum analog is

$$\hat{L}_z = \hat{x}\hat{p}_y - \hat{y}\hat{p}_x = -i\hbar \left( x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} \right)$$

This can be rewritten into spherical (or polar) coordinates and this leads to  $\hat{L}_z = -i\hbar \frac{d}{d\varphi}$ . Let us assume this form of  $\hat{L}_z$ , which will lead us to the Hamiltonian above.

### 13.3 Periodicity

Overall, the particle travels along the ring, in a wavelike form. The wavefunction should be a single-valued function, meaning that it should be a periodic function, since, for example, if we go around the ring at angle  $3\pi$ , it needs to have the same value as at the angle  $\pi$ , assuming we use the standard period from 0 to  $2\pi$ . This can be used as a boundary condition where

$$\psi(\varphi) = \psi(\varphi + 2\pi)$$

The boundary condition is periodic, but that is mathematically fine. Practically, we only deal with the first period of any periodic function anyways.

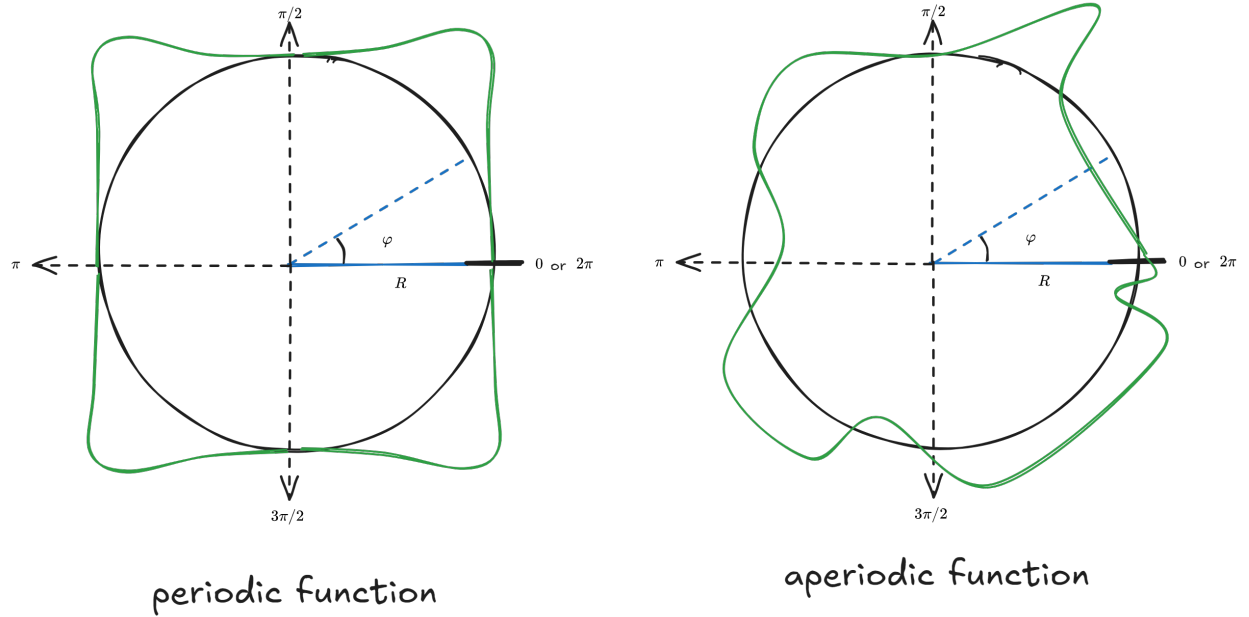


Figure 13.2

### 13.4 Solving

Now we attempt to solve the equation

$$-\frac{\hbar^2}{2mR^2} \frac{d^2\psi}{d\varphi^2} = E\psi(\varphi)$$

Notice that it is a second order differential equation with all constant terms, on both sides. Let's solve it by assuming we know the solution to be something in the form of  $e^{\lambda\varphi}$ .

$$\begin{cases} \frac{d^2\psi}{d\varphi} = \lambda^2 e^{\lambda\varphi} \\ \frac{d\psi}{d\varphi} = \lambda e^{\lambda\varphi} \\ \psi = e^{\lambda\varphi} \end{cases}$$

We substitute in this assumed solution to get

$$\begin{aligned} -\frac{\hbar^2}{2mR^2}\lambda^2 e^{\lambda\varphi} &= E e^{\lambda\varphi} \\ \text{dividing both sides by } e^{\lambda\varphi} & \\ -\frac{\hbar^2}{2mR^2}\lambda^2 &= E \end{aligned}$$

The assumed solution  $e^{\lambda\varphi}$  can be rewritten using the periodic boundary conditions

$$\begin{aligned} e^{\lambda\varphi} &= e^{\lambda(\varphi+2\pi)} \\ &= e^{\lambda\varphi+\lambda 2\pi} \\ e^{\lambda\varphi} &= e^{\lambda\varphi} e^{\lambda 2\pi} \quad \text{dividing both sides by } e^{\lambda\varphi} \\ 1 &= e^{\lambda 2\pi} \\ \implies \lambda &= 0 \end{aligned}$$

This is the only real solution, where  $\lambda = 0$ . To obtain more (complex) solutions, we assume that  $\lambda = ik$  where  $k \in \mathbb{R}$ . If we take the previous solution,  $\lambda = 0$ , then  $\lambda = 0 = ik \implies k = 0$ .

$$\begin{aligned} e^{\lambda 2\pi} &= e^{ik 2\pi} = 1 \\ \underbrace{\cos(2\pi k)}_{\text{even}} + \underbrace{i \sin(2\pi k)}_{=0, \sin(2\pi)=0} &= 1 \\ \cos(2\pi k) &= 1 \\ \implies k &= 0, \pm 1, \pm 2, \pm 3, \dots \end{aligned}$$

For each of these values of  $k$ , we have a valid periodic solution

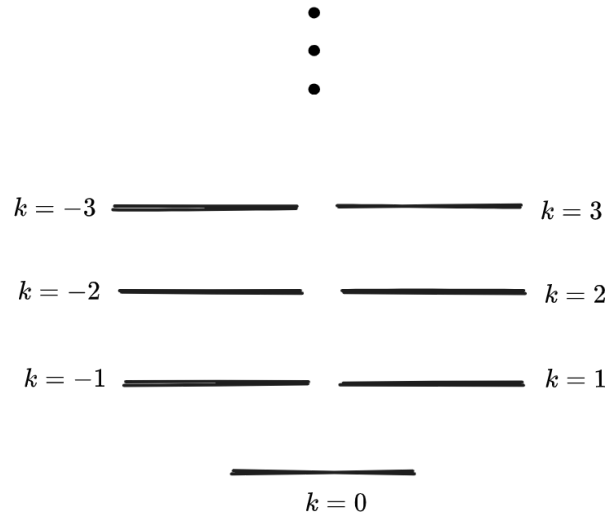
$$\psi_k(\varphi) = e^{ik\varphi}, k = 0, \pm 1, \pm 2, \pm 3, \dots$$

Let us emphasize that  $k = 0, \psi_0(\varphi) = e^{i0\varphi} = 1$  is a valid solution!

The corresponding energies are given by  $E = -\frac{\hbar^2}{2mR^2}\lambda^2, \lambda = ik$ ,

$$\begin{aligned} E_k &= -\frac{\hbar^2}{2mR^2}(ik)^2 \\ &= \frac{\hbar^2}{2mR^2}k^2 \end{aligned}$$

Note that  $(ik)^2 = (-ik)^2 = -k^2$ . We see that for every value  $E_k, k \neq 0$ , we have 2 solutions,  $e^{ik\varphi}, e^{-ik\varphi}$ . Each eigenvalue except  $E_0 = 0$  is two-fold degenerate.



**Figure 13.3**

It also follows that the general solution with energy  $E_k, k \neq 0$  can be written as

$$\begin{aligned} \psi_k(\varphi) &= c_1 e^{ik\varphi} + c_2 e^{-ik\varphi} \\ &= d_1 \cos(k\varphi) + d_2 \sin(k\varphi) \end{aligned}$$

The solutions for degenerate energy levels are not unique (general statement). We can express them in terms of real functions, cos and sin, *or* as plane waves. There are always two independent solutions (two arbitrary constants).

### 13.4.1 Generalization: Degenerate Eigenstates of the Schrödinger Equation are Not Unique

Assume we have

$$\begin{aligned}\hat{H}\psi_1 &= E\psi_1 \\ \hat{H}\psi_2 &= E\psi_2\end{aligned}$$

where  $\psi_1, \psi_2$  are different states of the same energy  $E$ , which is to say that the energy  $E$  is degenerate.

It then follows that any linear combination is also an eigenstate of  $\hat{H}$  with the *same* energy

$$\begin{aligned}\psi^{\text{new}} &= c_1\psi_1 + c_2\psi_2 \\ \hat{H}\psi^{\text{new}} &= \hat{H}(c_1\psi_1 + c_2\psi_2) = c_1E\psi_1 + c_2E\psi_2 \\ &= E(c_1\psi_1 + c_2\psi_2) \\ &= E\psi^{\text{new}}\end{aligned}$$

Hence  $\psi^{\text{new}}$  is an eigenstate with the same energy. This is the same reasoning we applied to the solution of particle in the box eigenstates:

$$\begin{aligned}c_1e^{ik\varphi} + c_2e^{-ik\varphi} &= c_1(\cos(k\varphi) + i\sin(\varphi)) + c_2(\cos(k\varphi) - i\sin(k\varphi)) \\ &= (c_1 + c_2)\cos(k\varphi) + i(c_1 - c_2)\sin(k\varphi) \\ &= d_1\cos(k\varphi) + d_2\sin(k\varphi)\end{aligned}$$

## 13.5 Normalization

$$\begin{aligned}
1 &= \int_0^{2\pi} \psi_k(\varphi) \psi_k^*(\varphi) d\varphi \\
&= \int_0^{2\pi} (c_1^* e^{-ik\varphi} c_1 e^{ik\varphi}) d\varphi \\
&= |c_1|^2 \int_0^{2\pi} \cancel{e^{-ik\varphi}} \overset{1}{e^{ik\varphi}} d\varphi \\
&= c_1^2 \int_0^{2\pi} 1 d\varphi \\
&= c_1^2 [\varphi]_0^{2\pi} \\
1 &= c_1^2 2\pi \\
c_1 &= \sqrt{\frac{1}{2\pi}}
\end{aligned}$$

Therefore, the normalized eigenstates for the particle in a box are  $\psi_k(\varphi) = \frac{1}{\sqrt{2\pi}} e^{ik\varphi}$ .

## 13.6 Summary of Particle on a Ring

We have the angular momentum Hamiltonian with corresponding Schrödinger equation

$$\hat{H}\psi = -\frac{\hbar^2}{2mR^2} \frac{d^2\psi}{d\varphi^2} = E\psi$$

With the periodic boundary condition  $\psi(\varphi) = \psi(\varphi + 2\pi)$ , we have the normalized solutions

$$\begin{aligned}
E_k &= \frac{\hbar^2}{2mR^2} k^2 \quad \text{where } k = 0, \pm 1, \pm 2, \pm 3, \dots \\
\psi_k(\varphi) &= \frac{1}{\sqrt{2\pi}} e^{ik\varphi}
\end{aligned}$$

# 14 Postulates of Quantum Mechanics and Hermitian Operators

At its core, quantum mechanics is simple. We have the nucleus and electrons. There are many specialty extensions to the theory, such as adding detail to understand nuclear effects (protons, neutrons) or field effects (electrical, magnetic, etc.), but we are dealing with the simplest and most core theory.

As quantum mechanics is a mathematical theory, it has **postulates**. A postulate is similar to an **axiom**, which is an assumption of the system that is unprovable. We prove things in a specified system, using the rules and restrictions that come from our inherent assumptions, and to prove our axioms using our axioms would be circular logic.

An example of a mathematical theory that has axioms is Euclidean geometry. Euclidean geometry is what models the physical space of our world. Any point in Euclidean space can be found by the number of coordinates that make up the space, e.g. 2 coordinates in 2D space, 3 in 3D space, etc. An example of an axiom (one of many) in this system is that only one line can connect any two points. Together with all the axioms we can derive all the properties of a system.

If a theory broke an axiom, it wouldn't necessarily be wrong, but a new set of axioms would have to apply to define this new system. We might have non-Euclidean geometry in this case.

## 14.1 Postulates of Quantum Mechanics

Similar to the mathematical axioms, everything is derived from the postulates, but importantly in contrast, *everything must be verified and hold up experimentally*.

If we end up deriving something in conflict with experimental evidence, that would indicate a problem with the theory. The theory would need to be changed, extended, or replaced. So far quantum mechanics (and its extensions such as quantum field theory) have held up.



## 14.2 Measurement

Quantum mechanics is a statistical theory. We can calculate possible outcomes and their probabilities. We have something called a measurement. In this measurement, we measure a quantity  $A$ . Because the predictions we make in quantum mechanics are statistical, we should make a large set of measurements on the same (or similar) systems. This is the analogue of rolling a die many times to get a statistical probability. If we measure a quantity  $A$ , this quantity is associated with an operator  $\hat{A}$  in quantum mechanics. As an operator it has eigenvalues (more details to come)

$$\hat{A}\phi_n(x) = a_n\phi_n(x)$$

These eigenvalues are the only possible outcomes we observe. Every measurement on an individual system yields one of the possible eigenvalues. If we do this for a large number of identical systems (an ensemble), the probabilities are described by the wavefunction that describes the *ensemble*. My notion that I replace a quantum system by an ensemble of identical systems is up to interpretation, because different people use different language. Everybody agrees that the possible outcomes of experiment are determined by the quantity that is measured:  $A$  or  $\hat{A}$ , while the probabilities to get a particular outcome are determined by the wavefunction  $\psi$ . The notion of probability used in these notes implies multiple measurements and the collecting of statistics over individual measurements, and this leads to the notion of the ensemble described by the wavefunction  $\psi$ .

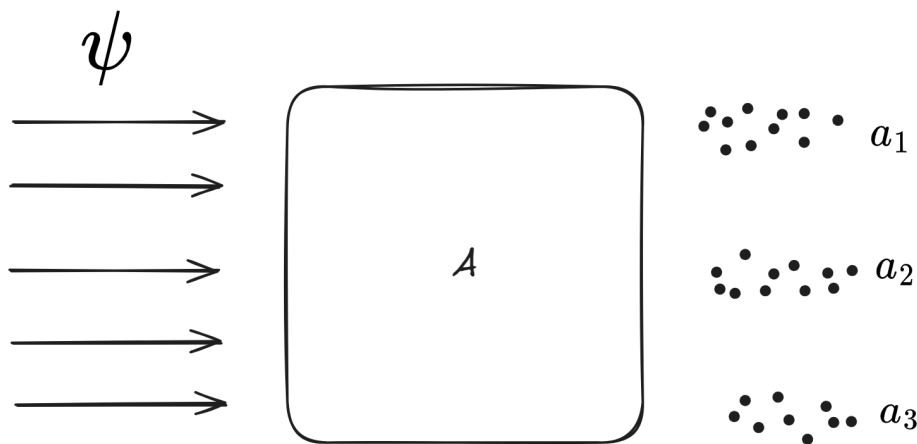


Figure 14.1

## 14.3 Hermiticity

Operators in quantum mechanics that describe properties that can in principle be measured are described by linear operators that are Hermitian. Such operators have an associated domain of integration. In 1-dimension we have the definite integral  $\int_a^b(\dots)dx$ . We will discuss the full mathematical details shortly. Let us summarize the main properties.

### 14.3.1 Useful properties of Hermitian Operators

Let us assume the operator  $\hat{A}$  with integration domain  $[a, b]$  is Hermitian and has eigenfunctions:

$$\hat{A}\phi_n(x) = a_n\phi_n$$

Then the following properties will be true

1. The eigenvalues  $a_n$  are all *real* (possibly degenerate)
2. The eigenfunctions can be chosen to be *orthonormal*

$$\int_a^b \phi_m^*(x)\phi_n(x)dx = \delta_{mn}$$
$$\delta_{mn} = \begin{cases} 1 & \text{if } m = n \\ 0 & \text{if } m \neq n \end{cases}$$

We will often use the Kronecker delta  $\delta_{mn}$  in what follows.

3. The full set of (orthonormal) eigenfunctions is *complete*. This means that any (proper, normalizable) arbitrary function  $\psi(x)$  can be written as a linear combination of eigenfunctions.

$$\psi(x) = \sum_n c_n \phi_n(x) \quad c_n = \int_a^b \phi_n^*(x)\psi(x)dx \quad (14.1)$$

A linear operator can be **Hermitian**. This class of operators is well behaved and has many nice properties, making this classification very important. If a operator is Hermitian, there are many conditions that hold.

#### 14.3.1.1 1<sup>st</sup> Condition

All eigenvalues are real. This means that our eigenvalues correspond to physical values.

Note: This does not imply that any real eigenvalue must be from a Hermitian operator.

### 14.3.1.2 2<sup>nd</sup> Condition

We can choose our eigenfunctions to be **orthonormal** to the integration domain.

- The general integration domain is denoted  $A$ , described by bounds  $a$  and  $b$ . This *domain is essential to the definition of a Hermitian operator*.
- The integration can be thought as an inner product. We will not use this language much in this class.

An orthonormal function is a function that is both **orthogonal** (perpendicular) to whatever we are comparing them to and normalized (equal to 1). Our eigenfunctions are orthogonal with respect to each other.

#### Inner Products



An inner product operation between two functions (vectors) results in a scalar.

It is a generalization of the dot product, working for more than just vectors in Euclidean space, but similar mental intuition can be used.

Consider two vectors  $\vec{u}$  and  $\vec{v}$ , where

$$\vec{u} = (u_1, u_2), \vec{v} = (v_1, v_2)$$

The dot product between them is

$$\vec{u} \cdot \vec{v} = u_1 v_1 + u_2 v_2$$

The dot product only works for real, Euclidean spaces. The inner product works for vectors in real or complex spaces and will output a real or complex number.

The inner product is notated  $\langle u, v \rangle$  (math convention) or  $\langle u | v \rangle$  (quantum/physics convention, known as Dirac or bra-ket notation). It can also be notated as an integral, especially in the case of two functions.

$$\int_a^b u(x)v(x) dx$$

#### 14.3.1.2.1 Orthogonalization

In order to orthogonalize something, we have to use a process called **Gram-Schmidt**. With any Hermitian operator, all of our eigenfunctions and values will retain their “eigenness” after this process.

By having our functions be orthonormal, the following definition holds

$$\int_A \phi_m^*(x) \phi_n(x) dx = \begin{cases} 1 & \text{if } m = n \\ 0 & \text{if } m \neq n \end{cases}$$

When indices aren't equal, the inner product of any two orthogonal functions will equal 0. When the indices are equal, because the functions are normalized, the inner product will equal 1.

This is called the **Kronecker delta**, denoted  $\delta_{mn}$ . It is a function of two variables, in this case  $n$  and  $m$ , which are also our indices. Any function that fits this property where the inner product (represented by the integration in this case) is equal to 1 if the variables are equal and equal to 0 if they aren't, is an example of the Kronecker delta.

#### 14.3.1.2.2 Gram-Schmidt

The main requirement of Gram-Schmidt is that the set of vectors being orthogonalized must be **linearly independent**. Despite orthogonalizing the vectors, the exact space spanned by the vectors doesn't change. Any vector possible to express in the old set must be able to be expressed in the new set.

For example, just because we change our coordinate system from Cartesian  $(x, y, z)$  to spherical coordinates doesn't mean we're changing the space itself. We can express all of three-dimensional space in either system.

### Linear Independence



Linear independence means that a vector can't be made as a linear combination of the others. To be a linear combination, we can add vectors, subtract vectors, scale vectors by a real number, and any combination of those operations. If any vector can be made from any combination of the others using those operations, the set of vectors are not linearly independent!

For example, if we have the vectors

$$(2, 0), (-4, 0)$$

the set is *linearly dependent* because the vectors are just scalar multiples of each other.

If we have another set,

$$(1, 0, 0), (0, 1, 0), (0, 0, 1), (2, 2, 2)$$

This set is also *linearly dependent* because the last vector is a linear combination of the earlier three vectors.

Note that linear independence is not always possible to see by inspection.

For example, the set  $(3, 1, 2), (-4, 6, 7), (2, 8, 9)$  is *linearly independent* and the set  $(2, 1, 2), (3, 5, 3), (0, -7, 0)$  is *linearly dependent*.

These concepts may be familiar from linear algebra, and they also apply in quantum mechanics. The vectors are now functions. This is a useful analogy and tool, but we will not do the precise math in this class.

### 14.3.1.3 3<sup>rd</sup> Condition

The complete set of (orthonormal) eigenfunctions of a Hermitian operator is complete. This means that any arbitrary wavefunction  $\psi$  (with appropriate conditions) is **complete**.

What this means is that any arbitrary wavefunction can be expressed as a linear combination of eigenstates  $(\phi_n)$ , as in

$$\psi(x) = \sum_n c_n \phi_n(x)$$

What to make our coefficient? Using the orthonormality of the  $\phi_n$  we can prove that the coefficient  $c_n$  for a arbitrary (given) function  $\psi(x)$  are given by<sup>1</sup>

$$c_n = \int_A \phi_n^*(x) \psi(x) dx$$

We will come back to this later, but it is a good choice for the function to behave the way you want it.

---

<sup>1</sup>Any linear combination of eigenstates is unique (by orthonormality), equivalent to the linear independence of the eigenfunctions  $\phi_n$ , therefore we can “guess” a formula for our coefficients. This formula is not a true guess, but is derived from more complex math.

### What if our operator wasn't Hermitian?



For example, if our operator wasn't Hermitian, we could run into a scenario where we need to orthogonalize our eigenfunctions, but after orthogonalization they would no longer be eigenfunctions and we would be completely stuck without a path forward. Disaster! We avoid such things by keeping it Hermitian.

### 14.3.2 Formal Definition of Hermiticity

A linear operator  $\hat{A}$  is Hermitian *if and only if* for any arbitrary pair of functions  $f(x)$  and  $g(x)$  that fit within the domain of the operator (by satisfying the boundary conditions)

$$\int_a^b f^*(x)(\hat{A}g(x)) dx = \int_a^b (\hat{A}f(x))^* g(x) dx$$

This can be extended to other dimensions (2/3-dimensional, time, etc.), but we will stick with the one spatial dimension for now, for the sake of clarity. The integration over the specific domain is a part of the full definition of the Hermitian operator  $\hat{A}$ .

Note that order does not matter within the integral, and that the functions are designated arbitrarily.

### 14.3.3 Examples of Hermitian Operators

The followings operators are Hermitian, some with conditions:

The potential energy operator, when the potential energy is a *real function*

$$\hat{V} \mapsto V(x)\psi(x)$$

The momentum operator, due to the factor of  $i$

$$\hat{p}_x \mapsto -i\hbar \frac{d}{dx}$$

The kinetic energy operator, which makes sense when it is derived from the momentum operator

$$\hat{T} = -\frac{\hbar^2}{2m} \left( \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) = -\frac{\hbar^2}{2m} \nabla^2$$

The total energy operator, the Hamiltonian

$$\hat{H} = \hat{T} + \hat{V}$$

### 14.3.3.1 Proofs of Hermiticity

Note that these proofs are for the sake of example, not memorization.

#### 14.3.3.1.1 Potential Energy

We use the formal definition for our proof listed above. Let  $\hat{V} = V(x)$  notation wise, for simplicity.

$$\begin{aligned}\int_a^b f^*(x)(\hat{V}g(x)) dx &= \int_a^b (V(x)f(x))^*g(x) dx? \\ &= \int_a^b \underbrace{V^*(x)}_{=V(x)} f^*(x)g(x) dx \\ &= \int_a^b V(x)f^*(x)g(x) dx \\ &= \int_a^b f^*(x)(V(x)g(x)) dx \\ \int_a^b f^*(x)(V(x)g(x)) dx &= \int_a^b (V(x)f(x))^*g(x) dx\end{aligned}$$

Our potential energy is Hermitian! This only works when  $V(x)$  is real. The conjugate of any real function is simply that function, and that applies to our operator derived from that function as well, so with this condition  $V^*(x) = V(x)$ , or  $\hat{V}^* = \hat{V}$ .

#### 14.3.3.1.2 Momentum

To prove:

$$\hat{p}_x = -i\hbar \frac{d}{dx}$$

$$\begin{aligned}\int_a^b f^*(x)(\hat{p}_x g(x)) dx &= \int_a^b f^*(x) \left( -i\hbar \frac{d}{dx} g(x) \right) dx \\ &= \int_a^b (\hat{p}_x f(x))^* g(x) dx = \int_a^b \left( -i\hbar \frac{d}{dx} f(x) \right)^* g(x) dx\end{aligned}$$

To proceed we will use integration by parts on

$$\int_a^b f^*(x) \left( -i\hbar \frac{dg}{dx} \right) dx$$

To do the integration by parts we choose  $u = f^*(x)$  and  $dv = \frac{dg}{dx}$ . Therefore,  $du = \frac{df^*}{dx} dx$  and  $v = g(x)$ . From there we can use our standard integration by parts formula ( $\int_a^b v du =$

$$[uv]_a^b - \int_a^b v \, du),$$

$$\begin{aligned} \int_a^b f^*(x) \left( -i\hbar \frac{dg}{dx} \right) dx &= [(-i\hbar) f^*(x) g(x)]_a^b - \int_a^b (-i\hbar) g(x) \frac{df^*}{dx} dx \\ &= [(-i\hbar) f^*(x) g(x)]_a^b + \int_a^b i\hbar \frac{df^*}{dx} g(x) dx \\ &= \underbrace{[(-i\hbar) f^*(x) g(x)]_a^b}_{\text{Wronskian}} + \int_a^b \left( -i\hbar \frac{df}{dx} \right)^* g(x) dx \\ \int_a^b f^*(x) \left( -i\hbar \frac{dg}{dx} \right) dx &= \int_a^b \left( -i\hbar \frac{df^*}{dx} \right) g(x) dx \quad \text{if} \quad [(-i\hbar) f^*(x) g(x)]_a^b = 0 \end{aligned}$$

We can see that the momentum operator is only Hermitian if that extra term, the **Wronskian**,  $[(-i\hbar) f^*(x) g(x)]_a^b$  is equal to 0. This is always true in quantum mechanics, which seeks to model physical situations and therefore all of our results should be physically meaningful. While theoretically it could be equal to something other than 0, any results would no longer carry physical meaning, so we ignore such cases.

Mathematically, this is justified by our boundary conditions, which assure that all results remain physically meaningful. For example, our particle in the box has boundary conditions  $f(0) = f(a) = 0$  and  $g(0) = g(a) = 0$ , or our particle on ring where  $\psi(\varphi) = \psi(\varphi + 2\pi)$ .

The imaginary  $-i$  in the above calculation is important. Otherwise, the operator  $\hbar \frac{d}{dx}$  would not be Hermitian because of the minus sign from the integration-by-parts.



# 15 Proving Hermitian Properties

## 15.1 Summary of Hermitian Properties

A Hermitian operator is some linear operator  $\hat{A}$ , with specific integration domain with bounds  $[a, b]$ , defined for any “true”, “proper” functions  $f(x)$  and  $g(x)$ .

An operator  $\hat{A}$  is Hermitian if and only if, for any “proper” functions  $f(x)$ ,  $g(x)$ , the following equation is always true:

$$\int_a^b f^*(x)(\hat{A}g(x)) dx = \int_a^b (\hat{A}f(x))^* g(x) dx$$

That is, we can always move  $\hat{A}$  over from acting on  $g(x)$  to acting on  $f(x)$  *inside* of the complex conjugate.

Such operators have eigenfunctions and eigenvalues, with some nice properties:

- All eigenvalues are real
- We can choose eigenfunctions to be orthonormal (orthogonal and normalized)

—

$$\int_a^b \phi_n^*(x)\phi_m^*(x) dx = \begin{cases} 1 & \text{if } m = n \\ 0 & \text{if } m \neq n \end{cases} = \delta_{nm} \quad (\text{Kronecker delta})$$

- Any wavefunction  $\psi(x)$  can be written as a linear combination of eigenstates, or in more formal words, the set of eigenfunctions for any Hermitian operator is complete

—

$$\psi(x) = \sum_n c_n \phi_n(x)$$

## 15.2 Proof and Discussion of Hermitian Properties

### 15.2.1 Real Eigenvalues

Invoking the definition of a Hermitian operator, with some operator  $\hat{A}$  with eigenvalues  $a_n$ ,

$$\begin{aligned} \int_a^b \phi_n^*(x)(\hat{A}\phi_n(x)) dx &= \int_a^b (\hat{A}\phi_n(x))^* \phi_n(x) dx \\ &= \int_a^b (a_n \phi_n(x))^* \phi_n(x) dx \\ &= a_n \int_a^b \phi_n^*(x) \phi_n(x) dx &= a_n^* \int_a^b \phi_n^* \phi_n(x) dx \end{aligned}$$

We get to a standstill, requiring two important definitions to continue. First, the inner product of a complex function and its conjugate is equal to its absolute square,  $\int_a^b \phi_n^*(x) \phi_n(x) dx = \int_a^b |\phi_n(x)|^2 dx$ . Second, this absolute square is equal to some positive real constant  $C$ , depending on the function. It can't be zero, as the function would be zero everywhere and can't be normalized then.

We can rewrite accordingly

$$\begin{aligned} a_n \int_a^b |\phi_n|^2 dx &= a_n^* \int_a^b |\phi_n|^2 dx \\ \implies a_n C &= a_n^* C \\ a_n C - a_n^* C &= 0 \\ C(a_n - a_n^*) &= 0 \end{aligned}$$

Because the  $C$  is equal to some nonzero constant,  $(a_n - a_n^*) = 0$ , hence  $a_n = a_n^*$ , which says that  $a_n$  must be real. This conclusion makes sense logically, as the conjugate of any real number is just that same real number.

#### Why can't our eigenfunction be 0? Another explanation



If an eigenfunction could be 0, every single operator would have it as an eigenfunction. An eigenfunction of 0 could have every possible eigenvalue associated with it. These conditions mean it is not meaningful to consider as an eigenfunction, and therefore we exclude the 0 eigenfunction/vector.

Mathematically, in the language of linear algebra: If we have an matrix  $\hat{A}$  and 0-value eigenvector  $\vec{0}$ ,  $\hat{A}\vec{0} = b\vec{0}$  for any constant  $b$ , and so any constant  $b$  is

an eigenvalue of  $\vec{0}$ .

## 15.2.2 Orthonormal Eigenfunctions

If we have two functions  $\phi_n(x)$  and  $\phi_m(x)$  with different eigenvalues  $a_n$  and  $a_m$  respectively, they must be orthonormal to each other.

### 15.2.2.1 Normality

The normalized part is simple, as long as we have appropriate coefficients both functions are easily normalized, but the orthogonality requires a proof.

### 15.2.2.2 Orthogonality

We use the Kronecker delta definition as a core part of the derivation. We assume that  $a_n \neq a_m$  with our two eigenfunctions having different eigenvalues (see degeneracy for the contrary case), and  $a_n^* = a_n$ , justified by the earlier real eigenvalues proof.

$$\begin{aligned}
 \int_a^b \phi_n^*(x)(\hat{A}\phi_m(x)) dx &= \int_a^b (\hat{A}\phi_n(x))^* \phi_m(x) dx \\
 \int_a^b \phi_n^*(x)(a_m\phi_m(x)) dx &= \int_a^b (a_n\phi_n(x))^* \phi_m(x) dx \\
 a_m \int_a^b \phi_n^*(x)\phi_m(x) dx &= \underbrace{a_n^*}_{a_n} \int_a^b \phi_n^*(x)\phi_m(x) dx \\
 \implies \underbrace{(a_n - a_m)}_{\neq 0 \because a_n \neq a_m} \underbrace{\int_a^b \phi_n^*(x)\phi_m(x) dx}_{=\delta_{nm}=0} &= 0
 \end{aligned}$$

Therefore, if we have two eigenfunctions  $\phi_n(x)$  and  $\phi_m(x)$  with different eigenvalues  $a_n \neq a_m$ , the two functions must be orthogonal to each other.

## 15.2.3 Degeneracy

We have some special proofs for degeneracy, where two different eigenfunctions have the same eigenvalue. Mathematically this is written as

$$\begin{aligned}
 \hat{A}\phi_n(x) &= a\phi_n(x) \\
 \hat{A}\phi_m(x) &= a\phi_m(x)
 \end{aligned}$$

If this is the case, then the linear combination is also composed of the same separate eigenfunctions and shared eigenvalues. This relies on the operator being linear, a key condition of being Hermitian.

$$\begin{aligned}
\hat{A}\psi(x) &= \hat{A}(c_n\phi_n(x) + c_m\phi_m(x)) \\
&= c_n\hat{A}\phi_n(x) + c_m\hat{A}\phi_m(x) \quad (\text{linear!}) \\
&= c_na\phi_n(x) + c_ma\phi_m(x) \\
a\psi(x) &= a(c_n\phi_n(x) + c_m\phi_m(x))
\end{aligned}$$

### 15.2.3.1 Orthonormality

If we have two degenerate eigenfunctions, we can make them orthonormal.

#### 15.2.3.1.1 Normality

We first normalize them with some appropriate coefficient.

#### 15.2.3.1.2 Orthogonality

Now we do orthogonality. Because they aren't orthogonal, the Kronecker delta doesn't apply and the inner product is equal to some constant  $C$ .

$$\int_a^b \phi_n^*(x)\phi_m(x) dx = C$$

We always orthogonalize our functions in relation to one another. We set a “reference function”, which is the first function and remains unmodified. The next function should be orthogonal to that one, and the next orthogonal to the two before it, etc. In this case  $\phi_n(x)$  is our reference function, remaining unchanged, while we create a new  $\phi_m(x)$ .

$$\phi_m^{\text{new}}(x) = \phi_m(x) - C\phi_n(x)$$

This is a form of the Gram-Schmidt process, and consequently the equation above has a familiar form to the classic formula, where  $C$  involves the projection of  $\phi_n(x)$  onto  $\phi_m(x)$ .

Substituting back into the above inner product,

$$\begin{aligned}\int_a^b \phi_n^*(x)(\phi_m(x) - C\phi_n(x)) \, dx &= \int_a^b \underbrace{(\phi_n^*(x)\phi_m(x))}_C - C \underbrace{(\phi_n^*(x)\phi_n(x))}_1 \, dx \\ &= C - C(1) \\ &= 0\end{aligned}$$

The inner product now equals 0, meaning the functions are now orthogonal to each other!

## Gram-Schmidt Process



If we have a set of linearly independent vectors  $\vec{v}_1, \vec{v}_2, \dots$  we can orthogonalize them. This new set is called  $\vec{u}_1, \vec{u}_2, \dots$ . We set the first vector, as our reference vector.

$$\vec{u}_1 = \vec{v}_1$$

For the second, we need to calculate the projection of some  $\vec{v}$  onto some  $\vec{u}$ . The projection is the “shadow” of  $\vec{v}$  on  $\vec{u}$ . The projection is given by the formula

$$\text{proj}_{\vec{u}}(\vec{v}) = \frac{\vec{v} \cdot \vec{u}}{|\vec{u}|^2} \vec{u}$$

We can then calculate the new second vector, where

$$\vec{u}_1 = \vec{v}_2 - \text{proj}_{\vec{u}_1}(\vec{v}_2)$$

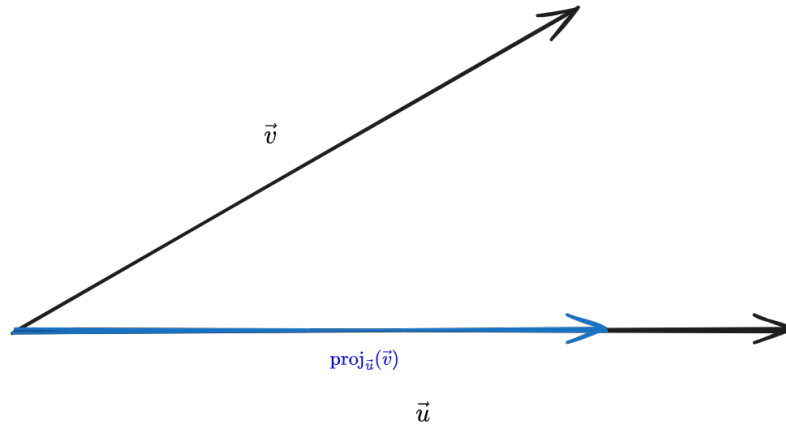


Figure 15.1

### 15.2.4 Orthonormal Set of Eigenfunctions

We can now say that the eigenfunctions of an Hermitian operator can always be chosen to be an orthonormal set of functions:

$$\int \phi_n(x)^* \phi_m(x) dx = \delta_{nm}$$

if eigenvalues are different then functions have to be orthogonal. if eigenvalues are the same we can make linear combinations that are orthonormal.

### 15.2.5 Completeness

The statement that any proper function can be expressed as a linear combination of eigenstates

$$\psi(x) = \sum_n c_n \phi_n(x)$$

is hard to prove. However it can be shown how one can obtain the expansion coefficients as

$$c_m = \int_a^b \phi_m^*(x) \psi(x) dx$$

To derive this we substitute  $\psi(x) = \sum_n c_n \phi_n(x)$  to get

$$\begin{aligned} c_m &= \int_a^b \phi_m^*(x) \left( \sum_n c_n \phi_n(x) \right) \\ &= \sum_n c_n \underbrace{\left( \int_a^b \phi_m^*(x) \phi_n(x) dx \right)}_{\delta_{nm}} \\ c_m &= \sum_n c_n \delta_{nm} \end{aligned}$$

A key subtlety of this derivation is the rearrangement of the integral and summation. This is only possible knowing we have a sum that converges nicely (in particular, it must be finite), which is an assumption that we must make for our quantum math to always correspond to physical situations. This and other derivations are actually very hard results in abstract math and would require pages of derivation without assumptions like this.

The index  $n$  here has possible values  $n = 1, 2, 3, \dots$  and  $m$  is an arbitrary but specific value out of that set.  $m$  could be 1, or 65, or 400, but is only ever equal to a specific value. The Kronecker delta lets us “pick out” a specific  $m$  from the sum. This is justified

by expanding the sum.

$$\begin{aligned}
c_m &= \sum_n c_n \delta_{nm} \\
&= c_1 \delta_{1m} + c_2 \delta_{2m} + \cdots + c_{m-1} \delta_{m-1,m} + c_m \delta_{mm} + c_{m+1} \delta_{m+1,m} + \cdots \\
&= 0 + 0 \cdots + 0 + c_m \cdot (1) + 0 + \cdots \\
&= c_m
\end{aligned}$$

This can also be understood by taking the integration  $\sum_n c_n \int_a^b \phi_m^*(x) \phi_n(x) dx$  and not explicitly invoking the Kronecker delta. For simplicity, let us choose a value of  $m$  to be 5.

$$\begin{aligned}
\sum_n c_n \int_a^b \phi_m^*(x) \phi_n(x) dx &= c_1 \int_a^b \phi_{m=5}^*(x) \phi_{n=1}(x) dx + c_2 \int_a^b \phi_5^*(x) \phi_2(x) dx \\
&\quad + c_3 \int_a^b \phi_5^*(x) \phi_3(x) dx + c_4 \int_a^b \phi_5^*(x) \phi_4(x) dx + c_5 \int_a^b \phi_5^*(x) \phi_5(x) dx \\
&= c_5 \int_a^b \phi_5^*(x) \phi_5(x) dx = c_5(1) = c_5
\end{aligned}$$

### 15.2.5.1 Normalization

Completeness itself is difficult to prove, but has an important consequence that relates highly to probability. Using the relationship  $\psi(x) = \sum_n c_n \phi_n(x)$  and the relationship implied by the former  $\psi^*(x) = \sum_m c_m^* \phi_m^*(x)$

$$\begin{aligned}
\int_a^b \psi^*(x) \psi(x) dx &= \int_a^b \left( \sum_m c_m^* \phi_m^*(x) \right) \left( \sum_n c_n \phi_n(x) \right) dx \\
&= \sum_m c_m^* \sum_n c_n \int_a^b \phi_m^*(x) \phi_n(x) dx \\
&= \sum_m c_m^* \sum_n c_n \delta_{nm} \\
&= \sum_m c_m^* (0 + \cdots + 0 + c_m + 0 + \cdots) \\
&= \sum_m c_m^* c_m \\
&= \sum_m |c_m|^2
\end{aligned}$$

For a normalized wavefunction, the absolute square of the coefficients is 1, or  $\sum_m |c_m|^2 = 1$ . This allows us to interpret the absolute square as a probability, where  $|c_m|^2 = P_m$ .

## 15.3 Probability

### 15.3.1 Calculation of Moments

We can express the expectation values of the operator as  $\hat{A} = \sum_m P_m a_m$  where  $c_m^* c_m = P_m$ . As a probability  $P_m \geq 0$ . Similarly  $\langle \hat{A}^2 \rangle = \sum_m P_m a_m^2$ . These are proven in the class notes and next lecture.

Note here that  $m$  as our index is an arbitrary choice. Depending on how you initially assign variables and indices, it could be  $P_n, P_k$ , etc.

### 15.3.2 Interpretation

Interpreting the absolute square of the coefficients as a probability allows us to get statistics for a discrete set of outcomes: the eigenvalues of  $\hat{A}$ . Each outcome/eigenvalue has a probability associated with it.

These eigenvalues are we actually observe when measuring the quantity associated with the operator. The probability corresponds to the probability of observing that specific outcome. Any non-eigenvalues are not observed.

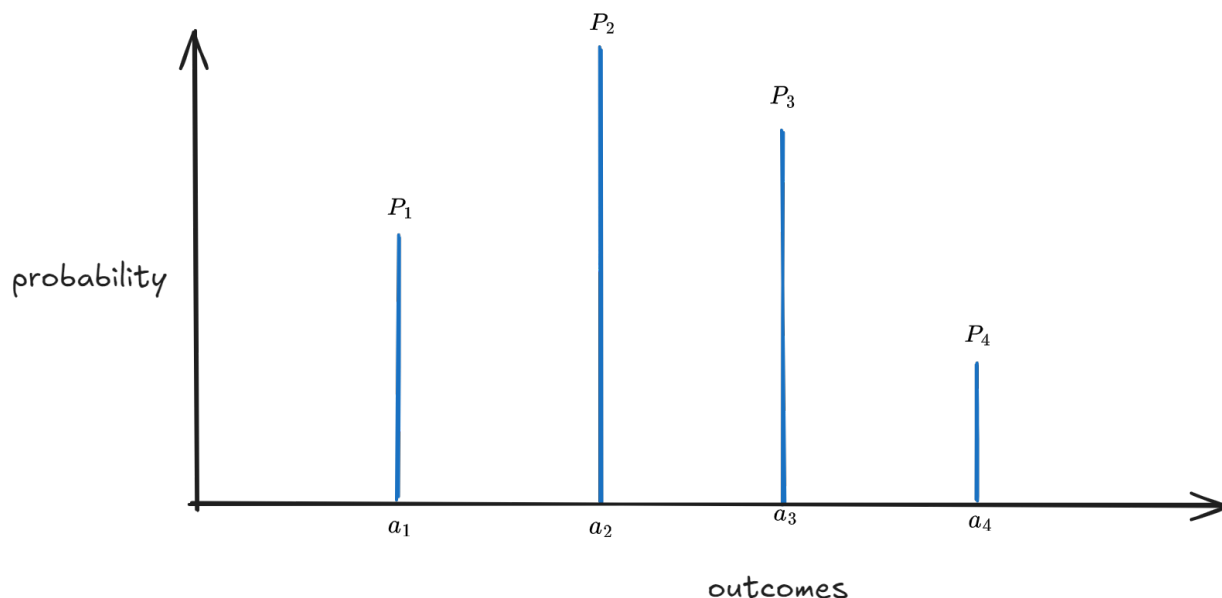


Figure 15.2



# 16 Probability and Measurement

## 16.1 Proving Moments

We start with an arbitrary wavefunction  $\psi(x)$  and arbitrary Hermitian operator  $\hat{A}$ .

$$\begin{aligned}
 \langle \hat{A} \rangle &= \int_a^b \psi^*(x) (\hat{A} \psi(x)) dx \\
 &= \int_a^b \left( \sum_m c_m^* \phi_m^*(x) \right) \left( \hat{A} \sum_n c_n \phi_n(x) \right) dx \\
 &= \int_a^b \left( \sum_m c_m^* \phi_m^*(x) \right) \left( \sum_n a_n c_n \phi_n(x) \right) dx \\
 &= \sum_m c_m^* \sum_n c_n a_n \int_a^b \phi_m^*(x) \phi_n(x) dx \\
 &= \sum_m c_m^* \sum_n c_n a_n \delta_{nm} \\
 &= \sum_m c_m^* (a_1 c_1 \delta_{1m} + \dots + a_{m-1} c_{m-1} \delta_{m-1,m} + a_m c_m \delta_{mm} + a_{m+1} c_{m+1} \delta_{m+1,m} + \dots) \\
 &= \sum_m c_m^* (0 + \dots + 0 + a_m c_m + 0 + \dots) \\
 &= \sum_m \underbrace{c_m^* c_m}_{P_m} a_m \\
 \langle \hat{A} \rangle &= \sum_m P_m a_m
 \end{aligned}$$

A note on the proof above, while the choice of  $m$  or  $n$  as indices/variables is arbitrary, it is important that we use different symbols for  $\psi(x)$  and  $\psi^*(x)$ . This allows for the key step of using the Kronecker delta to unite the coefficients of the wavefunction and its conjugate.

The process is near identical for  $\langle \hat{A}^2 \rangle$ , except an additional step inside the integral where

$$\hat{A}(\hat{A}\psi(x)) = \hat{A}(a_n \psi(x)) = a_n(a_n \psi(x)) = a_n^2 \psi(x)$$

resulting in

$$\langle \hat{A}^2 \rangle = \sum_m P_m a_m^2$$

## 16.2 Measurement

We have some observable quantity  $A$ , which is associated with some Hermitian operator  $\hat{A}$ . When we measure this quantity we observe its eigenvalues  $a_n$ , each of which has an associated probability  $P_n$  which can be obtained from the absolute square of the appropriate coefficient of the linear combination of eigenstates, where  $|c_n|^2 = P_n$ . To measure and understand a system, we measure its **ensemble**.

An ensemble is a theoretical idea, where we have identical copies of a system that are each measured. The many copies of the system allow us to understand the full system. This idea is required for two reasons

1. Quantum mechanics is a probabilistic system. We can't understand something by just measuring it once or a few times. To understand probability we need many, many data points!
2. By measuring a quantum system we change it. This means that we can't just measure the same object repeatedly, because by measuring it's a new object each time.

While this is a theoretical ideal, we can still understand physically. If we measure many identically prepared electrons, we are measuring the ensemble of that electron (with some physical error).

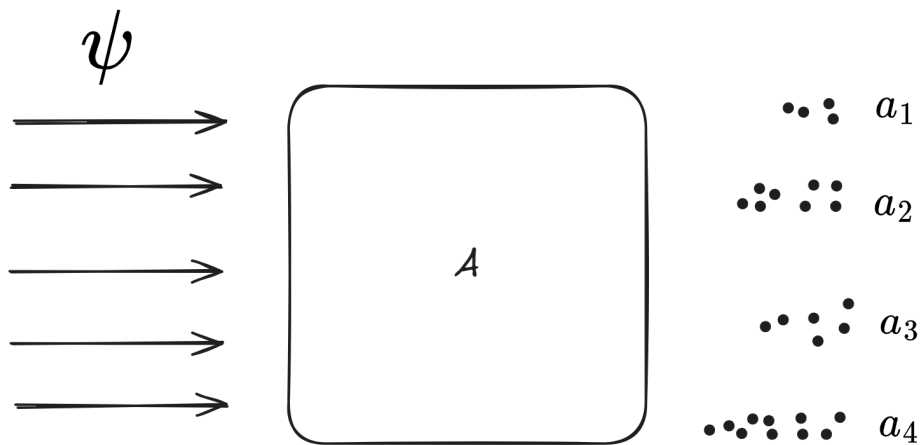
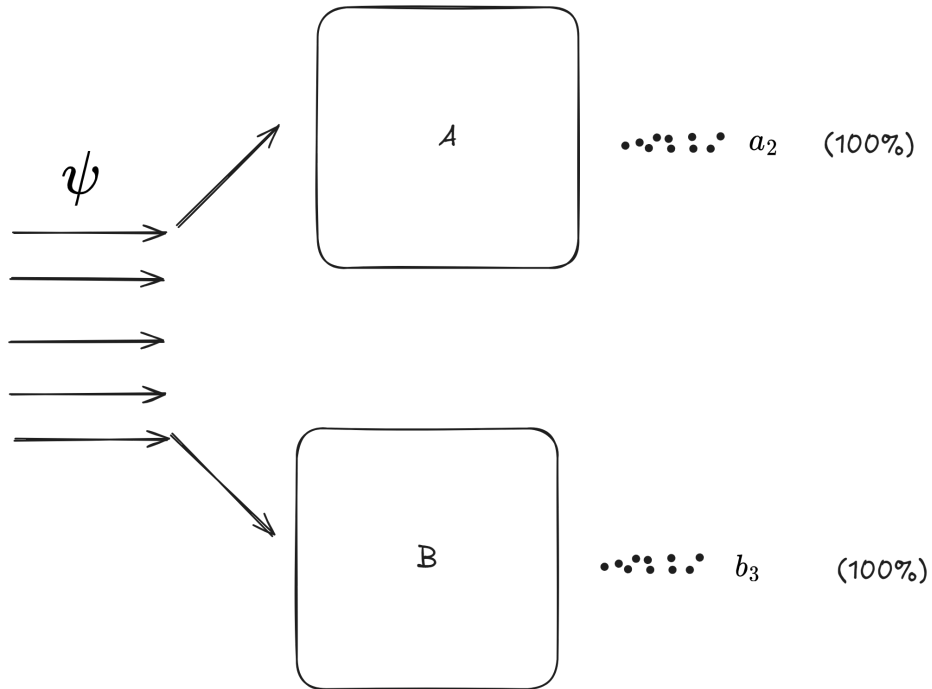


Figure 16.1

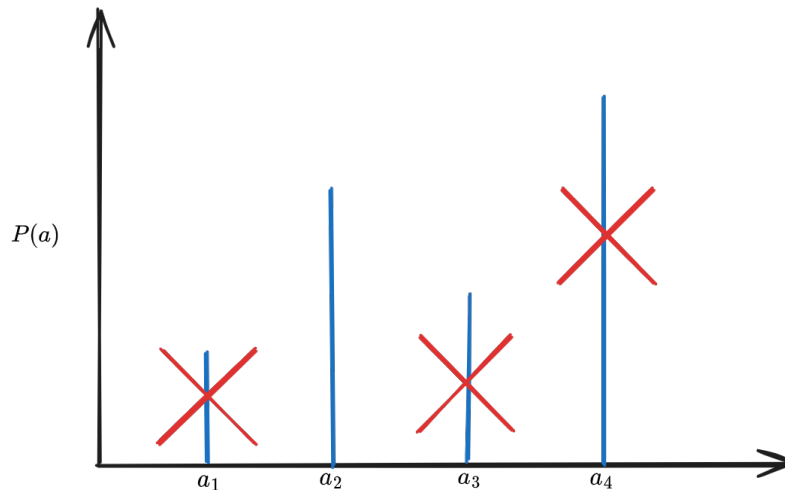
### 16.2.1 Controlling $\psi$

Upon measurement we could select only the systems that yielded for example  $a_2$  (or some other eigenvalue) and continue to do subsequent measurements with this ensemble. If we

measure  $\hat{A}$  on this new ensemble, they would result in a repeated value of 100%  $a_2$ . By doing this we are excluding the other probabilities.



**Figure 16.2**



**Figure 16.3**

For this new restricted system, the variance is 0. This makes sense, as we have a single sharp

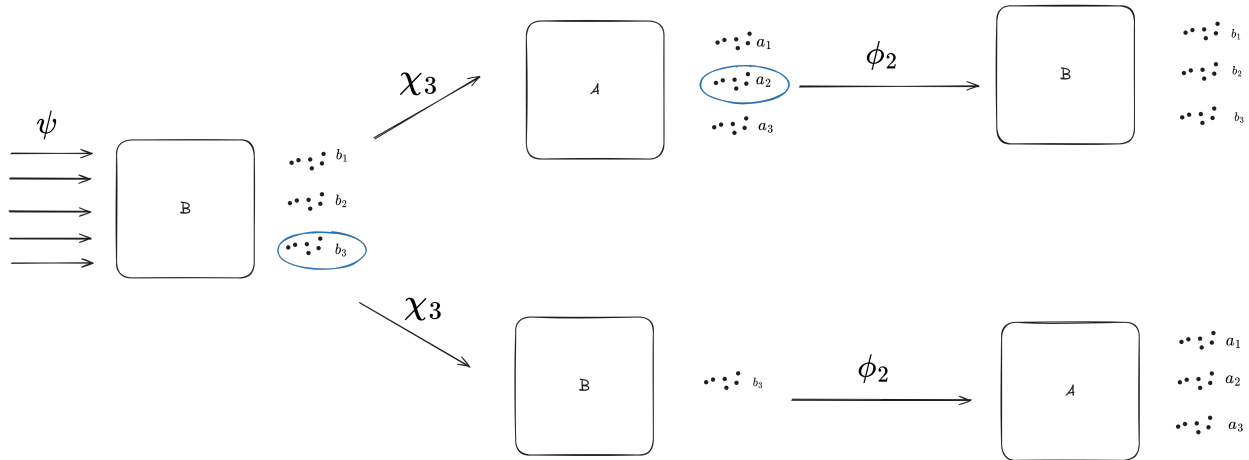
value. The expectation value is  $\langle \hat{A} \rangle = a_2$  and  $\langle \hat{A}^2 \rangle = a_2^2$ .

$$\sigma^2 = \langle \hat{A}^2 \rangle - \langle \hat{A} \rangle^2 = 0$$

$\psi(x)$  is composed of our eigenfunctions, but because we have only one eigenvalue  $a_2$  we need only care about  $\phi_{a_2}$  and  $c_{a_2}$ . The other coefficients are simply equal to 0, where  $|c_n|^2 = 0 \implies c_n = 0, n \neq 2$ .

### 16.2.2 Measuring Consecutively

Suppose we are measuring two quantities  $A$  and  $B$ . For the sake of clarity, the eigenfunctions of  $A$  are denoted  $a_n \phi_n$  and for  $B$  they are denoted  $b_k \chi_k$ . Suppose we want to measure  $a_2$  and  $b_3$ . Is it possible?



**Figure 16.4:** The eigenvalues  $b_1, b_2, b_3, \dots$  for the first and second measurements do not change, but their probabilities might. Similar for  $a_1, a_2, a_3, \dots$

We start by measuring the system  $B$ , restricting for  $b_3$  which has eigenfunction  $\chi_3$ . We then measure  $A$  restricted for  $a_2$ . We want to measure just  $b_3$  and then  $a_2$ , but instead we get a completely new set of outcomes. Why? Measuring  $A$  destroyed the outcomes of  $B$ .

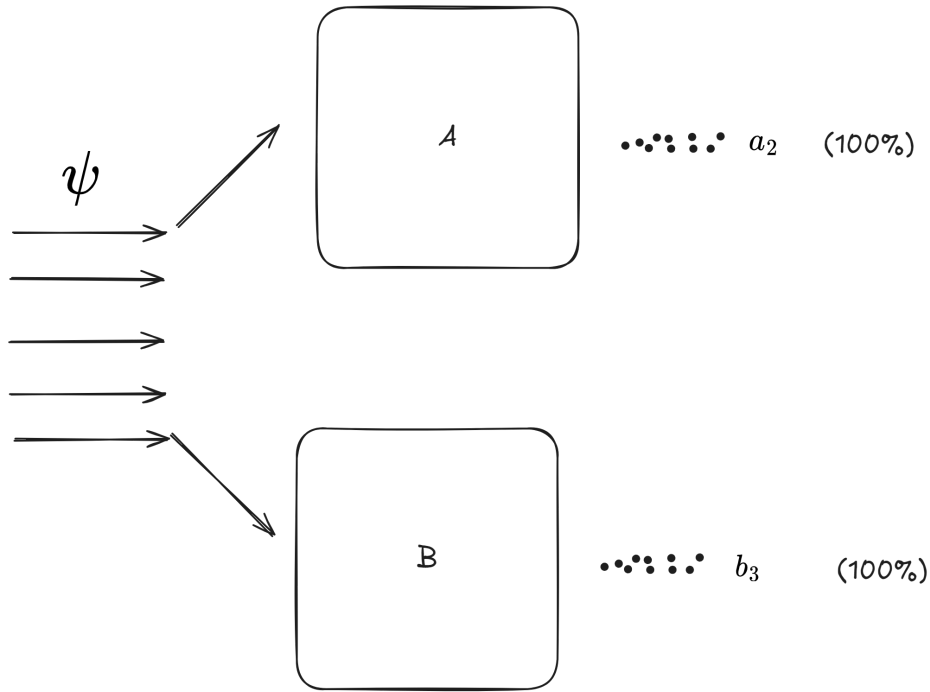
Sometimes in these experiments we could get sharp values for both. We will discuss this next. Spoiler alert: it can be done if the operators  $\hat{A}$  and  $\hat{B}$  commute:

$$\hat{A}\hat{B} = \hat{B}\hat{A}$$

# 17 Simultaneous Measurements, Heisenberg Uncertainty, and Dirac Delta

## 17.1 Simultaneous

Could we design an experiment in which we measure  $A$  and  $B$  on an ensemble  $\psi$ , and obtain sharp values for both  $A$  and  $B$ ? Can we measure two quantities of a wavefunction simultaneously?



**Figure 17.1**

Another way of phrasing it, is whether it's possible to obtain sharp values for  $A$  and  $B$ ? A sharp value means a variance of 0, where  $\sigma^2(A) = 0$  and  $\sigma^2(B) = 0$ .

This scenario exists if and only if  $\hat{A}$  and  $\hat{B}$  commute ( $[\hat{A}, \hat{B}] = 0$ ), in which case there exists a complete set of common eigenfunctions. Mathematically this is written as

$$\begin{aligned}\hat{A}\phi_{a_n, b_k} &= a_n \phi_{a_n, b_k} \\ \hat{B}\phi_{a_n, b_k} &= b_k \phi_{a_n, b_k}\end{aligned}$$

## 17.2 Example: Particle on a Ring

Suppose we are measuring energy and angular momentum of a particle. We could measure the same ensemble in different apparatuses.

It has energy described by the Hamiltonian

$$\hat{H} = -\frac{\hbar^2}{2mR^2} \frac{d^2}{d\varphi^2}$$

and angular momentum

$$\hat{L}_z = -i\hbar \frac{\partial}{\partial \varphi}$$

with eigenfunction

$$\psi_k(\varphi) = \frac{1}{\sqrt{2\pi}} e^{ik\varphi}$$

We can measure

$$\hat{H}\psi_k(\varphi) = \left( \frac{\hbar^2}{2mR^2} k^2 \right) \psi_k(\varphi)$$

$$\hat{L}_z\psi_k(\varphi) = (\hbar k)\psi_k(\varphi)$$

If  $k$  were restricted to something like  $k \pm 1$ , we would have one energy level with a degeneracy of 2 and two angular momentum levels.

## 17.3 General Proof

If the set of common eigenstates  $\phi_{a_n, b_k}(x)$  is complete, an arbitrary wavefunction is described by

$$\psi(x) = \sum_{a_n, b_k} c_{a_n, b_k} \phi_{a_n, b_k}(x)$$

Now we can evaluate  $\hat{A}\hat{B}\psi$  and the reverse order  $\hat{B}\hat{A}\psi$ . If they are the same it means that the operators commute ( $\psi(x)$  is arbitrary) Thus

$$\begin{aligned}
\hat{A}\hat{B}\psi(x) &= \hat{A} \sum_{a_n, b_k} c_{a_n, b_k} (\hat{B}\phi_{a_n, b_k}(x)) \\
&= \hat{A} \sum_{a_n, b_k} c_{a_n, b_k} b_k \phi_{a_n, b_k}(x) \\
&= \sum_{a_n, b_k} c_{a_n, b_k} b_k a_n \phi_{a_n, b_k}(x) \\
\hat{B}\hat{A}\psi(x) &= \hat{B} \sum_{a_n, b_k} c_{a_n, b_k} (\hat{A}\phi_{a_n, b_k}(x)) \\
&= \sum_{a_n, b_k} c_{a_n, b_k} a_n \hat{B}\phi_{a_n, b_k}(x) \\
&= \sum_{a_n, b_k} c_{a_n, b_k} a_n b_k \phi_{a_n, b_k}(x)
\end{aligned}$$

because we can reverse the order of numbers being multiplied ( $a_n b_k = b_k a_n$ ) these terms are the same. Therefore, since  $\psi$  is arbitrary,  $\hat{A}\hat{B} - \hat{B}\hat{A} = 0$ .

To summarize, if  $\hat{A}$  and  $\hat{B}$  have a complete set of common eigenfunctions, the operators commute.

The reverse is also true. If  $\hat{A}$  and  $\hat{B}$  commute they will have a complete set of common (orthonormal) eigenfunctions. We will skip the proof, as it is far more complicated.

## 17.4 Sharp Values for $\hat{A}$ and $\hat{B}$

If  $\hat{A}$  and  $\hat{B}$  commute, then we can generate an ensemble with both sharp values.

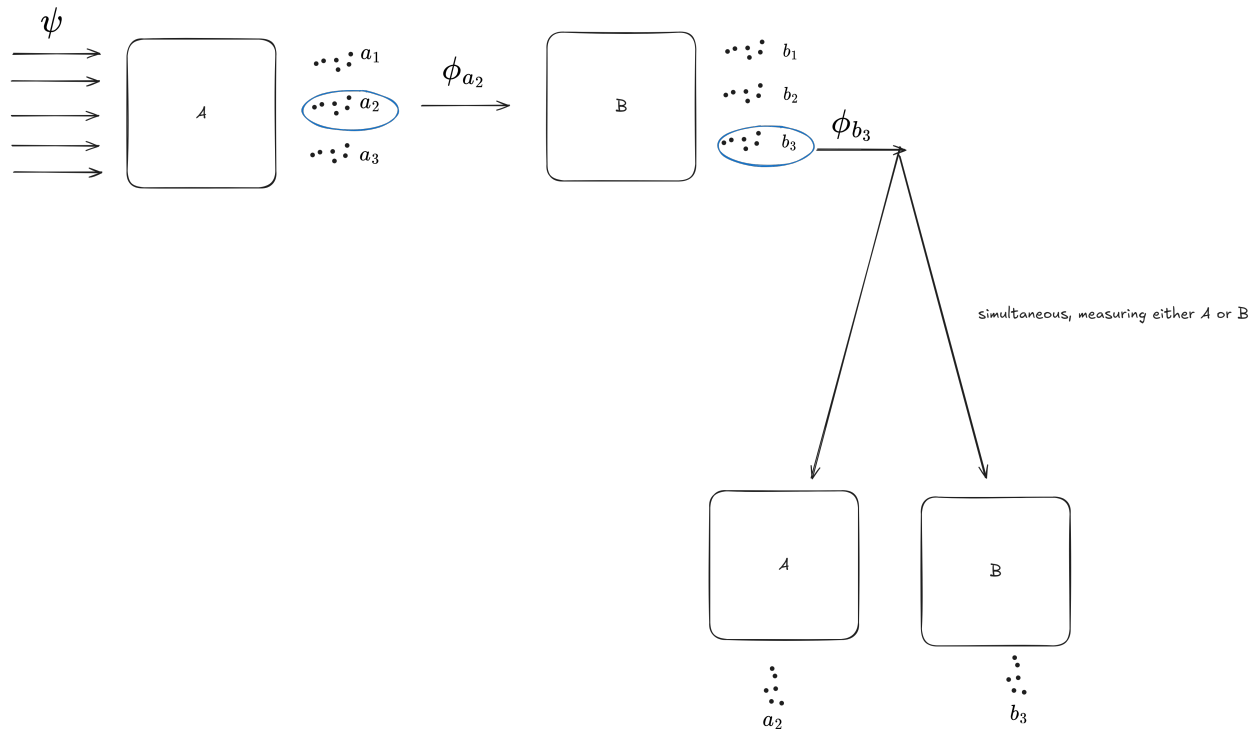


Figure 17.2

## 17.5 What if they don't commute?

Suppose  $\hat{A}$  and  $\hat{B}$  don't commute, the general case. What would happen if we tried to measure both quantities? We would get some amount of uncertainty.

We have the variance of the operators

$$\sigma_A^2 = \langle \hat{A}^2 \rangle - \langle \hat{A} \rangle^2$$

$$\sigma_B^2 = \langle \hat{B}^2 \rangle - \langle \hat{B} \rangle^2$$

The uncertainty of the operators is described by the standard deviations,  $\sigma$  of the two operators

$$\sigma_A \sigma_B \geq \frac{1}{2} \left| \int_a^b \psi^*(x) [\hat{A}, \hat{B}] \psi(x) dx \right|$$

This relation can be proved mathematically fairly easily, but we will skip the proof.

Note that the  $\frac{1}{2}$  coefficient helps bring this idea of uncertainty to correspond to the expected definitions of standard deviation and to precisely calculate this uncertainty, but is not necessary to understand the concept.



## 17.6 Heisenberg Uncertainty

Suppose we want to measure the position and momentum of a particle.

$$\begin{aligned}\hat{A} &= \hat{x} \\ \hat{B} &= -i\hbar \frac{d}{dx}\end{aligned}$$

Therefore

$$[\hat{x}, \hat{p}_x] = i\hbar$$

The uncertainty is described by

$$\sigma_x \sigma_{p_x} \geq \frac{1}{2} |i\hbar| = \frac{\hbar}{2} \quad (|i| = 1)$$

This is the famous Heisenberg uncertainty principle!<sup>1</sup>

In many textbooks it is still explained the way Heisenberg discussed this in his famous 1927 paper: by trying to measure both the position and momentum of a single electron. This is not a Hermitian operator in quantum mechanics, and so the theory does not support this kind of measurement.

Our interpretation in terms of measurements on an ensemble is more clear and consistent with the theory of quantum mechanics. There is much debate around this topic, so many professors, books, and even the top leaders in the field will all teach this differently!

## 17.7 Continuous Spectra

In the above analysis we silently assumed that an operator has discrete eigenvalues, such as separate real eigenvalues  $a_1, a_2, \dots$ . The first operator we considered, the momentum operator,  $\hat{p}_x = -i\hbar \frac{d}{dx}$  has a continuum of eigenvalues:

$$-i\hbar \frac{d}{dx} e^{ikx} = (\hbar k) e^{ikx}$$

In this equation  $k$  can be any real number. Therefore the eigenvalues are not separated real numbers, but a continuum. It is called a continuous spectra. These cases are mathematically

---

<sup>1</sup>Fun Fact: Heisenberg originally stated it as  $\sigma_x \sigma_{p_x} \geq \hbar$  before the equality was proved mathematically by Kennard in 1927.

more complicated. One aspect is that the eigenfunctions are *not* square-integrable

$$\int_{-\infty}^{+\infty} (e^{ikx})^* e^{ikx} dx = \int_{-\infty}^{+\infty} e^{-ikx} e^{ikx} dx = \int_{-\infty}^{+\infty} 1 dx \rightarrow \infty$$

This is a situation where we need more math than you'd like. Suffice it to say that it can be dealt with, and it uses that integrals

$$\int_{-\infty}^{+\infty} e^{-ikx} \psi(x) dx$$

are still finite and converge if  $\psi(x)$  is square-integrable. If the wavefunction of the system (ensemble) is described by such a  $\psi(x)$ , then the above integral defines a Fourier transform  $\tilde{\psi}(k)$ . The quantity

$$\tilde{\psi}^*(k) \tilde{\psi}(k)$$

defines the probability to find momentum  $\hbar k$ , in agreement with the usual postulates (up to an extra normalization of  $\frac{1}{2\pi}$ ).

### 17.7.1 Another Continuous Spectrum: Operator $\hat{x}$

Above we saw that the operator  $\hat{p}_x$  leads to some subtleties. There are even more issues when considering the position operator  $\hat{x}$ .

Consider the position operator  $\hat{x}$ , with some eigenvalue  $a$  with a corresponding eigenfunction  $f_a(x)$ .

$$\hat{x} f_a(x) = a f_a(x)$$

Because the position operator is just  $\hat{x} = x$ , i.e. multiply the function by  $x$

$$x f_a(x) = a f_a(x)$$

$$x f_a(x) - a f_a(x) = 0$$

$$(x - a) f_a(x) = 0$$

For  $x = a$ ,  $f_a(a)$  can be anything, but for values of  $x \neq a$  the function  $f_a(x)$  should be zero. So the eigenfunction  $f_a(x)$  is zero everywhere, except at the precise point  $a$ , where it can be anything. This is a strange beast and so it gets a special name.

### 17.7.1.1 Dirac Delta function

It is called the **Dirac delta**. This function is really a continuous analogue of the Kronecker delta

$$\delta_{mn} = \begin{cases} 1 & \text{if } m = n \\ 0 & \text{if } m \neq n \end{cases}$$

We saw before

$$\sum_m \delta_{mn} g(m) = g(n)$$

The function analogue satisfies

$$\delta(x) = \begin{cases} \infty & \text{if } x = 0 \\ 0 & \text{if } x \neq 0 \end{cases}$$

This definition has  $x$  centered specifically around 0, whereas before we had it centered around the arbitrary  $a$ . It could be rewritten more generally to be

$$\delta(x - a) = \begin{cases} \infty & \text{if } x = a \\ 0 & \text{if } x \neq a \end{cases}$$

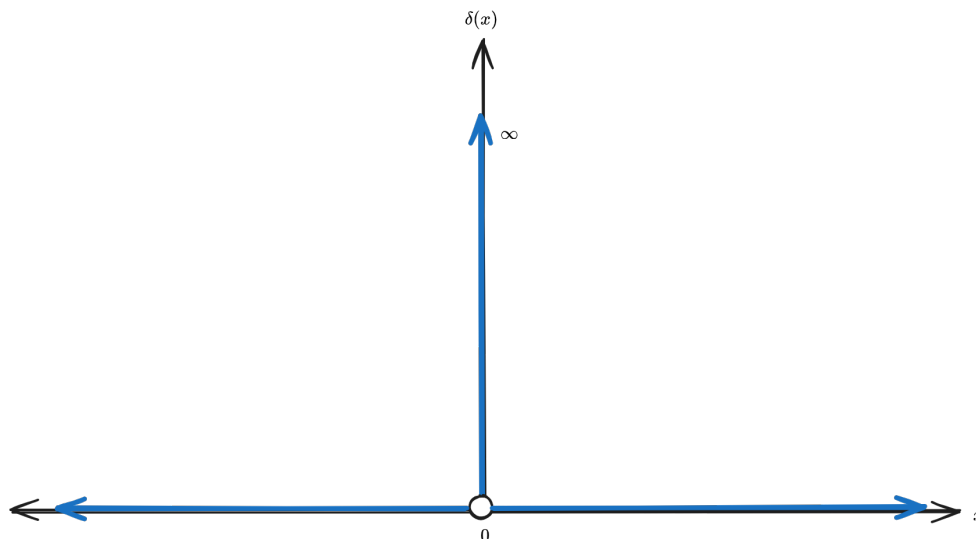
An additional important property of the Dirac delta is that when integrated against any “proper” function

$$\int_{-\infty}^{\infty} \delta(x - a) f(x) dx = f(a)$$

Just as the Kronecker delta “picks out” certain value of  $n$  of a discrete sum, the Dirac delta picks out a certain value of  $f(a)$  through a continuous sum (an integral). The Dirac delta function is not really a function. In mathematics it is called a distribution. It only consists of a limit, for example

$$\delta(x) = \lim_{\epsilon \rightarrow 0} \begin{cases} \frac{1}{2\epsilon} & : -\epsilon \leq x \leq \epsilon \\ 0 & : \text{elsewhere} \end{cases}$$

You can show that such a limit function has the properties of the Dirac delta function, and it looks like a spike with area 1.



**Figure 17.3:** The Dirac delta function. Sometimes known as the Dirac delta distribution as it is not truly a function, though it behaves like a function in most places. It is a set of functions with a shared limit, but it behaves like those functions on average. These functions are steeper and steeper impulse functions.

#### 17.7.1.1.1 Probabilities from Dirac Delta

The Dirac delta function is typically only used under an integral. The Dirac delta function is not square integrable, but it can be used to get the probability distribution to find the particle at position  $a$ , for a wavefunction described by  $\psi(x)$ . We calculate the probability amplitude, from the eigenfunction with eigenvalue  $a$ , such as  $\delta(x - a)$  (which is real):

$$c(a) = \int_{-\infty}^{+\infty} \delta(x - a)^* \psi(x) dx = \psi(a)$$

The probability to find the particle in the interval  $da$  around position  $a$  is then  $\psi^*(a)\psi(a)da$ . This is exactly the interpretation we gave of the meaning of the wavefunction in an earlier section. It is not a separate rule. It follows from the general postulate of measurement, using the eigenfunctions  $\delta(x - a)$ .

This section on operators with continuous eigenvalues is mathematically complicated. The reason they were discussed is because the operators  $\hat{x}, \hat{p}_x$  look so simple, but they have their complications.

# 18 Time Dependent Schrödinger Equation

## 18.1 Time Dependent Schrödinger Equation

The time dependent Schrödinger equation (TDSE) is one of the postulates of quantum mechanics, with the form

$$i\hbar \frac{\partial \psi(x, t)}{\partial t} = \hat{H} \psi(x, t)$$

The complete wave function needs to be specified at some starting (or reference) time  $t_0$

$$\psi(x, t_0)$$

which describes the position for all  $x$ . For all other times (not the starting condition) we can then use the TDSE to obtain the wave function at a later time. It can even be used in reverse and we can obtain the wave function at an earlier time than  $t_0$  too. It is important to realize that the wave function is fully determined at all times from this differential equation, once it is known at a specific time  $t_0$ . Probabilities and indeterminacy only enter when we want to observe properties, where then the previous discussion on measurement applies.

## 18.2 Special Case: $\hat{H}$ is Time Independent

For the first case, we consider when the Hamiltonian is time independent, with the one-dimensional form

$$\hat{H} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x) + \cancel{F(x, t)} \rightarrow 0$$

The field,  $F(x, t)$ , is assumed to be time independent. This assumption is fine for our purposes, but lasers and other time-dependent fields are excluded. This assumption will simplify the discussion, and is itself an important case in practice..

We try to arrive at the **special solution**

$$\psi_n(x, t) = c \phi_n(x) e^{-\frac{iE_n t}{\hbar}}$$

where  $c$  is some constant.

To do so we use the ubiquitous separation of variables. Let us designate two independent

functions  $\phi(x)$  and  $f(t)$ . Using the time independent Hamiltonian  $\hat{H}$

$$\begin{aligned} i\hbar \frac{\partial}{\partial t}(\phi(x)f(t)) &= (\hat{H}\phi(x))f(t) \\ i\hbar \frac{\partial f}{\partial t}\phi(x) &= (\hat{H}\phi(x))f(t) \\ i\hbar \frac{\partial f}{\partial t} \frac{1}{f(t)} &= \frac{\hat{H}\phi(x)}{\phi(x)} = E \end{aligned}$$

### 18.2.1 Special Solutions

Both sides should be constant, following the usual argument of separation of variables. With the hindsight of knowing it is constant, we call the constant: *energy*  $E$ .

We can rearrange  $\frac{\hat{H}\phi(x)}{\phi(x)} = E$  to get the classic time independent Schrödinger equation

$$\hat{H}\phi(x) = E\phi(x)$$

and the left side  $i\hbar \frac{\partial f}{\partial t} \frac{1}{f(t)} = E$  to get

$$i\hbar \frac{\partial f}{\partial t} = Ef(t)$$

We can solve for  $f(t)$  by integrating and using the separation of variables technique from ordinary differential equations:

$$\begin{aligned} i\hbar \frac{\partial f}{\partial t} &= Ef(t) \\ \frac{\partial f}{f(t)} &= \frac{E}{i\hbar} \\ \Rightarrow \int \frac{\partial f}{f(t)} &= \int \frac{E}{i\hbar} \\ \ln f(t) &= \frac{Et}{i\hbar} \\ \Rightarrow \ln f(t) &= \frac{Et}{i\hbar} \left( \frac{-i}{-i} \right) \\ \ln f(t) &= -\frac{iEt}{\hbar} \\ \Rightarrow f(t) &= e^{\frac{-iEt}{\hbar}} \end{aligned}$$

Technically there should be a  $+C$  after integrating that becomes a  $e^C$  after exponentiating, but we incorporate this into the constant of the special solution so it is ignored

here.

For good measure we can check that the solution  $f(t) = e^{-iEt/\hbar}$  indeed satisfies the differential equation. Checking is always easier than solving!

$$\begin{aligned} i\hbar \frac{\partial}{\partial t} \left( e^{-\frac{iEt}{\hbar}} \right) &= i\hbar \left( -\frac{iE}{\hbar} \right) e^{-\frac{iEt}{\hbar}} \\ &= E e^{-\frac{iEt}{\hbar}} \\ \hat{H} f(t) &= E f(t) \end{aligned}$$

To solve the classic Schrödinger equation we know that it has special solutions such that

$$\hat{H} \phi_n(x) = E \phi_n(x)$$

We then know that the combined special solutions are

$$\psi_n(x, t) = \phi_n(x) f(t) = c \phi_n(x) e^{-\frac{iE_n t}{\hbar}}$$

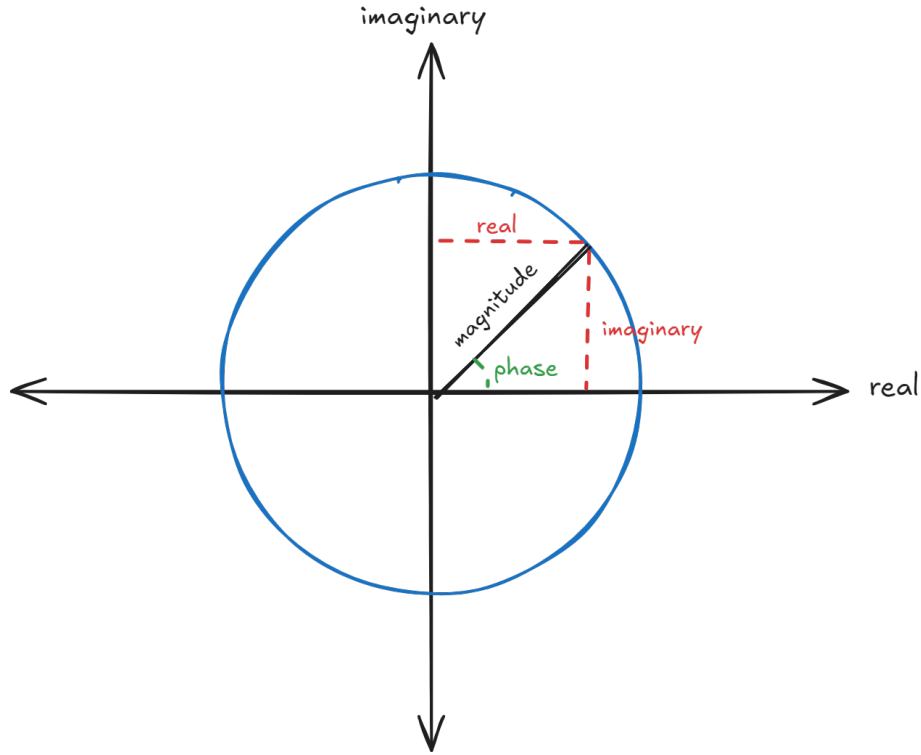
where  $c$  is some constant. An important part is that  $e^{-\frac{iE_n t}{\hbar}}$  is a **phase factor**. A (or more particularly an overall) phase factor has no direct physical meaning, as it will not change the expectation values or probability of a Hermitian operator.

### 18.2.1.1 Discussion of Special Solutions

The phase of the particle is described by  $f(t)$ , which is a complex number. It has a real phase and complex phase described by

$$\text{ph}(t) = \text{ph}_{\mathbb{R}}(t) + \text{ph}_i(t) = e^{-\frac{iE_n t}{\hbar}}$$

which can be modeled on the complex plane.



**Figure 18.1**

We know that a time-based periodic solution has a solution like the form  $e^{-i\omega t}$ , so it follows simply that  $\omega = \frac{E_n}{\hbar}$ . This angular frequency is just a version of energy governed by  $n^2$  (for the case of a particle on a box or ring), where the second angular frequency is four times as fast compared to the first, the third is nine times as fast as compared to the first, and so on.

Let's try and find the probability of the particle at a certain position now. We calculate the absolute square of the special solution. For the sake of simplicity, let  $c = 1$ .

The probability to find a particle in some interval  $[a, b]$  at time  $t$ :

$$\begin{aligned}
 P_n &= \int_a^b |\psi(x, t)|^2 dx = \int_a^b \psi^*(x, t) \psi(x, t) dx \\
 &= \int_a^b \phi_n^*(x) e^{\frac{iE_n t}{\hbar}} \phi_n(x) e^{-\frac{iE_n t}{\hbar}} dx \\
 &= \int_a^b \phi_n^*(x) \phi_n(x) e^{\frac{iE_n t}{\hbar} - \frac{iE_n t}{\hbar}} dx \\
 &= \int_a^b \phi_n^*(x) \phi_n(x) dx
 \end{aligned}$$



We see that the probability is independent of time.

The same will be true for any other property. *If wavefunction is an eigenstate of the Hamiltonian, the property does not depend on time.* For this reason these special solutions are called **stationary states**.

#### 18.2.1.1.1 Time Independent Operators

Suppose we have a time independent operator  $\hat{A}$  with eigenfunction  $\hat{A}\chi_n(x) = a_n\chi_n(x)$ . This means that the expectation value is also independent of time where

$$\langle \hat{A} \rangle_t = \langle \hat{A} \rangle_{t_0}$$

We can calculate the coefficients of the linear combination of eigenstates where

$$c_m(t) = \int_a^b \chi_m^*(x) \psi_n(x, t) dx$$

We can calculate the probability.

$$\begin{aligned} P_m &= c_m^*(t) c_m(t) \\ &= |c_m|^2 \end{aligned}$$

It is still a stationary state, so the phase factor (the only thing contributing time) cancels out and the resulting probability is independent of time.

These states where the probability to observe an eigenvalue is independent of time is the definition of a stationary state. The function itself is *not constant* but the *observables are*. The only thing that would change after adding time as variable is the wavefunction itself. We could plot this new time dependent wavefunction if we wanted, but we don't actually care about the wavefunction in and of itself.

## 18.3 General Solution

We know the special solution obtained from separation of variables is

$$\psi_n(x, t) = \phi_n(x) e^{-\frac{iE_n t}{\hbar}}$$

This is a solution for any value of  $n$ . Moreover,  $\hat{H}$  and  $i\hbar\frac{\partial}{\partial t}$  are linear operators. It follows that the general solution is a linear combination of our stationary states

$$\psi(x, t) = \sum_n c_n \phi_n(x) e^{-\frac{iE_n(t-t_0)}{\hbar}}$$

As the linear sum is combination of stationary states, where  $t = t_0 = 0$ ,

$$\psi(x, t_0) = \sum_n c_n \phi_n(x)$$

The coefficients  $c_n$  are determined by the (arbitrary, normalized) wave function at the initial time.

This linear combination satisfies the TDSE, as shown below. Remember that  $\hat{H}$  is independent of time. For simplicity we use  $t_0 = 0$  here,

$$\begin{aligned} i\hbar \frac{\partial \psi}{\partial t} &= \sum_n c_n \phi_n(x) \left( i\hbar \frac{-iE_n}{\hbar} e^{-\frac{E_n t}{\hbar}} \right) \\ &= \sum_n c_n \phi_n(x) \left( E_n e^{-\frac{iE_n t}{\hbar}} \right) \\ &= \sum_n c_n \left( \hat{H} \phi_n(x) \right) e^{-\frac{iE_n t}{\hbar}} \\ &= \hat{H} \sum_n c_n \phi_n(x) e^{-\frac{iE_n t}{\hbar}} \end{aligned}$$

Note:  $\hat{H}$  has to act on  $\phi_n$ . The order of terms is very important.

We can now determine all  $c_n$  from the wave function at the starting time  $t_0$  by using

$$c_n = \int_a^b \phi_n^*(x) \psi(x, t_0) dx$$

which allows us to calculate  $\psi(x, t)$  for all possible times as just as a combination of stationary states. This tends to be very simple in the grand scheme of things (and with some help from a computer).

In summary: How does one determine the coefficients  $c_n$ ? In quantum mechanics they are determined by the initial state (which is arbitrary in principle), at a choice of initial time ( $t_0 = 0$  here). We know a state  $\psi(x, t = 0)$  can always be written as  $\sum_n c_n \phi_n(x)$ . If we know the  $c_n$  at some time  $t_0$ , the coefficients at a later time are calculated as  $c_n e^{-iE_n(t-t_0)/\hbar}$ .

This is a mindbogglingly simple picture. Each state just moves with its own little clock! We can calculate it for today, and easy to calculate within a year from now!

## Classical vs. Quantum



In classical mechanics, quantities like position and momentum are curves/trajectories in phase space. To solve this we use many ordinary (one variable) differential equations, but in quantum we solve one partial (multivariable) differential equation with many linear terms. This allows us to solve many problems much easier, as to increase the complexity we just add more linear terms and computers are very good at solving linear terms.

## Example: Time Dependence of Non-Stationary States



Suppose we take an arbitrary linear combination of the two lowest energy states where

$$\psi(x, t) = c_1 \phi_1(x) e^{-\frac{iE_1 t}{\hbar}} + c_2 \phi_2(x) e^{-\frac{iE_2 t}{\hbar}}$$

the probability  $P_n(t) = |\psi(x, t)|^2$  would depend on time. Expanded below,

$$\begin{aligned} P_n(t) &= |c_1|^2 \phi_1^*(x) \phi_1(x) + |c_2|^2 \phi_2^*(x) \phi_2(x) \\ &+ c_1^* c_2 \phi_1^*(x) \phi_2(x) e^{-\frac{i(E_1 - E_2)t}{\hbar}} + c_2^* c_1 \phi_2^*(x) \phi_1(x) e^{-\frac{i(E_2 - E_1)t}{\hbar}} \end{aligned}$$

We have two terms with dependence on time, so this is not a stationary state.

We can calculate the time dependent coefficients of the general solution as well.

$$\begin{aligned} c_m(t) &= \int_a^b \phi_m^*(x) \psi(x, t) dx \\ &= \int_a^b \phi_m^*(x) \left( \sum_n c_n \phi_n(x) e^{-\frac{iE_n t}{\hbar}} \right) dx \\ &= \sum_n c_n e^{-\frac{iE_n t}{\hbar}} \underbrace{\int_a^b \phi_m^*(x) \phi_n(x) dx}_{\delta_{nm}} \\ &= \sum_n c_n e^{-\frac{iE_n t}{\hbar}} \delta_{nm} \\ &= c_m e^{-\frac{iE_m t}{\hbar}} \end{aligned}$$

Similarly, the expectation value of the operator would be time dependent.

$$\langle \hat{A}(t) \rangle = \int_a^b \psi^*(x, t) \hat{A} \psi(x, t) dx$$

But the energy  $\langle E \rangle$  or the probability to observe the energy  $P(E_m)$  would not change.

Working it out:

$$\begin{aligned}c_n &= \int_a^b \phi_n^*(x) \psi(x, t) \, dx \\&= \int_a^b \phi_n^*(x) \left( \sum_m c_m \phi_m(x) e^{-\frac{iE_m t}{\hbar}} \right) \, dx \\&= \int_a^b \left( \sum_m c_m \right) \phi_n^*(x) \phi_m(x) e^{-\frac{iE_m t}{\hbar}} \, dx \\&= \sum_m c_m \delta_{nm} e^{-\frac{iE_m t}{\hbar}} \\&= c_n e^{-\frac{iE_n t}{\hbar}}\end{aligned}$$

If we then take the absolute square, the phase factor cancels out and it is equivalent to  $t = t_0 = 0$ .

$$|c_n|^2 = |c_n(t_0)|^2$$

## Time Dependent Expectation Values



The full result of any expectation value that *is* time dependent is rather complicated, with multiple unwieldy inner product terms and later invoking the general product of 2 Hermitian operators, a nontrivial step.

A sketch of the proof involves taking the derivative of the operator's expectation inner product and invoking the conjugate of the Hamiltonian, which can be substituted variously into the 3 product rule terms.

This derivation is connected to the (general) Ehrenfest theorem, which relates the expectation value of any Hermitian operator and the expectation of the commutator of that operator with the Hamiltonian of the system.

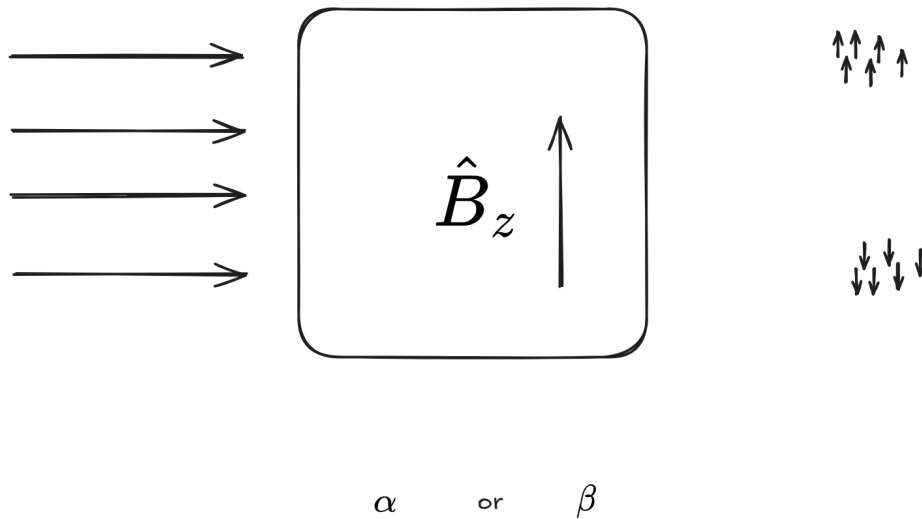
Basically: *Expectation values from quantum mechanics obey classical relations*. For example, it relates the position and momentum operators, and scalar potential with force.

For a full derivation, see the course notes. This was not discussed in class, and will not be on the midterm or final.

# 19 Spin and Quantum Entanglement

## 19.1 Measuring Spin

Suppose we measure the spin of a particle. We would measure that property in an inhomogeneous magnetic field,  $\hat{B}_z$  where we have two possible outcomes:  $\alpha$  and  $\beta$  spin. One opposes the magnetic field and the other is in a line.



**Figure 19.1**

Spin is a form of angular momentum, so we can relate the  $x, y, z$  as we can angular momentum. By convention, the  $\alpha$  and  $\beta$  or up and down spin functions are the eigenfunctions of the  $\hat{S}_z$  operator. The spin operators satisfy the same commutation relations as the angular momentum operators.

$$[\hat{s}_x, \hat{s}_y] = i\hbar\hat{s}_z$$

$$[\hat{L}_x, \hat{L}_y] = i\hbar\hat{L}_z$$

Importantly, they do not commute. therefore it is not possible to create (a complete set of) eigenfunctions that are eigenfunctions of both  $\hat{S}_z$  and  $\hat{S}_x$ . In our discussion of measurements we looked at measuring two non-commuting operators  $A$  and  $B$  in sequence. These spin operators are an ideal experiment (because they commute with  $\hat{H}$ ), and this simplifies

the consideration of time evolution. However, spin is a hard variable for us to understand. Below I will discuss an analogy with lunchboxes.

## 19.2 Lunch Box Analogy

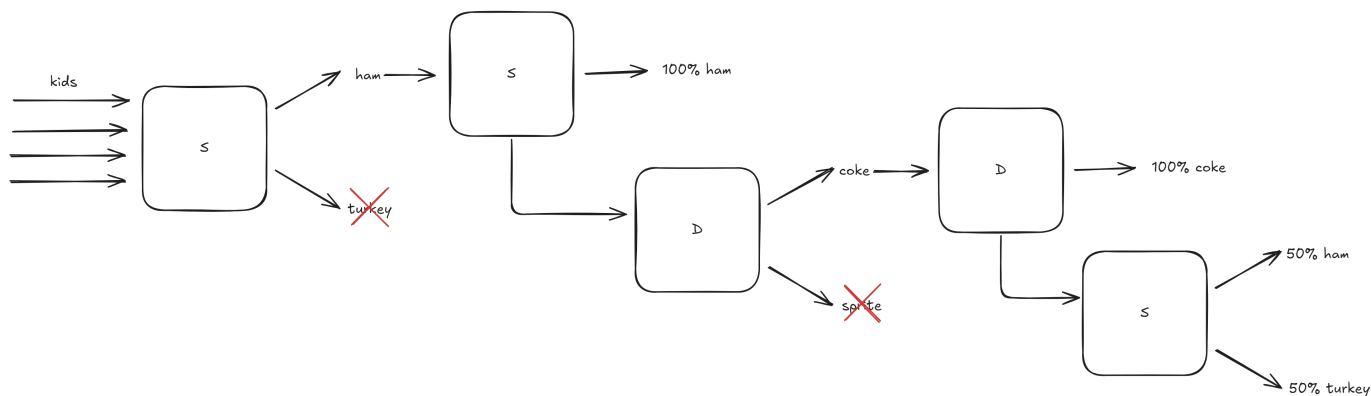
Suppose we have a group of schoolkids with the following items for lunch in their lunchbox:

- ham or turkey sandwich
- coke or sprite for their drinks
- apple or orange for their fruit

For a complete narrative of the lunchbox analogy see the additional lecture notes on Learn (optional, for fun).

In brief, we want to select the set of schoolkids that all have a ham sandwich and a coke, but following the rules of quantum measurement.

The diagram below is a sketch of what happens: Ask the question to the whole class, and only continue doing the experiment with the kids that have a ham sandwich, then only continue with those with a coke, and so on.



**Figure 19.2**

We first ask the class (the quantum ensemble) about their sandwich. The kids with ham can stay, the kids with turkey can go home. Next we ask the remaining kids about their drink. The kids with a coke stay, those with sprite can go home. In the classical world each kid would now have a ham sandwich and a coke in their lunchbox. But it is different in the quantum world, if we assume that the “sandwich and drink” questions do not commute (like  $\hat{S}_x$  and  $\hat{S}_y$ ). We cannot create an ensemble where every kid has a ham sandwich and a

coke drink. If we would ask these remaining kids what sandwiches they have again (after the drink question), the result is 50 % turkey and 50 % ham. This goes to the heart of some of the strange results from Quantum mechanics. The math of this is not so hard to understand, but the results are hard to intuit.

The key is that we must ask one question at a time of the ensemble. There is a famous case **Bell's inequality** that shows the quantum world disobeys some correlations between 3 observables, which are logical from a classical perspective. The experimental verification of these notions has been important to verify the validity of quantum mechanics and received the Nobel Prize in Physics for 2022.

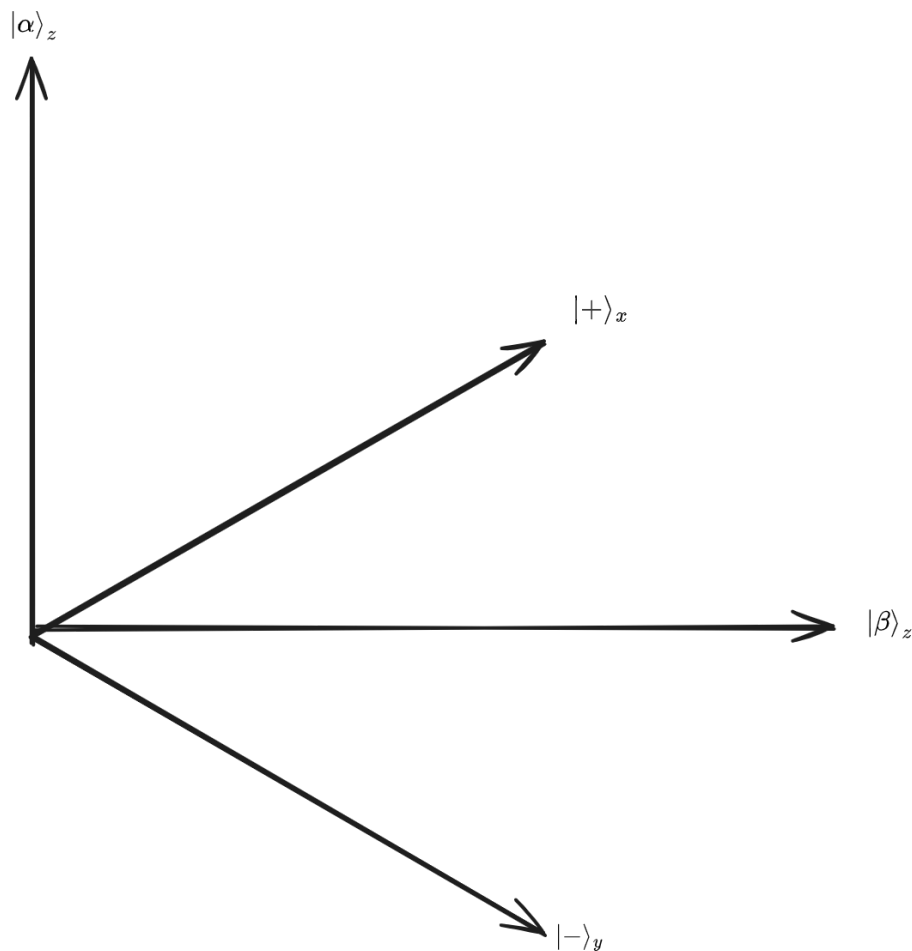
## 19.3 Spin

In this section we will use spin to explain how quantum mechanics describes these kind of results. We have two possible spin-eigenstates for the operator  $\hat{S}_z$

$$\begin{aligned}\hat{s}_z |\alpha\rangle &= \left(\frac{1}{2}\hbar\right) |\alpha\rangle \\ \hat{s}_z |\beta\rangle &= \left(-\frac{1}{2}\hbar\right) |\beta\rangle\end{aligned}$$

If we consider the operator  $\hat{S}_x$ , the eigenvalues are the same, but the eigenstates are different.

$$\begin{aligned}\hat{s}_x \frac{1}{\sqrt{2}}(|\alpha\rangle + |\beta\rangle) &= \left(\frac{1}{2}\hbar\right) |+\rangle_x \\ \hat{s}_x \frac{1}{\sqrt{2}}(|\alpha\rangle - |\beta\rangle) &= \left(-\frac{1}{2}\hbar\right) |-\rangle_x\end{aligned}$$



**Figure 19.3**

If we measure  $\hat{S}_x$  first and continue with the particles that have eigenvalue  $+\frac{1}{2}\hbar$ , the ensemble is described by the eigenstate

$$\frac{1}{\sqrt{2}}(|\alpha\rangle + |\beta\rangle) = |+\rangle_x$$

If we would measure  $\hat{S}_x$  again, only  $+\frac{1}{2}\hbar$  would be observed (100%). However, if we would measure  $\hat{S}_z$  we get a 50% probability for either  $+\frac{1}{2}\hbar$  or  $-\frac{1}{2}\hbar$ . As indicated in the figure there is not a state that is both an eigenfunction of  $\hat{S}_z$  and  $\hat{S}_x$ . This implies one cannot not create an ensemble that has sharp (100%) outcomes for both  $\hat{S}_z$  and  $\hat{S}_x$ . The math is not hard to understand. It is weird when we translate it in terms of lunchboxes.



## 19.4 Quantum Entanglement

In 1935 Einstein, Podolsky, and Rosen developed a paradox. Suppose we have a wavefunction where

$$|\psi\rangle = \alpha_1\beta_2 - \beta_1\alpha_2$$

To understand the particle we need two spin measurements.

We have two particles, both made and prepared in the same way, say in Waterloo. The spin of each particle (1 or 2) is then measured in two far away locations, say in Tokyo and London. The experiment is such that a large number of particles can be sent this way. Upon measurement the observer in Tokyo would find up or down spin with equal probability 50%. The same is true for the observer of particle 2, very far away in London. However, if the observers compare notes they find an exact anti-correlation for each particle pair. This relates to the two components of the wave function.

$$\alpha_1\beta_2 \rightarrow (\text{up } 1, \text{down } 2)\beta_1\alpha_2 \rightarrow (\text{down } 1, \text{up } 2)$$

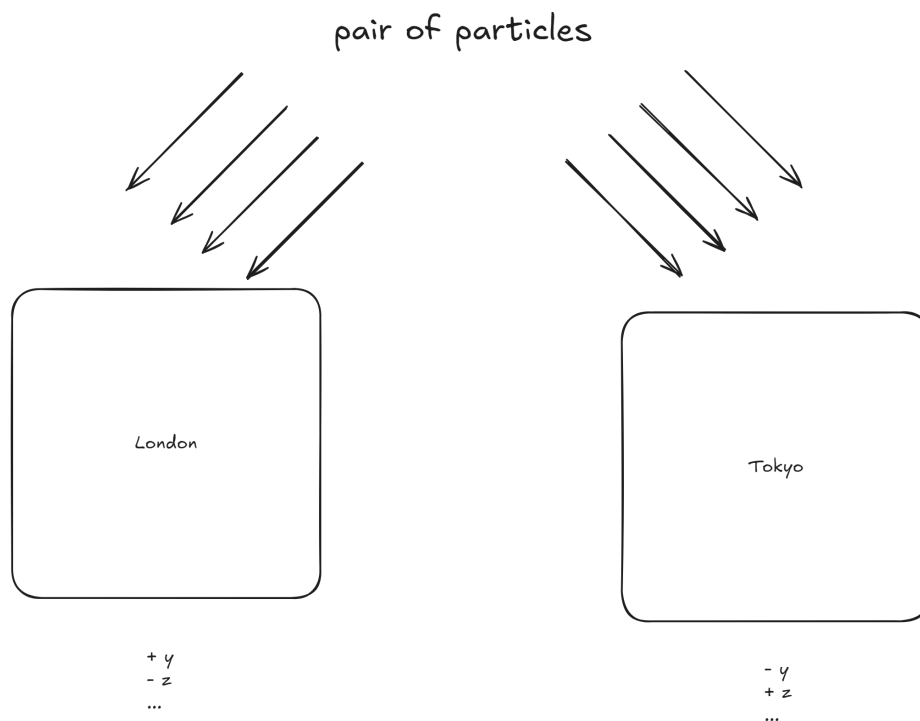
This would not be so remarkable. It might be comparable to have a pair of gloves. One glove travels to Tokyo, the other to London. If the glove in Tokyo is a right hand, then London gets the left hand glove. There is something special about the wave function. It is the same for every spin direction  $\theta$ :

$$|\psi\rangle = |+\rangle_\theta(1)|-\rangle_\theta(2) - |-\rangle_\theta(1)|+\rangle_\theta(2)$$

Hence the experimentalists might select a predetermined set of settings/directions for their magnetic fields

$$(x, y, z), (x + z), x + y + z, \dots$$

and upon comparing notes they would find exact anticorrelations. Think about this as an experiment with two coins. You flip the coin in Tokyo, say it lands on heads. The person Tokyo calls their partner in London and says, flip your coin. I am sure it is going to land tails. They can repeat this experiment many times and they are right 100% of the time. This experiment is similar to the one that received the Nobel prize. The decision which direction of spin to measure was done faster than the speed of light to avoid any type of communication between the quantum particles. It kept working.



**Figure 19.4**

## 19.5 Schrödinger's Cat

Suppose we have some quantum process, such as nuclear decay (with some probability), that prompts the release of poison in a room containing a cat. Is the cat alive or dead? We do not know until observing, but in the process of observing a state must be chosen.

We can write an sketch of wavefunctions to explain this process and its outcomes.

$$\begin{aligned}\psi_{\text{nucleus}} &= \psi_{\text{original}} + \psi_{\text{decay}} \\ \psi_{\text{cat}} \cdot \psi_{\text{nucleus}} &= \psi_{\text{alive}} \cdot \psi_{\text{original}} + \psi_{\text{dead}} \cdot \psi_{\text{decay}}\end{aligned}$$

We can add another variable, you.

$$\psi_{\text{you}} \cdot \psi_{\text{cat}} \cdot \psi_{\text{nucleus}} = \psi_{\text{happy}} \cdot \psi_{\text{alive}} \cdot \psi_{\text{original}} + \psi_{\text{sad}} \cdot \psi_{\text{dead}} \cdot \psi_{\text{decay}}$$

In one interpretation, we can think of this as a “Humble Theory”, that Schrödinger’s cat is not about one cat, but about many identical cats in identical rooms. We can use quantum mechanics to understand the probabilities of many cats being alive and/or dead. It is a sad experiment.

There are many strange aspects about quantum mechanics that are hard to understand if we connect it to the world as we know it. The mathematics is quite clear. The interpretation is hard, and leads to lots of discussion. The ensemble presentation advocated here is a kind of minimal interpretation that agrees with observations, but leaves open questions. It avoids a lot of the very strange aspects of quantum mechanics. Let me note here that in particular the entanglement property and the nature of superpositions can be put to use. That is at the heart of the great interest in quantum computing.

# 20 Hamiltonian of a Molecule, Born-Oppenheimer Approximation, and Potential Energy Surfaces

The Hamiltonian of some molecule is composed of multiple terms. The molecule (for chemists) consists of nuclei (point charges) and electrons. The motion of these particles and their Coulombic interactions enter the quantum mechanical Hamiltonian.

## 20.1 Kinetic Energy

Consider nuclei denoted  $\alpha$ , where each nucleus  $\alpha$  has some charge  $q_\alpha = z_\alpha e$ , where  $e$  is the elementary charge. It has a set of coordinates in three-dimensional space that describe its position,  $\vec{R}_\alpha = (x, y, z)_\alpha$ . It has some nuclear mass  $M_\alpha$ . We also have electrons with position  $\vec{r}_i = (x, y, z)_i$  and (consistent) mass  $m_e$ . With these in mind, we can start with some basic definitions.

The momentum of our particle is the same standard form, but has additional directions to account for all of space. Keeping in mind that  $\hat{p}_{x,\alpha} = i\hbar \frac{\partial}{\partial x_\alpha}$  Starting with nucleus  $\alpha$ , the nuclear kinetic energy term is

$$\begin{aligned}\hat{T}_{N,\alpha} &= \frac{1}{2M_\alpha}(\hat{p}_{x,\alpha}^2 + \hat{p}_{y,\alpha}^2 + \hat{p}_{z,\alpha}^2) \\ &= -\frac{\hbar^2}{2M_\alpha} \left( \frac{\partial^2}{\partial x_\alpha^2} + \frac{\partial^2}{\partial y_\alpha^2} + \frac{\partial^2}{\partial z_\alpha^2} \right) \\ &\equiv -\frac{\hbar^2}{2M_\alpha} \vec{\nabla}_\alpha^2\end{aligned}$$

where  $\vec{\nabla}$  is the gradient operator, sometimes called the total (spatial) derivative. To take the “square” of the gradient, we must use the dot product, which produces a scalar sum. Therefore this  $\vec{\nabla}^2$  is considered to be a scalar operator.

The gradient has all component partial derivatives contained within it. As a vector it points in the direction of maximum change, accounting for all variables.<sup>1</sup>

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<sup>1</sup> $\nabla^2$ , or  $\vec{\nabla}^2$  is often called the Laplacian or Laplace operator. Mathematicians denote the Laplacian by  $\Delta$  instead of  $\vec{\nabla}^2$ .

This is similar to the electron term, where

$$\hat{T}_{e,i} = -\frac{\hbar^2}{2m_e} \vec{\nabla}_i^2$$

### 20.1.1 Multiple Atoms/Electrons

The definition of kinetic energy for multiple atoms is easily extended. We just sum the kinetic energy operator over the entire set of atoms in our molecule. If we had H<sub>2</sub> we would have two terms, if we had C<sub>6</sub>H<sub>6</sub> we would have twelve terms.

$$\hat{T}_N = \sum_{\alpha} -\frac{\hbar^2}{2M_{\alpha}} \vec{\nabla}_{\alpha}^2$$

In this multiple atom scenario,  $\alpha$  becomes the index of atoms. Similarly, for electrons

$$\hat{T}_e = \sum_i -\frac{\hbar^2}{2m_e} \vec{\nabla}_i^2$$

## 20.2 Potential

The Coulomb relationship is well known. If we have particles 1 and 2, it's simply

$$V_{1,2} = \frac{q_1 q_2}{4\pi\epsilon_0 |\vec{r}_1 - \vec{r}_2|}$$

where  $\epsilon_0$  is the electric permittivity value, a measure of how easily charge moves depending on what material it's in. Here it will be the value in vacuum (a constant).

### 20.2.1 Nuclear–Nuclear

The potential energy between two nuclei is

$$\hat{V}_{NN} = \frac{1}{2} \sum_{\alpha \neq \beta} \frac{z_{\alpha} z_{\beta} e^2}{4\pi\epsilon_0 |\vec{R}_{\alpha} - \vec{R}_{\beta}|}$$

The potential energy between two electrons is

$$\hat{V}_{ee} = \frac{1}{2} \sum_{i \neq j} \frac{e^2}{4\pi\epsilon_0 |\vec{r}_i - \vec{r}_j|}$$

Note that the 1/2 coefficient is to simply exclude double counting. If we have electron 3 and 4, that has the same energy as 4 and 3 and so on. Similarly for the nuclei.

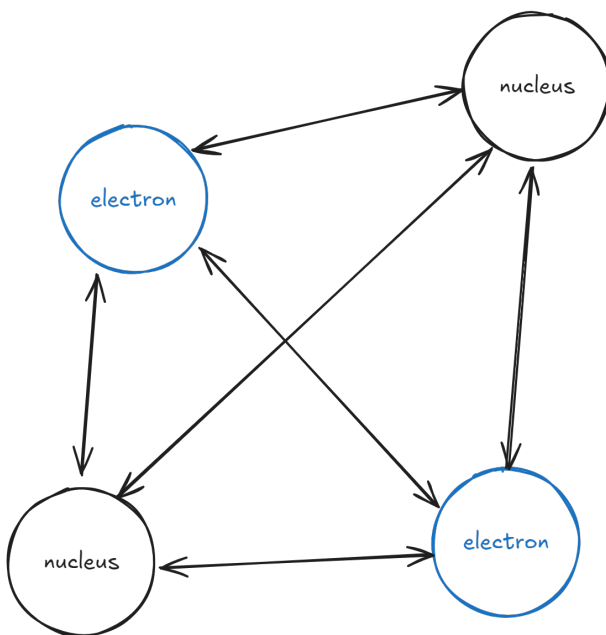
The potential energy between an electron and a nucleus is

$$\hat{V}_{Ne} = - \sum_{i,\alpha} \frac{z_\alpha e^2}{4\pi\epsilon_0 |\vec{r}_i - \vec{R}_\alpha|}$$

## 20.3 Combining Terms

Now we can construct a Hamiltonian with all interaction terms.

$$\hat{H} = \hat{T}_N + \hat{T}_e + \hat{V}_{Ne} + \hat{V}_{ee} + \hat{V}_{NN}$$



**Figure 20.1:** Counting the arrows (potential energy interactions) and circles (kinetic energy individuals), we have 10 terms for  $H_2$ .

The Hamiltonian depends on all the particles in the system, which are each described positionally by their own set of coordinates. Likewise, the wavefunction depends on the Cartesian coordinates of all the particles in the molecule (nuclei and electrons)

$$\psi(\underbrace{\vec{r}_1, \vec{r}_2, \dots, \vec{r}_K}_{\text{electrons}}, \underbrace{\vec{R}_1, \vec{R}_2, \dots, \vec{R}_M}_{\text{nuclei}})$$

For every nucleus and electron we have 3 Cartesian coordinates. If we have  $\text{H}_2$ , with 2 atoms and 2 electrons, we have 12 Cartesians ( $4 \times 3 = 12$ ). For benzene with 12 atoms and 42 electrons, 162 Cartesian coordinates ( $54 \times 3 = 162$ ), where 36 of those are nuclear terms. That's a lot of terms and very hard to solve, and benzene is a simple and relatively small molecule.

It will be clear that the Schrödinger equation is very hard to solve exactly. This should not be surprising as we have, in principle, described how to solve for the wavefunction of any molecule with our simple prescription above. We can make progress by making approximations, that in turn will yield essential insights.

## 20.4 Born-Oppenheimer Approximation

To make the Hamiltonian easier to solve, we solve the problem in two steps. In the first step we “fix” the nuclear configuration in space. This allows us to

1. Solve for electron energy electrons first.
2. Solve for nuclear energy afterwards, using the Schrödinger equation.

If we take the earlier Hamiltonian and exclude the kinetic energy of the nuclei

$$\hat{H}_{\text{el}} = \cancel{\hat{T}_N} + \hat{T}_e + \hat{V}_{Ne} + \hat{V}_{ee} + \hat{V}_{NN}$$

We now can describe our new way wavefunction as

$$\hat{H}_{\text{el}} = \frac{\psi(\{\vec{r}\}, \vec{R}_{\text{fixed}})}{E_{\text{el}}\{R_{\text{fixed}}\}\psi(\{\vec{r}\}, \vec{R}_{\text{fixed}})}$$

We now have a reduced Hamiltonian. For benzene we go from 162 to 126 Cartesians, an improvement. It will turn out that for electrons we can make very efficient approximations in terms of orbitals (that depend on the coordinates of 1 electron = 3 Cartesian coordinates).

It is often said in the textbooks that the Born-Oppenheimer approximation (the two-step solution) works because the relative time scales of nuclei vs. electrons, as kinetic energy is based on speed. Nuclei are so much larger and move so much slower, that in comparison to the electrons, they basically don't move and have negligible kinetic energy. The electrons instantaneously follow the motion of the nuclei. There is clear difficulty with this argument. If we are solving the time-independent Schrödinger equation, there is no time, and no (obvious) motion. The argument is really that the energy differences for different electronic states

(electronic excitations, around 4 eV) are much larger than nuclear excitations (translations, rotations vibrations). The largest vibrational energies are about  $4000\text{cm}^{-1} \approx 0.5\text{eV}$ , and they are hard to excite. We solve for the most important part first (large energies), and then add on the more subtle motion (energetically speaking).

## 20.5 Potential Energy Surfaces

We can calculate something called a **potential energy surface** (PES). It is analogous to a potential energy curve, seen before in reactions dynamics, but has a greater number of coordinates.

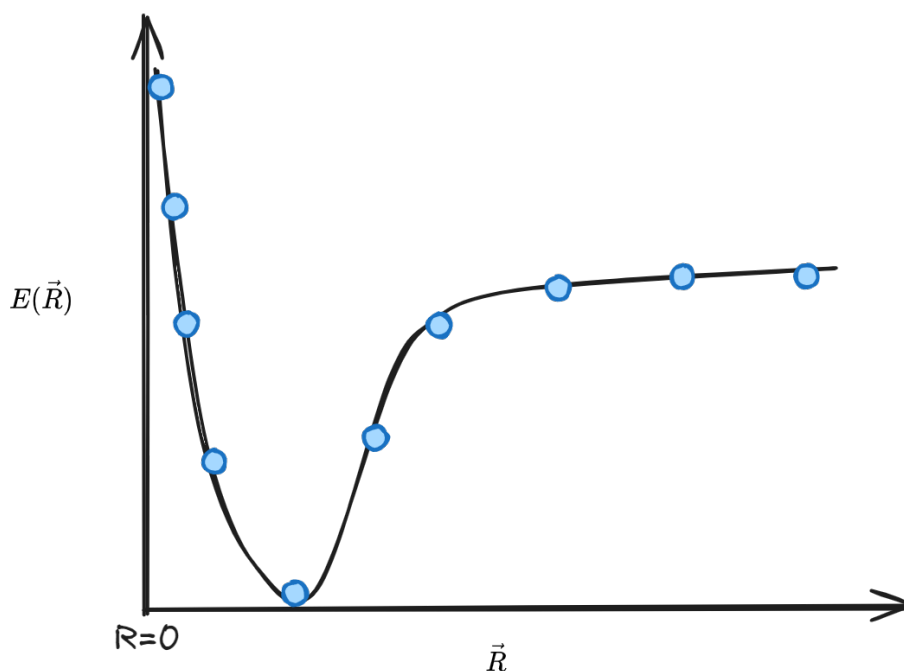


Figure 20.2

To calculate a PES, we would fix the nuclear coordinates in space and then calculate our electron energy. We would do this over many points, to accurately describe the molecule. We calculate one point, then tweak the nuclear coordinates slightly, then repeat.

So if we were to calculate the potential energy surface of benzene, we would have 36 nuclear Cartesians we could adjust. We could create a “grid” of Cartesians, the foundation of our potential energy surface. But is this feasible?

Not quite. If we take the example of a water molecule, where we could model it in relative coordinates as having  $R_1, R_2, \theta$  (3 Cartesians), a grid of length 10 would result in



$10^3$  points.

If we were to do the same with benzene, we would have  $10^{36}$  points... which is still not feasible. The feasibility depends on the resolution and complexity of the molecule.

However, conceptually the potential energy is very important. Minima on the potential energy surface define stable molecular structures. Saddle points define transition states. The language of how we talk about molecules is rooted in the potential energy surface and hence the Born-Oppenheimer approximation.

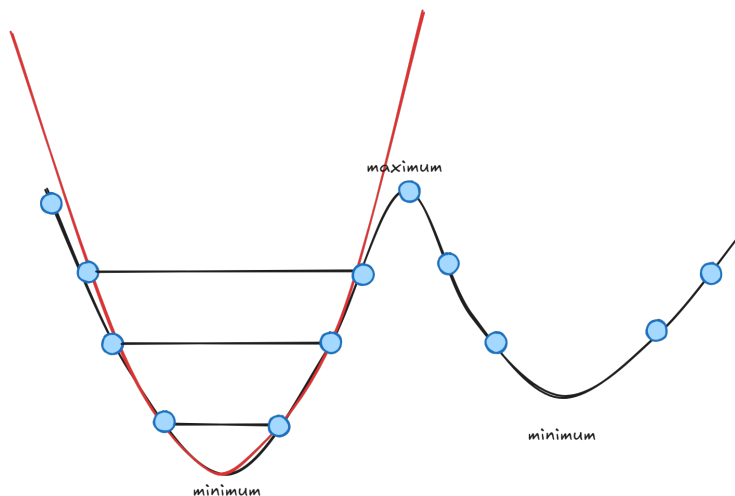
### 20.5.1 More Approximations

Practically,  $E_{\text{el}}(\{\vec{R}_{\text{fixed}}\})$  is never calculated. Instead we find the minimum on the potential energy surface, which can be optimized more easily. At this minimum we find some  $\vec{R}_{\text{eq}}$  value. Every minimum defines a stable molecular structure. It could be the case that a nuclear configuration describes multiple molecules, A and B for example, separated by a large distance. Hence, we can identify molecular identities (which nuclei from the molecule). This defines the masses of the molecule, and that is all we need to calculate translational energies. From the equilibrium we can also deduce the moments of inertia (see later chapter), and this suffices to determine approximate energy levels of molecular rotations. This is called the rigid rotor approximation.

In addition to the minimum, we can also obtain the value of the potential energy at the minimum, which provides the so-called electronic energy. Finally, one can calculate the second derivatives of the energy at the equilibrium structure. There are  $(3N) \cdot (3N)$  second derivatives that are organized in a matrix, the so-called of the system. A diagonalization of the Hessian gives access to vibrational energy levels. This is referred to as the harmonic approximation. In each (normal mode) direction the potential energy surface is approximated by a parabola.

These approximations may seem very crude: we approximate the full potential energy surface into a number of distinct harmonic surfaces, and solve the quantum mechanical problem for these much simpler problems. However, the results of these calculations are very useful. The hardest part is how to solve for the electronic energy accurately. This is the basis for thermochemistry, and the theory of chemical equilibrium. If we include transition states using the same harmonic oscillator/rigid rotor framework we can get good approximations to (elementary) rates of reaction. These topics are dealt with in great detail in statistical mechanics. The approximations lead to useful chemical concepts, and moreover they yield results that are sufficiently accurate to describe experiments.

Say we have the molecule  $\text{H}-\text{C}\equiv\text{N}$ . We know it can react to form  $\text{C}\equiv\text{N}-\text{H}$ . The potential energy curve could look like this, where each level is a harmonic oscillation of the nucleus.



**Figure 20.3:** Pictorial representation (in 1-dimension) of potential energy with two minima and a transition state. The harmonic approximation for the lowest minimum is indicated by the red line. We could make a similar plot for the second minimum. These harmonic basins accurately represent molecules and their nuclear motion in rigid rotor/harmonic oscillator approximation.

One can design more accurate treatments, calculating full potential energy surfaces for small molecules. The results from such calculations are not always easy to interpret as the solution to the Schrödinger equation does not directly indicate structural features, such as bonds, which can make interpretation of the results difficult. One would have to analyze the corresponding wave functions, possible superpositions.

In the remainder of this chapter we will analyze the various aspects of nuclear motion. The problem of solving for the electronic energy at a particular nuclear configuration is very complicated. We will only scratch the surface of this problem in later chapters.

## 21 Discussion of Vibrational Motion Using the Harmonic Oscillator Model

### 21.1 Summary

The simplest molecular system that shows vibrational motion is a diatomic molecule. We will analyze this in detail.

Diatomics have a relatively simple form. We can model them as a one-dimensional harmonic oscillator, where the Hamiltonian is

$$\hat{H} = -\frac{\hbar^2}{2\mu} \frac{d^2\psi}{dx^2} + \frac{1}{2}kx^2$$

and the Schrödinger equation can be solved relatively simply. We use the wavefunction  $\psi(x) = e^{-\alpha x^2/2}$ , with the parameter  $\alpha$  to represent the ground state wave function. We will introduce a ladder operator formulation to describe excited states of the harmonic oscillator, and this leads to equidistant energy levels, characteristic of vibrations in the harmonic oscillator approximation. Polyatomic molecules are straightforward to understand, simply extending to the use of multiple harmonic oscillators as the number of bonds increase. This will be discussed in subsequent lectures, as this is a very useful model, used widely in practical calculations.

### 21.2 Harmonic Oscillator Model for Diatomic molecules

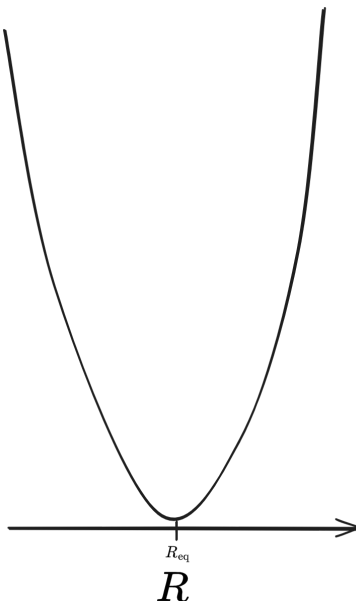
The potential of a diatomic molecule AB only depends on the internuclear distance  $R$ . We can find the minimum of the potential at  $R = R_{eq}$ , and define the distance from this equilibrium position as

$$x = R - R_{eq}$$

Around the minimum, we can model the potential energy as a quadratic/parabola. Because we expand the potential at  $x = 0, R = R_{eq}$ , there is no linear term in  $x$ , and the potential form is

$$V(x) = \frac{1}{2}kx^2$$

which is the quadratic or harmonic approximation.



**Figure 21.1**

There is a constant  $k$  which is called the force constant and it is the second derivative of the true potential at equilibrium

$$k = \left[ \frac{\partial^2}{\partial R^2} V(R) \right]_{R=R_{eq}}$$

The units of  $k$  are  $\text{N m}^{-1}$ .

The Hamiltonian that describes the vibrational motion for a diatomic is

$$\hat{H} = \underbrace{-\frac{\hbar^2}{2\mu} \frac{d^2}{dx^2}}_{\text{kinetic}} + \underbrace{\frac{1}{2} k x^2}_{\text{potential}}$$

where  $\mu$  is the reduced mass.

$$\mu = \frac{m_A m_B}{m_A + m_B}$$

The occurrence of the reduced mass and a single degree of vibrational freedom for a diatomic can be understood from classical mechanics. We will consider this more deeply when looking at rotational motion for a diatomic in the next chapter.

This Schrödinger Equation  $\hat{H}\psi(x) = E\psi(x)$  is not easy to solve. We will use hindsight

and assume that the (unnormalized) ground state has the form

$$\psi(x) = e^{-\alpha x^2/2}$$

Here  $\alpha$  is a parameter to be determined. We can use this, and its derivatives in our Hamiltonian, to solve the Schrödinger equation.

$$\begin{cases} \psi(x) = e^{-\alpha x^2/2} \\ \frac{d\psi}{dx} = -\alpha x e^{-\alpha x^2/2} \\ \frac{d^2\psi}{dx^2} = -\alpha e^{-\alpha x^2/2} + \alpha^2 x^2 e^{-\alpha x^2/2} \end{cases}$$

Be sure to review your knowledge of taking derivatives. Here we used the chain rule and the product rule. If we now substitute this into the Hamiltonian

$$\begin{aligned} \hat{H}\psi(x) &= -\frac{\hbar^2}{2\mu}(-\alpha e^{-\alpha x^2/2} + \alpha^2 x^2 e^{-\alpha x^2/2}) + \frac{1}{2}kx^2(e^{-\alpha x^2/2}) \\ &= \frac{\hbar^2}{2\mu}\alpha e^{-\alpha x^2/2} - \frac{\hbar^2}{2\mu}\alpha^2 x^2 e^{-\alpha x^2/2} + \frac{1}{2}kx^2 e^{-\alpha x^2/2} \\ &= \left( \frac{\hbar^2}{2\mu}\alpha - \frac{\hbar^2}{2\mu}\alpha^2 x^2 + \frac{1}{2}kx^2 \right) e^{-\alpha x^2/2} \end{aligned}$$

we see with the context of the Schrödinger equation (where the eigenvalues of the Hamiltonian are energy) that

$$E = \frac{\hbar^2}{2\mu}\alpha - \frac{\hbar^2}{2\mu}\alpha^2 x^2 + \frac{1}{2}kx^2$$

It is important to realize that  $E$  is a constant: It does not depend on the variable  $x$ . The force constant  $k$  is just a symbolic number, determined by the problem. The real problem is: we need to solve for the parameter  $\alpha$ , such that  $E$  will be a constant.

Let us consider a general procedure to solve the problem

$$\begin{aligned} \hat{H}\psi(x) &= E\psi(x) \\ \left( \frac{\hbar^2}{2\mu}\alpha - \frac{\hbar^2}{2\mu}\alpha^2 x^2 + \frac{1}{2}kx^2 \right) e^{-\alpha x^2/2} &= E e^{-\alpha x^2/2} \\ \implies \frac{\hbar^2}{2\mu}\alpha - \frac{\hbar^2}{2\mu}\alpha^2 x^2 + \frac{1}{2}kx^2 &= E \end{aligned}$$

From here we can equate terms. This is standard technique used when solving differential equations, most often when solving by trial solution (our trial solution was  $\psi(x) = e^{-\alpha x^2/2}$ ).

We do this by matching up the different powers of our variable on the left and right side. Here we use that  $E$  is a constant!

$$\frac{\hbar^2}{2\mu}\alpha^2 - \frac{\hbar^2}{2\mu}\alpha^2 x^2 + \frac{1}{2}kx^2 = 0x^2 + 0x + E$$

This allows us to form a system of equations. Starting with the  $x^2$  terms:

$$\begin{aligned} -\frac{\hbar^2}{2\mu}\alpha^2 x^2 + \frac{1}{2}kx^2 &= 0 \\ \frac{1}{2}kx^2 &= \alpha^2 x^2 \frac{\hbar^2}{2\mu} \\ kx^2 &= \alpha^2 x^2 \frac{\hbar^2}{\mu} \\ \alpha^2 &= \frac{k\mu}{\hbar^2} \\ \alpha &= \frac{\sqrt{k\mu}}{\hbar} \end{aligned}$$

We've now solved for  $\alpha$ ! We can rewrite it to a more convenient form, where

$$\alpha = \frac{\sqrt{\mu k}}{\hbar} = \frac{\mu\omega}{\hbar}$$

For the sake of the future, we now figure out the units of  $\alpha$ .

$$\begin{aligned} \alpha &= \frac{\mu\omega}{\hbar} \\ [\alpha] &= \frac{\text{kg} \cdot \text{s}^{-1}}{\text{J s}} \\ &= \frac{\text{s}^2}{\text{kg m}^2 \text{s}^2} \frac{\text{kg}}{\text{s}^2} \\ &= \frac{1}{\text{m}^2} \end{aligned}$$

We can use this in the next equation to write the full equation for energy.

$$\begin{aligned}\frac{\hbar^2}{2\mu}\alpha &= E \\ \frac{\hbar^2}{2\mu} \frac{\sqrt{k\mu}}{\hbar} &= E \\ \frac{\hbar}{2} \underbrace{\sqrt{\frac{k}{\mu}}}_{\omega} &= E \\ E_{\text{gs}} &= \frac{1}{2}\hbar\omega\end{aligned}$$

Here  $\omega = \sqrt{\frac{k}{\mu}}$  is the natural frequency, often denoted  $\omega_0$ . The units of  $\omega$  are

$$(\text{N m}^{-1} \text{kg}^{-1})^{1/2} = (\text{kg m s}^{-2} \text{m}^{-1} \text{kg}^{-1}) = \text{s}^{-1}$$

As noted before  $k$  (units  $\text{N m}^{-1}$ ) represents the spring constant, but in the context of a nonliteral spring, we often call it the **force constant**.

## 21.2.1 Nondimensionalizing and Standard Form of the Harmonic Oscillator

### Nondimensionalizing



Nondimensionalizing is a standard technique used in differential equations. We do it to get rid of units and simplify the expression. We can agree that the equation

$$\frac{dV}{d\tau} = 1 - V^2$$

looks much simpler to solve than

$$\frac{dv}{dt} = mg - \beta v^2$$

but they are the same core equation! We nondimensionalize a differential equation based on the base units (mass, length, charge, etc.) by devising a characteristic quantity to divide our original quantity by and then substitute in. Importantly, to do so we need to adjust the differential, based on the chain rule.

Often, nondimensionalizing requires trial and error to decide what constants to use and how to arrange them. We decide based on our units. If we have

length, we must figure out a characteristic length. For time, we must figure out a characteristic time, and similarly for other quantities.

For example, for the above equation we define a characteristic velocity  $v_c$ , then use it to define a dimensionless velocity  $V$  where  $V = v/v_c$ . We do the same for time, with  $t_c$  and  $\tau$  where  $\tau = t_c/\tau$ . We then apply the chain rule to get the adjusted  $\frac{dV}{d\tau}$  instead of  $\frac{dv}{dt}$ .

To make the equation simpler, we introduce a parameter, the dimensionless length  $q$ . Through some math we can arrive at the standard form of the equation, used in the literature<sup>1</sup>,

$$\psi_{\text{gs}}(q) = e^{-q^2/2}$$

To start we define variables. We have regular length  $x$ , with units m. Our dimensionless length,  $q$ , has no units. The characteristic length should be able to cancel out the units of the length. To this end, we take advantage of our  $\alpha$  variable. It has units  $\text{m}^{-2}$ , which is perfect.

Therefore,  $q$  is

$$q = \sqrt{\alpha}x$$

We can easily verify these units, because if we make a mistake here, it will propagate to the entire equation.

$$q = \sqrt{\alpha}x$$

$$[q] = \sqrt{\frac{1}{\text{m}^2}}\text{m} = 1$$

Knowing a value for  $q$ , it is easy to obtain the earlier wavefunction. Substitution is simple knowing what  $q$  is.

More complicatedly, we are going to rewrite the vibrational motion Hamiltonian. Its original form is restated below, but the final form will be revealed at the end.

$$\hat{H}(x) = -\frac{\hbar^2}{2\mu} \frac{d^2}{dx^2} + \frac{1}{2}kx^2$$

---

<sup>1</sup>The contents inside a function like sine, cosine, exponential, logarithm, etc. must be dimensionless. We cannot have  $e^{(4\text{ }\mu\text{m})}$ , for example.



We already have nondimensional length, but we must apply this to our differential operators. We take advantage of the chain rule to form an equation for our updated derivative operator.

$$\frac{d}{dq} = \frac{dx}{dq} \frac{d}{dx}$$

We rearrange our nondimensional length equation to get

$$x = \frac{q}{\sqrt{\alpha}}$$

We can now take the derivative on both sides of this equation of this equation to get

$$\frac{dx}{dq} = \frac{1}{\sqrt{\alpha}}$$

We substitute this into the chain rule

$$\frac{d}{dq} = \frac{1}{\sqrt{\alpha}} \frac{d}{dx} \implies \frac{d}{dx} = \sqrt{\alpha} \frac{d}{dq}$$

We can take the second derivative, only possible because  $\alpha$  is a constant independent of  $q$ ,

$$\begin{aligned} \frac{d}{dx} &= \sqrt{\alpha} \frac{d}{dq} \\ \frac{d}{dx} \left( \frac{d}{dx} \right) &= \frac{d}{dx} \left( \sqrt{\alpha} \frac{d}{dq} \right) \\ \frac{d}{dx} \left( \frac{d}{dx} \right) &= \sqrt{\alpha} \frac{d}{dq} \left( \sqrt{\alpha} \frac{d}{dq} \right) \\ \frac{d^2}{dx^2} &= \alpha \frac{d^2}{dq^2} \end{aligned}$$

First we will deal with the  $\frac{1}{2}kx^2$  term, as it does contain a differential. We need to equate  $\alpha$  and  $k$ , so we can clean up easily.

$$\begin{aligned} \alpha &= \frac{\sqrt{\mu k}}{\hbar} \\ \alpha^2 &= \frac{\mu k}{\hbar^2} \\ k &= \frac{\alpha^2 \hbar^2}{\mu} \end{aligned}$$

We now substitute this value of  $k$  and  $x = \frac{q}{\sqrt{\alpha}}$  into the Hamiltonian term. We also take

advantage of  $\alpha = \mu\omega/\hbar$ .

$$\begin{aligned}\frac{1}{2}kx^2 &= \frac{1}{2}\left(\frac{\alpha^2\hbar^2}{\mu}\right)\left(\frac{q^2}{\alpha}\right) \\ &= \frac{1}{2}\frac{\hbar^2\alpha}{\mu}q^2 \\ &= \frac{1}{2}\frac{\hbar^2}{\mu}\left(\frac{\mu\omega}{\hbar}\right)q^2 \\ &= \frac{1}{2}\hbar\omega q^2\end{aligned}$$

We now deal with the other Hamiltonian term, using our earlier second derivative,

$$\begin{aligned}-\frac{\hbar^2}{2\mu}\frac{d^2}{dx^2} &= -\frac{\hbar^2}{2\mu}\alpha\frac{d^2}{dq^2} \\ &= -\frac{\hbar^2}{2\mu}\frac{\mu\omega}{\hbar^2}\frac{d^2}{dq^2} \\ &= -\frac{1}{2}\hbar\omega\frac{d^2}{dq^2}\end{aligned}$$

The Hamiltonian is then

$$\begin{aligned}\hat{H}(q) &= -\frac{1}{2}\hbar\omega\frac{d^2}{dq^2} + \frac{1}{2}\hbar\omega q^2 \\ &= \hbar\omega\left(-\frac{1}{2}\frac{d^2}{dq^2} + \frac{1}{2}q^2\right) \\ &= \frac{1}{2}\hbar\omega\left(-\frac{d^2}{dq^2} + q^2\right)\end{aligned}$$

where  $\omega = \sqrt{\frac{k}{\mu}}$  (still). Much cleaner! Every harmonic oscillator has this same form, only the parameter  $\omega$  is specific for each Hamiltonian.

## 21.3 Ladder Operators

Our goal will be to derive excited states of the harmonic oscillator Hamiltonian and the corresponding eigenvalues. We will use a very powerful operator technique and the use of commutation relations. This is a state of the art procedure in quantum theory, and very beautiful. It is also fairly abstract. Let us try to understand.

We define two operators, ones that we designate  $\hat{b}^\dagger$  (sometimes called  $b^+$ , but the latter is more prone to reading errors) and  $\hat{b}$ . They are called **ladder operators**, because they

will allow us “ladder” up or down to find different energy eigenstates.

Using the same dimensionless  $q$  we discussed earlier,

$$\begin{aligned}\hat{b}^\dagger &= \frac{1}{\sqrt{2}} \left( q - \frac{d}{dq} \right) \\ \hat{b} &= \frac{1}{\sqrt{2}} \left( q + \frac{d}{dq} \right)\end{aligned}$$

where  $1/\sqrt{2}$  is a normalizing factor. An important identity of these ladder operators is obtained by multiplying them together

$$\begin{aligned}\hbar\omega\hat{b}^\dagger\hat{b} &= \frac{1}{\sqrt{2}}\frac{1}{\sqrt{2}}\hbar\omega\left(q - \frac{d}{dq}\right)\left(q + \frac{d}{dq}\right) \\ &= \frac{1}{2}\hbar\omega\left(q - \frac{d}{dq}\right)q + \left(q - \frac{d}{dq}\right)\frac{d}{dq} \\ &= \frac{1}{2}\hbar\omega\left(q^2 - \underbrace{\frac{d}{dq}q + q\frac{d}{dq}}_{[q, \frac{d}{dq}]} - \frac{d^2}{dq^2}\right) \\ &= \frac{1}{2}\hbar\omega\left(q^2 - \frac{d^2}{dq^2} + \left[q, \frac{d}{dq}\right]\right) \\ &= \underbrace{\frac{1}{2}\hbar\omega\left(q^2 - \frac{d^2}{dq^2}\right)}_{\hat{H}} + \frac{1}{2}\hbar\omega\left[q, \frac{d}{dq}\right] \\ &= \hat{H} + \frac{1}{2}\hbar\omega\left[q, \frac{d}{dq}\right]\end{aligned}$$

If we work out the value of the commutator

$$\left[q, \frac{d}{dq}\right]f(q) = q\frac{d}{dq}f(q) - \frac{d}{dq}qf(q) = q\frac{df}{dq} - f(q) - \frac{df}{dq}q = -f(q)$$

so that  $[q, \frac{d}{dq}] = -1$ , the equation is now

$$\begin{aligned}\hbar\omega\hat{b}^\dagger\hat{b} &= \hat{H} - \frac{1}{2}\hbar\omega \\ \hat{H} &= \hbar\omega\hat{b}^\dagger\hat{b} + \frac{1}{2}\hbar\omega \\ \hat{H} &= \hbar\omega\left(\hat{b}^\dagger\hat{b} + \frac{1}{2}\right)\end{aligned}$$

It follows we have expressed  $\hat{H}$  in terms of the ladder operators, rather than  $q, \frac{d}{dq}$ . To proceed we will develop some useful results.

### 21.3.1 Some Identities

We use some important identities, starting with  $[\hat{b}, \hat{b}^\dagger]$ .

$$\begin{aligned} [\hat{b}, \hat{b}^\dagger] &= \frac{1}{2} \left[ q + \frac{d}{dq}, q - \frac{d}{dq} \right] \\ &= \frac{1}{2} \left( \underbrace{[q, q]}_0 - \underbrace{\left[ \frac{d}{dq}, \frac{d}{dq} \right]}_0 + \left[ \frac{d}{dq}, q \right] - \left[ q, \frac{d}{dq} \right] \right) \\ &= \frac{1}{2} (1 - (-1)) \\ &= 1 \end{aligned}$$

$$\left[ \underbrace{q}_{\text{red}}, \underbrace{\frac{d}{dq}}_{\text{green}}, \underbrace{q}_{\text{blue}}, \underbrace{\frac{d}{dq}}_{\text{orange}} \right] = \frac{1}{2} \left( \underbrace{[q, q]}_{\text{red, blue}} - \underbrace{\left[ \underbrace{q}_{\text{red}}, \underbrace{\frac{d}{dq}}_{\text{orange}} \right]}_{\text{red, orange}} + \underbrace{\left[ \underbrace{\frac{d}{dq}}_{\text{green}}, q \right]}_{\text{green, blue}} - \underbrace{\left[ \underbrace{\frac{d}{dq}}_{\text{green}}, \underbrace{\frac{d}{dq}}_{\text{orange}} \right]}_{\text{green, orange}} \right)$$

Figure 21.2

Logically, any commutator commutes with itself ( $[\hat{b}^\dagger, \hat{b}^\dagger] = [\hat{b}, \hat{b}] = 0$ ), so many terms cancel out. This works by expanding the commutator, as seen in the figure above.

If we were to expand  $[\hat{b}^\dagger, \hat{b}]$  similarly, it would be

$$[\hat{b}^\dagger, \hat{b}] = -1$$

### 21.3.2 Use of the Ladder Operators to Construct New Eigenfunctions

Suppose that we have some known eigenstate of the Hamiltonian,  $\psi_n(q)$ , where

$$\hat{H}\psi_n(q) = E_n\psi_n(q)$$

We now make a claim that our ladder operators can generate new eigenfunctions of  $\hat{H}$ , with eigenvalues

$$\begin{aligned}\hat{H}(\hat{b}^\dagger\psi_n(q)) &= (E_n + \hbar\omega)(\hat{b}^\dagger\psi_n(q)) = E_{n+1}\psi_{n+1}(q) \\ \hat{H}(\hat{b}\psi_n(q)) &= (E_n - \hbar\omega)(\hat{b}\psi_n(q)) = E_{n-1}\psi_{n-1}(q)\end{aligned}$$

If we can prove this, this would be a powerful tool. With one eigenstate we can find all others with simple operations! Acting with  $\hat{b}^\dagger$  raises the energy by  $\hbar\omega$ , and this can be used repeatedly! Likewise acting by  $\hat{b}$  creates a new eigenstate with an new energy lowered by  $-\hbar\omega$

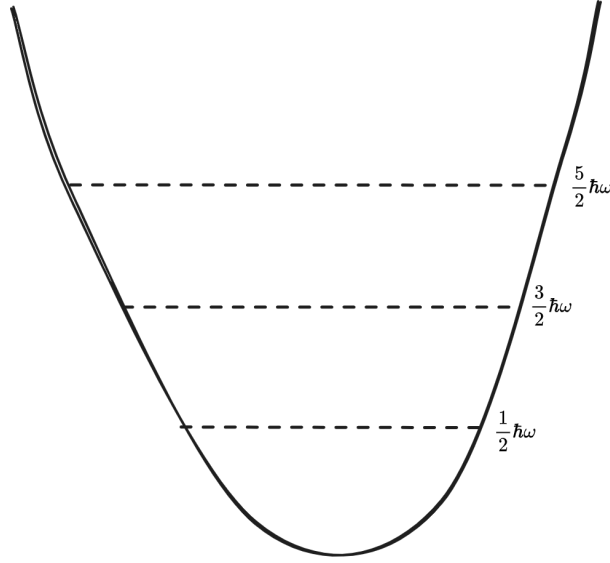
**Table 21.1:** Action of ladder operators on known eigenfunction  $\psi_n$  with energy  $E_n$

| Wavefunction                | Energy               |
|-----------------------------|----------------------|
| $(\hat{b}^\dagger)^2\psi_n$ | $E_n + 2\hbar\omega$ |
| $\hat{b}^\dagger\psi_n$     | $E_n + \hbar\omega$  |
| $\psi_n$                    | $E_n$                |
| $\hat{b}\psi_n$             | $E_n - \hbar\omega$  |
| $(\hat{b})^2\psi_n$         | $E_n - 2\hbar\omega$ |

This lowering process has to stop, when we reach the ground state, which has energy  $\frac{1}{2}\hbar\omega$ . It follows that the energies are given by the general formula

$$E_n = (n + \frac{1}{2})\hbar\omega$$

The equation above can be deduced from the basic lowering and raising formulas.



**Figure 21.3:** Energy levels of harmonic oscillator hamiltonian

Before proving these results, we use an important commutator identity, which is a version of something called the **Jacobi identity**.

$$\begin{aligned}
 [\hat{A}\hat{B}, \hat{C}] &= \hat{A}[\hat{B}, \hat{C}] + [\hat{A}, \hat{C}]\hat{B} \\
 (\hat{A}\hat{B})\hat{C} - \hat{C}(\hat{A}\hat{B}) &= \hat{A}(\hat{B}\hat{C} - \hat{C}\hat{B}) + (\hat{A}\hat{C} - \hat{C}\hat{A})\hat{B} \\
 \hat{A}\hat{B}\hat{C} - \hat{C}\hat{A}\hat{B} &= \hat{A}\hat{B}\hat{C} - \hat{A}\hat{C}\hat{B} + \hat{A}\hat{C}\hat{B} - \hat{C}\hat{A}\hat{B} \\
 \hat{A}\hat{B}\hat{C} - \hat{C}\hat{A}\hat{B} &= \hat{A}\hat{B}\hat{C} - \hat{C}\hat{A}\hat{B}
 \end{aligned}$$

### 21.3.2.1 Proving Ladder Operators Relations

Preemptively, we are dropping the  $(q)$  from  $\psi_n$ , for notational convenience.

If we have the commutator

$$\begin{aligned}
 [\hat{H}, \hat{b}^\dagger]\psi_n &= \hat{H}\hat{b}^\dagger\psi_n - \hat{b}^\dagger\hat{H}\psi_n \\
 \hat{H}\hat{b}^\dagger\psi_n &= [\hat{H}, \hat{b}^\dagger]\psi_n + \hat{b}^\dagger\hat{H}\psi_n \\
 \hat{H}\hat{b}^\dagger\psi_n &= [\hat{H}, \hat{b}^\dagger]\psi_n + E_n\hat{b}^\dagger\psi_n
 \end{aligned}$$

We start with expanding the commutator term, using the previous definition of the Hamil-

tonian involving ladder operators.

$$\begin{aligned}
[\hat{H}, \hat{b}^\dagger] &= \left[ \hbar\omega \left( \hat{b}^\dagger \hat{b} - \frac{1}{2} \right), \hat{b}^\dagger \right] \\
&= \hbar\omega \left[ \hat{b}^\dagger \hat{b} - \frac{1}{2}, \hat{b}^\dagger \right] \\
&= \hbar\omega \left( [\hat{b}^\dagger \hat{b}, \hat{b}^\dagger] + \left[ -\frac{1}{2}, \hat{b}^\dagger \right] \right) \\
&= \hbar\omega \underbrace{\left( \hat{b}^\dagger [\hat{b}, \hat{b}^\dagger] + [\hat{b}^\dagger, \hat{b}^\dagger] \hat{b} \right)}_{\text{Jacobi}} \\
&= \hbar\omega (\hat{b}^\dagger (1)) \\
&= \hbar\omega \hat{b}^\dagger
\end{aligned}$$

Note that any (linear) operator will commute with a constant.

Now substituting this back into the earlier equation

$$\begin{aligned}
\hat{H} \hat{b}^\dagger \psi_n &= (\hbar\omega \hat{b}^\dagger) \psi_n + E_n \hat{b}^\dagger \psi_n \\
&= (\hbar\omega + E_n) \hat{b}^\dagger \psi_n
\end{aligned}$$

The proof of the other identity is similar.

Our upwards ladder operator  $\hat{b}^\dagger$  raises the energy by  $\hbar\omega$ . This will go on forever. There are infinite eigenstates we can obtain! The downwards ladder operator  $\hat{b}$  lowers the energy by  $\hbar\omega$ , until it reaches the ground state energy of  $E_{\text{gs}} = \frac{1}{2}\hbar\omega$ . There are no further states in between and  $\hbar\omega$  is the smallest possible increment.

What if we were at our minimum and tried to lower the energy further using a ladder operator?

$$\hat{H} \hat{b} \psi_{\text{gs}} = (E_{\text{gs}} - \hbar\omega) \hat{b} \psi_{\text{gs}}$$

This seemingly creates another state that has even lower energy than the ground state. This is impossible. The only way out is that

$$\hat{b} \psi_{\text{gs}} = 0$$

If this is true we can also evaluate

$$\hat{H} \psi_{\text{gs}} = \hbar\omega (\hat{b}^\dagger \hat{b} + 1/2) \psi_{\text{gs}} = (0 + 1/2) \hbar\omega \psi_{\text{gs}} = \frac{1}{2} \hbar\omega \psi_{\text{gs}}$$

This is the correct result!

We can also evaluate  $\hat{b}\psi_{\text{gs}}$  explicitly using the  $p, q$ -representation:

$$\begin{aligned}\hat{b}\psi_{\text{gs}} &= \frac{1}{\sqrt{2}}\left(q + \frac{d}{dq}\right)\psi_{\text{gs}} \\ &= \frac{1}{\sqrt{2}}\left(q + \frac{d}{dq}\right)e^{-q^2/2} = \frac{1}{\sqrt{2}}(q - q)e^{-q^2/2} = 0\end{aligned}$$

Everything checks out.

Other eigenstates (not normalized) can be found explicitly by laddering up, starting from the ground state:

$$\begin{aligned}\hat{b}^\dagger\psi_{\text{gs}} &= \frac{1}{\sqrt{2}}\left(q - \frac{d}{dq}\right)\psi_{\text{gs}} \\ &= \frac{1}{\sqrt{2}}(q + q)e^{-q^2/2} = 2qe^{-q^2/2}\end{aligned}$$

This process can be continued, for example

$$\begin{aligned}\hat{b}^\dagger\psi_1 &= \frac{1}{\sqrt{2}}\left(q - \frac{d}{dq}\right)(qe^{-q^2/2}) \\ &= (q^2 - 1 + q^2)e^{-q^2/2} = (2q^2 - 1)e^{-q^2/2}\end{aligned}$$

Note that we simplified the result for  $\psi_1(q)$ , removing the factor of 2. Our states are not normalized in this discussion and hence an overall constant does not matter.

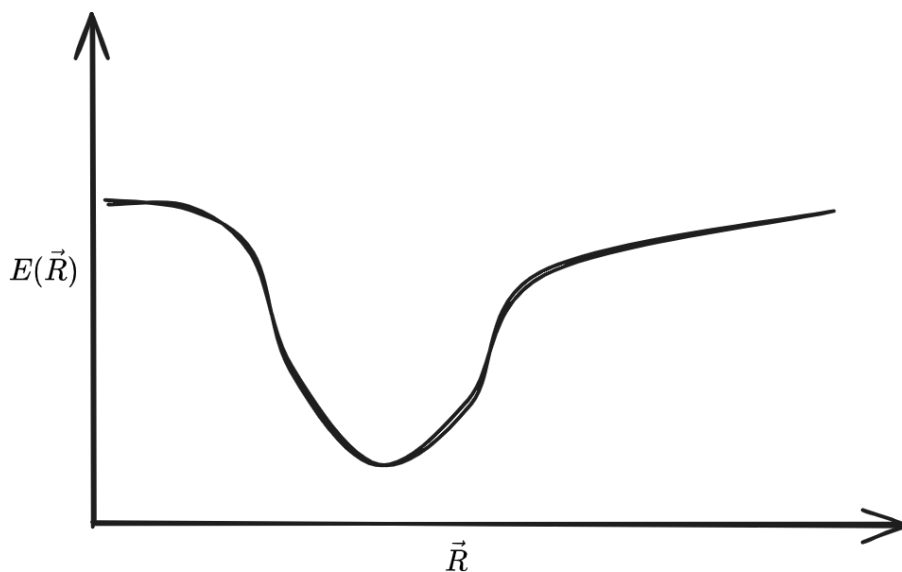
I hope you can appreciate the power of this operator approach to finding all eigenstates and eigenvalues for the harmonic oscillator. This approach is widely used in quantum chemistry! There are powerful ways to extend this (e.g. defining normalized states and matrix elements), but this suffices as an introduction.



## 22 2-Body Problem and Spherical Coordinates

### 22.1 2-Body Problem

Imagine a solar system, composed of a planet and sun. We call this a “2-body problem”, as they affect each other. Newton was the first to tackle this problem, and it is taught in classical mechanics courses. The two-body problem is also of central concern in quantum mechanics as these are the simplest examples of interacting systems, and the way to solve them creates insights that are relevant for much more complicated systems.



**Figure 22.1:** This is not a perfect graph. As we get farther radial distances, bonds might break and the energy system will change. We model energy as only dependent on distance though, for simplicity.

More relevant to us chemists, there are two main 2-body problems we care about. The first problem concerns nuclear motion of a diatomic molecule, where the two bodies are each atom, such as  $\text{H}_2$ ,  $\text{N}_2$ , etc. The other main problem is a hydrogen-like atom, with a nucleus and one electron, such as  $\text{H}$ ,  $\text{He}^+$ ,  $\text{Li}^{2+}$ , etc.

## 22.2 Classical Energy

Note: All vectors are indicated by bolding the letter, as in  $\vec{r} = \mathbf{r}$ . This is a common convention in mathematics.

### 22.2.1 Energy and Momentum

For a 2-body system, with a central potential that only depends on the distance between the particles, we know the energy can be written as a sum of terms where

$$E = \frac{\mathbf{p}_1^2}{2m_1} + \frac{\mathbf{p}_2^2}{2m_2} + V(|\mathbf{r}_1 - \mathbf{r}_2|)$$

We know that momentum has the time-derivative form (using Newton's compact notation of time-derivatives) of

$$\begin{aligned}\mathbf{p}_1 &= m_1 \frac{d\vec{r}}{dt} = \underbrace{m_1 \dot{\vec{r}}}_{\text{Newton's}} \\ \mathbf{p}_2 &= m_2 \frac{d\vec{r}}{dt} = m_2 \dot{\vec{r}}\end{aligned}$$

### 22.2.2 Defining Relative/Simpler Variables

Rewriting energy,

$$E = \frac{1}{2}m_1\dot{\mathbf{r}}_1^2 + \frac{1}{2}m_2\dot{\mathbf{r}}_2^2 + V(|r|)$$

where  $r = |\mathbf{r}_1 - \mathbf{r}_2|$  (magnitude of distance) and  $\mathbf{r} = \mathbf{r}_2 - \mathbf{r}_1$  (relative vector distance), for notational simplicity.

If we now define the center of mass, using the standard formula for a 2-body system,

$$\mathbf{R}_{\text{CoM}} = \frac{m_1\mathbf{r}_1 + m_2\mathbf{r}_2}{m_1 + m_2}$$

we can also obtain the momentum of the center of mass, and define the relative momentum

$$\begin{aligned}\mathbf{P}_{\text{CoM}} &= M\dot{\mathbf{R}}_{\text{CoM}} = m_1\dot{\mathbf{r}}_1 + m_2\dot{\mathbf{r}}_2 \\ \mathbf{p} &= \mu\dot{\mathbf{r}} = \mu(\dot{\mathbf{r}}_2 - \dot{\mathbf{r}}_1)\end{aligned}$$

where  $\mu$  is the reduced mass with the standard formula  $\mu = \frac{m_1 m_2}{m_1 + m_2}$  and  $M = m_1 + m_2$

### 22.2.3 Rewriting

We can now rewrite the energy term, in a form that matches the radial form of the potential and also simplifies the solution,

$$\begin{aligned}
 E_{\text{kin}} &= \frac{1}{2}m_1\dot{\mathbf{r}}_1^2 + \frac{1}{2}m_2\dot{\mathbf{r}}_2^2 \\
 &= \frac{1}{2M}\mathbf{P}_{\text{CoM}}^2 + \frac{1}{2\mu}\mathbf{p}^2 \\
 &= \frac{1}{2M}\mathbf{P}_{\text{CoM}}^2 + \frac{1}{2}\mu\dot{\mathbf{r}}^2 \\
 &= \frac{1}{2}\left(\frac{1}{M}\mathbf{P}_{\text{CoM}}^2 + \mu\dot{\mathbf{r}}^2\right)
 \end{aligned}$$

We wish to show that the top equation is equal to the bottom equation. We do this stepwise, starting from the bottom.

$$\begin{aligned}
 E_{\text{kin}} &= \frac{1}{2}\left(\frac{1}{M}\mathbf{P}_{\text{CoM}}^2 + \mu\dot{\mathbf{r}}^2\right) \\
 &= \frac{1}{2}\left(\frac{1}{m_1 + m_2}(m_1\dot{\mathbf{r}}_1 + m_2\dot{\mathbf{r}}_2)^2 + \frac{m_1m_2}{m_1 + m_2}(\dot{\mathbf{r}}_1 - \dot{\mathbf{r}}_2)^2\right) \\
 &= \frac{1}{2}\left(\frac{1}{m_1 + m_2}(m_1^2\dot{\mathbf{r}}_1^2 + m_2^2\dot{\mathbf{r}}_2^2 + 2m_1m_2\dot{\mathbf{r}}_1\dot{\mathbf{r}}_2) + \frac{m_1m_2}{m_1 + m_2}(\dot{\mathbf{r}}_1^2 + \dot{\mathbf{r}}_2^2 - 2\dot{\mathbf{r}}_1\dot{\mathbf{r}}_2)\right) \\
 &= \frac{1}{2}\frac{1}{m_1 + m_2}(m_1^2\dot{\mathbf{r}}_1^2 + m_2^2\dot{\mathbf{r}}_2^2 + \cancel{2m_1m_2\dot{\mathbf{r}}_1\dot{\mathbf{r}}_2} + m_1m_2\dot{\mathbf{r}}_1^2 + m_1m_2\dot{\mathbf{r}}_2^2 - \cancel{2m_1m_2\dot{\mathbf{r}}_1\dot{\mathbf{r}}_2}) \\
 &= \frac{1}{2}\frac{1}{m_1 + m_2}(m_1^2\dot{\mathbf{r}}_1^2 + m_2^2\dot{\mathbf{r}}_2^2 + m_1m_2\dot{\mathbf{r}}_1^2 + m_1m_2\dot{\mathbf{r}}_2^2) \\
 &= \frac{1}{2}\frac{1}{m_1 + m_2}((m_1^2 + m_1m_2)\dot{\mathbf{r}}_1^2 + (m_2^2 + m_1m_2)\dot{\mathbf{r}}_2^2) \\
 &= \frac{1}{2}\frac{1}{\cancel{m_1} + m_2}(m_1(\cancel{m_1} + m_2)\dot{\mathbf{r}}_1^2 + m_2(\cancel{m_2} + m_1)\dot{\mathbf{r}}_2^2) \\
 &= \frac{1}{2}(m_1\dot{\mathbf{r}}_1^2 + m_2\dot{\mathbf{r}}_2^2)
 \end{aligned}$$

This derivation, which was done by Newton, is valid for classical mechanics, and shows that each form of the classical kinetic energy is equivalent, and the definitions are all consistent. In order to define the Hamiltonian in quantum mechanics we want to use the kinetic energy in terms of momenta (which then become operators  $i\hbar\nabla$ ). Therefore we will use

$$E_{\text{kin}} = \frac{\mathbf{P}_{\text{COM}}^2}{2M} + \frac{\mathbf{p}^2}{2\mu}$$

## 22.3 Hamiltonian and Schrödinger Equation

Denoting  $E_{\text{kin}}$  as  $T$ , to fit the quantum mechanical convention we get the standard quantum expression for  $\hat{H} = \hat{T} + \hat{V}$

$$\begin{aligned}\hat{H} &= \frac{\mathbf{P}_{\text{CoM}}^2}{2M} + \frac{\mathbf{p}^2}{2\mu} + V(r) \\ &= -\frac{\hbar^2}{2M} \nabla_{\text{CoM}}^2 - \frac{\hbar^2}{2\mu} \nabla^2 + V(r)\end{aligned}$$

The wavefunction is expressed in terms of  $\mathbf{R}_{\text{CoM}}$  and  $\mathbf{r}$  and the time-independent Schrödinger equation is given by

$$\left[ \frac{\mathbf{P}_{\text{CoM}}^2}{2M} + \frac{\mathbf{p}^2}{2\mu} + V(r) \right] \Psi(\mathbf{R}_{\text{CoM}}, \mathbf{r}) = E \Psi(\mathbf{R}_{\text{CoM}}, \mathbf{r})$$

By rewriting the Schrödinger equation in terms of the new coordinates we can solve it using separation of variables. This would not work in the original particle coordinates.

Using the basic separation of variables approach the wavefunction  $\Psi(\mathbf{R}_{\text{CoM}}, \mathbf{r})$  is written as a product

$$\Psi(\mathbf{R}_{\text{CoM}}, \mathbf{r}) = \phi(\mathbf{R}_{\text{CoM}}) \psi(\mathbf{r})$$

Applying the usual separation of variables techniques we get,

$$\frac{-\frac{\hbar^2}{2M} \nabla_{\text{CoM}}^2 \phi(\mathbf{R}_{\text{CoM}})}{\phi(\mathbf{R}_{\text{CoM}})} + \frac{\left( -\frac{\hbar^2}{2\mu} \nabla^2 + V(r) \right) \psi(\mathbf{r})}{\psi(\mathbf{r})} = E$$

We have two terms that each have to be constant. The first is translational energy for the center-of-mass motion. The second is the energy we are actually interested in, for the relative motion of the nuclei.

### 22.3.1 Translational Energy

Interestingly, the translational term is simply a free particle moving in 3-dimensions. To solve it we put the particle in a (large) 3-dimensional box, of macroscopic dimensions (to normalize solutions). The solutions are simply those of the particle in a box problem, with energy  $E_{\text{T}} = \frac{\hbar^2 \pi^2}{2ML^2} (n_x^2 + n_y^2 + n_z^2)$ .  $L$  is large (for example, 1 dm or 1 L of volume), and the energy spacing is very small then. This leads to almost continuous energies. This is the quantum foundation for use in statistical mechanics, for gases, and fairly simple analysis

leads to the ideal gas law  $PV = nRT$ ! We consider this problem solved here.

### 22.3.2 Relative Motion

The second term is more interesting for quantum mechanics.

$$-\frac{\hbar^2}{2\mu}\nabla^2\psi(\mathbf{r}) + V(r)\psi(\mathbf{r}) = E_{\text{rel}}\psi(\mathbf{r})$$

In what follows we will just call this relative energy  $E$ , and this equation will need significant work to solve it. Importantly, because the potential term only depends on the magnitude  $r$  and the kinetic term is expressed in terms of the 3-dimensional gradient, it will be useful to use spherical coordinates. This will lead to more separation of variables. The name of the game!

## 22.4 Spherical Coordinates

Spherical coordinates are used often, when problems have spherical symmetry in 3-dimensions. They were employed by Newton, when describing planetary motion. Polar coordinates are the cousin for 2-dimensional problems and are easier to visualize and use. Integrating in spherical coordinates is often significantly easier than Cartesian coordinates. Integrating in 3-dimensions using Cartesian coordinates is often a frustrating exercise in finding complicated bounds and many, many square roots and impossible integrals. Let us commence.

We define 3 spherical coordinates. Instead of  $x, y, z$  we define  $r, \theta, \varphi$ .

$$\begin{cases} x = r \sin \theta \cos \varphi \\ y = r \sin \theta \sin \varphi \\ z = r \cos \theta \end{cases}$$

The magnitude of the radius is

$$r = \sqrt{x^2 + y^2 + z^2}$$

The **azimuthal angle**, denoted  $\varphi$ , is calculated

$$\varphi = \arctan\left(\frac{y}{x}\right)$$

It has bounds from  $[0, 2\pi]$ .

The **polar angle**, denoted  $\theta$ , is calculated

$$\theta = \arccos \left( \frac{z}{\sqrt{x^2 + y^2 + z^2}} \right)$$

It has bounds  $[0, \pi]$

We can integrate spherically

$$\int_{-\infty}^{\infty} dx \int_{-\infty}^{\infty} dy \int_{-\infty}^{\infty} dz f(x, y, z) = \int_0^{\infty} r^2 dr \int_0^{\pi} \sin \theta d\theta \int_0^{2\pi} d\varphi F(r, \theta, \varphi)$$

The math in red is a conversion between differentials, requiring the use of the chain rule. It is annoying to do, but intuitively it makes a lot of sense. When we integrate in 3-dimensional Cartesian, we are summing up little cubes of constant volume. When we integrate spherically, we are summing up little wedges, that change in size as they get farther from the origin. The extra factor  $r^2 \sin \theta$  is called the Jacobian determinant of the transformation, similar to a  $u$ -substitution factor when changing variables during integration (one cannot just naively substitute  $du = dx$ , the chain rule must be used).

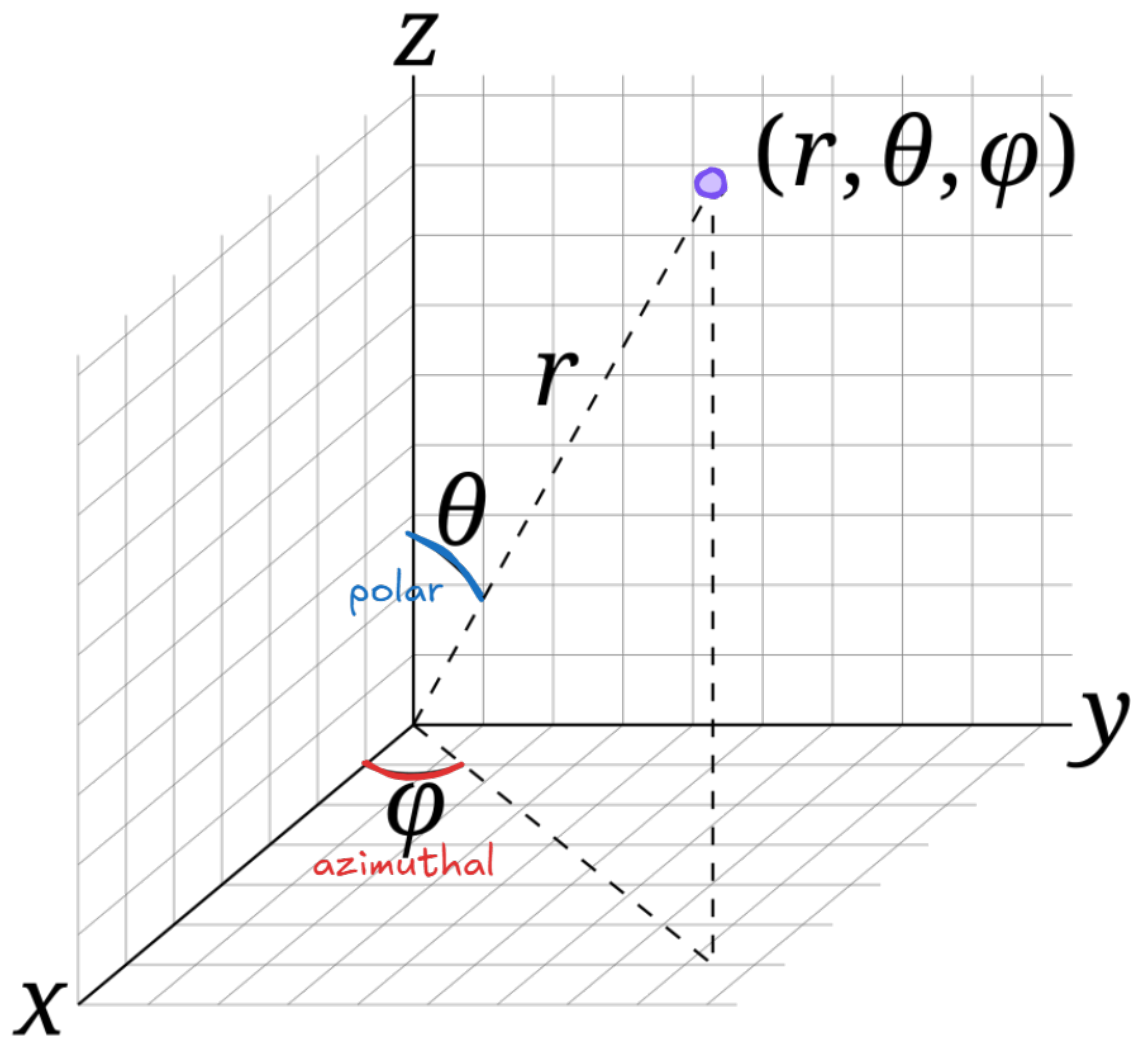
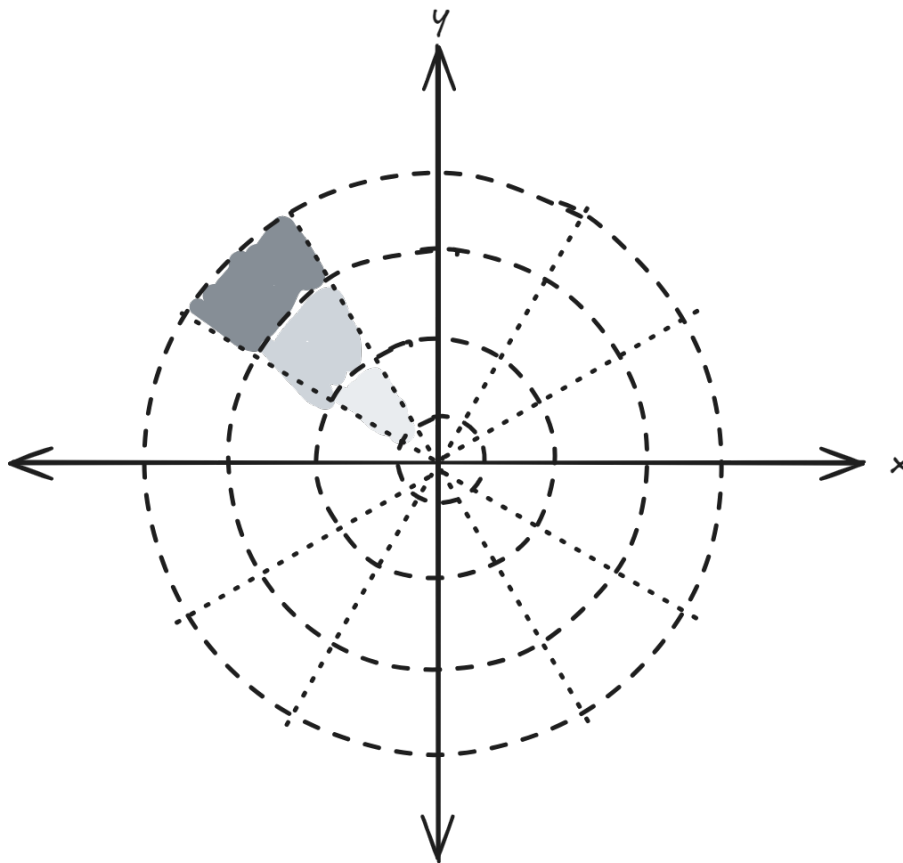


Figure 22.2



**Figure 22.3:** This wedge size difference is sketched in polar coordinates. We see that the wedges closer to the center are smaller, and those farther are bigger. Even with infinitesimal wedges, this still holds (hard to wrap your mind around, but true). Of course, in 3-dimensions, our wedges now have an extra dimension but the same idea holds.

Converting between coordinate systems is difficult and long, but requires only basic trigonometry, algebra, and the chain rule! So simple... (not).

### Spherical Coordinates Conventions



Warning! Physics and mathematics generally use opposite conventions for the angle, where the meaning of  $\varphi$  and  $\theta$  are switched, when following the North American standard. It does not matter what you use *if* you stay consistent, for example in the integration bounds and trigonometric conversions. If this stresses you out, just use the system shown in class.

Additionally, the order for radius, azimuthal, and polar, which is stan-



dard for math, is often (but not always) opposite in physics to keep the system righthanded. Don't read  $(r, \theta, \varphi)$  and think you know which standard you're dealing with without checking.

The math convention is based on the idea that spherical coordinates are an extension of polar coordinates. In polar coordinates, we have an angle  $\theta$  on the  $xy$  plane, so if we need to add another angle by adding another axis, we call it  $\varphi$ . The physics convention is that the polar angle,  $\theta$  in their system, is the more physically significant angle and so you end up writing it in calculations more, so it gets to be  $\theta$ .

This is helpful to keep in mind when reading other sources. For example, calculus resources (where this topic is often first introduced) will use the math conventions, but chemistry or physics resources will use the physics convention, research papers will be up to personal or journal preference, and different countries use different conventions... So make sure you know what they mean.

## 23 Angular Hamiltonian and Spherical Harmonics

### 23.1 Converting Hamiltonian

We can rewrite the Hamiltonian in terms of spherical coordinates

$$\begin{aligned}-\frac{\hbar^2}{2\mu}\nabla^2 &= -\frac{\hbar^2}{2\mu}\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{\hat{L}^2}{2\mu r^2} \\ &= -\frac{\hbar^2}{2\mu}\left(\frac{2}{r}\frac{\partial}{\partial r} + \frac{\partial^2}{\partial r^2}\right) + \frac{\hat{L}^2}{2\mu r^2}\end{aligned}$$

where

$$\hat{L}^2 = -\hbar^2\left(\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\varphi^2}\right)$$

which is the total angular momentum operator, squared.

The derivation is very long, particularly for the Laplacian<sup>1</sup>.

#### 23.1.1 Chain Rule Sketch

A preliminary sketch of doing this involves the chain rule, where

$$\begin{aligned}\frac{\partial}{\partial x}f(r, \theta, \varphi) &= \left(\frac{\partial f}{\partial r}\frac{\partial r}{\partial x}\right) + \left(\frac{\partial f}{\partial\theta}\frac{\partial\theta}{\partial x}\right) + \left(\frac{\partial f}{\partial\varphi}\frac{\partial\varphi}{\partial x}\right) \\ \Rightarrow \frac{\partial}{\partial x} &= \left(\frac{\partial r}{\partial x}\frac{\partial}{\partial r}\right) + \left(\frac{\partial\theta}{\partial x}\frac{\partial}{\partial\theta}\right) + \left(\frac{\partial\varphi}{\partial x}\frac{\partial}{\partial\varphi}\right)\end{aligned}$$

This allows us to express the  $x$  differential in terms of our new coordinates. This process is similar for the the differentials of  $y$  and  $z$ .

From the equations  $r = \sqrt{x^2 + y^2 + z^2}$ ,

$$\frac{\partial r}{\partial x} = \frac{\partial\sqrt{x^2 + y^2 + z^2}}{\partial x} = \frac{1}{2}\frac{1}{\sqrt{x^2 + y^2 + z^2}}2x$$

---

<sup>1</sup>[https://digitalcommons.lib.uconn.edu/cgi/viewcontent.cgi?article=1034&context=chem\\_educ](https://digitalcommons.lib.uconn.edu/cgi/viewcontent.cgi?article=1034&context=chem_educ) this is a full paper on the very topic with over 70 equations... let's leave it to the experts ☺

Using the substitution  $x = r \cos \theta \sin \varphi$ , we can express everything in terms of spherical coordinates.

### 23.1.2 Angular Momentum Operator

The total angular momentum is composed of its parts,

$$\hat{L}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2$$

The  $z$  axis is chosen as our “special” axis so the form of  $\hat{L}_z$  is particularly important

$$\hat{L}_z = -i\hbar \left( x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} \right) = -i\hbar \frac{\partial}{\partial \varphi}$$

$\hat{L}$  only depends on the angular coordinates  $\theta$  and  $\varphi$ , and  $\hat{L}_z$  only depends on  $\varphi$ . The commutator relation between  $\hat{L}^2$  and  $\hat{L}_z$  is very important for derivations and will be used later, when solving the angular Schrödinger equation, where

$$[\hat{L}^2, \hat{L}_z] = 0$$

which means they commute and as a consequence, have common eigenfunctions.

## 23.2 Angular Solutions

This will be revisited in Chapter 7, but for now they can be listed. These angular solutions are called the **spherical harmonics**,  $Y_l^m(\theta, \varphi)$ , with eigenvalues

$$\begin{aligned}\hat{L}^2 Y_l^m(\theta, \varphi) &= l(l+1)\hbar^2 Y_l^m(\theta, \varphi) \\ \hat{L}_z Y_l^m(\theta, \varphi) &= m\hbar Y_l^m(\theta, \varphi)\end{aligned}$$

It is best to think of these as groups of functions for specific  $l$ , where  $l$  takes on the whole number values of  $l = 0, 1, 2, \dots$ . For a given  $l$  we have a group of functions that all share the same  $l$ , while  $m$  has values  $m = -l, -l+1, \dots, l-1, l$ . There are  $2l+1$  functions that all have the same eigenvalue for  $\hat{L}^2$  (degenerate), but different eigenvalues for  $\hat{L}_z$ .

Notice a pattern... they look awfully similar to certain quantum numbers. If we make a table, we can figure this out:

**Table 23.1**

| $Y_l^m$                              | # Functions | Degeneracy | Orbitals                                       |
|--------------------------------------|-------------|------------|------------------------------------------------|
| $l = 0, m = 0$                       | 1           | 1          | $s$                                            |
| $l = 1, m = -1, 0, 1$                | 3           | 3          | $p_x, p_y, p_z$                                |
| $l = -2, m = -2, -1, 0, 1, 2$        | 5           | 5          | $d_{xy}, d_{xz}, d_{yz}, d_{z^2}, d_{x^2-y^2}$ |
| $l = -3, m = -3, -2, -1, 0, 1, 2, 3$ | 7           | 7          | $f$ orbitals                                   |

We see the number of functions/degeneracy is equal to  $2l+1$ . We see that the degeneracy is equal to the number of orbitals, where we have one  $s$  orbital, three  $p$ , and five  $d$  and so on.

### 23.3 Exact Solution

Using the spherical harmonics, the solution to the Schrödinger equation for diatomics can be written in a product form

$$\Psi(r, \theta, \varphi) = f(r)Y_l^m(\theta, \varphi)$$

We now apply this solution to the equation

$$\underbrace{\left(-\frac{\hbar^2}{2\mu} \left(\frac{2}{r} \frac{\partial}{\partial r} + \frac{\partial^2}{\partial r^2}\right) + \frac{\hat{L}^2}{\mu r^2} + V(r)\right)}_{\hat{H}} f(r)Y_l^m(\theta, \varphi) = E f(r)Y_l^m(\theta, \varphi)$$

$$\left(-\frac{\hbar^2}{2\mu} \left(\frac{2}{r} \frac{\partial}{\partial r} + \frac{\partial^2}{\partial r^2}\right) + \frac{l(l+1)\hbar^2}{2\mu r^2} + V(r)\right) f(r)\cancel{Y_l^m(\theta, \varphi)} = E f(r)\cancel{Y_l^m(\theta, \varphi)}$$

$$\left(-\frac{\hbar^2}{2\mu} \left(\frac{2}{r} \frac{\partial}{\partial r} + \frac{\partial^2}{\partial r^2}\right) + \frac{l(l+1)\hbar^2}{2\mu r^2} + V(r)\right) f(r) = E f(r)$$

This is the 1-dimensional radial equation, dependent on some value of  $l$ . It will be used for radial solutions of the hydrogen atom, later. The potential for diatomics can be complicated and the 1-dimensional radial Schrödinger Equation would have to be solved numerically. However, as we saw in the chapter on vibrations, the potential can to good approximation be treated as a quadratic function. That leads to further simplifications!

## 23.4 Harmonic Oscillator and Rigid Rotor

The radial solution *is* the harmonic oscillator, but disguised.

### 23.4.1 $r$ to $x$

Let

$$rf(r) = g(r)$$

This is a trick to simplify normalization, which will be evident later on. Let

$$g(r) = g(x)$$

where  $x$  is a relative coordinate, used to adjust radial distance into a more useful form, as in

$$r = x + r_{\text{eq}}$$

With this formula,  $x$  is the displacement from the equilibrium distance, the minimum of the potential energy surface. This makes sense, as a rigid rotor would have some sort of rest position, and we would be deviating from that slightly. At the equilibrium distance,  $x = 0$ .

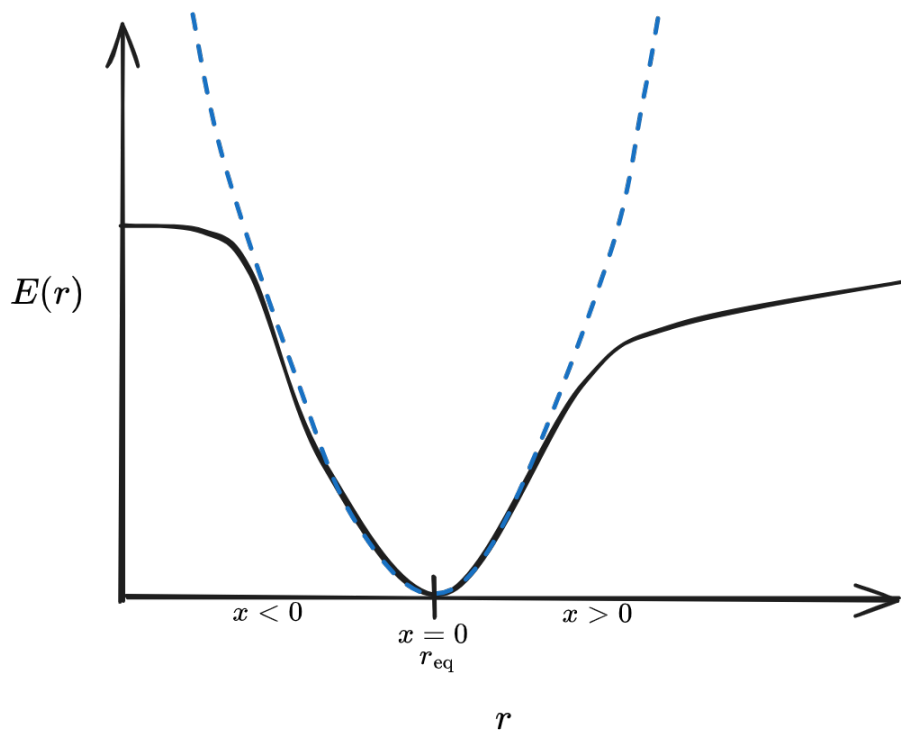


Figure 23.1

Everything can be rewritten in terms of this deviation, including

$$V(r) \rightarrow V(x) = \frac{1}{2}kx^2$$

Therefore the Hamiltonian is (with some conversion work not shown)

$$\left( -\frac{\hbar^2}{2\mu} \frac{d^2}{dx^2} + \frac{l(l+1)\hbar^2}{2\mu(r_{\text{eq}} + x)^2} + V(x) \right) g(x) = Eg(x)$$

The normalization goes from

$$\left( \int_0^\infty r^2 |f(r)|^2 dr \rightarrow \int_0^\infty |g(r)|^2 dr \rightarrow \int_{-\infty}^\infty |g(x)|^2 dx \right) = 1$$

### 23.4.2 $l$ to $J$

As we are explicitly talking about rotations of the rigid rotor, we change from  $l$  (angular momentum quantum number) to  $J$  (rotational quantum number). This is just a change in notation from math to physics of rotations.

There are two options to solve. The first option is to keep  $V(x)$  as the full potential and solve for a specific value of  $J$ . This is most easily done numerically on a computer.

The second (more common) approach is to make the approximation that  $V(x) = \frac{1}{2}kx^2$  where  $\omega = \sqrt{\frac{k}{\mu}}$  and make the additional approximation that  $r_{\text{eq}} + x = R_{\text{eq}}$ , so the second term becomes a constant with a specific value of  $J$ .

$$\left( -\frac{\hbar^2}{2\mu} \frac{d^2}{dx^2} + \frac{J(J+1)\hbar^2}{2\mu R_{\text{eq}}^2} + \frac{1}{2}kx^2 \right) g(x) = Eg(x)$$

Let  $B = \frac{\hbar^2}{2\mu R_{\text{eq}}^2}$ , the so-called rotational constant, to clean up the equation, so it becomes

$$-\frac{\hbar^2}{2\mu} \frac{d^2}{dx^2} + BJ(J+1) + \frac{1}{2}kx^2 = Eg(x)$$

The differential equation in  $x$  is just the Harmonic oscillator, which we solved in the previous chapter. To highlight we are discussing vibrations we rename the quantum number  $n$  in  $E_n = (n + \frac{1}{2})\hbar\omega$  to the vibrational quantum number  $v = 0, 1, 2, \dots$

### 23.4.3 Final Rovibrational Energies

If we were to solve this equation for a specific value of  $J$  we would have two energy terms, rotational and vibrational. The energies would look like

$$E_{\text{rot}} = BJ(J + 1)$$
$$E_{\text{vib}} = \left(v + \frac{1}{2}\right) \hbar\omega$$

The combined energy is then

$$E_{v,J} = \left(v + \frac{1}{2}\right) \hbar\omega + BJ(J + 1)$$

with the vibrational,  $v$ , and rotational,  $J$ , quantum numbers. In spectroscopy we often convert energy differences to wave numbers (of the corresponding light emitted or absorbed in the transition). The first term is in the order of  $1000 \text{ cm}^{-1}$  and the second  $0.1 \text{ cm}^{-1}$ .

This is a simplified model. If we were to make the approximation  $B$  depend on  $v$ , the computation would become more accurate and more complicated. For diatomics it is possible to obtain more accurate solutions, and it is necessary to get a good description of experiment.

However, the rigid rotor and harmonic oscillator model also works for polyatomic molecules. For larger molecules it is very hard to improve on this description and it is fairly accurate, in particular for purposes of thermochemistry. This is the real reason that this model is so widely used and important.

## 24 Rovibrational Energy

### 24.1 Overview

As (somewhat) derived earlier, we have the rovibrational energy

$$E_{v,J'} = \left(v + \frac{1}{2}\right) \hbar\omega + BJ'(J' + 1)$$

where  $B = \frac{\hbar^2}{2\mu R_{\text{eq}}^2}$  and  $\omega = \sqrt{\frac{k}{\mu}}$  and  $\mu = \frac{m_1 m_2}{m_1 + m_2}$ . The degeneracy of each energy level is  $2J' + 1$ .

The equation is using slightly different notation, with  $J$  becoming  $J'$ . This is because we often substitute in formulas involving  $J$  when dealing with spectral transitions, such as  $J' = J + 1$ , which is more comprehensible than  $J = J + 1$ .

An alternate way of thinking about  $B$  is

$$B = \frac{\hbar^2}{2I}$$

where  $I$  is the **moment of inertia**, and is equal to

$$I = \mu R_{\text{eq}}^2$$

For the rovibrational wavefunction, it has the product form

$$\Psi(r, \theta, \varphi) = f_v(r) Y_J^{M_J}(\theta, \varphi)$$

similar to the previously discussed harmonics  $Y_l^m(\theta, \varphi)$ . The quantum number  $M_J$  has possible values

$$M_J = -J, \dots, J, 2J + 1$$

in comparison to the possible values for  $m$ ,

$$m = -l, -l + 1, \dots, l - 1, l$$



## 24.2 Solving for Energy

By convention, we most often report ‘energy’ in units of  $\text{cm}^{-1}$ , as per standard spectroscopy convention. Constants also come in units of wavenumber.

Wavenumber in this convention may have the variable  $\tilde{k}$

$$\Delta E = h\nu = \frac{hc}{\lambda} = hc\tilde{k}$$

To convert energy to wavenumber simply divide by Planck’s constant and the speed of light, in centimeters per second

$$\tilde{E}[\text{cm}^{-1}] = \frac{E}{hc} = \left(v + \frac{1}{2}\right)\tilde{\omega} + \tilde{B}J'(J' + 1)$$

where

$$\tilde{\omega}[\text{cm}^{-1}] = \frac{\hbar\omega}{hc} = \frac{\omega}{2\pi c}$$

and the constant  $\tilde{B}$  is

$$\tilde{B}[\text{cm}^{-1}] = \frac{B}{hc} = \frac{\hbar^2}{hc2I} = \frac{\left(\frac{h}{2\pi}\right)^2}{hc2I} = \frac{h^2}{4\pi^2 c 2I} = \frac{h}{8\pi^2 c}$$

We emphasize that one should use  $c \approx 3 \times 10^{10} \text{cm s}^{-1}$  in these equations to get the proper value for the constants in  $\text{cm}^{-1}$ .

## 24.3 Possible Transitions

Based on something called a **selection rule**, only certain transitions are allowed, based on specific absorptions of energy. These are a little complicated and left unexplained, but we will go over two possible transitions that are allowed. Below, the (qualitative) diagram shows which transitions are allowed

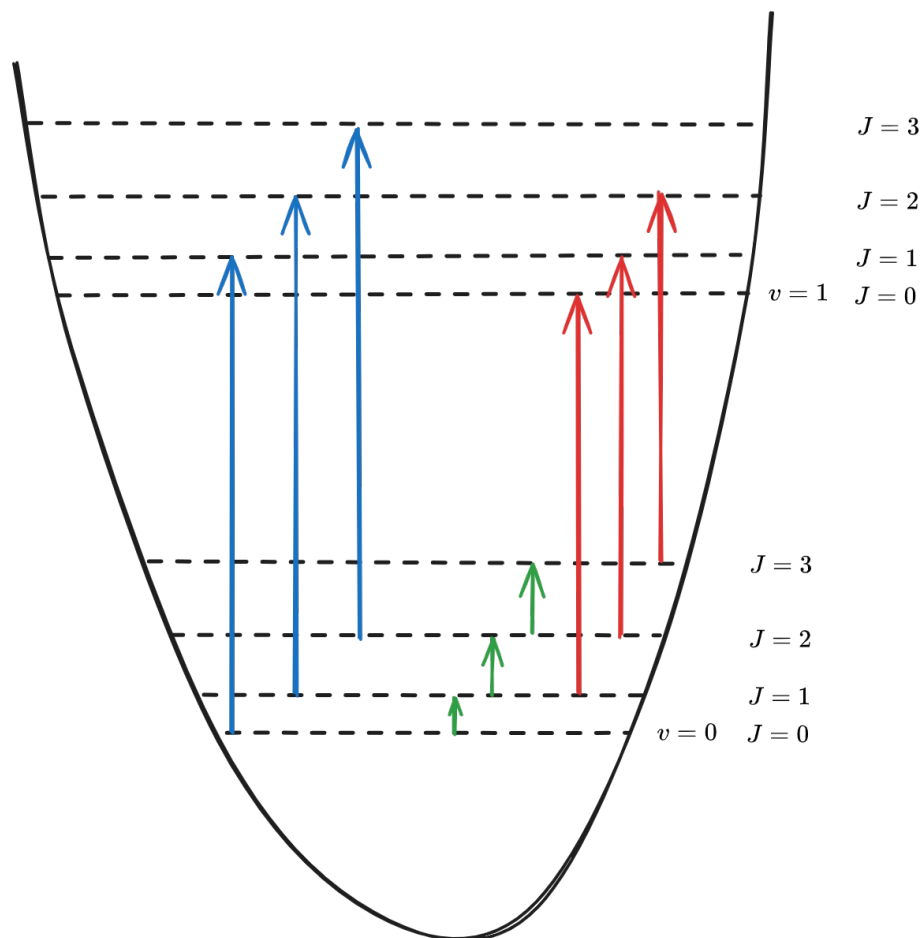


Figure 24.1

### 24.3.1 Pure Rotational

An allowed transition is a pure rotation, where  $\Delta J = -1$  or  $1$ . Because we are staying in the ground vibrational level,  $\Delta v = 0$ , we use the rotational constant of the lowest vibrational state  $\tilde{B}_0$ . The transitions are indicated by the green lines in the spectrum.

The formula for this for the increase in  $J$  is derived, with a higher energy of  $J' = J + 1$

and a lower energy of  $J' = J$ .

$$\begin{aligned}
\Delta \tilde{E} &= \tilde{E}_{J+1} - \tilde{E}_J \\
&= \left( \left( \frac{1}{2} + 0 \right) \tilde{\omega} + \tilde{B}_0(J+1)(J+1+1) \right) - \left( \left( \frac{1}{2} + v \right) \tilde{\omega} + \tilde{B}_0 J(J+1) \right) \\
&= \frac{1}{2} \tilde{\omega} + \tilde{B}_0(J^2 + 2J + J + 2) - \frac{1}{2} \tilde{\omega} - \tilde{B}_0(J^2 + J) \\
&= \tilde{B}_0(J^2 + 3J + 2) - \tilde{B}_0(J^2 + J) \\
&= 2\tilde{B}_0 J + 2\tilde{B}_0 \\
&= 2\tilde{B}_0(J+1)
\end{aligned}$$

This leads to set of equidistant lines in the spectrum. As  $J = 0, 1, 2, \dots$  we get a set of lines of  $2\tilde{B}, 4\tilde{B}, 6\tilde{B}, \dots$  that are equivalent.

### 24.3.2 Rovibrational

The other allowed transitions are  $\Delta v = 1$ , so an increase in vibrational energy, and an increase or decreased by 1 in the rotational energy level, so  $\Delta J = -1$  or 1. Making the assumption that  $\tilde{B}_0 \approx \tilde{B}_1$ , we can derive the  $J \rightarrow J+1$  transition,

$$\begin{aligned}
\Delta E &= \tilde{E}_{1,J+1} - \tilde{E}_{0,J} \\
&= \left( \left( \frac{1}{2} + 1 \right) \tilde{\omega} + \tilde{B}(J+1)(J+1+1) \right) - \left( \left( \frac{1}{2} + 0 \right) \tilde{\omega} + \tilde{B} J(J+1) \right) \\
&= \tilde{\omega} + 2\tilde{B}(J+1)
\end{aligned}$$

The other  $J \rightarrow J-1$  energy difference is derived similarly

$$\begin{aligned}
\Delta E &= E_{1,J-1} - E_{0,J} \\
&= \tilde{\omega} - 2\tilde{B}J
\end{aligned}$$

### Rovibrational Terminology



Spectroscopists commonly use the terms *P-branch*, *Q-branch*, and *R-branch*. Each one refers to a set of allowed rovibrational transitions involving  $J$ .

The P-branch has  $\Delta J = -1$ , the Q-branch  $\Delta J = 0$ , and R-branch  $\Delta J = +1$ . To remember, the numerical and alphabetical orders both go from low to high.

### 24.3.2.1 Not Assuming Constants are the Same

In reality the  $B$ -constants are different for the various vibrational states. By taking different constants, we could account for the rotational term in the Hamiltonian that looks like  $\frac{J(J+1)\hbar^2}{2\mu(x+R_e)^2}$ . The average distance in the excited vibrational state is larger, and as a consequence  $B_1 < B_0$ . If we don't assume  $\tilde{B}_0 \approx \tilde{B}_1$ , then a more accurate formula for  $\Delta v = 1$  and  $\Delta J = 1$  is derived.

$$\begin{aligned}\Delta\tilde{E} &= \tilde{E}_{1,J+1} - \tilde{E}_{0,J} \\ &= \left( \left( \frac{1}{2} + 1 \right) \tilde{\omega} + \tilde{B}_1(J+1)(J+1+1) \right) - \left( \left( \frac{1}{2} + 0 \right) \tilde{\omega} + \tilde{B}_0 J(J+1) \right) \\ &= \tilde{\omega} + 2\tilde{B}_1 + (3\tilde{B}_1 - \tilde{B}_0)J + (\tilde{B}_1 - \tilde{B}_0)J^2\end{aligned}$$

Importantly because we have the relation  $\tilde{B}_1 < \tilde{B}_0$ , the rightmost term will be negative since  $\tilde{B}_1 - \tilde{B}_0 < 0$ . The effect of this is that lines in the spectrum get closer together for the so-called R-branch  $v = 0, J \rightarrow v = 1, J + 1$ . They can even start to run “backwards”

The decrease in rotational energy levels is derived similarly,

$$\begin{aligned}\Delta\tilde{E} &= \tilde{E}_{1,J-1} - \tilde{E}_{0,J} \\ &= \tilde{\omega} - (\tilde{B}_1 + \tilde{B}_0)J + (\tilde{B}_1 - \tilde{B}_0)J^2\end{aligned}$$

The relation  $\tilde{B}_1 < \tilde{B}_0$  still holds. Now in the so-called P-branch, where  $v = 0, J \rightarrow v = 1, J - 1$ , the energy spacing increases with increasing  $J$ .

There is a nice problem on the assignments that explores these complications.

### 24.3.3 Low Temperatures

While the energies in quantum mechanics do not depend on temperature the intensities of the lines do depend on temperature and at low temperature many transitions are no longer observed, simplifying the spectrum.

In the above picture we assume that initially molecules are populating the rotational levels, as we excite from different states. This is true at higher temperatures. The population of the states is given by the Boltzmann populations (for further discussion, continue on to a statistical mechanics course).

Mathematically this is written as

$$\rho(v, J) \propto (2J + 1)e^{-E/k_{\text{B}}T}$$

As we get closer to 0 K, we start to see more of the molecules populate the lowest energy state  $E_{0,0}$ . Theoretically, if we were to get to absolute zero we would see the entire population in the ground state, and therefore we would have only transitions from the ground state. This transition would be only possible to see using excitation spectroscopy, of course. The pure rotational transition only has a magnitude of  $2\tilde{B}$ , while the rovibrational transition only has a magnitude of  $\hbar\omega + 2\tilde{B}$  where  $\Delta J = 1$  and  $\Delta v = 1$  in the final state.

# 25 Rotational, Vibrational, Translational Degrees of Freedom for Polyatomic Molecules

## 25.1 Overview

The harmonic oscillator model can be applied for any molecule. The starting point is to find the nuclear position of the atoms, using the Born-Oppenheimer approximation where  $\hat{T}_N$  is neglected. This is done using the electronic Schrödinger equation, that can be solved at each nuclear configuration to give the potential energy surface

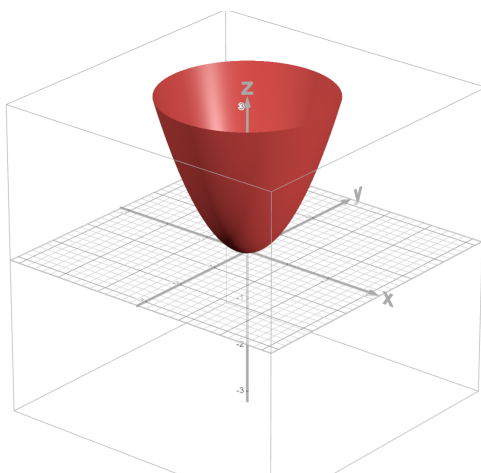
$$\hat{H}(\{\vec{R}\}) = E^{\text{el}}(\{\vec{R}\})\psi(\{\vec{R}\})$$

We do this by finding the minimum in  $3N$  dimensions (with  $N$  atoms),

$$\frac{\partial E^{\text{el}}}{\partial R_{\alpha,k}} = 0$$

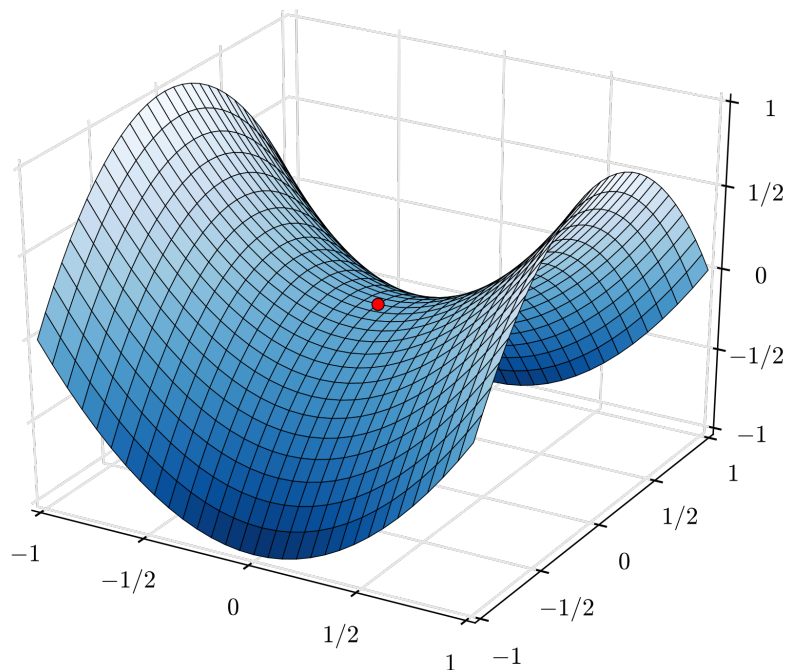
where  $k = x, y, z$  and  $\alpha$  is referring to each nucleus,  $3N$  components in total. The derivative being 0 indicates an extremum. This can be either a minimum, maximum or even a saddle point. There are of course further calculus details to assure we find actually find a minimum.

You might ask how many dimensions we are dealing with. For every atom, we have an equilibrium position  $R_{\text{eq}}$  and hence  $3N$  coordinates for the complete molecule.



**Figure 25.1:** A multivariable function with a clear minimum.

Conceptually it is not difficult, and this process is very similar to finding a minimum in one dimension, but with further dimensions we do run into some unique problems. For example, we could have a saddle point, a point that is both a minimum and maximum at the same time (in different directions). Despite this complication, the overall concept of extrema in multivariable functions is very similar to the the single variable functions we are more used to.



**Figure 25.2:** Saddle-point. For a more everyday comparison, imagine the shape of a Pringle chip. Image made by [Nicoguardo](#), licensed under [CC BY 3.0](#).

## 25.2 Energy Types and Degrees of Freedom

The potential energy (the electronic energy) does not change if we translate the molecule as a whole. However, we can associate kinetic energy with translating molecules (e.g. in a gas). This is referred to as the translational energy and there are 3 degrees of freedom associated with translation (in  $x, y, z$  directions). This is a good reminder that a change does not necessarily change the energy of the molecule, depending on the type of energy and specifics of the change.

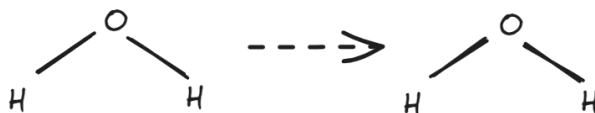


Figure 25.3

Likewise, a molecule can rotate as a whole, and the potential energy (i.e. electronic energy  $E^{el}(\vec{R})$ ) will not change. However there is kinetic energy associated with the rotational motion of the molecule, and we have quantum rotational energy levels (see earlier). In the rigid rotor approximation they are determined by the equilibrium nuclear configuration of the molecule  $\vec{R}_{eq}$ .

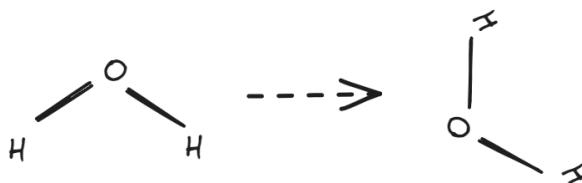


Figure 25.4

General molecules (like  $\text{H}_2\text{O}$ ) can rotate in 3 independent directions and they have 3 rotational degrees of freedom. Linear molecules are a special class. If they are rotated along their axis the geometry does not change. Like a rod, they have only 2 rotational degrees of freedom. All molecules (even atoms, which cannot rotate or vibrate) have 3 degrees of translational freedom.

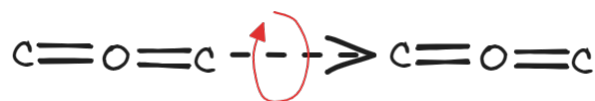


Figure 25.5

The number of vibrational degrees of freedom is simply  $3N$ , the number of translational and rotational degrees of freedom. It is different for linear and nonlinear molecules, as summarized in the table below.



**Table 25.1**

|           | Translational | Rotational | Vibrational |
|-----------|---------------|------------|-------------|
| Linear    | 3             | 2          | $3N - 5$    |
| Nonlinear | 3             | 3          | $3N - 6$    |

### 25.2.1 Discussion on IR/Raman Spectroscopy

An important note about the vibrational degrees of freedom is that there are different classes: for example, asymmetric and symmetric vibrations. This can be observed physically in spectroscopy, where IR spectroscopy shows mostly asymmetric vibrations and Raman spectroscopy shows mostly symmetric vibrations.

In IR spectroscopy, as the energy of the light matches some specific bond vibrational frequency, the bond will vibrate and the charge distribution changes. This produces a dipole moment. The charge distribution can only change if we have an asymmetric vibration, so a molecule like  $\text{O}=\text{C}=\text{O}$  has many inactive vibrations.

Polarizability is the tendency for charge to be distorted in an electric field. Imagine the shape of an electron cloud and how easily it can be stretch or squashed. The Raman laser photons interact with molecules, with most photons elastically colliding, retaining the same energy. A small amount of those photons will inelastically collide. This lost energy will transfer to and excite the molecules into a higher energy virtual state, with the vibration causing a change in polarizability, which allows for the emitted photon to be changed in energy relative to the incident energy (usually lower energy), or a Raman scattered photon. This vibration changes the polarizability, with more polar bonds being less changed and having their photon energy be less changed. Thus Raman is more sensitive to symmetric and less polar bonds asymmetric bonds, opposite to IR.

## 25.3 Quantum Chemistry Calculations and Nuclear Motion

Using the rigid rotor and harmonic oscillator approaches the computational protocols are fairly straightforward and are readily carried out in computer programs. The start of a calculation is to optimize to a stationary point on the potential energy surface, either a minimum or a transition state (first-order saddle point). Thus we obtain a molecular structure. The

type of nuclei determine the molecular mass (molecule A and molecule B, for example), and this yields translational energies and corresponding thermal properties (free energies). The position of the nuclei in the optimized configuration, labeled

$$R_{\alpha,j}, \alpha = x, y, z, j = 1, N_{\text{atom}}$$

determine molecular structure and from this one can calculate the moment-of-inertia tensor (a  $3 \times 3$  symmetric matrix) using computational software

$$I_{\alpha,\beta} = \delta_{\alpha\beta} \sum_j m_j (R_{x,j}^2 + R_{y,j}^2 + R_{z,j}^2) - \sum_j R_{\alpha,j} R_{\beta,j}$$

This tensor defines the rotational problem for the molecule. In the practice of thermochemistry it suffices to find the eigenvalues of this matrix leading to moments of inertia. There are three for non-linear molecules,  $I_a, I_b, I_c$ , and just one for a linear molecule. There are easy formulas derived from statistical mechanics to calculate thermal properties from just these values. That covers the translational and rotational motion as used in most thermochemistry applications. Spectroscopy can be more complicated.

To actually solve this equation we would diagonalize  $I_{\alpha,\beta}$  and end up with three eigenvalues.

## Diagonal Definition



A matrix can be nondiagonal. For example:

$$\begin{bmatrix} 1 & -4 & 0 \\ 2 & -2 & 7 \\ 1 & 8 & -3 \end{bmatrix}$$

Some matrices are symmetric, which means the matrix is equal to its transposed version. Symmetric matrices always have real eigenvalues. To transpose a matrix, take the first column and make it the first row, make the second column the second row, and so on. Most matrices in physics are nondiagonal, but symmetric.

$$\begin{bmatrix} 1 & 2 & -3 \\ 2 & 0 & 4 \\ -3 & 4 & -7 \end{bmatrix}$$

Some special matrices are diagonal, or have all zeroes except down the diagonal,

starting from the top left. They have many important properties and make many calculations significantly easier. For example:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

Some matrices are antidiagonal. A key property is that multiplying two antidiagonal matrices together will produce a diagonal one. For example:

$$\begin{bmatrix} 0 & 0 & -3 \\ 0 & -1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

The eigenvalues of a diagonal matrix are just the entries on the diagonal. Therefore, diagonalizing, or taking a nondiagonal matrix and converting it into a diagonal one, is an important initial step in many calculations. Many but not all matrices can be diagonalized. We work with matrices that are able to be diagonalized in quantum mechanics.

## Tensor Definition



A tensor is a mathematical object. It has a very complicated definition, but in simple words you can think of it as something that mathematically groups different related quantities together into one total object. In this case, we group together the three quantities of different rotations.

These are useful, because they allow us to keep track of how the quantities transform no matter what type of coordinates we're working with. We can represent our coordinates in spherical or Cartesian, and any transformation stays consistent.

### 25.3.1 Vibrations

The calculation of vibrational energies is most complicated. In an electronic structure calculation one would calculate the  $3N \times 3N$  matrix of second derivatives, at the prior located equilibrium geometry

$$K_{\alpha i, \beta j} = \frac{\partial^2 E(\vec{R}_{\text{eq}})}{\partial R_{\alpha, i} \partial R_{\beta, j}}$$

One then defines the so-called mass-weighted Hessian

$$\tilde{K}_{\alpha i, \beta j} = \sqrt{m_i} K_{\alpha i, \beta j} \sqrt{m_j}$$

The eigenvalues of this matrix provide the square of vibrational frequencies,  $\omega_i^2$ , while the corresponding eigenvectors describe the collective motion associated with the eigenvalues. These eigenvectors are specifically called the **normal modes** of vibration.

Three eigenvalues of  $\tilde{\mathbf{K}}$  are zero and they describe overall translations. Another 3 eigenvalues (2 for linear molecules) are zero and the eigenvectors describe (infinitesimal) overall rotations. That leaves  $(3N-6)$  vibrational frequencies (and  $(3N-5)$  for a linear molecule). If a structure corresponds to a saddle point (transition state) exactly one of the eigenvalues is imaginary (negative square), and the corresponding eigenvector reflects a reaction coordinate.

### 25.3.2 Vibrational Hamiltonian

Each normal mode and vibrational frequency defines a 1-dimensional harmonic oscillator Hamiltonian. Placing these in a matrix, called the Hessian<sup>1</sup>, and then diagonalizing it, we can obtain a complete and independent/non-coupled set of vibrational eigenstates. This is possible by building on just our knowledge of the 1-dimensional harmonic oscillator. For each mode we can define ladder operators  $\hat{b}_i, \hat{b}_i^\dagger$

$$\begin{aligned}\hat{b}_i &= \frac{1}{\sqrt{2}} \left( q_i + \frac{d}{dq_i} \right) \\ \hat{b}_i^\dagger &= \frac{1}{\sqrt{2}} \left( q_i - \frac{d}{dq_i} \right)\end{aligned}$$

The full vibrational Hamiltonian is then

$$\hat{H} = \hbar\omega_1 \left( \hat{b}_1^\dagger \hat{b}_1 + \frac{1}{2} \right) + \hbar\omega_2 \left( \hat{b}_2^\dagger \hat{b}_2 + \frac{1}{2} \right) + \cdots + \hbar\omega_k \left( \hat{b}_k^\dagger \hat{b}_k + \frac{1}{2} \right)$$

The energy eigenvalue is the sum over the individual eigenvalues

$$E_{n_1, n_2, n_3} = \hbar\omega_1 \left( n_1 + \frac{1}{2} \right) + \hbar\omega_2 \left( n_2 + \frac{1}{2} \right) + \cdots + \hbar\omega_k \left( n_k + \frac{1}{2} \right)$$

while the wave functions would be products of the individual harmonic oscillator functions,

---

<sup>1</sup>The Hessian is also a matrix of 2<sup>nd</sup> order partial derivatives of a function.

that can be expressed in terms of the ladder operators

$$\Psi_{n_1, n_2, \dots, n_k} \rightarrow (\hat{b}_1^\dagger)^{n_1} (\hat{b}_2^\dagger)^{n_2} \dots (\hat{b}_k^\dagger)^{n_k} |0, 0, \dots, 0\rangle$$

The notation is a bit complicated, but in essence the result is very simple. It all follows from a separation of variables in a suitable (normal mode) coordinate system!

The energy of the ground state is then

$$E_{0,0,0} = \frac{1}{2} \hbar (\omega_1 + \omega_2 + \dots \omega_k)$$

The compact and general vibrational ground state energy is expressed as

$$E_{zp} = \sum_i \frac{1}{2} \hbar \omega_i$$

where  $E_{zp}$  stands for the famous zero point energy. This sets a reasonable convention, where the ground state is designated as having 0 energy. This is then our baseline to compare energy levels of excited states.

The main take away of this section is that the description of nuclear motion is surprisingly simple in the widely used independent particle, rigid rotor, harmonic oscillator approximation. It is sufficiently accurate to describe thermal contributions to energies and other properties for many applications. All of this is the basis for statistical mechanics. A statistical mechanical calculation will start by using electronic structure calculations to determine intramolecular forces, which can be very time consuming if one does not use a lookup table, then use statistical techniques for bulk scale intermolecular forces. The models we discussed are not simply toy models to get us started in quantum mechanics. They are used in state of the art quantum chemistry calculations.

Let me emphasize once more that none of the quantum mechanical energy levels depend on temperature. The population of the energy levels and quantum states depends on temperature, and this is the subject of statistical mechanics, not quantum mechanics.

## 26 Angular Momentum Theory

### 26.1 The Purpose of This Theory

Angular momentum theory is a powerful tool. It lets us explain and predict a number of chemical phenomena, such as:

- Rotational spectra, using the angular solution to the Schrödinger equation,  $Y_l^m(\theta, \varphi)$
- Lets us understand the orbitals we chemists know and love,  $s, p, d, f$
- Gives meaning to electron spin, and the distinction between up and down spin
- Mathematically derive nuclear spin, as seen in NMR
- The triplet states of oxygen, important for its unusual radical stability
- ... And more!

### 26.2 Useful Preliminary Definitions

Total angular momentum:

$$\hat{L}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2$$

Angular momentum<sup>1</sup>:

$$\hat{L}_x = \hat{y}\hat{p}_z - \hat{z}\hat{p}_y$$

$$\hat{L}_y = \hat{z}\hat{p}_x - \hat{x}\hat{p}_z$$

$$\hat{L}_z = \hat{x}\hat{p}_y - \hat{y}\hat{p}_x$$

Momentum:

$$\hat{p}_x = -i\hbar \frac{\partial}{\partial x}$$

$$\hat{p}_y = -i\hbar \frac{\partial}{\partial y}$$

$$\hat{p}_z = -i\hbar \frac{\partial}{\partial z}$$

---

<sup>1</sup>These can be derived using the cross product. An interesting fact is that this relation (or versions of it) hold across mathematics as a whole and are tied to how our coordinate system is defined.

Ladder operators:

$$\begin{aligned}\hat{L}_+ &= \hat{L}_x + i\hat{L}_y \\ \hat{L}_- &= \hat{L}_x - i\hat{L}_y\end{aligned}$$

### 26.2.1 Useful Commutation Relations Between Position and Momentum Operators in 3D

Let us denote  $r_\alpha \rightarrow \hat{x}, \hat{y}, \hat{z}$  and  $\hat{p}_\alpha \rightarrow \hat{p}_x, \hat{p}_y, \hat{p}_z$ . In the following we will use greek symbols  $\alpha, \beta$  to represent any cartesian direction  $\alpha = x, y, z, \beta = x, y, z$ . We will evaluate commutators by acting on a general function  $f(x, y, z)$ .

Two position or two momentum operators always commute,

$$\begin{aligned}[\hat{r}_\alpha, \hat{r}_\beta] &= 0 \\ [\hat{p}_\alpha, \hat{p}_\beta] &= 0\end{aligned}$$

For example

$$\begin{aligned}[\hat{x}, \hat{y}]f(x, y, z) &= xyf(x, y, z) - yxf(x, y, z) = 0 \\ [\hat{p}_x, \hat{p}_y]f(x, y, z) &= (-i\hbar)^2 \left( \frac{\partial f(x, y, z)}{\partial y} x - \frac{\partial f(x, y, z)}{\partial y} x \right) = 0\end{aligned}$$

where we use the fact that partial derivatives commute and therefore the order we take them in does not matter.

We also know from previous discussions that

$$\begin{aligned}[\hat{x}, \hat{p}_x] &= i\hbar \\ \text{likewise, } [\hat{y}, \hat{p}_y] &= [\hat{z}, \hat{p}_z] = i\hbar\end{aligned}$$

If the coordinates are *different* position and momentum commute, e.g.

$$\begin{aligned}[\hat{x}, \hat{p}_y]f(x, y, z) &= -i\hbar \left( x \frac{\partial f(x, y, z)}{\partial y} - \frac{\partial x f(x, y, z)}{\partial y} \right) \\ &= -i\hbar \left( x \frac{\partial f(x, y, z)}{\partial y} - x \frac{\partial f(x, y, z)}{\partial y} \right) = 0\end{aligned}$$

Together all of these relations can be summarized as

$$\begin{aligned}[\hat{r}_\alpha, \hat{p}_\beta] &= i\hbar\delta_{\alpha,\beta} \\ [\hat{r}_\alpha, \hat{r}_\beta] &= 0 \\ [\hat{p}_\alpha, \hat{p}_\beta] &= 0\end{aligned}$$

The Kronecker delta  $\delta_{\alpha,\beta}$  signifies that the contribution is non-zero only if the directions are the same ( $\alpha = \beta$ ). This is a powerful way to express many relations as  $\alpha = x, y, z, \beta = x, y, z$ .

### 26.2.2 Results That Can Be Derived From This

Angular momentum commutation relations:

$$\begin{aligned}[\hat{L}_x, \hat{L}_y] &= i\hbar\hat{L}_z \\ [\hat{L}_y, \hat{L}_z] &= i\hbar\hat{L}_x \\ [\hat{L}_z, \hat{L}_x] &= i\hbar\hat{L}_y\end{aligned}$$

$\hat{L}^2$  expressed in terms of ladder operators (two ways):

$$\begin{aligned}\hat{L}^2 &= \hat{L}_-\hat{L}_+ + \hat{L}_z^2 + \hbar\hat{L}_z \\ \hat{L}^2 &= \hat{L}_+\hat{L}_- + \hat{L}_z^2 - \hbar\hat{L}_z\end{aligned}$$

Commutation of  $\hat{L}^2$  and directional angular momentum:

$$\begin{aligned}[\hat{L}^2, \hat{L}_x] &= 0 \\ [\hat{L}^2, \hat{L}_y] &= 0 \\ [\hat{L}^2, \hat{L}_z] &= 0\end{aligned}$$

Commutation of  $\hat{L}^2$  and ladder operators:

$$\begin{aligned}[\hat{L}^2, \hat{L}_-] &= 0 \\ [\hat{L}^2, \hat{L}_+] &= 0\end{aligned}$$



Commutation of  $\hat{L}_z$  with ladder operators:

$$\begin{aligned}[\hat{L}_z, \hat{L}_+] &= \hbar \hat{L}_+ \\ [\hat{L}_z, \hat{L}_-] &= -\hbar \hat{L}_-\end{aligned}$$

## 26.3 Interlude: Useful Identities with Commutators)

We will do quite some manipulations involving commutators. We will not act on functions, but manipulate the commutators as operator expressions. All of this becomes fairly abstract. We will proof each type of manipulation.

### 26.3.1 Swapped Order

Fairly simple, it is

$$[\hat{A}, \hat{B}] = -[\hat{B}, \hat{A}]$$

#### 26.3.1.1 Proof

$$\begin{aligned}[\hat{A}, \hat{B}] &= \hat{A}\hat{B} - \hat{B}\hat{A} \\ &= -(\hat{B}\hat{A} - \hat{A}\hat{B}) \\ &= -[\hat{B}, \hat{A}]\end{aligned}$$

### 26.3.2 Constant: Scalar That Can Be Pulled Out

If we have some scalar constant multiplied to one of the terms in the commutator we can pull it out front, or vice versa. Importantly, the constant can be pulled out front even if it is only multiplied to one term of the commutator.

$$\begin{aligned}[c\hat{A}, \hat{B}] &= c[\hat{A}, \hat{B}] \\ [\hat{A}, c\hat{B}] &= c[\hat{A}, \hat{B}] \\ [c\hat{A}, d\hat{B}] &= cd[\hat{A}, \hat{B}] \\ [c\hat{A}, c\hat{B}] &= c^2[\hat{A}, \hat{B}]\end{aligned}$$

### 26.3.2.1 Proof

$$\begin{aligned}[c\hat{A}, \hat{B}] &= c\hat{A}\hat{B} - \hat{B}c\hat{A} \\ &= c(\hat{A}\hat{B} - \hat{B}\hat{A}) \\ &= c[\hat{A}, \hat{B}]\end{aligned}$$

### 26.3.3 Operator Commutation

For example

$$\begin{aligned}\text{If } [\hat{A}, \hat{C}] = 0 \text{ and } [\hat{B}, \hat{C}] = 0 \text{ then} \\ [\hat{A}\hat{C}, \hat{B}] = \hat{C}[\hat{A}, \hat{B}] = [\hat{A}, \hat{B}]\hat{C}\end{aligned}$$

The significance is that an operator that commutes with everything else can be treated as a number. Very useful.

### 26.3.4 Simplification of Commutators: Products of Operators

An important commutator identity is the Jacobi identity.

$$[\hat{A}\hat{B}, \hat{C}] = \hat{A}[\hat{B}, \hat{C}] + [\hat{A}, \hat{C}]\hat{B}$$

Similarly,

$$[\hat{A}, \hat{B}\hat{C}] = [\hat{A}, \hat{B}]\hat{C} + \hat{B}[\hat{A}, \hat{C}]$$

For a commutator with 4 components, as in  $[\hat{A}\hat{B}, \hat{C}\hat{D}]$ , we can use these two identities consecutively.

$$\begin{aligned}[\hat{A}\hat{B}, \hat{C}\hat{D}] &= \hat{A}[\hat{B}, \hat{C}\hat{D}] + [\hat{A}, \hat{C}\hat{D}]\hat{B} \\ &= \hat{A}[\hat{B}, \hat{C}]\hat{D} + \hat{A}\hat{C}[\hat{B}, \hat{D}] + [\hat{A}, \hat{C}]\hat{D}\hat{B} + \hat{C}[\hat{A}, \hat{D}]\hat{B}\end{aligned}$$

As we can see, we end up with four commutators after expansion. While not too hard, the math can get tedious fast.

### 26.3.4.1 Proof

$$\begin{aligned}
[\hat{A}\hat{B}, \hat{C}] &= \hat{A}\hat{B}\hat{C} - \hat{C}\hat{A}\hat{B} \\
&= (\hat{A}\hat{B}\hat{C} - \hat{C}\hat{A}\hat{B}) + \underbrace{(\hat{A}\hat{C}\hat{B} - \hat{A}\hat{C}\hat{B})}_0 \\
&= (\hat{A}\hat{B}\hat{C} - \hat{A}\hat{C}\hat{B}) + (\hat{A}\hat{C}\hat{B} - \hat{C}\hat{A}\hat{B}) \\
&= \hat{A}(\hat{B}\hat{C} - \hat{C}\hat{B}) + (\hat{A}\hat{C} - \hat{C}\hat{A})\hat{B} \\
&= \hat{A}[\hat{B}, \hat{C}] + [\hat{A}, \hat{C}]\hat{B}
\end{aligned}$$

## 26.3.5 Simplification of Commutators: Sums of Operators

If we have addition in the commutator, it can be expanded.

$$\begin{aligned}
[\hat{A} + \hat{B}, \hat{C}] &= [\hat{A}, \hat{C}] + [\hat{B}, \hat{C}] \\
[\hat{A}, \hat{B} + \hat{C}] &= [\hat{A}, \hat{B}] + [\hat{A}, \hat{C}]
\end{aligned}$$

Similar to the multiplicative identity, we can do the same for 4 components

$$\begin{aligned}
[\hat{A} + \hat{B}, \hat{C} + \hat{D}] &= [\hat{A}, \hat{C} + \hat{D}] + [\hat{B}, \hat{C} + \hat{D}] \\
&= [\hat{A}, \hat{C}] + [\hat{A}, \hat{D}] + [\hat{B}, \hat{C}] + [\hat{B}, \hat{D}]
\end{aligned}$$

This property is the distributive property of commutators. operators act like numbers under addition , in essence because  $\hat{A} + \hat{B} = \hat{B} + \hat{A}$

### 26.3.5.1 Proof

$$\begin{aligned}
[\hat{A}, \hat{C} + \hat{B}] &= \hat{A}(\hat{C} + \hat{B}) - (\hat{C} + \hat{B})\hat{A} \\
&= \hat{A}\hat{C} + \hat{A}\hat{B} - \hat{C}\hat{A} - \hat{B}\hat{A} \\
&= (\hat{A}\hat{B} - \hat{B}\hat{A}) + (\hat{A}\hat{C} - \hat{C}\hat{A}) \\
&= [\hat{A}, \hat{B}] + [\hat{A}, \hat{C}]
\end{aligned}$$

## 26.4 Derivations of Angular Momentum Relations

### 26.4.1 Directional Angular Momentum

This relation can be derived using standard calculus

$$[\hat{L}_x, \hat{L}_y] = i\hbar \hat{L}_z$$

While the calculus method is certainly possible, it requires tedious use of the chain rule and a long page of algebra. We can make use of the multiplicative, constant, and additive identities instead.

$$\begin{aligned} [\hat{L}_x, \hat{L}_y] &= [\hat{y}\hat{p}_z - \hat{z}\hat{p}_y, \hat{z}\hat{p}_x - \hat{x}\hat{p}_z] \rightarrow \text{apply distributive property under addition} \\ &= [\hat{y}\hat{p}_z, \hat{z}\hat{p}_x] - [\hat{y}\hat{p}_z, \hat{x}\hat{p}_z] - [\hat{z}\hat{p}_y, \hat{z}\hat{p}_x] + [\hat{z}\hat{p}_y, \hat{x}\hat{p}_z] \rightarrow \text{pull out operators that commute with everything} \\ &= \hat{y}\hat{p}_x[\hat{p}_z, \hat{z}] - \hat{y}\hat{x}[\hat{p}_z, \hat{p}_z] - \hat{z}^2[\hat{p}_y, \hat{p}_x] + \hat{y}\hat{p}_x[\hat{z}, \hat{p}_z] \rightarrow \text{evaluate commutators} \\ &= -i\hbar \hat{y}\hat{p}_x + i\hbar \hat{x}\hat{p}_y = i\hbar(\hat{x}\hat{p}_y - \hat{y}\hat{p}_x) \\ &= i\hbar \hat{L}_z \end{aligned}$$

I think you can appreciate the compactness of the derivation. We made extensive use of our tricks with commutators to simplify things. You did the problem yourself using an explicit evaluation of the operators acting on a function. It was painful. Now we have a simple four-line derivation.

An identical process can be used to derive for  $[\hat{L}_y, \hat{L}_z]$  and  $[\hat{L}_z, \hat{L}_x]$ . It may be useful practice to do yourself, not looking at these notes.

### 26.4.2 Commutation of $\hat{L}^2$ with Directional Angular Momentum

We wish to derive the result

$$[\hat{L}^2, \hat{L}_z] = 0$$

We will use the multiplicative identities, as earlier

$$\begin{aligned} [\hat{L}^2, \hat{L}_z] &= [\hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2, \hat{L}_z] \\ &= [\hat{L}_x^2, \hat{L}_z] + [\hat{L}_y^2, \hat{L}_z] + [\hat{L}_z^2, \hat{L}_z] \\ &= \hat{L}_x[\hat{L}_x, \hat{L}_z] + [\hat{L}_x, \hat{L}_z]\hat{L}_x + \hat{L}_y[\hat{L}_y, \hat{L}_z] + [\hat{L}_y, \hat{L}_z]\hat{L}_y + \cancel{\hat{L}_z[\hat{L}_z, \hat{L}_z]} + \cancel{[\hat{L}_z, \hat{L}_z]\hat{L}_z} \end{aligned}$$

We will use the earlier directional angular momentum results for these remaining terms.

$$\begin{aligned}
[\hat{L}^2, \hat{L}_z] &= \hat{L}_x[\hat{L}_x, \hat{L}_z] + [\hat{L}_x, \hat{L}_z]\hat{L}_x + \hat{L}_y[\hat{L}_y, \hat{L}_z] + [\hat{L}_y, \hat{L}_z]\hat{L}_y \\
&= \hat{L}_x(-i\hbar\hat{L}_y) + (-i\hbar\hat{L}_y)\hat{L}_x + \hat{L}_y(i\hbar\hat{L}_x) + (i\hbar\hat{L}_x)\hat{L}_y \\
&= i\hbar(-\hat{L}_x\hat{L}_y - \hat{L}_y\hat{L}_x + \hat{L}_y\hat{L}_x + \hat{L}_x\hat{L}_y) \\
&= 0
\end{aligned}$$

Similarly we can obtain

$$\begin{aligned}
[\hat{L}^2, \hat{L}_x] &= 0 \\
[\hat{L}^2, \hat{L}_y] &= 0
\end{aligned}$$

### 26.4.3 $\hat{L}^2$ in Terms of Ladder Operators

We use the preliminary expansion of  $\hat{L}^2$  in terms its directional components.

$$\hat{L}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2$$

We then expand  $\hat{L}_-\hat{L}_+$ .

$$\begin{aligned}
\hat{L}_-\hat{L}_+ &= (\hat{L}_x - i\hat{L}_y)(\hat{L}_x + i\hat{L}_y) \\
&= (\hat{L}_x - i\hat{L}_y)\hat{L}_x + (\hat{L}_x - i\hat{L}_y)i\hat{L}_y \\
&= \hat{L}_x^2 - i\hat{L}_y\hat{L}_x + i\hat{L}_x\hat{L}_y + \hat{L}_y^2 \\
&= \hat{L}_x^2 + \hat{L}_y^2 + i(\hat{L}_x\hat{L}_y - \hat{L}_y\hat{L}_x) \\
&= \hat{L}_x^2 + \hat{L}_y^2 + i[\hat{L}_x, \hat{L}_y] \\
&= \hat{L}_x^2 + \hat{L}_y^2 + i(i\hbar\hat{L}_z) \\
&= \hat{L}_x^2 + \hat{L}_y^2 - \hbar\hat{L}_z
\end{aligned}$$

Therefore, to use  $\hat{L}_-\hat{L}_+$  to describe  $\hat{L}^2$ , we must add  $\hbar\hat{L}_z$  to cancel out that extra  $-\hbar\hat{L}_z$  term, and add  $\hat{L}_z^2$ .

$$\hat{L}^2 = \hat{L}_-\hat{L}_+ + \hat{L}_z^2 + \hbar\hat{L}_z$$

This can be done similarly to describe  $\hat{L}^2$  in terms of  $\hat{L}_+\hat{L}_-$ .

$$\hat{L}^2 = \hat{L}_+\hat{L}_- + \hat{L}_z^2 - \hbar\hat{L}_z$$

### 26.4.4 Commutation of $\hat{L}_z$ with Ladder Operators

We use the preliminary definition that expresses  $\hat{L}_+$  (and  $\hat{L}_-$ ) in terms of  $\hat{L}_x$  and  $\hat{L}_y$ .

$$\begin{aligned}
 [\hat{L}_z, \hat{L}_+] &= [\hat{L}_z, \hat{L}_x + i\hat{L}_y] \\
 &= [\hat{L}_z, \hat{L}_x] + [\hat{L}_z, i\hat{L}_y] \\
 &= [\hat{L}_z, \hat{L}_x] + i[\hat{L}_z, \hat{L}_y] \\
 &= i\hbar\hat{L}_y + i(-i\hbar\hat{L}_x) \\
 &= i\hbar\hat{L}_y + \hbar\hat{L}_x \\
 &= \hbar(\hat{L}_x + i\hat{L}_y) \\
 &= \hbar\hat{L}_+
 \end{aligned}$$

This can be done similar for

$$[\hat{L}_z, \hat{L}_-] = -\hbar\hat{L}_-$$

Both these relations will be useful. Note the different ordering of  $\hat{L}_+$  and  $\hat{L}_-$ .

### 26.4.5 Commutation of $\hat{L}^2$ with Ladder Operators

This is easy using the previous relation  $[\hat{L}^2, \hat{L}_z] = 0$ , with similar We use the same preliminary expansion of  $\hat{L}^2$  and expansion of the ladder operator.

$$\begin{aligned}
 [\hat{L}^2, \hat{L}_+] &= [\hat{L}^2, \hat{L}_x + i\hat{L}_y] \\
 &= [\hat{L}^2, \hat{L}_x] + i[\hat{L}^2, \hat{L}_y] \\
 &= 0 + 0 = 0
 \end{aligned}$$

This can be done similarly for

$$[\hat{L}^2, \hat{L}_-] = 0$$

## 26.5 Crux

So why does this all matter? A key result is that

$$[\hat{L}^2, \hat{L}_z] = 0$$

which means that they commute. Because they commute, this means that  $\hat{L}^2$  and  $\hat{L}_z$  have a set of common eigenfunctions. Let us denote these eigenfunctions as  $g_{a,b}$ .

$$\begin{aligned}\hat{L}^2 g_{a,b} &= a g_{a,b} \\ \hat{L}_z g_{a,b} &= b g_{a,b}\end{aligned}$$

In what follows we will derive all the possible eigenvalues  $a, b$  of the angular momentum operators. In addition we provide a recipe how to obtain the other eigenfunctions if one of them is known. We will use all of our preliminary work to make this quite doable. Before this we need one more ingredient.

### 26.5.1 Key Result $a \geq b^2$

A key result is that

$$a \geq b^2$$

This can be proved by using  $\hat{L}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2$  and the hermiticity of  $\hat{L}_x$  (and by extension  $\hat{L}_z, \hat{L}_y$ ).

While not doing the full proof, the logic is this: We use calculate the expectation value of  $\hat{L}_x^2$  as a Hermitian operator, using some arbitrary function angular function  $\psi(\theta, \varphi)$ .

This produces an absolute square, which by definition is nonnegative, so its only possible results are either 0 or some positive number. By identical logic, this also holds for  $\hat{L}_y^2$  and  $\hat{L}_z^2$ . We can then express the total angular momentum expectation as a sum, using linearity, as below

$$\langle \hat{L}^2 \rangle = \langle \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2 \rangle = \langle \hat{L}_x^2 \rangle + \langle \hat{L}_y^2 \rangle + \langle \hat{L}_z^2 \rangle$$

Picking a direction to focus on,  $\hat{L}_z$  we know that  $\langle \hat{L}^2 \rangle$  is either equal to  $\langle \hat{L}_z^2 \rangle$  (assuming both  $\langle \hat{L}_x^2 \rangle$  and  $\langle \hat{L}_y^2 \rangle$  are 0) or greater ( $\langle \hat{L}_x^2 \rangle$  and/or  $\langle \hat{L}_y^2 \rangle$  are positive). Mathematically this can be written as

$$\langle \hat{L}^2 \rangle \geq \langle \hat{L}_z^2 \rangle$$

It stands to reason then that this relationship applies to its eigenvalues. We know the eigenvalue of  $\hat{L}_z$  to be  $b$ , therefore the eigenvalue of  $\hat{L}_z^2$  is  $b^2$ . We know the eigenvalue of  $\hat{L}^2$  to be  $a$ , therefore

$$a \geq b^2$$

## 26.5.2 Crux: Using Ladder Operators to Create Other Eigenfunctions

If  $g_{a,b}$  is an eigenfunction of both  $\hat{L}^2$  and  $\hat{L}_z$

$$\hat{L}^2 g_{a,b} = a g_{a,b}$$

$$\hat{L}_z g_{a,b} = b g_{a,b}$$

Then so are the new function  $\hat{L}_+ g_{a,b}$  and  $\hat{L}_- g_{a,b}$ . The eigenvalue  $a$  does not change, but the eigenvalue  $b$  is raised by  $+\hbar$ , if  $\hat{L}_+$  is used, and lowered by  $-\hbar$  if  $\hat{L}_-$  is used. Alternatively a ladder operator may yield a zero result.

Let us write the detailed mathematical equations:

$$\hat{L}^2(\hat{L}_+ g_{a,b}) = a(\hat{L}_+ g_{a,b})$$

$$\hat{L}_z(\hat{L}_+ g_{a,b}) = (b + \hbar)(\hat{L}_+ g_{a,b})$$

and

$$\hat{L}^2(\hat{L}_- g_{a,b}) = a(\hat{L}_- g_{a,b})$$

$$\hat{L}_z(\hat{L}_- g_{a,b}) = (b - \hbar)(\hat{L}_- g_{a,b})$$

These relations are easy to proof (with our tools), and I will use one more very general operator relation (almost trivial, but *very* useful)

$$\hat{A}\hat{B} = [\hat{A}, \hat{B}] + \hat{B}\hat{A}$$

We can hence reverse the action of two operators by adding the commutator. This relation is valid when acting on any function. Let use it in the present context:

$$\begin{aligned} \hat{L}^2(\hat{L}_+ g_{a,b}) &= [\hat{L}^2, \hat{L}_+] g_{a,b} + \hat{L}_+(\hat{L}^2 g_{a,b}) \\ &= 0 + \hat{L}_+ a g_{a,b} = a(\hat{L}_+ g_{a,b}) \end{aligned}$$

$$\begin{aligned} \hat{L}_z(\hat{L}_+ g_{a,b}) &= [\hat{L}_z, \hat{L}_+] g_{a,b} + \hat{L}_+(\hat{L}_z g_{a,b}) \\ &= \hbar \hat{L}_+ g_{a,b} + b \hat{L}_+ g_{a,b} = (b + \hbar)(\hat{L}_+ g_{a,b}) \end{aligned}$$

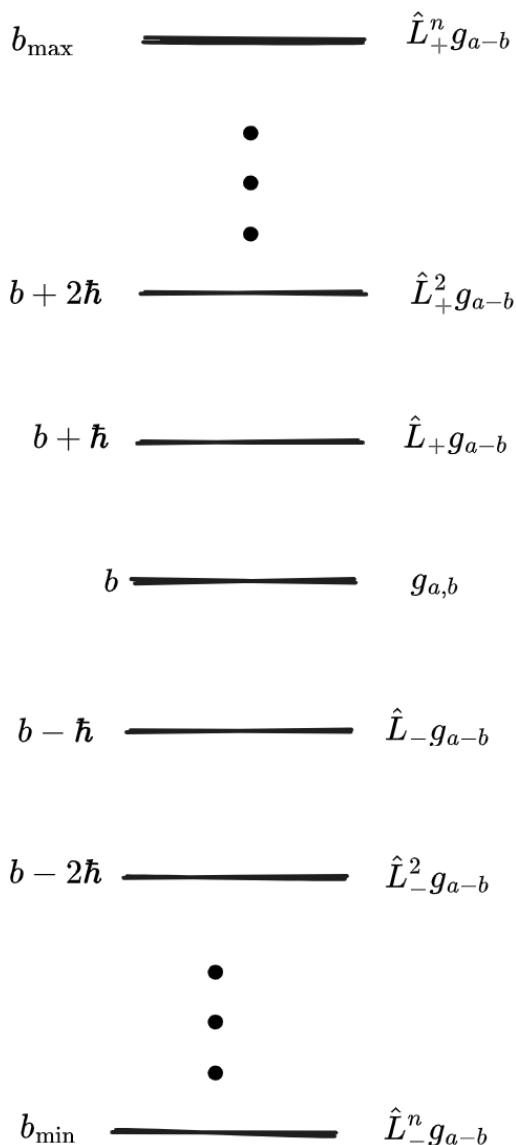
The key idea is to reverse the order of the operators, because they then act on an eigen-



function. The extra commutators evaluate to something simple and voila. The relations for  $\hat{L}_{-}g_{a,b}$  can be shown in a similar simple way. As a reminder we only used the previously derived relations

$$\begin{aligned}[\hat{L}_z, \hat{L}_+] &= \hbar \hat{L}_+ \\[\hat{L}_z, \hat{L}_-] &= -\hbar \hat{L}_- \\[\hat{L}^2, \hat{L}_+] &= 0 \\[\hat{L}^2, \hat{L}_-] &= 0\end{aligned}$$

We can see the usefulness of the ladder operators. They give us key insights in the eigenvalue patterns, and a protocol to create new eigenfunctions. See the figure below.



$$a \geq b^2$$

upper and lower bounded!

**Figure 26.1:** Action of repeated action of the ladder operators on a given function  $g_{a,b}$ . The eigenvalue  $a$  stays the same. The eigenvalue  $b$  increases by  $\hbar$  when laddering up, and decreases by  $-\hbar$  when laddering down. Both process have to terminate because  $b_{max}^2 \leq a$  and  $b_{min}^2 \leq a$ . In the figure all  $g_{a-b}$  should just be  $g_{a,b}$ . Sorry.

The  $a \geq b^2$  is *very* important in that it bounds the ladder operators. We can't just ladder up or down infinitely. If we were to ladder up on the maximum eigenvalue  $b_{max}$ , the next function generated would have to vanish. Similarly, the next function would vanish if the down ladder acted upon  $b_{min}$ .

Using this and multiple derived forms of our angular momentum operators, it will let

us derive the concept of orbitals as angular solutions of the Schrödinger equation... next class.

# 27 Understanding Angular Momentum and Orbitals

## 27.1 Quick Review

### 27.1.1 Useful Forms

Total angular momentum:

$$\begin{aligned}\hat{L}^2 &= \hat{L}_- \hat{L}_+ + \hat{L}_z^2 + \hbar \hat{L}_z \\ \hat{L}^2 &= \hat{L}_+ \hat{L}_- + \hat{L}_z^2 - \hbar \hat{L}_z\end{aligned}$$

### 27.1.2 Laddering

We know (from last lecture) that the eigenvalues of  $\hat{L}_z$  and  $\hat{L}^2$ , acting upon some function  $g_{a,b}$  can be written

$$\begin{aligned}\hat{L}^2 g_{a,b} &= a g_{a,b} \\ \hat{L}_z g_{a,b} &= b g_{a,b}\end{aligned}$$

We also know that the eigenfunctions  $\hat{L}_+ g_{a,b}$  and  $\hat{L}_- g_{a,b}$  when acted upon by  $\hat{L}_z$  and  $\hat{L}^2$  produce the following results

$$\begin{aligned}\hat{L}_z(\hat{L}_+ g_{a,b}) &= (b + \hbar) \hat{L}_+ g_{a,b} \\ \hat{L}_z(\hat{L}_- g_{a,b}) &= (b - \hbar) \hat{L}_- g_{a,b}\end{aligned}$$

$$\begin{aligned}\hat{L}^2(\hat{L}_+ g_{a,b}) &= a \hat{L}_+ g_{a,b} \\ \hat{L}^2(\hat{L}_- g_{a,b}) &= a \hat{L}_- g_{a,b}\end{aligned}$$

### 27.1.3 Key Result

The eigenvalues  $b$  are bounded by the key inequality

$$a \geq b^2$$

This bounding is an important concept. We can't just ladder up or down infinitely, there'll always be a maximum or minimum energy value.

This bounding results in functions vanishing when trying to go past the maximum or minimum, as in

$$\begin{aligned}\hat{L}_+ g_{a, b_{\max}} &= 0 \\ \hat{L}_- g_{a, b_{\min}} &= 0\end{aligned}$$

## 27.2 Bounds

### 27.2.1 Upper Bound

Let us designate  $b_{\max} = l\hbar$  where  $l$  is some arbitrary value that we can designate out of convenience. This is an okay assumption to make, as the natural units of angular momentum (at the atomic level) are expressed in terms of quantized  $\hbar$ .

Knowing the next function has to vanish if acting upon the maximum value, we can say

$$\hat{L}_+ g_{a, l\hbar} = 0$$

We can now find the eigenvalues  $\hat{L}^2$  using this  $b_{\max} = l\hbar$  convenience. We use the form of  $\hat{L}^2$  derived earlier that expresses it in terms of the ladder operators. We use the version that has  $\hat{L}_+$  acting first. The reason is that we know  $\hat{L}_+ g_{a, b_{\max}} = 0$ .

$$\hat{L}^2 = \hat{L}_- \hat{L}_+ + \hat{L}_z^2 + \hbar \hat{L}_z$$

We act on the eigenfunction  $g_{a,l\hbar}$

$$\begin{aligned}
\hat{L}^2 g_{a,l\hbar} &= (\hat{L}_- \hat{L}_+ + \hat{L}_z^2 + \hbar \hat{L}_z) g_{a,l\hbar} \\
&= \hat{L}_- \hat{L}_+ g_{a,l\hbar} + \hat{L}_z^2 g_{a,l\hbar} + \hbar \hat{L}_z g_{a,l\hbar} \\
&= (l^2 \hbar^2 + \hbar^2 l) g_{a,l\hbar} \\
&= l \hbar^2 (l + 1) g_{a,l\hbar}
\end{aligned}$$

We see that the eigenvalue of  $\hat{L}^2$  acting upon  $g_{a,l\hbar}$  in this circumstance is

$$a = l(l + 1) \hbar^2$$

We now have a defined upper bound, based entirely on our choice of  $l$ .

## 27.2.2 Lower Bound

A similar process can be done for  $\hat{L}_-$ . We use  $b_{\min} = -k\hbar$ , and knowing the next function has to vanish if acting upon the minimum value.

$$\hat{L}_- g_{a,-k\hbar} = 0$$

We use the alternate form of  $\hat{L}^2$  where  $\hat{L}_-$  acts first.

$$\hat{L}^2 = \hat{L}_+ \hat{L}_- + \hat{L}_z^2 - \hbar \hat{L}_z$$

We use this reverse order because now  $\hat{L}_- g_{a,b_{\min}} = 0$ . We derive

$$\begin{aligned}
\hat{L}^2 g_{a,k\hbar} &= (\hat{L}_+ \hat{L}_- + \hat{L}_z^2 - \hbar \hat{L}_z) g_{a,-k\hbar} \\
&= \hat{L}_+ \hat{L}_- g_{a,-k\hbar} + \hat{L}_z^2 g_{a,-k\hbar} - \hbar \hat{L}_z g_{a,-k\hbar} \\
&= \hat{L}_+(0) + \hat{L}_z^2 g_{a,-k\hbar} - \hbar \hat{L}_z g_{a,-k\hbar} \\
&= ((-k\hbar)^2 - \hbar(-k\hbar)) g_{a,-k\hbar} \\
&= (k^2 \hbar^2 + \hbar^2 k) g_{a,-k\hbar} \\
&= k \hbar^2 (k + 1) g_{a,-k\hbar}
\end{aligned}$$

We see then that the eigenvalue of is

$$a = k(k + 1) \hbar^2$$

### 27.2.3 Connecting the bounds

We have derived two formulas for the eigenvalue  $a$

$$\begin{aligned} a &= l\hbar^2(l+1) \\ a &= k\hbar^2(k+1) \end{aligned}$$

where  $-k \leq l$ . The simplest solution is the only consistent solution and we find

$$k = l$$

So the ladder operator  $\hat{L}_-$  can act on a state with  $b_{max} = l\hbar$  and it toggles down by units of  $\hbar$  to reach the value  $b_{min} = -l\hbar$ . One possibility is that  $l$  is an integer, e.g.  $l = 3, b = 3\hbar, 2\hbar, 1\hbar, 0, -1\hbar, -2\hbar, -3\hbar$ . There would be  $2l + 1$  functions (odd). There is another possibility, that  $l$  is half-integer, for example  $l = \frac{3}{2}, b = \frac{3}{2}\hbar, \frac{1}{2}\hbar, -\frac{1}{2}\hbar, -\frac{3}{2}\hbar$ . For such half-integer values there are an even number of functions, still  $2l + 1$ . These are the only possibilities as we need to go down in an integer (even or odd) number of steps from  $+l\hbar$  to  $-l\hbar$ . For example if  $l = \frac{3}{4}$ , we would get  $b = \frac{3}{4}\hbar, -\frac{1}{4}\hbar, -\frac{5}{4}\hbar, \dots$ , which does not fit, as we skipped  $-l\hbar$ .

## 27.3 Summary of angular momentum analysis

Using only the commutation relations of the angular momentum operators we get the following patterns of eigenvalues. The eigenvalues of the  $\hat{L}^2$  operator are  $l(l+1)\hbar^2$ , where  $l \geq 0$ , and  $l$  can be either an integer  $l = 0, 1, 2, 3, \dots$ , or  $l$  is half integer,  $l = \frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \dots$ . Each of the eigenvalues of  $\hat{L}^2$  has a degeneracy of  $2l + 1$ . The functions in a group of eigenfunctions corresponding to  $l$  differ by the eigenvalues of the  $\hat{L}_z$ -operator, which can take on the values  $m\hbar$ , where  $m = -l, -l+1, -l+2, \dots, l-1, l$ . There are  $2l + 1$  functions in the so-called multiplet. Let me repeat, we deduced this using only the commutation relations and definitions in the first section of this chapter. If an particular eigenfunction is known, the partners in the multiplet can be found by acting with the ladder operators  $\hat{L}_+$  and  $\hat{L}_-$ .

### 27.3.1 Meaning of it All

So we know  $l$  can be an integer or half-integer. Nicely,  $l$  being an integer describes eigenfunctions in terms of angular coordinates  $g_{l,m}(\theta, \varphi)$ . We have encountered these functions before as spherical harmonics, where we called them  $Y_m^l(\theta, \phi)$ . These functions describe the

rotational eigenfunctions of diatomic molecules. We will see they also describe the angular part of atomic orbitals!

If  $l$  is a half-integer we cannot create functions in angular coordinates. However, these half-integer quantum numbers turn out to correspond to spin. We are familiar with electron spin which can be up or down. It corresponds to  $g_{\frac{1}{2}, \frac{1}{2}}$  and  $g_{\frac{1}{2}, -\frac{1}{2}}$  angular momentum. Just two eigenfunctions! We arrived at this for free from our angular momentum analysis. As an example of a spin function corresponding to  $l = \frac{3}{2}$  consider the nuclear spin of a  $^{15}\text{N}$  nucleus. We call it an  $I = 3/2$  nucleus with 4 possible spin directions,  $-\frac{3}{2}, -\frac{1}{2}, \frac{1}{2}, \frac{3}{2}$

It is no coincidence that the variables  $l$  and  $m$  have been chosen, with previous classes introducing them as a quantum numbers that describe the orbitals of atoms. In essence, we have derived these quantum numbers. Let's work this out in table format.

**Table 27.1**

| $l$           | $\hat{L}^2 (\hbar^2)$<br>$\frac{a}{\hbar^2} = l(l+1)$ | $\hat{L}_z (\hbar)$<br>$\frac{b}{\hbar} = m$                                      | Degeneracy of $\hat{L}_z$ | Orbital | Spin State |
|---------------|-------------------------------------------------------|-----------------------------------------------------------------------------------|---------------------------|---------|------------|
| 0             | 0                                                     | 0                                                                                 | 1                         | $s$     | singlet    |
| 1             | 2                                                     | $-1, 0, 1$                                                                        | 3                         | $p$     | triplet    |
| 2             | 6                                                     | $-2, -1, 0, 1, 2$                                                                 | 5                         | $d$     | quintet    |
| 3             | 12                                                    | $-3, -2, -1, 0, 1, 2, 3$                                                          | 7                         | $f$     | septet     |
| $\frac{1}{2}$ | $\frac{3}{4}$                                         | $-\frac{1}{2}, \frac{1}{2}$                                                       | 2                         | -       | doublet    |
| $\frac{3}{2}$ | $\frac{15}{4}$                                        | $-\frac{3}{2}, -\frac{1}{2}, \frac{1}{2}, \frac{3}{2}$                            | 4                         | -       | quartet    |
| $\frac{5}{2}$ | $\frac{35}{4}$                                        | $-\frac{5}{2}, -\frac{3}{2}, -\frac{1}{2}, \frac{1}{2}, \frac{3}{2}, \frac{5}{2}$ | 6                         | -       | sextet     |

## 27.4 Final Remarks on Formal Angular Momentum Theory

We have derived the structure of angular momentum solutions, using only commutation relations. While we have used some other preliminary starting points to make our math



easier, *everything can be fundamentally derived from these commutation relations:*

$$\begin{aligned}[\hat{L}_x, \hat{L}_y] &= i\hbar \hat{L}_z \\[\hat{L}_y, \hat{L}_z] &= i\hbar \hat{L}_x \\[\hat{L}_z, \hat{L}_x] &= i\hbar \hat{L}_y \\ \hat{L}^2 &= \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2\end{aligned}$$

These operators are taken as the fundamental definition of angular momentum. The notion of spin is hard to understand from any physical model. However, one can define spin as some internal angular momentum with operators  $\hat{S}_x, \hat{S}_y, \hat{S}_z, \hat{S}^2 = \hat{S}_x^2 + \hat{S}_y^2 + \hat{S}_z^2$  that satisfy the usual commutation relations of angular momentum. That suffices to derive the structure of the eigenstates, which we interpret as electron or nuclear spin.

## 27.5 Angular Momentum in Spherical Form

Using an expanded definition of angular momentum, based on what we already know and not just the absolute fundamentals, we can write our operators in spherical coordinates.

### Converting to Spherical



To do this we would start by taking the standard orthonormal Cartesian basis and converting it to spherical.

We would then take derivatives of this converted basis, allowing us to convert our Cartesian differentials to spherical differentials.

We would convert the gradient and Laplacian to spherical.

We would then use the definition of angular momentum as the cross product of position and angular momentum.

All this allows to finally get expressions for  $\hat{L}^2$ , directionals  $\hat{L}_x, \hat{L}_y, \hat{L}_z$ , and ladder operators  $\hat{L}_+, \hat{L}_-$ . This is all a lot of work and extensive algebra so steps aren't shown<sup>a</sup>.

<sup>a</sup>See <https://www.ifi.unicamp.br/~maplima/fi001/2012/aula20b.pdf> for this process

We would get:

$$\begin{aligned}\hat{L}^2 &= -\hbar^2 \left( \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left( \sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \varphi^2} \right) \\ \hat{L}_z &= -i\hbar \frac{\partial}{\partial \varphi} \\ \hat{L}_+ &= \hbar e^{i\varphi} \left( \frac{\partial}{\partial \theta} + i \frac{\cos \theta}{\sin \theta} \frac{\partial}{\partial \varphi} \right) \\ \hat{L}_- &= \hbar e^{-i\varphi} \left( \frac{\partial}{\partial \theta} - i \frac{\cos \theta}{\sin \theta} \frac{\partial}{\partial \varphi} \right)\end{aligned}$$

We can also get equations for  $\hat{L}_x$ ,  $\hat{L}_y$  and are shown for informational purposes, but they are not needed in this course.

$$\begin{aligned}\hat{L}_x &= i\hbar \left( \sin \varphi \frac{\partial}{\partial \varphi} + \frac{\cos \theta}{\sin \theta} \cos \varphi \frac{\partial}{\partial \varphi} \right) \\ \hat{L}_y &= i\hbar \left( -\cos \varphi \frac{\partial}{\partial \theta} + \frac{\cos \theta}{\sin \theta} \sin \varphi \frac{\partial}{\partial \varphi} \right)\end{aligned}$$

### 27.5.1 Eigenfunctions of $\hat{L}_z$ , with respect to $\varphi$

We know the angular eigenfunctions have the form  $g_{l,m}(\theta, \varphi)$ , which has been relabeled as  $Y_l^m(\theta, \varphi)$ . This is the standard form in spherical coordinates, where these eigenfunctions are called the **spherical harmonics**. Just because we have called it something else, all the properties previously derived hold, and we can use them to derive further relations.

The spherical harmonics can be written as the product of two independent functions. Let us assume this here.

$$Y_l^m(\theta, \varphi) = f(\theta)g(\varphi)$$

We can easily find the eigenfunctions of  $\hat{L}_z$  by using our spherical form acting upon the angular solution. Conveniently, the spherical form of  $\hat{L}_z$  is only with respect to  $\varphi$ , so we need only use  $g(\varphi)$ .

$$\begin{aligned}\hat{L}_z g(\varphi) &= -i\hbar \frac{\partial}{\partial \varphi} g(\varphi) \\ &= (b)g(\varphi) \\ &= (m\hbar)g(\varphi)\end{aligned}$$

Using some mathematical intuition (and previous knowledge), we can reverse-engineer the

form for  $g(\varphi)$ . We know it to be

$$g(\varphi) = e^{im\varphi}$$

This is easy to check

$$\begin{aligned} -i\hbar \frac{\partial}{\partial \varphi} e^{im\varphi} &= -i\hbar(im)e^{im\varphi} \\ &= (m\hbar)e^{im\varphi} \end{aligned}$$

Where have we encountered this before? Particle on a ring, of course. Without rederiving it, we know that there is a periodic boundary condition where

$$\begin{aligned} g(\varphi) &= g(\varphi + 2\pi) \\ e^{im\varphi} &= e^{im(\varphi+2\pi)} \end{aligned}$$

This implies that  $e^{2im\pi} = \cos(2m\pi) + i\sin(2m\pi) = 1$ , and it follows that  $m$  should be an integer. This is true for all our functions where  $l$  is an integer. We noted before that these functions can be viewed as functions of  $\theta, \varphi$  on a sphere.

### 27.5.2 Difficulty to Represent Half-Integer Spin in Angular Coordinates

A key result of this periodic boundary condition is that  $m$  must be an integer. This is the reason that half-integer values of  $m$  (spin functions) can't be expressed in spherical coordinates of  $\theta$  and  $\varphi$ . If we were to try, we would get

$$\begin{aligned} -i\hbar \frac{\partial}{\partial \varphi} e^{i\frac{1}{2}\varphi} &= -i\hbar \left(i\frac{1}{2}\right) e^{i\frac{1}{2}\varphi} \\ &= \frac{1}{2}\hbar e^{i\frac{1}{2}\varphi} \end{aligned}$$

We have an eigenvalue  $\frac{1}{2}\hbar$ , which is acceptable, but if we now consider the periodic boundary condition, we would get

$$e^{i\frac{1}{2}(\varphi+2\pi)} = e^{i\pi} e^{i\frac{1}{2}\varphi} = -e^{i\frac{1}{2}\varphi}$$

If we rotate over a full circle this function would change sign. That is not allowed, and we never think of spin functions in some angular space. We think of spin as some abstract quantity, where we can still have eigenfunctions and eigenvalues, but these are all we have.

In summary: spin can't be represented by the particle on a ring model.

### 27.5.3 Discussion of $\theta$ Form of the Spherical Harmonics

We have already determined the  $\varphi$ -dependent part to be  $g(\varphi) = e^{im\varphi}$ . We move on to the  $\theta$ -dependent part.

While the function of  $\theta$  has been called the generic  $f(\theta)$ , we can write it to be something else.

$$\begin{aligned} Y_l^m(\varphi, \theta) &= f(\theta)g(\varphi) \\ &= P_l^m(\theta)e^{im\varphi} \end{aligned}$$

where  $P_l^m(\theta)$  are called the **associated Legendre polynomials**. We might be wondering, how do we get these? We use the ladder operators and their previously derived properties. The property of interest is the vanishing property, where the up ladder acting upon the maximum will produce 0 and the down ladder acting upon the minimum will also produce 0.

The function with the maximum eigenvalue of the spherical harmonics is denoted  $Y_l^l$ , so using our spherical ladder functions we get

$$\begin{aligned} \hat{L}_+ Y_l^l(\theta, \varphi) &= \hat{L}_+ P_l^l(\theta) e^{il\varphi} \\ &= \hbar e^{i\varphi} \left( \frac{\partial}{\partial \theta} + i \frac{\cos \theta}{\sin \theta} \frac{\partial}{\partial \varphi} \right) P_l^l(\theta) e^{il\varphi} = 0 \end{aligned}$$

From this we can deduce that (unnormalized, ignoring any constants) highest  $m$ -valued function of the  $Y_l^m(\theta, \varphi)$  multiplet is of the form

$$P_l^l(\theta) = \sin^l \theta$$

This can be verified.

$$\begin{aligned}
\hbar e^{i\varphi} \left( \frac{\partial}{\partial \theta} + i \frac{\cos \theta}{\sin \theta} \frac{\partial}{\partial \varphi} \right) \sin^l \theta e^{il\varphi} &= \hbar e^{i\varphi} \left( \frac{\partial}{\partial \theta} \sin^l \theta + i \frac{\cos \theta}{\sin \theta} \frac{\partial}{\partial \varphi} \sin^l \theta \right) e^{il\varphi} \\
&= \hbar e^{i\varphi} \left( \frac{\partial}{\partial \theta} \sin^l \theta e^{il\varphi} + i \frac{\cos \theta}{\sin \theta} \frac{\partial}{\partial \varphi} \sin^l \theta e^{il\varphi} \right) \\
&= \hbar e^{i\varphi} \left( l \sin^{l-1} \theta \cos \theta e^{il\varphi} + i \frac{\cos \theta}{\sin \theta} \sin^l \theta e^{il\varphi} (il) \right) \\
&= \hbar e^{i\varphi} (l \sin^{l-1} \theta \cos \theta e^{il\varphi} + i \cos \theta \sin^{l-1} e^{il\varphi} (il)) \\
&= \hbar e^{i\varphi} (l \sin^{l-1} \theta \cos \theta e^{il\varphi} - l \cos \theta \sin^{l-1} e^{il\varphi}) \\
&= \hbar e^{i\varphi} (0) \\
&= 0
\end{aligned}$$

We can now generate the other functions in the multiplet. A nice trick is that we can save ourselves some time and only generate down to  $m = 0$ . This is because

$$Y_l^m(\theta, \varphi) = P_l^{|m|}(\theta) e^{im\varphi}$$

This means that the part of the function with respect to  $\theta$  remains the same, no matter if we have  $+m$  or  $-m$ . This comes from our spherical form of  $\hat{L}^2$

$$\hat{L}^2 P_l^m(\theta, \varphi) e^{im\varphi} = -\hbar^2 \left( \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left( \sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \varphi^2} \right) P_l^m(\theta, \varphi) e^{im\varphi}$$

If we were to derive (not shown) the differential equation based by having  $\hat{L}^2$  act upon on  $P_l^m(\theta, \varphi) e^{im\varphi}$  with either  $-m$  or  $+m$ , we would get the same equation (ignoring any normalization or constants)

$$\left( \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left( \sin \theta \frac{\partial}{\partial \theta} \right) - \frac{m^2}{\sin^2 \theta} \right) P_l^{|m|}(\theta) = l(l+1) P_l^{|m|}(\theta)$$

The only instance of  $m$  in the equation are squared, so the sign doesn't matter at all.

### 27.5.3.1 Ladder Operator Construction of Spherical Harmonics

We use three key pieces of information.

- The highest  $m$ -valued function of the multiplet  $Y_l^m(\theta, \varphi)$  is of the form  $\sin^l \theta e^{il\varphi}$ .
- We can use our ladder operators in their spherical forms to calculate up to  $m = 0$ .

- We know that signed equations of  $m$  take the form  $P_l^{|m|} e^{im\varphi}$ .

Using this we can derive the equations of different orbitals such as  $p$ ,  $d$ ,  $f$ , etc.<sup>1</sup> To do so, we ignore any normalization or constants to make it easier, but the normalized forms can easily be found from a reference<sup>2</sup>.

We find the  $p$ -orbitals, as an example. We know the highest valued multiplet function is

$$Y_l^m(\theta, \varphi) = e^{i\varphi} \sin \theta$$

We then act upon it with our ladder operator  $\hat{L}_-$ .

$$\begin{aligned} \hat{L}_- Y_1^1 &= Y_1^0 \\ \hbar e^{i\varphi} \left( \frac{\partial}{\partial \theta} + i \frac{\cos \theta}{\sin \theta} \frac{\partial}{\partial \varphi} \right) \sin \theta e^{i\varphi} &= \hbar e^{-i\varphi} \left( \frac{\partial}{\partial \theta} \sin \theta e^{i\varphi} - i \frac{\cos \theta}{\sin \theta} \frac{\partial}{\partial \varphi} \sin \theta e^{i\varphi} \right) \\ &= \hbar e^{-i\varphi} \left( \cos \theta e^{i\varphi} - i \frac{\cos \theta}{\sin \theta} \sin \theta e^{i\varphi} (i) \right) \\ &= \hbar e^{-i\varphi} (\cos \theta e^{i\varphi} + \cos \theta e^{i\varphi}) \\ &= 2\hbar \cos \theta \\ &\sim \cos \theta \end{aligned}$$

We know the equation for  $m = -1$  to be  $e^{-i\varphi} \sin \theta$ . We now have the complete set.

### 27.5.3.2 Real Cartesian Orbitals

The spherical harmonics are complex functions in general, and they do not resemble the usual cartesian atomic orbitals,  $p_x$ ,  $d_{x^2-y^2}$ , etc. This form is related to the fact that we require the functions to be eigenfunction of  $\hat{L}_z$ , which gives the complex  $e^{im\varphi}$  form. We can take linear combinations of the spherical harmonics with same value of  $l$  and this will not change the eigenvalue of  $\hat{L}^2$ . In many applications that is all we care about, and so let us consider suitable real linear combinations. There is an obvious choice since  $Y_l^m(\theta, \varphi) = P_l^{|m|}(\theta) e^{im\varphi}$ , where the functions  $P_l^{|m|}(\theta)$  are real. We can define

$$\begin{aligned} Y_l^{m,\cos} &= \frac{1}{2}(Y_l^m + Y_l^{-m}) = P_l^{|m|}(\theta) \frac{1}{2}(e^{im\varphi} + e^{-im\varphi}) = P_l^{|m|}(\theta) \cos(m\varphi) \\ Y_l^{m,\sin} &= \frac{1}{2i}(Y_l^m - Y_l^{-m}) = P_l^{|m|}(\theta) \frac{1}{2i}(e^{im\varphi} - e^{-im\varphi}) = P_l^{|m|}(\theta) \sin(m\varphi) \end{aligned}$$

<sup>1</sup>In case you don't remember,  $s$  orbitals are just a sphere... we don't need to derive them.

<sup>2</sup>[https://en.wikipedia.org/wiki/Table\\_of\\_spherical\\_harmonics](https://en.wikipedia.org/wiki/Table_of_spherical_harmonics)

It is also the case that these functions have a nice representation in terms of Cartesian coordinates. The simplest example is the case of  $p$ -functions, where  $P_1^1(\theta) = \sin(\theta)$  and  $P_1^0(\theta) = \cos(\theta)$ . We will also use the known definition of spherical coordinates

$$x = r \sin \theta \cos \varphi$$

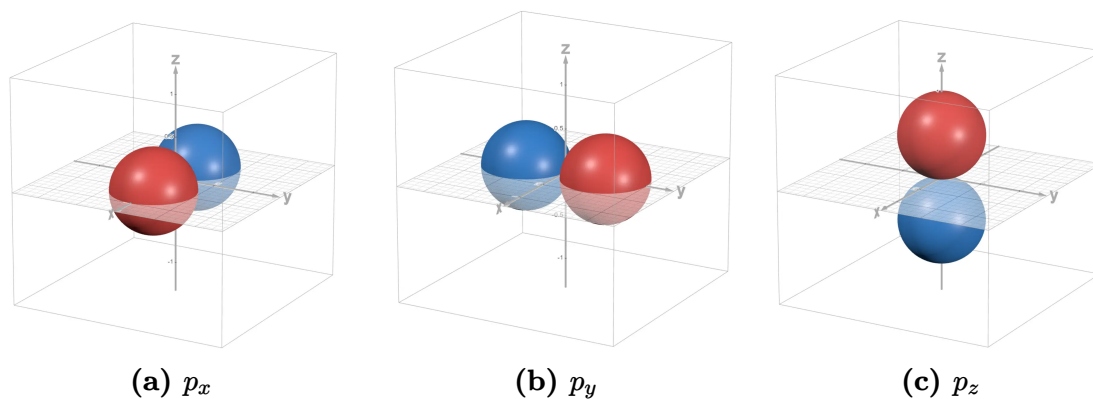
$$y = r \sin \theta \sin \varphi$$

$$z = r \cos \theta$$

We see we have the orbitals

$$\begin{aligned}\sin \theta \cos \varphi &= \frac{x}{r} \\ \sin \theta \sin \varphi &= \frac{y}{r} \\ \cos \theta &= \frac{z}{r}\end{aligned}$$

We have the standard forms of the  $p$ -orbitals. We can actually plot these and see our beautiful orbitals.



**Figure 27.1**

For the  $d$ -functions we have  $P_2^2(\theta) = \sin^2(\theta)$ ,  $P_2^1(\theta) = \cos(\theta) \sin(\theta)$ ,  $P_2^0(\theta) = (3 \cos^2(\theta) - 1)$ . It is a nice exercise in trigonometry to turn the angular functions into their Cartesian counterparts given the familiar  $d$ -functions:  $d_{xy}$ ,  $d_{xz}$ ,  $d_{yz}$ ,  $d_{x^2-y^2}$ ,  $d_{z^2}$ . This is also discussed in the extra lecture notes for the H-atom.

## 28 Radial Equation

### 28.1 Solving Radial Equations

For a one electron equation, we can solve it on paper with some (complicated) math. In the math world we call this an *analytic solution*.

For a two electron or more equation, the only practical way to solve it is using computers to calculate a numerical solution. This numerical solution can be absolutely excellent, but it is still numerical.

### 28.2 Introduction

We have already discussed the angular solutions to the Schrödinger equation. We know the angular function is called a spherical harmonic, and composed of a  $\theta$  and  $\varphi$  dependent parts. Therefore, we write the full radial and angular function as a product where

$$\psi(r, \theta, \varphi) = f(r)Y_l^m(\theta, \varphi)$$

The radial Hamiltonian has the form

$$\hat{H} = -\frac{\hbar^2}{2\mu} \frac{1}{r} \frac{d}{dr} - \frac{\hbar^2}{2\mu} \frac{d^2}{dr^2} + \frac{\hbar^2 l(l+1)}{2\mu r^2} + V(r)$$

where the potential energy term is

$$V(r) = -\frac{Ze^2}{4\pi\epsilon_0 r} \quad Z = 1 \text{ (hydrogen)}$$

We could then apply this Hamiltonian to the function  $f(r)$  to obtain a radial energy, as in

$$\hat{H}f(r) = \left( -\frac{\hbar^2}{2\mu} \frac{1}{r} \frac{d}{dr} - \frac{\hbar^2}{2\mu} \frac{d^2}{dr^2} + \frac{\hbar^2 l(l+1)}{2\mu r^2} + V(r) \right) f(r) = Ef(r)$$



### 28.2.1 Cleaning Up

While the previous form is possible to work with, it is a little cluttered. We can bring out a factor of  $\frac{\hbar^2}{2\mu}$ , which will allow us to clean up things nicely.

$$\begin{aligned}\frac{\hbar^2}{2\mu} \left( -\frac{2}{r} \frac{d}{dr} - \frac{d^2}{dr^2} + \frac{l(l+1)}{r^2} - \frac{2\mu}{\hbar^2} \left( \frac{e^2}{4\pi\epsilon_0 r} \right) \right) f(r) &= E f(r) \\ \left( -\frac{2}{r} \frac{d}{dr} - \frac{d^2}{dr^2} + \frac{l(l+1)}{r^2} - \frac{2}{a_0 r} \right) f(r) &= \frac{2\mu}{\hbar^2} E f(r) \\ \left( -\frac{2}{r} \frac{d}{dr} - \frac{d^2}{dr^2} + \frac{l(l+1)}{r^2} - \frac{2}{a_0 r} \right) f(r) &= \varepsilon f(r)\end{aligned}$$

In making this change, we made a number of key substitutions. The first is the incorporation of only a single physical constant, the Bohr radius  $a_0$ . The Bohr radius is a useful measure of distance at this scale, being the average distance between the nucleus and electron of a ground state hydrogen.

$$\frac{\mu e^2}{\hbar^2 4\pi\epsilon_0} = \frac{1}{a_0} \implies a_0 = \frac{\hbar^2 4\pi\epsilon_0}{\mu e^2}$$

Let me note that the Bohr radius  $a_0$  is a constant, obtained by setting  $\mu = m_e$ . For hydrogen like atoms  $\mu$  is close to  $m_e$ . We get simple and correct results, with this slight abuse of notation. The second is a new scaled energy,  $\varepsilon$ , where

$$\varepsilon = \frac{2\mu}{\hbar^2} E$$

After making these changes, we can easily apply  $f(r)$ .

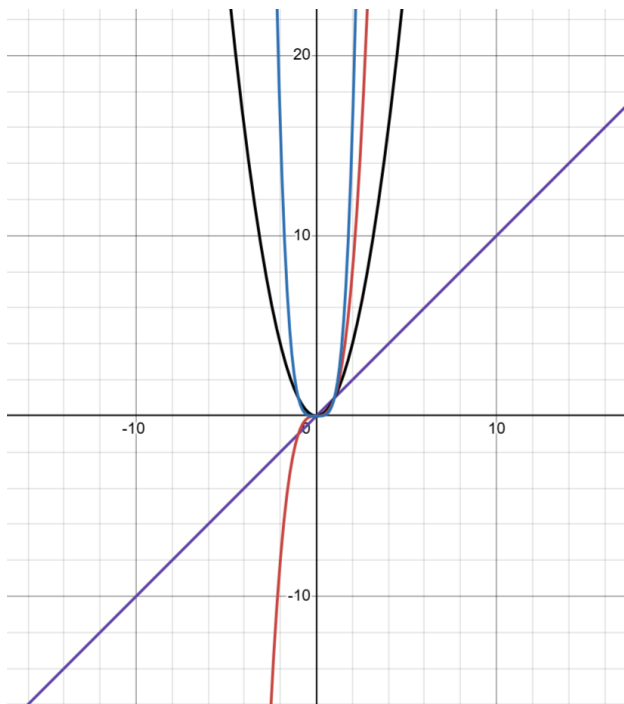
$$\begin{aligned}\left( -\frac{2}{r} \frac{d}{dr} - \frac{d^2}{dr^2} + \frac{l(l+1)}{r^2} - \frac{2}{a_0 r} \right) f(r) &= \varepsilon f(r) \\ -\frac{2}{r} \frac{df}{dr} - \frac{d^2 f}{dr^2} + \frac{l(l+1)}{r^2} f(r) - \frac{2}{a_0 r} f(r) &= \varepsilon f(r)\end{aligned}$$

### 28.2.2 Trial Solution

Without getting into the weeds, we have a differential equation here with two key properties. First, it is not homogeneous (equal to 0 on the right hand side), and instead on the right hand side we have some function. Second, we don't have constant terms. The only way to easily solve this problem is to use a trial solution. With some foresight we choose

$$f(r) = p(r)e^{-\alpha r}$$

where  $\alpha$  is some positive constant and  $p(r)$  is some polynomial function of the form  $p(r) = a + br + cr^2 + \dots$ .



**Figure 28.1:** A variety of polynomial functions.

If we take the derivative of  $f(r)$  we have to make use of the chain rule, getting

$$\begin{aligned}\frac{df}{dr} &= \frac{dp}{dr}e^{-\alpha r} - \alpha p(r)e^{-\alpha r} \\ &= \left(\frac{dp}{dr} - \alpha p(r)\right)e^{-\alpha r} \\ \frac{d^2f}{dr^2} &= \frac{d^2p}{dr^2}e^{-\alpha r} - \alpha e^{-\alpha r}\frac{dp}{dr} - \alpha e^{-\alpha r}\frac{dp}{dr} + \alpha^2 p(r)e^{-\alpha r} \\ &= \left(\frac{d^2p}{dr^2} - 2\alpha\frac{dp}{dr} + \alpha^2 p(r)\right)e^{-\alpha r}\end{aligned}$$

### 28.2.3 Substitute Trial solution into radial equation

We now can use this trial solution in our radial Schrödinger equation.

$$\begin{aligned} & -\frac{2}{r}\left(\frac{dp}{dr}-\alpha p(r)\right)e^{-\alpha r}-\left(\frac{d^2p}{dr^2}-2\alpha\frac{dp}{dr}+\alpha^2p(r)\right)e^{-\alpha r} \\ & +\left(\frac{l(l+1)}{r^2}p(r)\right)e^{-\alpha r}-\left(\frac{2}{a_0r}p(r)\right)e^{-\alpha r} \\ & =(\varepsilon p(r))e^{-\alpha r} \\ \\ \Rightarrow & -\frac{2}{r}\left(\frac{dp}{dr}-\alpha p(r)\right)-\left(\frac{d^2p}{dr^2}-2\alpha\frac{dp}{dr}+\alpha^2p(r)\right) \\ & +\frac{l(l+1)}{r^2}p(r)-\frac{2}{a_0r}p(r)=\varepsilon p(r) \\ \\ & -\frac{2}{r}\frac{dp}{dr}+\frac{2\alpha}{r}p(r)-\frac{d^2p}{dr^2}+2\alpha\frac{dp}{dr}-\alpha^2p(r)+\frac{l(l+1)}{r^2}p(r)-\frac{2}{a_0r}p(r) \\ & =\varepsilon p(r) \end{aligned}$$

### 28.2.4 Solutions

Before discussing the general solution, let us focus on some limited solutions. We can select by choosing a value for  $l$  and a value for  $p(r)$ .

#### 28.2.4.1 $s$ -Orbitals, Simplest Example

By definition,

$$l = 0$$

By choice,

$$\begin{aligned} p(r) &= 1 \\ \Rightarrow \frac{dp}{dr} &= 0 \\ \Rightarrow \frac{d^2p}{dr^2} &= 0 \end{aligned}$$

Then we would get

$$\frac{2\alpha}{r}-\alpha^2-\frac{2}{a_0r}=\varepsilon$$

As is common in differential equations, we match up powers of our variables to solve for generic constants, in this case  $\alpha$ .

We can rewrite to visually see this idea, though this step is not required

$$\left(2\alpha - \frac{2}{a_0}\right)r^{-1} + (-\alpha^2)r^0 = (0)r^{-1} + (\varepsilon)r^0$$

We can create two equations from this

$$\begin{aligned} 2\alpha - \frac{2}{a_0} &= 0 \\ 2\alpha &= \frac{2}{a_0} \\ \alpha &= \frac{1}{a_0} \end{aligned}$$

$$\begin{aligned} -\alpha^2 &= \varepsilon \\ \varepsilon &= -\frac{1}{a_0^2} \end{aligned}$$

Bringing back  $\varepsilon = \frac{2\mu}{\hbar^2}E \implies E = \frac{\hbar^2}{2\mu}\varepsilon$ , we can solve for energy

$$E = -\frac{\hbar^2}{2\mu a_0^2}$$

and get the complete function

$$f(r) = p(r)e^{-\alpha r} = e^{-\frac{1}{a_0}r}$$

This is the solution for the lowest  $s$ -orbital. It is the ground state solution.

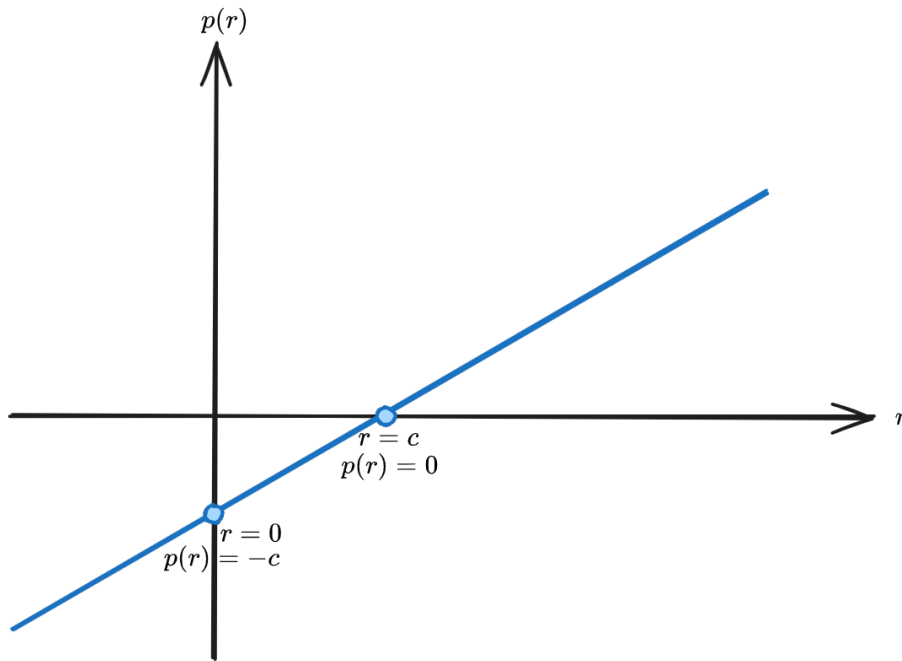
#### 28.2.4.2 $s$ -Orbitals, Another Example

By definition

$$l = 0$$

By choice

$$\begin{aligned}
 p(r) &= r - c \\
 \Rightarrow \frac{dp}{dr} &= 1 \\
 \Rightarrow \frac{d^2p}{dr^2} &= 0
 \end{aligned}$$



**Figure 28.2**

We get

$$\begin{aligned}
 -\frac{2}{r} + \frac{2\alpha}{r}(r - c) + 2\alpha - \alpha^2(r - c) - \frac{2}{a_0 r}(r - c) &= \varepsilon(r - c) \\
 -\frac{2}{r} + 2\alpha - \frac{2\alpha c}{r} + 2\alpha - \alpha^2 r + \alpha^2 c - \frac{2}{a_0} + \frac{2c}{a_0 r} &= \varepsilon r - \varepsilon c \\
 \left(-2 - 2\alpha c + \frac{2c}{a_0}\right)r^{-1} + (-\alpha^2)r + \left(4\alpha + \alpha^2 c - \frac{2}{a_0}\right) &= (\varepsilon)r + (-\varepsilon c)
 \end{aligned}$$

We get the equations and solve the equations starting with the highest power in  $r$ , and

working in sequence down to lower powers of  $r$ , using the solutions we found

$$\begin{aligned}
 -\alpha^2 &= \varepsilon \\
 \varepsilon &= -\frac{1}{4a_0^2} \\
 4\alpha + \alpha^2 c - \frac{2}{a_0} &= -\varepsilon c \rightarrow \text{substitute } \alpha \\
 4\alpha - \varepsilon c - \frac{2}{a_0} &= -\varepsilon c \\
 4\alpha - \frac{2}{a_0} &= 0 \\
 4\alpha &= \frac{2}{a_0} \\
 \alpha &= \frac{1}{2a_0} \\
 -2 - 2\alpha c + \frac{2c}{a_0} &= 0 \\
 -2 - 2\left(\frac{1}{2a_0}\right)c + \frac{2c}{a_0} &= 0 \\
 -2 - \frac{c}{a_0} + \frac{2c}{a_0} &= 0 \\
 \frac{-c + 2c}{a_0} &= 2 \\
 c &= 2a_0
 \end{aligned}$$

We get energy and function

$$\begin{aligned}
 E &= \frac{\hbar^2}{2\mu} \left( -\frac{1}{4a_0^2} \right) = -\frac{1}{4} \frac{\hbar^2}{2\mu a_0^2} \\
 f(r) &= (r - c)e^{-\alpha r} = (r - 2a_0)e^{-\frac{1}{2a_0}r}
 \end{aligned}$$

This can be done for further examples, such as  $p$ -orbitals (use  $l = 1$ ), but the process remains the same.

### 28.2.4.3 General Solution

#### 28.2.4.3.1 Highest Power

Let's designate the highest power in the polynomial  $p(r)$  to be  $m$ , so that we have

$$p(r) = c_0 + c_1r + c_2r^2 + \cdots + c_mr^m$$

We want to analyse the terms that involve the highest powers of  $r$ , which we derive from assuming

$$p(r) = r^m$$

Our equation becomes

$$\begin{aligned} -\frac{2}{r}mr^{m-1} + \frac{2\alpha}{r}r^m - m(m-1)r^{m-2} + 2\alpha mr^{m-1} + \frac{l(l+1)}{r^2}r^m - \frac{2}{a_0r}r^m - \alpha^2r^m &= \varepsilon r \\ -2mr^{m-2} + 2\alpha r^{m-1} - m(m-1)r^{m-2} + 2\alpha mr^{m-1} + l(l+1)r^{m-2} - \frac{2}{a_0}r^{m-1} - \alpha^2r^m &= \varepsilon r \end{aligned}$$

We get a system of equations, one for each term looking order-by-order.

For  $r^m$ :

$$-\alpha^2 = \varepsilon \quad (\text{always!})$$

For  $r^{m-1}$ :

$$\begin{aligned} 2\alpha + 2\alpha m - \frac{2}{a_0} &= 0 \\ 2\alpha(1+m) &= \frac{2}{a_0} \\ \alpha &= \frac{1}{a_0(1+m)} \end{aligned}$$

From these two consistent terms, energy is

$$\begin{aligned} \varepsilon &= -\frac{1}{a_0^2(1+m)^2} \\ E &= \frac{\hbar^2}{2\mu} \left( -\frac{1}{a_0^2(1+m)^2} \right) \end{aligned}$$

We know the function is

$$f(r) = r^m e^{-\frac{1}{a_0(1+m)}r}$$

This is all independent of  $l$ .

#### 28.2.4.3.2 $n$ , Principal Quantum Number

For convenience, we designate  $n = m + 1$ . This is the same  $n$  we recognize as the principal quantum number. We can write the above equations as

$$\begin{aligned}\alpha &= \frac{1}{a_0 n} \\ E_n &= -\frac{\hbar^2}{2\mu a_0^2} \frac{1}{n^2} \\ f_n(r) &= r^{n-1} e^{-\frac{1}{a_0 n} r} + \dots\end{aligned}$$

#### 28.2.4.3.3 Lowest Power

We designate the smallest power of  $r$  in the polynomial  $p(r)$  to be  $s$ , as in

$$p(r) = c_s r^s + c_{s+1} r^{s+1} + \dots + c_{m-1} r^{m-1} + c_m r^m$$

From the same equation we can obtain terms for  $s$ . We use the same equation as previously, but with respect to  $s$ , getting

$$\begin{aligned}-\frac{2}{r} m r^{s-1} + \frac{2\alpha}{r} r^s - s(s-1) r^{s-2} + 2\alpha s r^{s-1} + \frac{l(l+1)}{r^2} r^s - \frac{2}{a_0 r} r^s - \alpha^2 r^s &= \epsilon r \\ -2s r^{s-2} + 2\alpha r^{s-1} - s(s-1) r^{s-2} + 2\alpha s r^{s-1} + l(l+1) r^{s-2} - \frac{2}{a_0} r^{s-1} - \alpha^2 r^s &= \epsilon r\end{aligned}$$

We now focus on the lowest power of  $r$ ,  $r^{s-2}$ . We see that the terms are *dependent on  $l$*  unlike the highest power.

$$\begin{aligned}-2s - s(s-1) + l(l+1) &= 0 \\ -2s - s^2 + s + l^2 + l &= 0 \\ -s^2 - s + l^2 + l &= 0 \\ s^2 + s &= l^2 + l \\ \implies s &= l\end{aligned}$$

#### 28.2.4.4 Interpretation

If  $s = l$ , then depending on our chosen  $l$  value (which dictates the angular function), we will have a minimum power  $s$ . Of course our minimum is equal to or smaller than our maximum,



mathematically written as

$$s \leq m$$

$$l \leq n - 1$$

This means our smallest power determines what radial nodes are possible. The neat thing about this, is that we know these rules already.

For example: Suppose we have a  $d$ -orbital, which means  $l = 2$ . We know that the first  $d$ -orbital occurs when  $n = 3$  to form a  $3d$ . We can then have  $4d$ ,  $5d$ , etc. But let's prove this mathematically. We place this value of  $l$  in the inequality,

$$2 \leq n - 1$$

$$\implies n \geq 3$$

saying that  $n$  must be 3 or greater, restating exactly what we knew.

Similarly, the same rules that we already know, such as

$$\# \text{ nodes} = n - l - 1$$

still apply. This is because there are always  $l$  angular nodes.

Interestingly, the radial nodes can be solved for. They are non-zero roots of the polynomial  $p(r)$ .

## 28.2.5 Table of Solutions

**Table 28.1**

| $l$ | $s$ | $n$ | $m$ | Polynomial<br>$p(r)$ | Orbital<br>Name | # Angular<br>Nodes | # Radial<br>Nodes | Total<br>Nodes | Node<br>Degeneracy |
|-----|-----|-----|-----|----------------------|-----------------|--------------------|-------------------|----------------|--------------------|
| 0   | 0   | 1   | 0   | $a$                  | $1s$            | 0                  | 0                 | 0              | 1                  |
| 0   | 0   | 2   | 1   | $a + br$             | $2s$            | 0                  | 1                 | 1              | 2                  |
| 1   | 1   | 2   | 1   | $br$                 | $2p$            | 1                  | 0                 | 1              |                    |
| 0   | 0   | 3   | 2   | $a + br + cr^2$      | $3s$            | 0                  | 2                 | 2              | 3                  |
| 1   | 1   | 3   | 2   | $br + cr^2$          | $3p$            | 1                  | 1                 | 2              |                    |
| 2   | 2   | 3   | 2   | $cr^2$               | $3d$            | 2                  | 0                 | 2              |                    |

We see that the number of node degeneracy corresponds to the energy degeneracy (as discussed below).

These solutions correspond to the angular solutions which correspond to the Bohr energies observed experimentally in the foundation of quantum mechanics, and discussed at the start of the course.

## 28.3 Summary of Angular and Radial Solutions

We have a spherical wavefunction with angular and radial components

$$\psi(r, \theta, \varphi) = p_{n,l}(r) e^{-\frac{r}{na_0}} Y_{l,m}(\theta, \varphi)$$

where  $p_{n,l}(r)$  is a polynomial with form

$$p_{n,l}(r) = ar^l + br^{l+1} + \dots + vr^{n-2} + wr^{n-1}$$

with energy only dependent on the radial component and the principle quantum number  $n$ <sup>1</sup> where  $n$  can be

$$n = 1, 2, 3, \dots$$

The energy is

$$E(n) = -\frac{\hbar^2}{2\mu a_0^2} \frac{1}{n^2}$$

The angular component, while not affecting the energy, affects the shape of the orbitals with quantum numbers  $m$  and  $l$

$$l = 0, 1, 2, \dots$$

$$m = -l, -l+1, \dots, l-1, l$$

### 28.3.1 Energy Degeneracy

At the  $n = 1$  level we have a degeneracy of 1, as we only have the 1s (1 orbital). We see at the  $n = 2$  level, we have a degeneracy of 4 as we have 2s (1 orbital) and 2p (3 orbitals). At the  $n = 3$  level we have a degeneracy of 9 as we have 3s, (1) 3p (3), and 3d (5). Therefore,

---

<sup>1</sup>This is only true for one electron atoms but not true for multi-electron atoms in general. This has the consequence of making 2s, 2p degenerate, 3s, 3p, 3d degenerate, etc. *only for hydrogen* (or hydrogen-like atoms).

our total degeneracy based on  $n$  is

$$\text{degeneracy} = n^2$$

Mathematically this can be justified by remembering the previous angular degeneracy,  $2l+1$ . We sum this over all our possible values, starting from  $l = 0$  and going to some chosen  $n$ . Our sum is then<sup>2</sup>

$$\begin{aligned}\sum_{l=0}^{n-1} (2l+1) &= \sum_{l=0}^{n-1} (2l) + \sum_{l=0}^{n-1} (1) \\ &= 2 \sum_{l=0}^{n-1} (l) + n \\ &= 2 \left( \frac{1}{2} n(n-1) \right) + n \\ &= n^2 - n + n \\ &= n^2\end{aligned}$$

---

<sup>2</sup>Using common sum identities. If you don't know these off the top of your head, don't worry, but just absorb that we can justify these using simple counting, as before, and algebra of spin degeneracy and get the same result.

## 29 Variational Principle

### 29.1 Exact Solutions

There are very few exact solutions, so few that we can list them out

- 1 electron problems
- Angular momentum problems
- (Imperfect) models, such as
  - Harmonic oscillator
  - Particle in a box/ring

The question becomes: How do we calculate beyond these exact solutions? We use specific methods of approximation, such as Hartree-Fock, density functional theory, coupled cluster method, or configuration interaction. We can use software like Gaussian or Orca. The point is that we have many methods and theories to compute from.

We can compute energies, vibrations, thermochemical properties, and spectroscopy quite well. In terms of phases the easiest is gas phase, with solids more expensive, and liquids more approximate.

But how we do know that our approximations are valid? A key way is the **variational principle**, which allows us to continually optimize our results and is a foundation of quantum calculations.

### 29.2 Variational Principle

We have a normalized wavefunction, call it  $\psi$ . We can calculate the inner product by integrating, as we have done countless times before.

For the sake of convenience, we say we are integrating over  $V$ , meaning all space, with the variable chosen to be  $\tau$  and  $d\tau$ . We do this, because we could be integrating in Cartesian, spherical, or some other coordinate system. It doesn't matter which one, as the variational principle holds no matter our choice.

For a normalized function

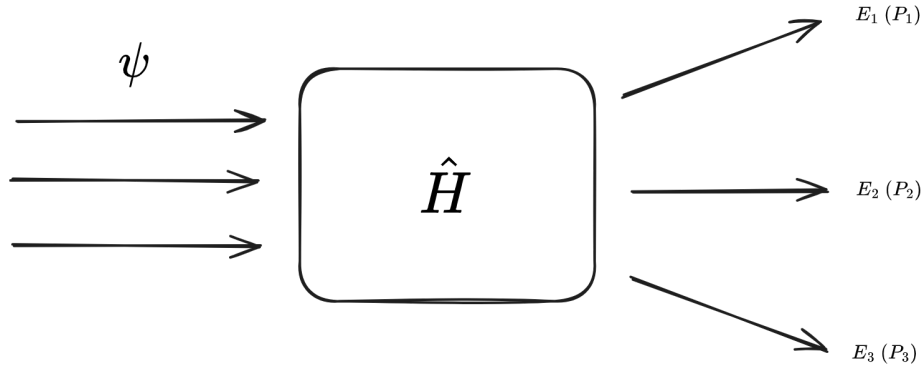
$$\int_V \psi^*(\tau)\psi(\tau) d\tau \equiv 1$$

We can calculate the expectation value of the operator acting upon the wavefunction. Importantly, when comparing this calculated value to the *exact ground state energy* it is always higher (or equal) to this lowest possible energy eigenvalue  $E_0$  (or  $E_{gs}$ ).

$$\langle \psi | \hat{H} | \psi \rangle = \int_V \psi^*(\tau)(\hat{H}\psi(\tau)) d\tau \geq E_0$$

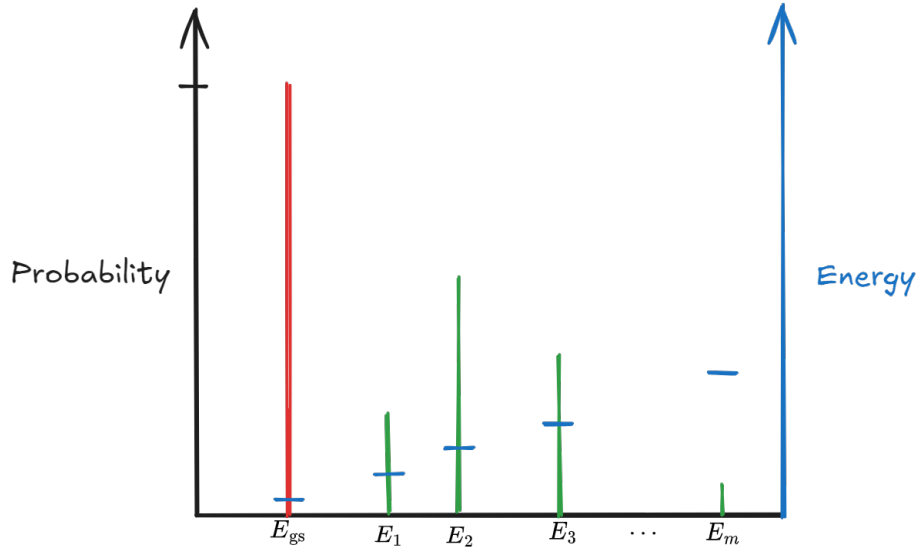
This is called the variational principle.

We can first get an intuitive sense of how to understand the variational principle. If we would measure the energy of an ensemble described by the arbitrary normalized wave function  $\Psi$ , one would always find one of the possible eigenvalues of  $\hat{H}$ ,  $E_m$ , with some probability  $p_m$ . The average energy would always be greater than or equal the lowest possible energy  $E_0$ . We know this average is the same as the expectation value, and this is the essence of the variational principle. We present a pictorial presentation below:



**Figure 29.1:** We know that each eigenstate energy has an associated probability

If we were to write out all these probabilities in sketch, we would get something like the following figure.



**Figure 29.2**

While the individual probabilities might be small, the individual energies will always be *greater than the true ground state energy*. So the weighted average energy will be either

- greater than the ground state energy (generally the case)
- equal to the ground state energy (when the only eigenfunction is equivalent to the ground state wavefunction and the only energy to appear is the ground state, which is a special case)

Let us also carry out the full formal proof. We know the Hamiltonian has orthonormal eigenfunctions

$$\hat{H}\phi_k(\tau) = E_k\phi_k(\tau)$$

$$\int_V \phi_m^*(\tau)\psi_k(\tau) d\tau = \delta_{mk}$$

and that any normalized wave function  $\Psi(\tau)$  can be expressed as a linear combination of the eigenstates

$$\psi(\tau) = \sum_k c_k \phi_k(\tau)$$

$$\sum_k c_k^* c_k = 1$$

Let us use this linear combination in the expectation value. In order to get correct

proofs you should always use different summation labels for different sums in an equation.

$$\begin{aligned}
\int_V \psi^*(\tau)(\hat{H}\psi(\tau)) \, d\tau &= \int_V \sum_m (c_m \phi_m(\tau))^* \left( \hat{H} \sum_k c_k \phi_k(\tau) \right) \, d\tau \quad \text{use linearity } \hat{H} \\
&= \int_V \sum_k c_m^* \phi_m^*(\tau) \left( E_k \sum_k c_k \phi_k(\tau) \right) \, d\tau \\
&= \sum_m c_m^* \sum_k c_k E_k \int \phi_m^*(\tau) \phi_k(\tau) \, d\tau \quad \text{use orthonormality } \phi_k \\
&= \sum_m c_m^* \sum_k c_k E_k \delta_{mk} \quad \text{use property of kronecker delta} \\
&= \sum_m c_m^* c_m E_m \\
&= \sum_m |c_m^*|^2 E_m \\
&= \sum_m P_m E_m, P_m \geq 0
\end{aligned}$$

Comparing  $E_0$  to  $E_m$ , we know that

$$E_m \geq E_0$$

Therefore

$$\sum_m P_m E_m \geq \sum_m P_m E_0 = E_0 \sum_m P_m = E_0$$

Hence, we have proven the variational principle

$$\int_V \psi^*(\tau)(\hat{H}\psi(\tau)) \, d\tau \geq E_0$$

This is a very useful concept, because we know that any guessed wave function (with some parameters) has an energy expectation value greater than the ground state, and as we optimize its parameters, the energy will decrease. The best value to approximate the ground state energy (and ground state) will be found by minimizing the energy expectation value.

### 29.2.1 Unnormalized Function

If we don't have a normalized function, we must find the normalization constant for our wavefunction by

$$\begin{aligned}\int_V \left( \frac{1}{N} \psi(\tau) \right)^* \frac{1}{N} \psi(\tau) d\tau &= \frac{1}{N^2} \int \psi^*(\tau) \psi(\tau) d\tau \equiv 1 \\ &= \int \psi^*(\tau) \psi(\tau) d\tau = N^2\end{aligned}$$

We can normalize the variational inequality by

$$\begin{aligned}\tilde{E}_0 &= \langle \psi | \hat{H} | \psi \rangle \geq E_0 \\ &= \frac{1}{N^2} \int_V \psi^*(\tau) \hat{H} \psi(\tau) d\tau \\ \tilde{E}_0 &= \frac{\int_V \psi^*(\tau) \hat{H} \psi(\tau) d\tau}{\int_V \psi^*(\tau) \psi(\tau) d\tau} \geq E_0\end{aligned}$$

This final statement is the classical full statement of the variational principle. It holds for any (square-integrable) wave function  $\psi(\tau)$ .

There are two important cases of variational optimization in the context of this class.

1. The trial wave function is expressed in terms of some parameters (for example, exponents in exponential functions). To minimize this we can take the derivative and minimize with respect to some parameter, often incorporated as part of the normalization constant. Let's call this parameter  $\alpha$ .

$$\frac{\partial}{\partial \alpha} \left( \frac{\langle \psi | \hat{H} | \psi \rangle}{\langle \psi | \psi \rangle} \right) = \frac{\partial \tilde{E}_0}{\partial \alpha} = 0$$

We can then calculate the energy with respect to these optimized parameters

$$E_{\psi_{\text{opt}}} = \frac{\langle \psi_{\text{opt}} | \hat{H} | \psi_{\text{opt}} \rangle}{\langle \psi_{\text{opt}} | \psi_{\text{opt}} \rangle}$$

We can also optimize with respect to multiple parameters, such as having an  $\alpha$  and  $\beta$ , which is a similar process.

2. The wavefunctions are expressed as a linear combination of known, fixed functions (so-called basis functions) denoted  $f_k(\tau)$

$$\psi(\tau) = \sum_k c_k f_k(\tau)$$



where the coefficients  $c_k$  are optimized to minimize the energy. For example, we could set  $f_k(\tau)$  to be a set of atomic orbitals. This application uses the so-called linear variational principle. It leads to a matrix eigenvalue problem and this can be very effectively solved by computers. The linear variational principle is widely used in quantum chemistry programs.

## 29.2.2 Examples

### 29.2.2.1 Unworked: Harmonic Oscillator

We know the harmonic oscillator Hamiltonian has the form

$$\hat{H} = \hbar\omega \left( -\frac{1}{2} \frac{d^2}{dx^2} + \frac{1}{2} x^2 \right)$$

We take a Gaussian trial function

$$\psi_{\text{trial}}(x) = e^{-\alpha x^2}$$

with one parameter  $\alpha$  that we seek to optimize.

As previously derived when discussing the harmonic oscillator, the exact ground state energy is

$$E_0 = \frac{1}{2} \hbar\omega$$

If we were to optimize  $\alpha$  using the variational principle, we would obtain  $\alpha = \frac{1}{2}$ . Previously we have shown that the exact ground state for the harmonic oscillator is  $e^{-\frac{1}{2}x^2}$ . Minimizing the energy for this trial wave function would give the exact result. The energy we would obtain would be

$$\tilde{E}_0 = \frac{1}{2} \hbar\omega$$

This means that our function has been perfectly optimized, obtaining the exact ground state energy.

### Gaussian Functions



A Gaussian function is a type of mathematical function, with the most basic form of

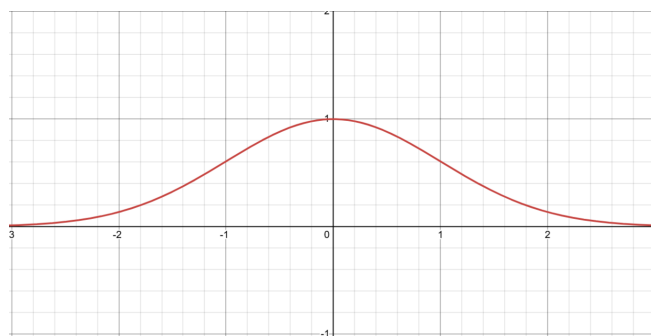
$$e^{-x^2}$$

A common parameterized version of this is

$$ae^{-\frac{(x-b)^2}{c^2}}$$

It is very hard to integrate on paper, often impossible without the use of special techniques but are very easy and stable to integrate for computers. They show up suddenly in math and science in many places.

One of the great innovations of the Gaussian program was the use of Gaussian orbitals to speed up calculations, though many pieces of software now use this technique.



**Figure 29.3**

### 29.2.2.2 Partially Worked: 1s Orbital

The radial Hamiltonian is

$$\hat{H} = -\frac{2}{r} \frac{d}{dr} - \frac{d^2}{dr^2} + \frac{l(l+1)}{r^2} - \frac{2}{a_0 r}$$

We know for a 1s orbital that  $l = 0$  so the Hamiltonian is now

$$\hat{H} = -\frac{2}{r} \frac{d}{dr} - \frac{d^2}{dr^2} - \frac{2}{a_0 r}$$

We use a trial function

$$\begin{aligned}\psi_{\text{trial}} &= e^{-\alpha r} \\ \frac{d\psi}{dr} &= -\alpha e^{-\alpha r} \\ \frac{d^2\psi}{dr^2} &= \alpha^2 e^{-\alpha r}\end{aligned}$$

We apply the Hamiltonian to our wavefunction. This can be done in the integral itself, but it's easier to just do it now

$$\begin{aligned}\hat{H}\psi &= \left( -\frac{2}{r} \frac{d}{dr} - \frac{d^2}{dr^2} - \frac{2}{a_0 r} \right) e^{-\alpha r} \\ &= \frac{2}{r} \alpha e^{-\alpha r} - \alpha^2 e^{-\alpha r} - \frac{2}{a_0 r} e^{-\alpha r} \\ &= \left( \frac{2}{r} \alpha - \alpha^2 - \frac{2}{a_0 r} \right) e^{-\alpha r}\end{aligned}$$

To apply the variational principle we should apply it to the original problem of the hydrogen atom in spherical coordinates.

The integral in spherical coordinates requires the inclusion of the Jacobian factor  $r^2 \sin \theta$ . We can integrate over the angles  $\phi, \theta$  as none of the functions here depend on angles.

$$\int_0^\infty r^2 dr \underbrace{\int_0^\pi \sin \theta d\theta \int_0^{2\pi} d\phi}_{4\pi} = 4\pi \int_0^\infty r^2 dr$$

The quantity to calculate simplifies to

$$\begin{aligned}\frac{\langle \psi | \hat{H} | \psi \rangle}{\langle \psi | \psi \rangle} &= \frac{4\pi \int_0^\infty r^2 \psi^*(r) (\hat{H}\psi(r)) dr}{4\pi \int_0^\infty r^2 \psi^*(r) \psi(r) dr} \\ &= \frac{\int_0^\infty r^2 (e^{-\alpha r})^* (\hat{H}(e^{-\alpha r})) dr}{\int_0^\infty r^2 (e^{-\alpha r})^* e^{-\alpha r} dr} \\ \tilde{E}_0(\alpha) &= \frac{\int_0^\infty r^2 (e^{-2\alpha r}) \left( -\frac{2}{r} \frac{d}{dr} - \frac{d^2}{dr^2} - \frac{2}{a_0 r} \right) dr}{\int_0^\infty r^2 e^{-2\alpha r} dr}\end{aligned}$$

This integral is complicated. If we were to compute the energy equation and then take the derivative and set it to 0

$$\frac{\partial \tilde{E}_0}{\partial \alpha} = 0$$

then solve for a value of  $\alpha$ , we would get

$$\alpha_{\text{opt}} = \frac{1}{a_0} \implies \alpha = \frac{\mu e^2}{4\pi \epsilon_0 \hbar^2}$$

We would then substitute this value into the energy equation, getting

$$\tilde{E}_0 = -\frac{1}{2} \frac{e^2}{4\pi \epsilon_0 a_0}$$

which is equal to the true ground state of

$$E_0 = -\frac{1}{2} \frac{e^2}{4\pi\epsilon_0 a_0}$$

We know this is the result we should get, as the exact ground state wave function has the form  $e^{-\alpha r}$ . This is an alternative way to obtain the exact solution. Let us now consider a more interesting example.

### 29.2.2.3 Unworked: 1s Orbital (Non-Trivial)

Suppose we repeated the above with (real) trial wavefunction

$$\psi_{\text{trial}}(r) = e^{-\alpha r^2}$$

We would get the variational equation

$$\begin{aligned} \tilde{E}_0(\alpha) &= \frac{\langle \psi | \hat{H} | \psi \rangle}{\langle \psi | \psi \rangle} \\ &= \frac{4\pi \int_0^\infty r^2 \psi^*(r) (\hat{H} \psi(r)) \, dr}{4\pi \int_0^\infty r^2 \psi^*(r) \psi(r) \, dr} \\ &= \frac{\int_0^\infty r^2 (e^{-\alpha r^2})^* (\hat{H}(e^{-\alpha r^2})) \, dr}{\int_0^\infty r^2 (e^{-\alpha r^2})^* e^{-\alpha r^2} \, dr} \\ &= \frac{\int_0^\infty r^2 e^{-\alpha r^2} (\hat{H}(e^{-\alpha r^2})) \, dr}{\int_0^\infty r^2 e^{-\alpha r^2} e^{-\alpha r^2} \, dr} \\ &\dots \\ &= \frac{3\hbar^2 \alpha}{2\mu} - \frac{2e\sqrt{\alpha}}{\epsilon_0 (2\pi)^{3/2}} \end{aligned}$$

We can take the derivative and set it to 0

$$\begin{aligned} \frac{\partial \tilde{E}_0}{\partial \alpha} &= \frac{3\hbar^2}{2\mu} - \frac{2e}{\epsilon_0 (2\pi)^{3/2} \sqrt{\alpha}} = 0 \\ \sqrt{\alpha} &= \frac{2\mu e^2}{3(2\pi)^{3/2} \epsilon_0 \hbar^2} \\ \alpha_{\text{opt}} &= \frac{\mu^2 e^4}{18\pi^3 \epsilon_0^2 \hbar^4} \end{aligned}$$

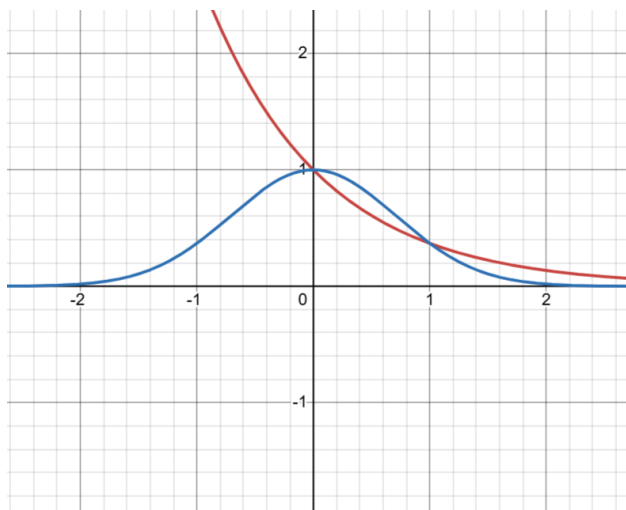
We then substitute this in to the energy equation

$$\begin{aligned}\tilde{E}_0(\alpha_{\text{opt}}) &= -\frac{4}{3\pi} \left( \frac{\mu e^4}{16\pi^2 \epsilon_0 \hbar^2} \right) \\ &= -0.424 \frac{e^2}{4\pi \epsilon_0 a_0} \\ E_0 &= -0.5 \frac{e^2}{4\pi \epsilon_0 a_0}\end{aligned}$$

We see that

$$\tilde{E}_0 > E_0$$

This is because the trial wavefunction is not exact for any value of  $\alpha$ .



**Figure 29.4:**  $e^{-x}$  compared to  $e^{-x^2}$ .

Gaussian functions are widely used in quantum chemistry. The reason is that it is often easy to calculate integrals over gaussian functions. We can expand orbitals as linear combinations of Gaussian functions with typically fixed exponents. The linear optimization is easier to do. This is the origin of the name Gaussian for one of the widely used quantum chemistry programs.

## 30 Variational Principle: More Concrete Examples

There are two common variants of problems regarding the variational principle found on the exam:

1. Make there be a parameter to optimize, such as the harmonic oscillator using differential equations
2.  $2 \times 2$  linear problems, using matrices. We will discuss this later.

### 30.1 Variational Principle

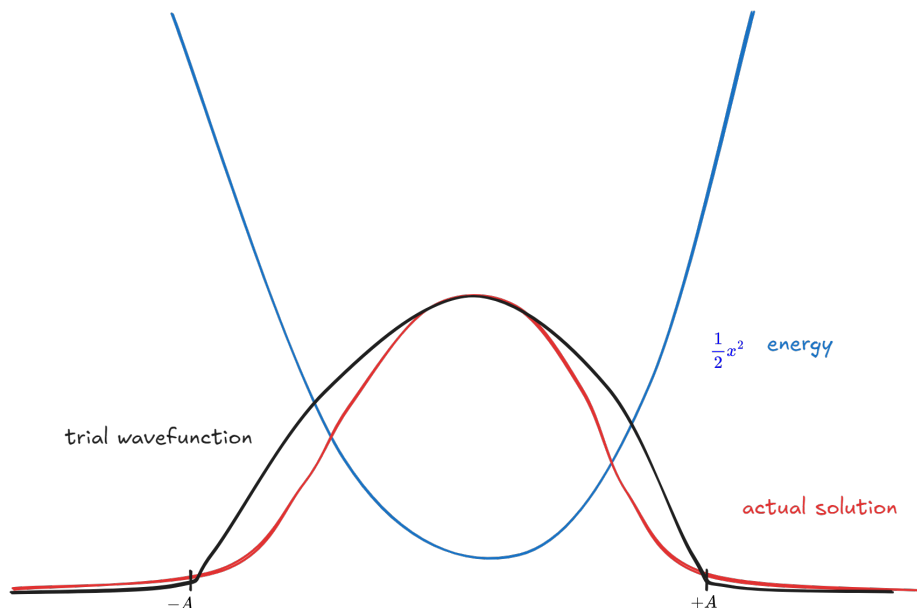
#### 30.1.1 Parameter

We have the harmonic oscillator Hamiltonian

$$\hat{H} = \hbar\omega \left( -\frac{1}{2} \frac{d}{dx^2} + \frac{1}{2} x^2 \right)$$

with (real) trial function and bounds

$$\begin{aligned} \psi(x) &= (A - x)(A + x) = A^2 - x^2, -A < x < A \\ \psi(x) &= 0, \text{ elsewhere} \end{aligned}$$



**Figure 30.1**

The goal of the variational principle is adjusting the trial wavefunction by minimizing energy. We do this by modifying our parameter, in this case  $A$ .

We have energy

$$\tilde{E} = \frac{\int_{-A}^A \psi(x) \hat{H} \psi(x) dx}{\int_{-A}^A \psi(x) \psi(x)} = \frac{N}{D} \geq E_{\text{gs}}$$

The denominator is

$$D = \int_{-A}^A (A^2 - x^2)(A^2 - x^2) dx = \int_{-A}^A (A^4 - 2A^2x^2 + x^4) dx$$

We might be given some information or identities to simplify integration, such as

$$\int_{-A}^A x^m = \left[ \frac{1}{m+1} x^{m+1} \right]_{-A}^A$$

where if  $m$  is even, the integral is equal to

$$\frac{1}{m+1} A^{m+1} - (-A)^{m+1} = \frac{2}{m+1} A^{m+1}$$

and if  $m$  is odd, the integral is equal to 0.

We integrate the denominator now

$$A^4(2A^1) - 2A^2 \left( \frac{2A^3}{3} \right) + \left( \frac{2A^5}{5} \right) = A^5 \left( 2 + \frac{4}{3} + \frac{2}{5} \right) = \frac{16}{15}A^5$$

We first apply the Hamiltonian to the function

$$\begin{aligned}\psi(x) &= A^2 - x^2 \\ \frac{d\psi}{dx} &= -2x \\ \frac{d^2\psi}{dx^2} &= -2\end{aligned}$$

We get

$$\begin{aligned}N &= \int_{-A}^A \hbar\omega(A^2 - x^2)(\hat{H}(A^2 - x^2)) dx \\ &= \hbar\omega \int_{-A}^A (A^2 - x^2) \left( -\frac{1}{2} \frac{d}{dx^2} + \frac{1}{2}x^2 \right) (A^2 - x^2) dx \\ &= \hbar\omega \int_{-A}^A (A^2 - x^2) \left( -\frac{1}{2}(-2) + \frac{1}{2}x^2(A^2 - x^2) \right) dx \\ &= \hbar\omega \int_{-A}^A (A^2 - x^2) \left( 1 + \frac{1}{2}A^2x^2 - \frac{1}{2}x^4 \right) dx \\ &= \hbar\omega \int_{-A}^A \left( A^2 + \frac{1}{2}A^4x^2 - \frac{1}{2}A^2x^4 - x^2 - \frac{1}{2}A^2x^4 + \frac{1}{2}x^6 \right) dx \\ &= \hbar\omega \int_{-A}^A \left( A^2 + \frac{1}{2}A^4x^2 - A^2x^4 - x^2 + \frac{1}{2}x^6 \right) dx \\ &= \hbar\omega \left[ A^2(2A) + \frac{1}{2}A^4 \left( \frac{2A^3}{3} \right) - A^2 \left( \frac{2A^5}{5} \right) - \left( \frac{2A^3}{3} \right) + \frac{1}{2} \left( \frac{2A^7}{7} \right) \right] \\ &= \hbar\omega \left( 2A^3 + \frac{A^7}{3} - \frac{2A^7}{5} - \frac{2A^3}{3} + \frac{A^7}{7} \right) \\ &= \hbar\omega \left( \frac{4}{3}A^3 + \frac{8}{105}A^7 \right)\end{aligned}$$

We combine the numerator and denominator

$$\begin{aligned}\frac{N}{D} &= \hbar\omega \frac{15}{16A^5} \left( \frac{4}{3}A^3 + \frac{8}{105}A^7 \right) \\ &= \hbar\omega \left( \frac{5}{4}A^{-2} + \frac{1}{14}A^2 \right)\end{aligned}$$



We can now take the derivative with respect to  $A$ .

$$\begin{aligned}
 \frac{\partial E}{\partial A} &= \hbar\omega \left( \frac{5}{4}(-2A^{-3}) + \frac{1}{14}(2A) \right) = 0 \\
 &= \hbar\omega \left( -\frac{5}{2}A^{-3} + \frac{1}{7}A \right) \\
 \frac{0}{\hbar\omega} &= -\frac{5}{2}A^{-3} + \frac{1}{7}A \\
 \frac{1}{7}A &= \frac{5}{2}A^{-3} \\
 A^4 &= \frac{35}{2} \\
 A_{\text{opt}} &= \sqrt[4]{\frac{35}{2}}
 \end{aligned}$$

If we input this back into the energy equation

$$\begin{aligned}
 E &= \hbar\omega \left( \frac{5}{4} \left( \sqrt[4]{\frac{35}{2}} \right)^{-2} + \frac{1}{14} \left( \sqrt[4]{\frac{35}{2}} \right)^2 \right) \\
 &= \hbar\omega(\sim 0.598) > \hbar\omega(0.5)
 \end{aligned}$$

We see that the energy is greater than the ground state energy, as expected.

# 31 Linear Variational Principle

## 31.1 The Essence

As mentioned when introducing the variational principle, we can construct a **linear variational problem**. To do we have a fixed basis set composed of (orthonormal) basis functions  $\{f_n\}$ . Our trial wavefunction is constructed from that basis set, as in

$$\psi(\tau) = \sum_n c_n f_n(\tau)$$

As it is a variational problem, we go in with the goal to minimize energy. We do this by optimizing our parameters, the coefficients of the trial wavefunction. We can have many parameters because it is a linear problem, something computers are great at dealing with.

Practically, on paper we will deal only with  $2 \times 2$  problems with real functions and coefficients, but the idea remains the same no matter the scale of the problem.

## 31.2 Deriving the Linear Form

We restate the variational problem to derive the linear form

$$\tilde{E}(\psi) = \frac{\int_V \psi^*(\tau)(\hat{H}\psi(\tau)) d\tau}{\int_V \psi^*(\tau)\psi(\tau) d\tau} = \frac{N}{D}$$

We designate the numerator and denominator  $N$  and  $D$  to deal with them separately. Importantly, throughout the derivations we must use different indices when subbing in the summed trial function into the same integral. This is key. As an additional reminder, all coefficients and basis functions are real.

Note: The  $(\tau)$  is dropped for cleanliness, as it is generic and not important.

The numerator:

$$\begin{aligned}
N &= \int_V \psi^* (\hat{H} \psi) \, d\tau \\
&= \int_V \left( \sum_n c_n f_n \right)^* \left( \hat{H} \left( \sum_m c_m f_m \right) \right) \, d\tau \\
&= \sum_n c_n \sum_m c_m \underbrace{\int_V f_n \hat{H} f_m \, d\tau}_{H_{nm}} \\
&= \sum_{n,m} c_n c_m H_{nm}
\end{aligned}$$

The denominator:

$$\begin{aligned}
D &= \int_V \psi^* \psi \, d\tau \\
&= \int_V \left( \sum_n c_n f_n \right)^* \left( \sum_m c_m f_m \right) \, d\tau \\
&= \sum_n c_n \sum_m c_m \int_V f_n^* f_m \, d\tau \\
&= \sum_n c_n \sum_m c_m \delta_{nm} \\
&= \sum_n c_n^2
\end{aligned}$$

Therefore our variational expression is

$$\frac{\sum_{n,m} c_n H_{nm} c_m}{\sum_n c_n^2}$$

### 31.2.1 Derivative

We can take the derivative for each  $k$  (a chosen coefficient), and set it to zero to do our optimization

$$\frac{\partial}{\partial c_k} \left( \frac{N}{D} \right) = \frac{\frac{\partial N}{\partial c_k} D - N \frac{\partial D}{\partial c_k}}{D^2} = 0$$

Therefore, as only the numerator of the derivative can contribute to the above expression being 0,

$$\begin{aligned}
\frac{\partial N}{\partial c_k} D - N \frac{\partial D}{\partial c_k} &= 0 \\
\frac{\partial N}{\partial c_k} D &= N \frac{\partial D}{\partial c_k} \\
\frac{\partial N}{\partial c_k} &= \frac{N}{D} \frac{\partial D}{\partial c_k} \\
\frac{\partial N}{\partial c_k} &= \tilde{E} \frac{\partial D}{\partial c_k} \\
\frac{\partial N}{\partial c_k} - \tilde{E} \frac{\partial D}{\partial c_k} &= 0
\end{aligned}$$

We note that

$$\frac{\partial c_n}{\partial c_k} = \delta_{nk}$$

We justify this idea with the following example

$$\begin{aligned}
\frac{\partial x}{\partial x} &= 1 \\
\frac{\partial x}{\partial y} &= 0 \\
\frac{\partial y}{\partial y} &= 1 \\
\frac{\partial y}{\partial x} &= 0
\end{aligned}$$

### 31.2.1.1 Numerator

We take the derivative of the numerator

$$\begin{aligned}
\frac{\partial N}{\partial c_k} &= \frac{\partial}{\partial c_k} \sum_{n,m} c_n c_m H_{nm} \\
&= \sum_{n,m} (\delta_{nk} H_{nm} c_m + c_n H_{nm} \delta_{mk}) \\
&= \sum_m \sum_n \delta_{nk} H_{nm} c_m + \sum_n \sum_m c_n H_{nm} \delta_{mk} \\
&= \sum_m H_{km} c_m + \sum_n c_n H_{nk}
\end{aligned}$$

### 31.2.1.2 Denominator

We take the derivative of the denominator

$$\begin{aligned}
\frac{\partial D}{\partial c_k} &= \frac{\partial}{\partial c_k} \sum_n c_n^2 \\
&= \sum_n 2c_n \delta_{nk} \\
&= 2c_k \\
&= c_k + c_k
\end{aligned}$$

### 31.2.1.3 Combination

We can now combine these two derivatives by  $\frac{\partial N}{\partial c_k} - \tilde{E} \frac{\partial D}{\partial c_k} = 0$

$$\sum_m (H_{km} c_m - \tilde{E} c_k) + \sum_n (c_n H_{nk} - \tilde{E} c_k)$$

Since  $H_{kn} = H_{nk}$  and the indices used in the equations are arbitrary, this is simply twice of the same expression. Therefore, we get

$$2 \sum_m (H_{km} c_m - \tilde{E} c_k) = 0 \implies \sum_m (H_{km} c_m - \tilde{E} c_k) = 0$$

This can be represented as a matrix eigenvalue equation.

## 31.2.2 Matrix Form

This is a matrix eigenvalue equation of

$$\sum_m H_{km} c_m = \tilde{E} c_k$$

with a square matrix of  $B \times B$  size.

$$\underbrace{\begin{bmatrix} H_{11} & H_{12} & \dots & H_{1B} \\ H_{21} & H_{22} & & \\ \vdots & & \ddots & \\ H_{B1} & & & H_{BB} \end{bmatrix}}_{\mathbf{H}} \underbrace{\begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_B \end{bmatrix}}_{\mathbf{c}} = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_B \end{bmatrix} \tilde{E}$$

We end up getting a matrix eigenvalue equation that looks like the Schrödinger equation

$$\mathbf{H}\mathbf{c} = \tilde{E}\mathbf{c}$$

where the bold denotes a matrix/vector. This matrix Hamiltonian is classified as a **complex Hermitian matrix**. Such matrices have real eigenvalues.

To optimize this matrix we can let  $\mathbf{c}$  be dependent on some parameter  $\lambda$ , so the equation can be denoted

$$\mathbf{H}\mathbf{c}_\lambda = \mathbf{c}_\lambda \tilde{E}_\lambda$$

We can rearrange this equation to solve for the characteristic polynomial

$$\begin{aligned}\mathbf{H}\mathbf{c} - \tilde{E}\mathbf{c} &= 0 \\ (\mathbf{H} - \mathbb{1}\tilde{E})\mathbf{c} &= 0\end{aligned}$$

where  $\mathbb{1}$  is the identity matrix<sup>1</sup>, often denoted  $I$  as well. This form makes it clear that  $\tilde{E}$  are the eigenvalues. The only nontrivial solution for this is when the matrix is not invertible, which is equivalent to the determinant of the matrix being 0. Therefore

$$\det(\mathbf{H} - \mathbb{1}\tilde{E})\mathbf{c} = 0$$

For  $2 \times 2$  matrices we can solve such problems by hand, for example on an exam! In practice computers are used to calculate the matrix elements of the Hamiltonian matrix and to do the subsequent diagonalization. The biggest calculations in quantum chemistry might use dimensions of  $10^9$  (for full configuration interaction calculations).

### 31.2.2.1 Deep Connection to Early Quantum Mechanics

We have done quantum mechanics with differential equations, using the Schrödinger equation. Using the linear variational principle we have arrived at a matrix eigenvalue problem as an approximate method using a finite set of basis functions. When he was 23 years old, Heisenberg invented an earlier version of quantum mechanics, that he called matrix mechanics. It is the same as this approach, except as an exact approach it used infinite basis sets, and matrices of infinite dimension. This was rather mindboggling at the time. There was a big controversy between wave mechanics (Schrödinger) and matrix mechanics (Heisenberg).

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<sup>1</sup>The identity matrix is equivalent to the 1 of regular numbers, called either **unity** or **identity**. We just have it here to assure that our energy is properly in matrix form, otherwise we'd end up with a matrix minus a scalar, which doesn't make sense.

Dirac reconciled the two approaches and showed them to be two viewpoints that were actually doing the same problem. For us today it is easiest to think of Heisenberg as a basis expansion, akin to the linear variational principle. Both viewpoints and the difference in representation (Dirac) are used extensively in the field of quantum theory.

### 31.2.2.2 $2 \times 2$ Problem: Discussion of General Case

Assume we have Hamiltonian

$$\hat{H} = \hat{H}_0 + \hat{V}$$

Let us consider as basis functions known orthonormal eigenstates of hamiltonian  $\hat{H}_0$

$$\hat{H}_0\phi_1 = E_1\phi_1$$

$$\hat{H}_0\phi_2 = E_2\phi_2$$

We can form the matrix of  $\hat{H}_0$  directly, without further calculations

$$\mathbf{H} = \mathbf{H}_0 = \begin{bmatrix} E_1 & 0 \\ 0 & E_2 \end{bmatrix}$$

We can also calculate the matrix elements of the matrix  $\mathbf{V}$

$$\mathbf{H} = \mathbf{V} = \begin{bmatrix} V_{11} & V_{12} \\ V_{21} & V_{22} \end{bmatrix}$$

For this we generally need to calculate the integrals

$$V_{ij} = \int_V \phi_i^*(\tau) \hat{V} \phi_j(\tau) d\tau$$

Assume we have calculated all matrix-elements (annoying integrals) and we have the full Hamiltonian matrix

$$\mathbf{H} = \mathbf{H}_0 + \mathbf{V} = \begin{bmatrix} E_1 + V_{11} & V_{12} \\ V_{21} & E_2 + V_{22} \end{bmatrix}$$

We can generalize this further, taking advantage of Hermiticity for

$$\begin{aligned} V_{12} &= V_{21}^* \equiv v \\ V_{12} &= V_{21} \quad (\text{real}) \end{aligned}$$

which allows us to form the matrix  $A$

$$A = \begin{bmatrix} a & v \\ v & b \end{bmatrix}$$

This is a general form of a symmetric 2x2 matrix. We would solve the characteristic polynomial as

$$\begin{aligned} \det(A - \mathbb{1}E) &= \begin{vmatrix} a - E & v \\ v & b - E \end{vmatrix} = (a - E)(b - E) - v^2 \\ &= E^2 - Ea - Eb + ab - v^2 \\ &= E^2 - E(a + b) + (ab - v^2) \end{aligned}$$

We use the quadratic formula  $\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$  to solve for two possible values of  $E$

$$\begin{aligned} E &= \frac{(a + b) \pm \sqrt{(-(a + b))^2 - 4(ab - v^2)}}{2} \\ &= \frac{(a + b) \pm \sqrt{a^2 + 2ab + b^2 - 4ab + 4v^2}}{2} \\ &= \frac{(a + b) \pm \sqrt{a^2 - 2ab + b^2 + 4v^2}}{2} \end{aligned}$$

Further simplifications occur assuming that  $a = b$ , we can solve to get

$$\begin{aligned} E &= \frac{(a + a) \pm \sqrt{a^2 - 2aa + a^2 + 4v^2}}{2} \\ &= \frac{2a \pm \sqrt{4v^2}}{2} \\ &= \frac{2a \pm 2v}{2} \\ E &= a + v, a - v \end{aligned}$$



To obtain the first eigenvector, we substitute the first value back into the matrix

$$\begin{aligned} A - \mathbb{1}E &= \begin{bmatrix} a - (a + v) & v \\ v & a - (a + v) \end{bmatrix} \\ &= \begin{bmatrix} -v & v \\ v & -v \end{bmatrix} \end{aligned}$$

We use some generic vector, call it  $\mathbf{w}$  and multiply it into one of the rows. It doesn't matter which one we choose, as both are linearly dependent.

$$\begin{aligned} (A - \mathbb{1}E)\mathbf{w} &= \begin{bmatrix} -v & v \\ v & -v \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = 0 \\ \implies -vw_1 + vw_2 &= 0 \\ vw_2 &= vw_1 \\ w_2 &= w_1 \end{aligned}$$

We calculate the first eigenvector

$$\mathbf{w} = \begin{bmatrix} w_1 \\ w_1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} w_1 \sim \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

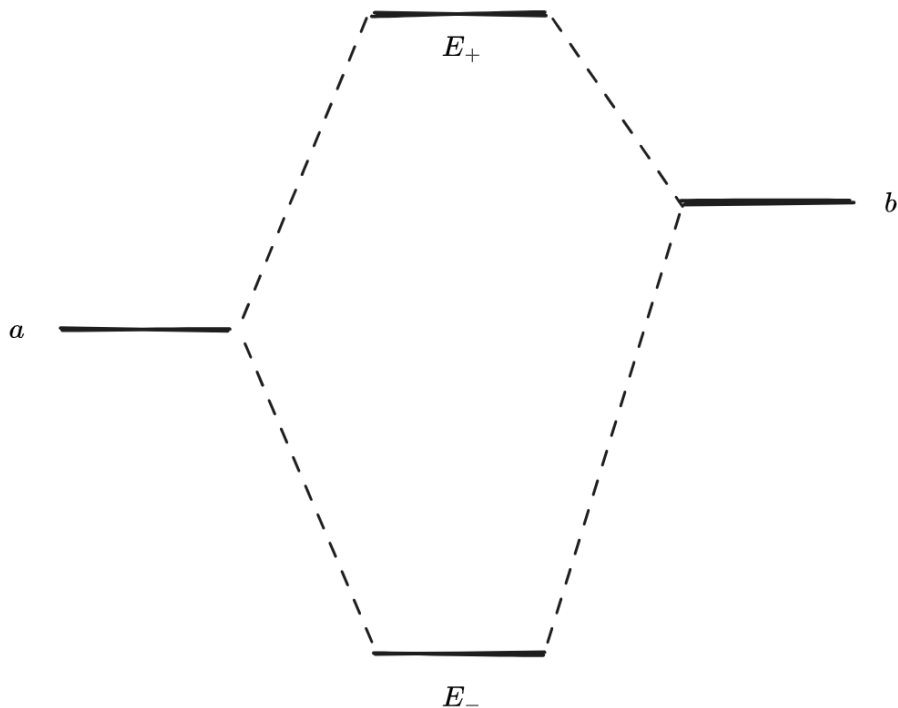
The first eigenvector is then

$$\tilde{\mathbf{E}}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Similarly, the other eigenvector is

$$\tilde{\mathbf{E}}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Knowing  $a = b$ , we can draw this out. We have something called **diagonally degenerate**, meaning that things are the same along the diagonal. This is not generally true. We have two other cases, one where  $a < b$  and the other  $b > a$ . The eigenvectors can also be obtained, but this is less instructive. Leave it to computers to calculate! Here is a graphical representation of how energy levels of  $\hat{H}_0$  get perturbed by  $\hat{V}$ . One level always goes up the other goes down. As a quick sanity check, it is always true (for  $2 \times 2$  problems) that  $E_+ + E_- = a + b$ , which can be useful.



**Figure 31.1:** The  $a$  and  $b$  orbitals are unequal in energy, in the general case.

### 31.2.3 Complicated Specific Example: Particle on a Ring with a Complex Perturbation

Let us assume the Hamiltonian is given as

$$\hat{H} = \hat{H}_0 + \hat{V}$$

where  $\hat{H}_0$  is some known Hamiltonian. It can be from the harmonic oscillator, particle in a box, particle on a ring, or any other known example.

As basis functions we take two known eigenfunctions

$$\hat{H}_0\phi_1 = E_1\phi_1$$

$$\hat{H}_0\phi_2 = E_2\phi_2$$

where  $\phi_1$  and  $\phi_2$  are orthonormal.

We can form the matrix

$$\begin{array}{ccc} \phi_1 & \phi_2 & \hat{H} \\ \begin{bmatrix} E_1 & 0 \\ 0 & E_1 \end{bmatrix} & & \begin{bmatrix} \phi_1 \\ \phi_2 \end{bmatrix} \end{array}$$

### 31.2.3.1 Calculation of Matrix Elements

#### Matrix Labeling



Elements of a matrix are labeled  $a_{ij}$  where  $i$  is the row number and  $j$  is the column number. This is standard. While technically this doesn't matter as long as you are consistent, using the standard is best to avoid confusion in problems.

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

We have the Hamiltonian

$$\hat{H}_0 = -\frac{\hbar^2}{2mR^2} \frac{d^2}{d\varphi^2}$$

and perturbation

$$\hat{V} = \gamma L_z = -i\hbar\gamma \frac{\partial}{\partial\varphi}$$

where  $\gamma$  is some known, fixed parameter.

We have basis functions that are orthonormal eigenfunctions of  $\hat{H}_0$  (which we know based on our understanding of previous examples)

$$\begin{cases} \phi_1 = \frac{1}{\sqrt{\pi}} \cos \varphi \\ \phi_2 = \frac{1}{\sqrt{\pi}} \sin \varphi \end{cases}$$

#### 31.2.3.1.1 Hamiltonian

We have done this problem before and know the energy for particle on a cylinder to be

$$E_k = \frac{\hbar^2}{2mR^2} k^2$$

We choose  $k = -1, 1$ , which produces degenerate energy solutions of

$$E_1 = E_2 = \frac{\hbar^2}{2mR^2}$$

Therefore our matrix must be

$$H = \begin{bmatrix} \frac{\hbar^2}{2mR^2} & 0 \\ 0 & \frac{\hbar^2}{2mR^2} \end{bmatrix}$$

### 31.2.3.1.2 Worked Out: Hamiltonian Math

If we know the functions are orthonormal eigenfunctions of  $\hat{H}_0$ , one can (and should) just write down the Hamiltonian matrix. We can also verify explicitly<sup>2</sup>: The integral formula for the matrix elements of  $H_{ij}$  is

$$H_{ij} = \int_0^{2\pi} \phi_i(\varphi) \hat{H}_0 \phi_j(\varphi) d\varphi$$

Using this, we calculate

$$\begin{aligned} H_{11} &= \int_0^{2\pi} \phi_1(\varphi) \hat{H}_0 \phi_1(\varphi) d\varphi \\ &= \int_0^{2\pi} \left( \frac{1}{\sqrt{\pi}} \cos \varphi \right) \left( -\frac{\hbar^2}{2mR^2} \frac{d^2}{d\varphi^2} \left( \frac{1}{\sqrt{\pi}} \cos \varphi \right) \right) d\varphi \\ &= \frac{1}{\pi} \int_0^{2\pi} (\cos \varphi) \left( \frac{\hbar^2}{2mR^2} \cos \varphi \right) d\varphi \\ &= \frac{\hbar^2}{2mR^2\pi} \int_0^{2\pi} \cos^2 \varphi d\varphi \\ &= \frac{\hbar^2}{2mR^2\pi} \left[ \cos \varphi \sin \varphi + \frac{\varphi}{2} \right]_0^{2\pi} \\ &= \frac{\hbar^2}{2mR^2\pi} (\pi) \\ &= \frac{\hbar^2}{2mR^2} \end{aligned}$$

We now calculate for  $H_{12}$

$$\begin{aligned} H_{12} &= \int_0^{2\pi} \phi_1(\varphi) H_0 \phi_2(\varphi) d\varphi \\ &= \frac{1}{\pi} \int_0^{2\pi} (\cos \varphi) \left( -\frac{\hbar^2}{2mR^2} \frac{d^2}{d\varphi^2} (\sin \varphi) \right) d\varphi \\ &= \frac{\hbar^2}{2mR^2\pi} \int_0^{2\pi} \cos \varphi \sin \varphi d\varphi \\ &= \frac{\hbar^2}{2mR^2\pi} [\sin^2 \varphi]_0^{2\pi} \\ &= 0 \end{aligned}$$

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<sup>2</sup>Not needed on the exam.

We now calculate for  $H_{21}$

$$\begin{aligned}
H_{12} &= \int_0^{2\pi} \phi_2(\varphi) H_0 \phi_1(\varphi) d\varphi \\
&= \frac{1}{\pi} \int_0^{2\pi} (\sin \varphi) \left( -\frac{\hbar^2}{2mR^2} \frac{d^2}{d\varphi^2} (\cos \varphi) \right) d\varphi \\
&= \frac{\hbar^2}{2mR^2\pi} \int_0^{2\pi} \sin \varphi \cos \varphi d\varphi \\
&= 0
\end{aligned}$$

We now calculate for  $H_{22}$

$$\begin{aligned}
H_{22} &= \int_0^{2\pi} \phi_2(\varphi) H_0 \phi_2(\varphi) d\varphi \\
&= \frac{1}{\pi} \int_0^{2\pi} \sin \varphi \left( -\frac{\hbar^2}{2mR^2} \frac{d^2}{d\varphi^2} (\sin \varphi) \right) d\varphi \\
&= \frac{\hbar^2}{2mR^2\pi} \int_0^{2\pi} \sin^2 \varphi d\varphi \\
&= \frac{\hbar^2}{2mR^2\pi} \left[ \frac{-\sin(2x) - 2x}{4} \right]_0^{2\pi} \\
&= \frac{\hbar^2}{2mR^2\pi} (\pi) \\
&= \frac{\hbar^2}{2mR^2}
\end{aligned}$$

### 31.2.3.1.3 Potential Energy

The matrix elements also need to be calculated for the (remaining) part of the potential, and here one has to calculate explicitly. The integral formula for the matrix elements of  $V_{ij}$  is

$$V_{ij} = \int_0^{2\pi} \phi_i(\varphi) \hat{V} \phi_j(\varphi) d\varphi$$

We now calculate  $V_{11}$

$$\begin{aligned}
V_{11} &= \int_0^{2\pi} \phi_1(\varphi) \hat{V} \phi_1(\varphi) d\varphi \\
&= \int_0^{2\pi} \left( \frac{1}{\sqrt{\pi}} \cos \varphi \right) \left( -i\hbar\gamma \frac{d}{d\varphi} \left( \frac{1}{\sqrt{\pi}} \cos \varphi \right) \right) d\varphi \\
&= \frac{i\hbar\gamma}{\pi} \int_0^{2\pi} \cos \varphi \sin \varphi d\varphi \\
&= \frac{i\hbar\gamma}{\pi} [\sin^2 \varphi]_0^{2\pi} \\
&= 0
\end{aligned}$$

We now calculate  $V_{12}$

$$\begin{aligned}
V_{12} &= \int_0^{2\pi} \phi_1(\varphi) \hat{V} \phi_2(\varphi) d\varphi \\
&= \frac{1}{\pi} \int_0^{2\pi} \cos \varphi \left( -i\hbar\gamma \frac{d}{d\varphi} (\sin \varphi) \right) d\varphi \\
&= -\frac{i\hbar\gamma}{\pi} \int_0^{2\pi} \cos^2 \varphi d\varphi \\
&= -\frac{i\hbar\gamma}{\pi} \left[ \cos \varphi \sin \varphi + \frac{\varphi}{2} \right]_0^{2\pi} \\
&= -\frac{i\hbar\gamma}{\pi} (\pi) \\
&= -i\hbar\gamma
\end{aligned}$$

We now calculate  $V_{21}$

$$\begin{aligned}
V_{12} &= \int_0^{2\pi} \phi_2(\varphi) \hat{V} \phi_1(\varphi) d\varphi \\
&= \frac{1}{\pi} \int_0^{2\pi} \sin \varphi \left( -i\hbar\gamma \frac{d}{d\varphi} (\cos \varphi) \right) d\varphi \\
&= \frac{i\hbar\gamma}{\pi} \int_0^{2\pi} \sin^2 \varphi d\varphi \\
&= \frac{i\hbar\gamma}{\pi} \left[ \frac{\sin(2x) + 2x}{4} \right]_0^{2\pi} \\
&= \frac{i\hbar\gamma}{\pi} (\pi) \\
&= i\hbar\gamma
\end{aligned}$$

We now calculate  $V_{22}$

$$\begin{aligned}
V_{12} &= \int_0^{2\pi} \phi_2(\varphi) \hat{V} \phi_2(\varphi) d\varphi \\
&= \frac{1}{\pi} \int_0^{2\pi} \sin \varphi \left( -i\hbar\gamma \frac{d}{d\varphi} (\sin \varphi) \right) d\varphi \\
&= -\frac{i\hbar\gamma}{\pi} \int_0^{2\pi} \sin \varphi \cos \varphi d\varphi \\
&= 0
\end{aligned}$$

Therefore, we get the matrix

$$V = \begin{bmatrix} 0 & -i\hbar\gamma \\ i\hbar\gamma & 0 \end{bmatrix}$$

#### 31.2.3.1.4 Eigenvalues

First we obtain our full Hamiltonian matrix by adding  $H + V$

$$\mathbf{H} = H + V = \begin{bmatrix} \frac{\hbar^2}{2mR^2} & -i\hbar\gamma \\ i\hbar\gamma & \frac{\hbar^2}{2mR^2} \end{bmatrix}$$

We obtain our eigenvalues by constructing the characteristic polynomial

$$\begin{aligned}
\det(\mathbf{H} - \mathbb{1}E) &= \begin{vmatrix} \frac{\hbar^2}{2mR^2} - E & -i\hbar\gamma \\ i\hbar\gamma & \frac{\hbar^2}{2mR^2} - E \end{vmatrix} = 0 \\
&= \left( \frac{\hbar^2}{2mR^2} - E \right) \left( \frac{\hbar^2}{2mR^2} - E \right) - (-i\hbar\gamma)(i\hbar\gamma) = 0 \\
&= \left( \frac{\hbar^2}{2mR^2} - E \right)^2 - \hbar^2\gamma^2 = 0 \\
&= \left( \frac{\hbar^2}{2mR^2} - E \right)^2 = \hbar^2\gamma^2 \\
&= \frac{\hbar^2}{2mR^2} - E = \pm\hbar\gamma \\
E &= \frac{\hbar^2}{2mR^2} \mp \hbar\gamma
\end{aligned}$$

We have two eigenvalues then

$$E = \frac{\hbar^2}{2mR^2} - \hbar\gamma, \frac{\hbar^2}{2mR^2} + \hbar\gamma$$

To find the eigenfunction we input these into the matrix. We use some generic vector  $\mathbf{v}$  to assist in getting the eigenvector.

$$\begin{aligned}
\mathbf{H} - \mathbb{1}E &= \begin{bmatrix} \frac{\hbar^2}{2mR^2} - \left(\frac{\hbar^2}{2mR^2} - \hbar\gamma\right) & -i\hbar\gamma \\ i\hbar\gamma & \frac{\hbar^2}{2mR^2} - \left(\frac{\hbar^2}{2mR^2} - \hbar\gamma\right) \end{bmatrix} = 0 \\
&= \begin{bmatrix} \hbar\gamma & -i\hbar\gamma \\ i\hbar\gamma & \hbar\gamma \end{bmatrix} = 0 \\
(\mathbf{H} - \mathbb{1}E)\mathbf{v} &= \begin{bmatrix} \hbar\gamma & -i\hbar\gamma \\ i\hbar\gamma & \hbar\gamma \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0 \\
\implies \hbar\gamma v_1 - i\hbar\gamma v_2 &= 0 \\
\hbar\gamma v_1 &= i\hbar\gamma v_2 \\
v_1 &= iv_2
\end{aligned}$$

Therefore the first eigenvector is

$$\mathbf{E}_1 = \begin{bmatrix} i \\ 1 \end{bmatrix}$$

We get the other eigenvector by a similar process

$$\begin{aligned}
\mathbf{H} - \mathbb{1}E &= \begin{bmatrix} \frac{\hbar^2}{2mR^2} - \left(\frac{\hbar^2}{2mR^2} + \hbar\gamma\right) & -i\hbar\gamma \\ i\hbar\gamma & \frac{\hbar^2}{2mR^2} - \left(\frac{\hbar^2}{2mR^2} + \hbar\gamma\right) \end{bmatrix} = 0 \\
&= \begin{bmatrix} -\hbar\gamma & -i\hbar\gamma \\ i\hbar\gamma & -\hbar\gamma \end{bmatrix} = 0 \\
(\mathbf{H} - \mathbb{1}E)\mathbf{v} &= \begin{bmatrix} -\hbar\gamma & -i\hbar\gamma \\ i\hbar\gamma & -\hbar\gamma \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0 \\
\implies -\hbar\gamma v_1 - i\hbar\gamma v_2 &= 0 \\
-\hbar\gamma v_1 &= i\hbar\gamma v_2 \\
v_1 &= -iv_2
\end{aligned}$$

Therefore the second eigenvector is

$$\mathbf{E}_2 = \begin{bmatrix} -i \\ 1 \end{bmatrix}$$

Note: Eigenvectors can look slightly different, depending on how we do the rearrangement and substitution of  $v_1$  and  $v_2$ . The key is that they are the *same vectors*, except for scaling by some (possibly complex) coefficient.



For example, we could have  $\mathbf{E}_2$  be

$$\mathbf{E}_2 = \begin{bmatrix} -4i \\ 4 \end{bmatrix}$$

where we have just multiplied a 4 into it. Often we prefer for the first component to be 1, for simplicity. With that in mind we could also have

$$\mathbf{E}_2 = (-i) \begin{bmatrix} -i \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ i \end{bmatrix}$$

which is how the professor's notes wrote it. Additionally, the assignment of eigenvector 1 and 2 is arbitrary. I have labeled it based on the order I did the math in, but I very well could have done it the other way.

Similarly

$$\mathbf{E}_1 = (-i) \begin{bmatrix} i \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -i \end{bmatrix}$$

### 31.2.3.2 Eigenfunctions

We take the dot product of our eigenvectors with the original functions  $\phi$ . We neglect the original normalization factor for simplicity. We would have to renormalize our functions anyways.

$$\begin{bmatrix} 1 \\ -i \end{bmatrix} \cdot \begin{bmatrix} \cos \varphi \\ \sin \varphi \end{bmatrix} = \cos \varphi - i \sin \varphi = e^{-i\varphi}$$

We do the same for the other eigenvector

$$\begin{bmatrix} 1 \\ i \end{bmatrix} \cdot \begin{bmatrix} \cos \varphi \\ \sin \varphi \end{bmatrix} = \cos \varphi + i \sin \varphi = e^{i\varphi}$$

We then multiply each eigenfunction with its *matching* eigenvalues

$$E_1 e^{-i\varphi} = \left( \frac{\hbar^2}{2mR^2} - \hbar\gamma \right) e^{-i\varphi}$$

$$E_2 e^{i\varphi} = \left( \frac{\hbar^2}{2mR^2} + \hbar\gamma \right) e^{i\varphi}$$

### 31.3 Further Discussion

Our initial Hamiltonian was given by

$$\hat{H} = \hat{H}_0 + \hat{V} = \frac{-\hbar^2}{2mR^2} \frac{d^2}{d\varphi^2} - i\hbar\gamma \frac{d}{d\varphi}$$

We took as our sample eigenstate

$$c_1 \left( \frac{1}{\sqrt{\pi}} \cos(\varphi) \right) + c_2 \left( \frac{1}{\sqrt{\pi}} \sin(\varphi) \right)$$

and found the solutions  $e^{i\varphi}$  and  $e^{-i\varphi}$ . You can easily verify that these are the exact solutions to this problem by substituting them into the differential equations, with the exact eigenvalues. We could have written the Hamiltonian differently as

$$\hat{H} = \hat{H}_0 + \gamma \hat{L}_z$$

by using the angular momentum operator. The eigenfunctions  $\phi_k = e^{ik\varphi}$  are eigenfunctions of  $\hat{H}_0, \hat{L}_z, \hat{H}$ . The physical realization of such an operator is through a magnetic field  $B$ . We put this in the parameter  $\gamma$ . This example is in many respects quite advanced. We obtained a complex Hamiltonian matrix, but it is a Hermitian matrix satisfying  $H_{ji} = H_{ij}^*$ , which has real eigenvalues, but complex eigenvectors. This example is really quite complicated.

## 32 Atomic Term Symbols

### 32.1 Introduction

Suppose we have a carbon atom, which has 6 electrons of course.

We know it has orbitals

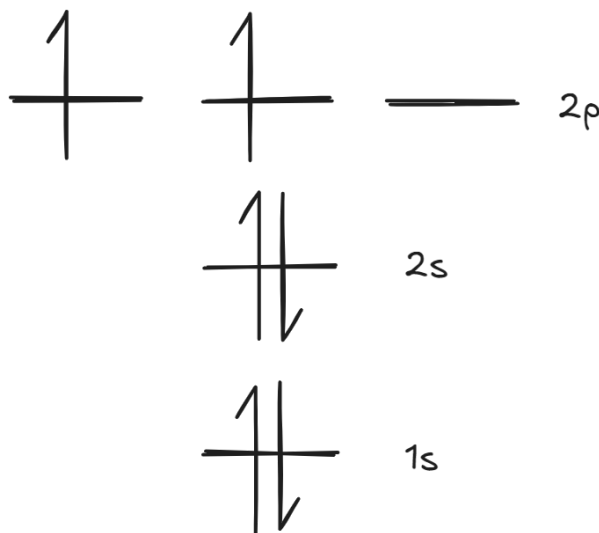


Figure 32.1

The total energy of this configuration in a simple one-body approximation is the sum of occupied orbital energies

$$E_{\text{gs}}^{\text{C}} = 2\varepsilon_{1s} + 2\varepsilon_{2s} + 2\varepsilon_{2p}$$

Given we have to fill  $1s$  and  $2s$  fully, and both are filled, we only have one possible configuration for each (shown). But for  $2p$  we have multiple configurations. We can calculate this easily. We have 6 possible spin-orbitals and 2 electrons and there are “6 choose 2”<sup>1</sup> ways to

<sup>1</sup>The general term for this calculation is “ $n$  choose  $k$ ” (produces a combination). Another similar calculation is “ $n$  pick  $k$ ” (produces a permutation). They are different calculations! Don’t choose the wrong button on your calculator if you’re not calculating manually. We use a choose calculation because our electrons are indistinguishable from each other and it does not matter which order we place them. Something like guessing a password is a permutation, where order matters.

define a particular orbital configuration.

$$\binom{6}{2} = \frac{6!}{2!(6-2)!} = 15 \text{ states}$$

A consideration here is the degeneracy. If the Hamiltonian is simply a sum of the 1-electron Hamiltonians, as in

$$\hat{H} = \hat{h}$$

then all the energy states are degenerate. If were to consider the potential energy, with its electron repulsion then the Hamiltonian would be

$$\hat{H} = \hat{h} + \hat{V}$$

then the energy states *would not be degenerate*.

## 32.2 Configuration Interaction

If were to construct a Hamiltonian matrix, as in the linear variational principle with wavefunction  $\psi = \sum_k c_k \phi_k$ , with 6 electrons we would have 18 coordinates ( $6 \times 3 = 18$ ). Assuming we were to express the 15 states in a matrix, we would have a  $15 \times 15$  matrix, with 18 dimensional integrals...

Now these integrals can be reduced to 3 dimensional (1 particle) or 6 dimensional (2 particle) integrals, it would still be quite a feat and not possible on paper. This method is called the **Configuration Interaction** method, basically applied linear variational principle. This calculation can be done using Quantum Chemistry Software.

This method would be used to determine degeneracy with or without considering the  $V_{ee}$  interaction term. This describes what needs to be done in principle.

### 32.2.1 Predicting Eigenvalue Patterns From Considerations of Angular Momentum and Spin operators

Atoms have spherical symmetry, and in addition there is spin symmetry. We can use this symmetry to predict degeneracy patterns of the energy levels and get a deeper insight.

### 32.2.2 Eigenvalues of $\hat{L}_z$ , $\hat{S}_z$

If we use orbitals that are eigenstates of  $l_z$ , with  $m_l = -l \cdots +l$  and  $s_z$  with eigenvalues  $\frac{1}{2}, -\frac{1}{2}$ , using  $\hbar = 1$  in atomic units, one can easily predict the eigenvalue of the operators  $\hat{L}_z^{\text{tot}}$  and  $\hat{S}_z^{\text{tot}}$ . We do this by summing the  $m_l$  values ( $\dots, -1, 0, 1, \dots$ ) and summing the  $m_s$  values ( $-\frac{1}{2}, \frac{1}{2}$ ). Given we have 6 electrons, it should be obvious that we sum for each of the 6 orbitals.

$$\hat{L}_z^{\text{tot}} = l_z^{(1)} + l_z^{(2)} + \cdots + l_z^{(6)}$$

Similarly, for spin

$$\hat{S}_z^{\text{tot}} = s_z^{(1)} + s_z^{(2)} + \cdots + s_z^{(6)}$$

For spin we have

$$\begin{aligned}\hat{S}_z \varphi_\alpha &= \frac{1}{2} \hbar \varphi_\alpha \\ \hat{S}_z \varphi_\beta &= -\frac{1}{2} \hbar \varphi_\beta\end{aligned}$$

### 32.2.3 Commutation relations with Molecular Hamiltonian

If we neglect the effects of **spin-orbit coupling** ( a small effect due to relativity, which we will consider again later) we have the following commutation relations due to the spherical symmetry of the atomic potential. The commutation relations with spin are general for all molecular (non-relativistic) Hamiltonians.

$$\begin{aligned}[\hat{H}, \hat{L}^2] &= 0 \\ [\hat{H}, \hat{L}_z] &= 0 \\ [\hat{H}, \hat{S}^2] &= 0 \\ [\hat{H}, \hat{S}_z] &= 0\end{aligned}$$

Also the (total) spin operators commute with the (total) angular momentum operators. This means that the Hamiltonian has a shared set of eigenfunctions with each of these operators. This means that every eigenstate can be characterized in terms of  $L, M_L, S, M_S$ . In bra-ket notation this dependence would be written as

$$|L, M_L, S, M_S\rangle$$

where  $M_L = -L, \dots, L$  and  $M_S = -S, \dots, S$ . Moreover since the operators  $\hat{L}_\pm$  and  $\hat{S}_\pm$  commute also with  $\hat{H}$ , states that differ only in  $M_L$  or  $M_S$  value have the same energy. The energy will only depend on  $L$  and  $S$ , so the different values of  $M_L$  and  $M_S$  will lead to energy degeneracy. Our overall degeneracy with state having  $L, S$  then will be

$$\text{degeneracy} = (2L + 1)(2S + 1)$$

For total angular momentum of the many-electron states, we use capital letter  $S, L, P, D, F, G, \dots$  to indicate  $L = 0, 1, 2, 3, 4$ . This is in analogy to the angular names of orbitals of  $\hat{L}_z$   $s, p, d, f, g, \dots$ . All of these naming conventions are rooted in long ago history. We use the angular momentum letter, combined with the spin degeneracy  $(2S+1)$  to write an **atomic term symbol**. This term symbol corresponds to a specific set of degenerate states that all have the same energy eigenvalue (neglecting spin-orbit coupling)

For example, we could have the term symbol  $^3P$ , which indicates  $L = 1$  (and therefore degeneracy  $2(1)+1 = 3$ ) and  $S = 1$  (with degeneracy 3), with a total degeneracy of  $3 \times 3 = 9$ . We will discuss examples of how to use this effectively below. Let us first complete the story of the naming conventions of atomic term symbols and consider spin-orbit coupling.

### 32.2.3.1 Inclusion of Spin-Orbit Coupling

If we include the effects of spin-orbit coupling, then the operators  $\hat{L}_z$  and  $\hat{S}_z$  no longer commute with  $\hat{H}$ . In this case, we can not label the eigenstates with the quantum numbers  $M_L, M_S$ . Instead we can define **total angular momentum operators**  $\hat{J} = \hat{L} + \hat{S}$  and the corresponding operators called  $\hat{J}^2$  and  $\hat{J}_z$ . The operators  $\hat{L}^2, \hat{S}^2, \hat{J}^2, \hat{J}_z$  all commute and can be used to characterize degenerate multiplets. The fundamental quantum number is  $J$ , while the values for  $M_J = -J, \dots, J$ , creating  $(2J+1)$  states in each multiplet. These eigenstates of  $\hat{J}^2$ ,

*hat* $J_z$  are constructed from linear combinations of states in the  $^{2S+1}L$  multiplet. We label this dependency in bra-ket notation as.

$$|n, L, S, J, M_J\rangle$$

The types of resulting multiplets given values of  $L, S$  can simply be obtained as

$$J = |L + S|, \dots, |L - S|$$

and

$$M_J = -J, \dots, J$$

with degeneracy

$$\text{degeneracy} = 2J + 1$$

These results can be derived from angular momentum coupling. In a given J-multiplet the energies are degenerate, but the degeneracy is lifted in the presence of a magnetic field, which splits the levels. This is known as the Zeeman effect.

We indicate this possible degeneracy with a subscript on the complete atomic term symbol

$$^{2S+1}L_J$$

For a given value of  $L, S$  the various J-levels are ordered according to

1. If the shell is less than half-filled, the smallest value of  $J$  is more stable
2. If the shell is half-filled or more, the highest value of  $J$  is more stable
3. If the shell is half filled, for  $L=0$ , there is only one J level

If we were to take the  $^3P$  level ( $L = 1, S = 1$ ) in the carbon atom (with 2 electrons in 2p orbitals) we would have  $J$  values of

$$J = |1 + 1|, \dots, |1 - 1| \\ \implies J = 2, 1, 0$$

with degeneracy ( $2J + 1$ ) of 5, 3, and 1 respectively. We would have term symbols

$$^3P_2, ^3P_1, ^3P_0$$

With 2 electrons in the shell, the lowest value of J ( $J=0$ ) is the lowest energy term.

In spectroscopy, the splitting of energy levels induced by  $J$  would be small, about the order of wavenumbers for light atoms. Levels that differ in  $S$  or  $L$  are about 1 eV apart, which is about  $8000 \text{ cm}^{-1}$ . While the splitting of levels with different J is very small, it is very important in technology. It is the basis for storage devices and a lot of electronics.

The full explanation of atomic energy levels is complicated. Even today it is hard to calculate atomic energy levels accurately because there can be so many close lying states, split by small energy differences. We can understand the patterns from the angular and Spin symmetry analysis, but this does not provide the quantitative energy splittings. In

the following we will discuss a simple recipe to get at important parts of the multiplet information. A question like this is always on the exam, and you can prepare yourself well for it, working through the examples below.

## 32.3 Recipe

You are given a particular electronic configuration, e.g.  $[Ar]4s13d7$  for the Fe atom (complicated example), and you can obtain the following information

1. Determine the total number of microstates in this configuration (using factorials), by considering the open shell and number of electrons available.
2. Determine the  $M_S = S, M_L = L$  values for the state of maximum  $M_S$  and of these maximum  $M_L$ . This is always a single well-defined configuration. There are in total  $(2S+1)(2L+1)$  states associated with these values of  $S, L$ . You can find the corresponding  $J$ -values and list all term values associated with this set of states. One of these term symbols will be the ground state of this electronic configuration.
3. Determine the  $M_S = S, M_L = L$  values for the state of maximum  $M_L$  and of these maximum  $M_S$ . This is always a single well-defined configuration. There are in total  $(2S+1)(2L+1)$  states associated with these values of  $S, L$ . You can find the corresponding  $J$ -values and list all term values associated with this set of states and their degeneracies  $(2J+1)$ .
4. Determine the remaining number of states in the electronic configuration. These states are typically linear combinations of determinants and we will not consider this further here.
5. Sketch the found energy levels, paying attention to relative gaps between  $J$  (order of wavenumbers) versus  $L$  and  $S$  (about  $1 \text{ eV} \approx 8000 \text{ cm}^{-1}$ ).

## 32.4 Fully Worked Example: Carbon

### 32.4.1 Orbital Filling/Open Shell

We fill  $1s$  and  $2s$  fully, and fill  $2p$  according to the recipe above.



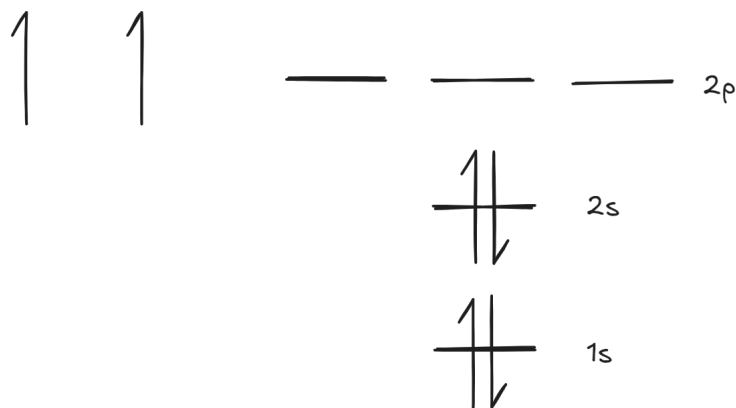


Figure 32.2

### 32.4.2 Number of Microstates

With only two electrons available for the  $2p$  level and 6 possible orbitals, we calculate the number of microstates to be

$$\binom{6}{2} = \frac{6!}{2!(6-2)!} = 15$$

### 32.4.3 Maximizing $M_S$

We fill the  $2p$  levels as to maximize  $M_S$  (put electrons with spin-up as much as possible). Of these we also want to maximize  $M_L$  (fill levels from left to right, labeling as  $m_l = 1, 0, -1$

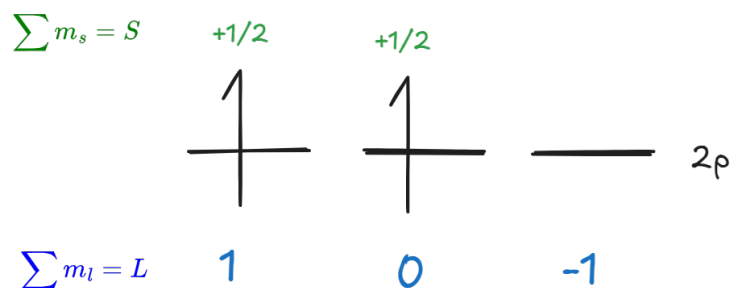


Figure 32.3

This configuration has the maximum spin, calculated as

$$M_S = \sum m_s = \frac{1}{2} + \frac{1}{2} = 1, S = M_S = 1$$

We calculate the angular momentum for this configuration

$$M_L = \sum m_l = 1 + 0 = 1, L = M_L = 1$$

Our degeneracy is then

$$\text{degeneracy} = (2L + 1)(2S + 1) = (2(1) + 1)(2(1) + 1) = (3)(3) = 9$$

Therefore we would have spin degeneracy  $2(1) + 1 = 3$  and orbital name  $P$ . This would result in atomic term symbol of

$$^3P$$

We would calculate the possible  $J$  values

$$\begin{aligned} J &= |L + S|, \dots, |L - S| \\ &= |1 + 1|, \dots, |1 - 1| \\ &= 2, 1, 0 \end{aligned}$$

We would then have multiplets

$$^3P_2, ^3P_1, ^3P_0$$

We can calculate the degeneracy of  $J$  for each value of  $J$ , assuming no splitting

$$\begin{aligned} \text{degeneracy} &= 2J + 1 \\ &= 2(2) + 1 &= 5 \\ &= 2(1) + 1 &= 3 \\ &= 2(0) + 1 &= 1 \\ & &= 9 \end{aligned}$$

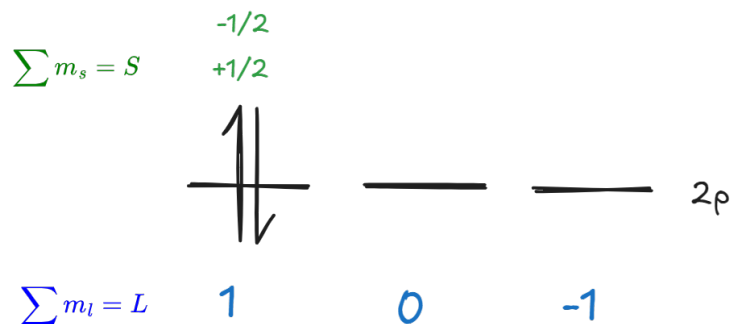
We see that adding the degeneracy of  $J$  results in the same value as multiplying the degeneracy of  $S$  and  $L$ . This is *always true* and is a useful check that you've done things correctly. Here  $5 + 3 + 1 = 9$ . Since we have less than half-filled case the energy ordering is  $^3P_0 < ^3P_1 < ^3P_2$

Of the 15 states, we have accounted for 9 of them.

### 32.4.4 Maximizing $M_L$

In the next configuration we try to maximize  $M_L$ , filling in orbitals from the left side onwards.

We first put in  $\alpha$ -spin the  $\beta$ -spin electrons in each  $m_l$  level



**Figure 32.4**

We calculate  $S$  through  $M_S$ :

$$M_S = \sum m_s = \frac{1}{2} - \frac{1}{2} = 0, S = M_S = 0$$

We calculate  $L$

$$M_L = \sum m_l = 1 + 1 = 2, L = M_L = 2$$

We have degeneracy

$$\text{degeneracy} = (2L + 1)(2S + 1) = (2(2) + 1)(2(0) + 1) = (5)(1) = 5$$

We would have atomic term symbol

$$^1D$$

We calculate  $J$

$$\begin{aligned} J &= |L + S|, \dots, |L - S| \\ &= |2 + 0|, \dots, |2 - 0| \\ &= 2 \end{aligned}$$

We would have a multiplet

$$^1D_2$$

with  $J$  degeneracy of

$$\text{degeneracy} = 2(2) + 1 = 5$$

which matches up with our previous degeneracy calculation.

This accounts for 5 energy levels.

### 32.4.5 Accounting Energy Levels

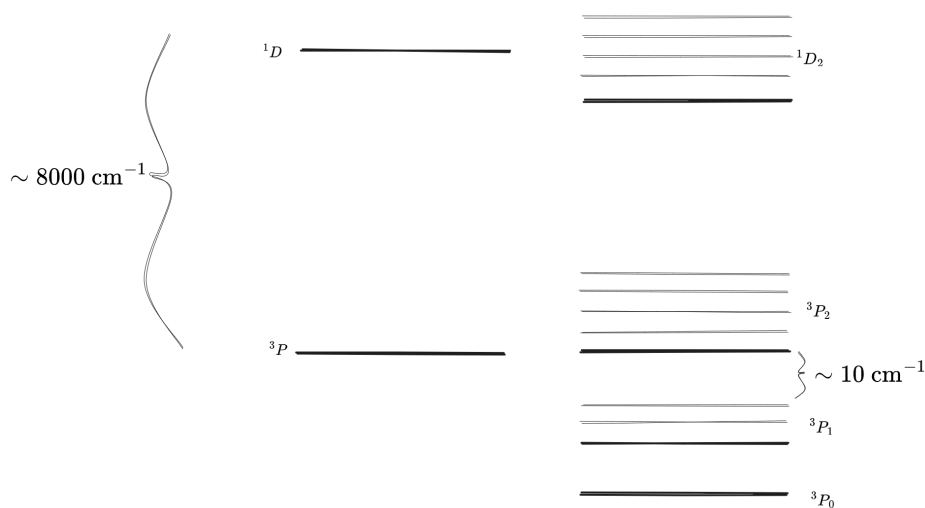
We have  $9 + 5 = 14$  energy levels accounted for. We are missing 1. This missing energy level has only one state and it has to be  $^1S_0$ . This is a complicated linear combination of states. Here we can know the term symbol by guessing.

### 32.4.6 Listing first two term symbols

We have term values with degeneracy indicated in parentheses

$$^3P_2(5), ^3P_1(3), ^3P_0(1), ^1D_2(5), ^1S_0(1)$$

### 32.4.7 Sketching



**Figure 32.5:** The  $^1S$  level should not be drawn. It is higher than the other 2 levels.  $^3P_0$  is the ground state. Separation between individual levels is about  $10 \text{ cm}^{-1}$ , and separation between the  $P$  and  $D$  states is about  $1000 \text{ cm}^{-1}$

## 33 Additional Atomic Term Examples

### 33.1 Nitrogen

#### 33.1.1 Filling Orbitals/Open Shell

We fill the lower orbitals  $1s$  and  $2s$  fully.

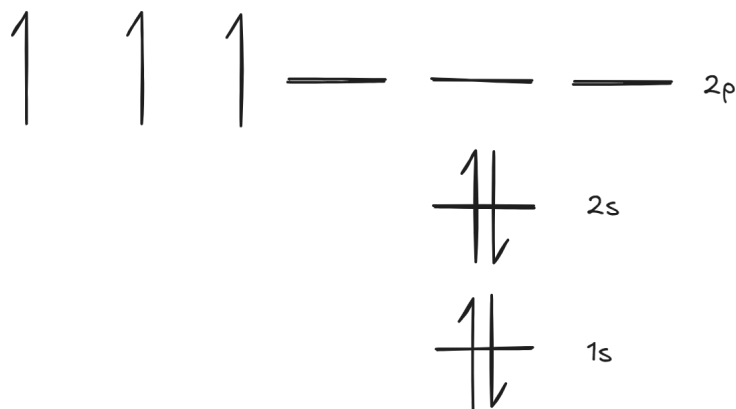


Figure 33.1

We see that  $2p$  is our open shell.

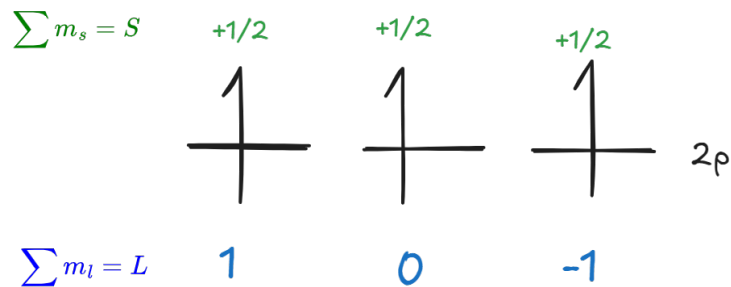
#### 33.1.2 Number of Microstates

The number of microstates is “6 choose 3”

$$\binom{6}{3} = \frac{6!}{3!(6-3)!} = 20$$

#### 33.1.3 Maximizing $M_S$

We fill to maximize  $M_S$



**Figure 33.2**

We obtain maximum  $M_S$

$$M_S = \sum m_s = \frac{1}{2} + \frac{1}{2} + \frac{1}{2} = \frac{3}{2}, S = M_S = \frac{3}{2}$$

We obtain maximum  $M_L$

$$M_L = \sum m_l = 1 + 0 - 1 = 0, L = M_L = 0$$

We calculate degeneracy

$$\text{degeneracy} = (2(0) + 1) \left( 2 \left( \frac{3}{2} \right) + 1 \right) = (1)(4) = 4$$

We calculate possible  $J$  values

$$\begin{aligned} J &= |L + S|, \dots, |L - S| \\ &= \left| \frac{3}{2} + 0 \right|, \dots, \left| \frac{3}{2} - 0 \right| \\ &= \frac{3}{2} \end{aligned}$$

We calculate and check that  $J$  degeneracy is equal to the previous degeneracy

$$\text{degeneracy} = 2 \left( \frac{3}{2} \right) + 1 = 4$$

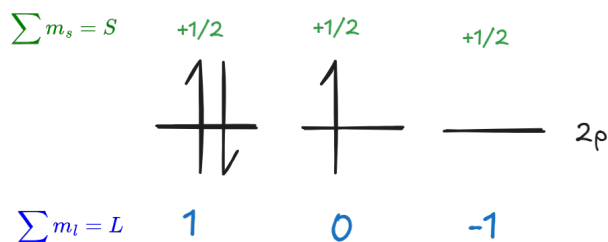
We assign the term symbol

$$^4S_{3/2}$$

We have accounted for 4 of the 20 possible states.

### 33.1.4 Maximizing $M_L$

We fill to maximize  $M_L$



**Figure 33.3**

We calculate  $S$

$$M_S = \sum m_s = \frac{1}{2} - \frac{1}{2} + \frac{1}{2} = \frac{1}{2}, S = M_S = \frac{1}{2}$$

We calculate  $L$

$$M_L = \sum m_l = 1 + 1 + 0 = 2, M_L = L = 2$$

We calculate the degeneracy

$$\text{degeneracy} = (2(2) + 1) \left( 2 \left( \frac{1}{2} \right) + 1 \right) = (5)(2) = 10$$

We calculate possible values of  $J$

$$\begin{aligned}
 J &= |L + S|, \dots, |L - S| \\
 &= \left| 2 + \frac{1}{2} \right|, \dots, \left| 2 - \frac{1}{2} \right| \\
 &= \frac{5}{2}, \frac{3}{2}
 \end{aligned}$$

We calculate degeneracies for each value of  $J$  and check total degeneracy

$$\begin{aligned}
 \text{degeneracy} &= 2J + 1 \\
 2 \left( \frac{5}{2} \right) + 1 &= 6 \\
 2 \left( \frac{3}{2} \right) + 1 &= 4 \\
 \text{total} &= 10
 \end{aligned}$$

We assign term symbols.

$$^2D_{5/2}, ^2D_{3/2}$$

We have accounted for 10 out of the 20 possible states with this attempt.

### 33.1.5 Accounting

We have accounted for  $4 + 10 = 14$  possible states. That means we are missing 6 states. This suffices. You might guess that the remaining term symbol is  $^2P$  (6 states), with term symbols  $^2P_{3/2}, ^2P_{1/2}$

### 33.1.6 Finding Missing State: Thought Exercise

We can work backwards. We are missing 6 states, so our total degeneracy given our choice of placing electrons should be 6.

There is no secret method for this, and it is very possible to end up not being able to figure it out. This is why we don't do this, but just to show how we arrived at the earlier term symbols as a thought exercise. We sketch out some electron configurations and quickly calculate their degeneracies until we end up with something reasonable.

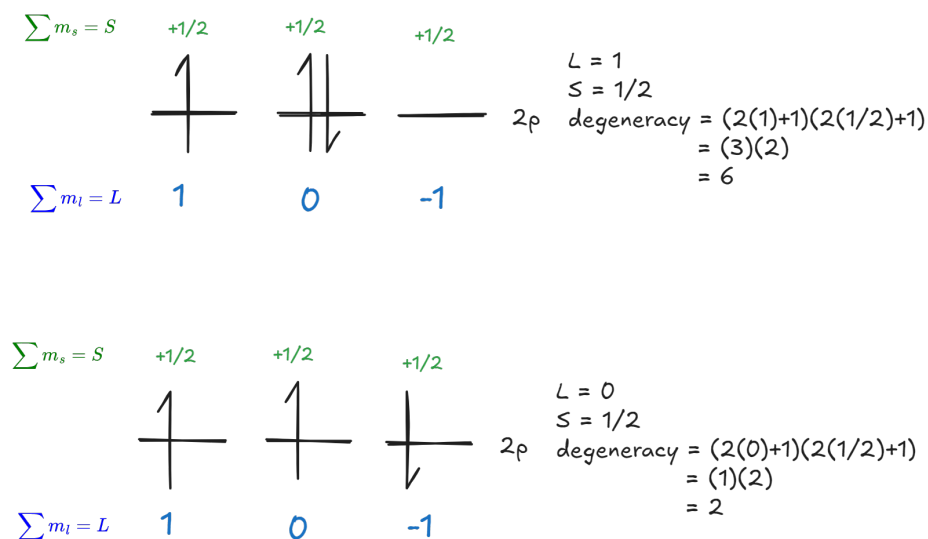


Figure 33.4

As can be seen from the calculations in the image, the top configuration is reasonable with a degeneracy of 6.



We calculate the possible  $J$  values

$$\begin{aligned} J &= |L + S|, \dots, |L - S| \\ &= \left|1 + \frac{1}{2}\right|, \dots, \left|1 - \frac{1}{2}\right| \\ &= \frac{3}{2}, \frac{1}{2} \end{aligned}$$

We calculate the  $J$  degeneracies and check

$$\begin{aligned} \text{degeneracy} &= 2J + 1 \\ &= 2\left(\frac{3}{2}\right) + 1 = 4 \\ &= 2\left(\frac{1}{2}\right) + 1 = 2 \\ &= 6 \end{aligned}$$

We assign the atomic term symbols

$${}^2P_{1/2}, {}^2P_{3/2}$$

### 33.1.7 Listing Out

We have term symbols and their degeneracies

$${}^4S_{3/2}(4), {}^2D_{5/2}(6), {}^2D_{3/2}(4), {}^2P_{3/2}(4), {}^2P_{1/2}(2)$$

for a total of 20 states.

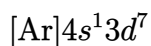
## 33.2 Iron

### 33.2.1 Filling Orbitals/Open Shell

Using a periodic table, we know the electron configuration of iron to be

$$[\text{Ar}]4s^23d^6$$

But we run into a common scenario with transition metals! The  $3d$  and  $4s$  energy levels are similar in energy and we will consider



and have two open shells, increasing the complexity.

### 33.2.2 Number of Microstates

The number of microstates, accounting for both the  $4s$  and  $3d$  orbitals is

$$\binom{2}{1} \binom{10}{7} = \frac{2!}{1!(2-1)!} \frac{10!}{7!(10-7)!} = 2 \cdot 120 = 240$$

### 33.2.3 Maximizing $M_S$

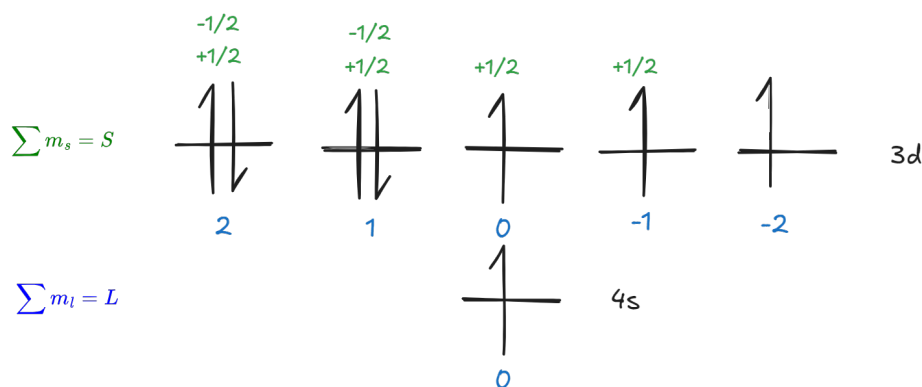


Figure 33.5

We calculate  $S$

$$M_S = \sum m_s = 6 \left( \frac{1}{2} \right) + 2 \left( -\frac{1}{2} \right) = 3 - 1 = 2, S = M_S = 2$$

We calculate corresponding highest  $M_L$

$$M_L = \sum m_l = 2(2) + 2(1) + 2(0) - 1 - 2 = 4 + 2 - 1 - 2 = 3, L = M_L = 3 \rightarrow F$$

We calculate the degeneracy

$$\text{degeneracy} = (2(3) + 1)(2(2) + 1) = (7)(5) = 35$$

We calculate possible values of  $J$

$$\begin{aligned} J &= |L + S|, \dots, |L - S| \\ &= |3 + 2|, \dots, |3 - 2| \\ &= 5, 4, 3, 2, 1 \end{aligned}$$

We can calculate and check the degeneracies of the  $J$  values

$$\begin{aligned} \text{degeneracy} &= 2J + 1 \\ 2(5) + 1 &= 11 \\ 2(4) + 1 &= 9 \\ 2(3) + 1 &= 7 \\ 2(2) + 1 &= 5 \\ 2(1) + 1 &= 3 \\ \text{total} &= 35 \end{aligned}$$

We assign term symbols

$${}^5F_5, {}^5F_4, {}^5F_3, {}^5F_2, {}^5F_1$$

### 33.2.4 Maximizing $M_L$

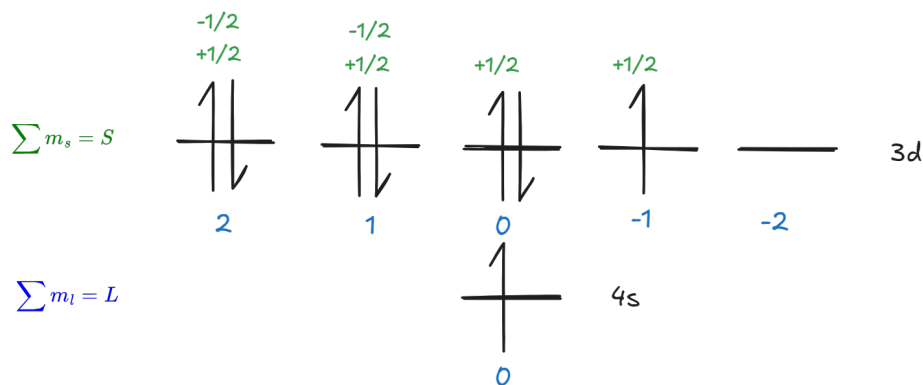


Figure 33.6

We calculate  $M_L$

$$M_L = \sum m_l = 2(2) + 2(1) + 3(0) - 1 = 4 + 2 - 1 = 5, L = M_L = 5(S, P, D, F, G, H) \rightarrow H$$

We calculate  $S$

$$M_S = \sum m_s = 5\left(\frac{1}{2}\right) + 3\left(-\frac{1}{2}\right) = 1, S = M_S = 1$$

We calculate the degeneracy

$$\text{degeneracy} = (2(5) + 1)(2(1) + 1) = (11)(3) = 33$$

We calculate possible  $J$  values

$$\begin{aligned} J &= |L + S|, \dots, |L - S| \\ &= |5 + 1|, \dots, |5 - 1| \\ &= 6, 5, 4 \end{aligned}$$

We calculate the  $J$  degeneracies

$$\begin{aligned} \text{degeneracy} &= 2J + 1 \\ &= 2(6) + 1 = 13 \\ &= 2(5) + 1 = 11 \\ &= 2(4) + 1 = 9 \\ &= 33 \end{aligned}$$

We assign term symbols

$${}^3H_6, {}^3H_5, {}^3H_4$$

We have accounted for 33 of the possible 240 microstates.

### 33.2.5 Accounting

We have  $240 - 35 - 33 = 172$  states accounted for. This is too large to do by trial and error. So we are left with the microstates we have calculated.

### 33.2.6 Listing Out

We list out all term symbols and their degeneracies. The shell is more than half-filled and so the highest J-levels are lowest in energy.

$${}^5F_5(11), {}^5F_4(9), {}^5F_3(7), {}^5F_2(5), {}^5F_1(3), {}^3H_6(13), {}^3H_5(11), {}^3H_4(9)$$

### 33.2.7 Sketching

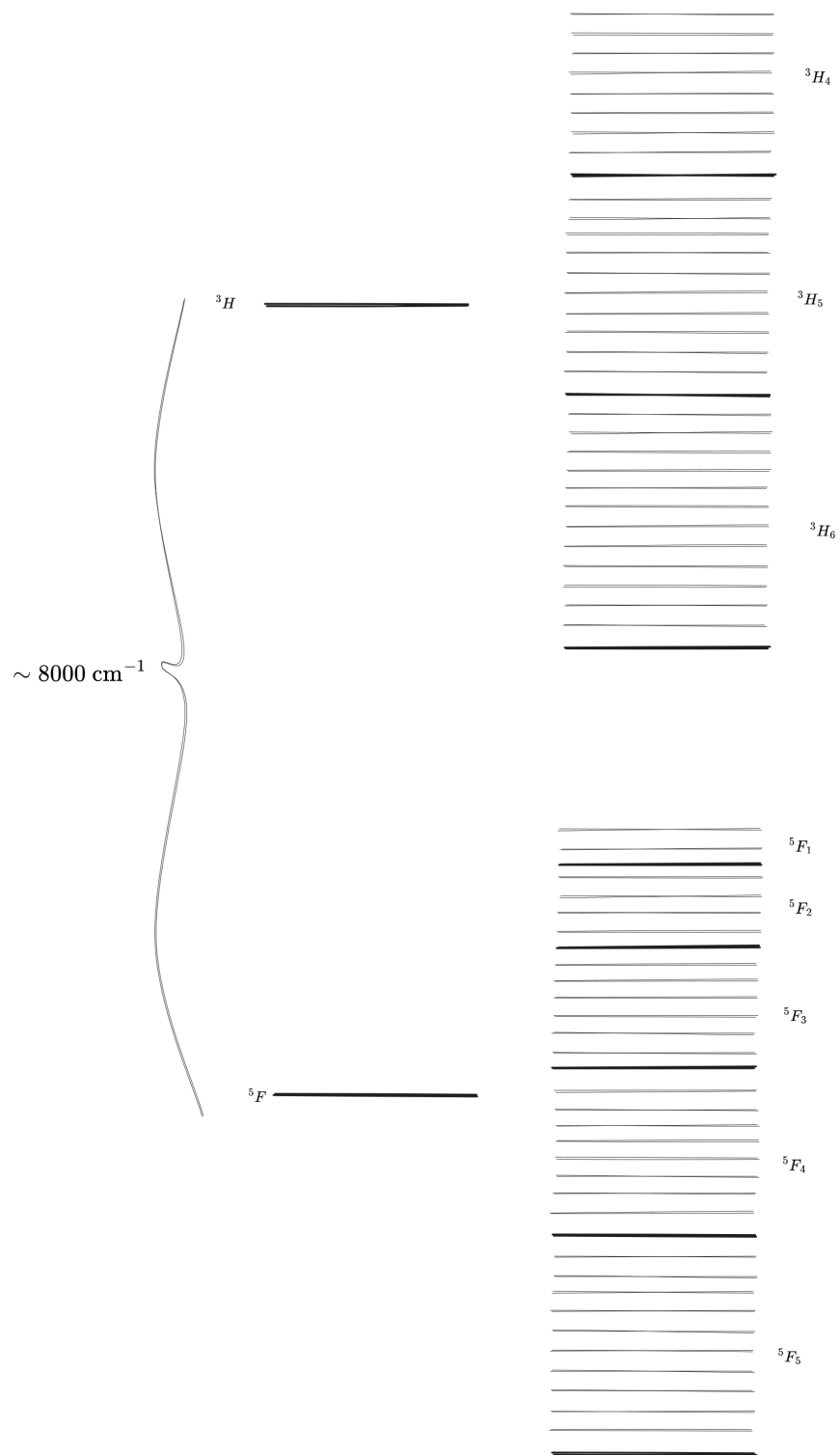


Figure 33.7